

Sample Problem 11.1

The position of a particle moving along a straight line is defined by the relation $x = t^3 - 6t^2 - 15t + 40$, where x is expressed in feet and t in seconds. Determine (a) the time at which the velocity is zero, (b) the position and distance traveled by the particle at that time, (c) the acceleration of the particle at that time, (d) the distance traveled by the particle from $t = 4$ s to $t = 6$ s.

STRATEGY: You need to use the basic kinematic relationships between position, velocity, and acceleration. Because the position is given as a function of time, you can differentiate it to find equations for the velocity and acceleration. Once you have these equations, you can solve the problem.

MODELING and ANALYSIS: Taking the derivative of position, you obtain

$$x = t^3 - 6t^2 - 15t + 40 \quad (1)$$

$$v = \frac{dx}{dt} = 3t^2 - 12t - 15 \quad (2)$$

$$a = \frac{dv}{dt} = 6t - 12 \quad (3)$$

These equations are graphed in Fig. 1.

a. Time at Which $v = 0$. Set $v = 0$ in Eq. (2) for

$$3t^2 - 12t - 15 = 0 \quad t = -1 \text{ s} \quad \text{and} \quad t = +5 \text{ s} \quad \blacktriangleleft$$

Only the root $t = +5$ s corresponds to a time after the motion has begun: for $t < 5$ s, $v < 0$ and the particle moves in the negative direction; for $t > 5$ s, $v > 0$ and the particle moves in the positive direction.

b. Position and Distance Traveled When $v = 0$. Substitute $t = +5$ s into Eq. (1), yielding

$$x_5 = (5)^3 - 6(5)^2 - 15(5) + 40 \quad x_5 = -60 \text{ ft} \quad \blacktriangleleft$$

The initial position at $t = 0$ was $x_0 = +40$ ft. Since $v \neq 0$ during the interval $t = 0$ to $t = 5$ s, you have

$$\text{Distance traveled} = x_5 - x_0 = -60 \text{ ft} - 40 \text{ ft} = -100 \text{ ft}$$

$$\text{Distance traveled} = 100 \text{ ft in the negative direction} \quad \blacktriangleleft$$

c. Acceleration When $v = 0$. Substitute $t = +5$ s into Eq. (3) for

$$a_5 = 6(5) - 12 \quad a_5 = +18 \text{ ft/s}^2 \quad \blacktriangleleft$$

d. Distance Traveled from $t = 4$ s to $t = 6$ s. The particle moves in the negative direction from $t = 4$ s to $t = 5$ s and in the positive direction from $t = 5$ s to $t = 6$ s; therefore, the distance traveled during each of these time intervals must be computed separately.

From $t = 4$ s to $t = 5$ s: $x_5 = -60$ ft

$$x_4 = (4)^3 - 6(4)^2 - 15(4) + 40 = -52 \text{ ft}$$

(continued)

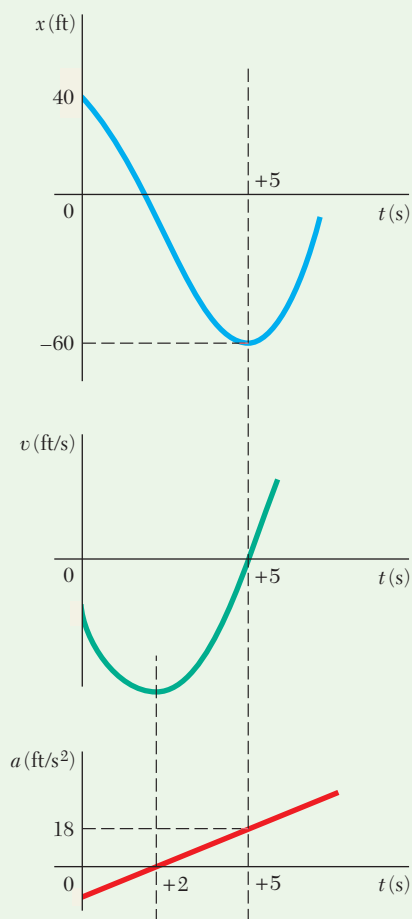


Fig. 1 Motion curves for the particle.

$$\begin{aligned}\text{Distance traveled} &= x_5 - x_4 = -60 \text{ ft} - (-52 \text{ ft}) = -8 \text{ ft} \\ &= 8 \text{ ft in the negative direction}\end{aligned}$$

$$\text{From } t = 5 \text{ s to } t = 6 \text{ s: } x_5 = -60 \text{ ft}$$

$$x_6 = (6)^3 - 6(6)^2 - 15(6) + 40 = -50 \text{ ft}$$

$$\begin{aligned}\text{Distance traveled} &= x_6 - x_5 = -50 \text{ ft} - (-60 \text{ ft}) = +10 \text{ ft} \\ &= 10 \text{ ft in the positive direction}\end{aligned}$$

Total distance traveled from $t = 4 \text{ s}$ to $t = 6 \text{ s}$ is $8 \text{ ft} + 10 \text{ ft} = 18 \text{ ft}$

REFLECT and THINK: The total distance traveled by the particle in the 2-second interval is 18 ft, but because one distance is positive and one is negative, the net change in position is only 2 ft (in the positive direction). This illustrates the difference between total distance traveled and net change in position. Note that the maximum displacement occurs at $t = 5 \text{ s}$, when the velocity is zero.

Sample Problem 11.2

You throw a ball vertically upward with a velocity of 10 m/s from a window located 20 m above the ground. Knowing that the acceleration of the ball is constant and equal to 9.81 m/s^2 downward, determine (a) the velocity v and elevation y of the ball above the ground at any time t , (b) the highest elevation reached by the ball and the corresponding value of t , (c) the time when the ball hits the ground and the corresponding velocity. Draw the v - t and y - t curves.

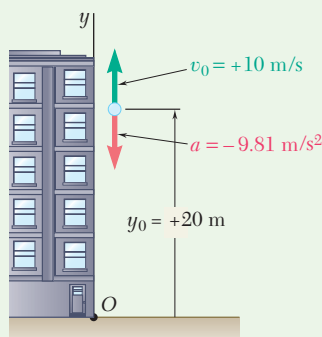


Fig. 1 Acceleration, initial velocity, and initial position of the ball.

STRATEGY: The acceleration is constant, so you can integrate the defining kinematic equation for acceleration once to find the velocity equation and a second time to find the position relationship. Once you have these equations, you can solve the problem.

MODELING and ANALYSIS: Model the ball as a particle with negligible drag.

a. Velocity and Elevation. Choose the y axis measuring the position coordinate (or elevation) with its origin O on the ground and its positive sense upward. The value of the acceleration and the initial values of v and y are as indicated in Fig. 1. Substituting for a in $a = dv/dt$ and noting that, when $t = 0$, $v_0 = +10 \text{ m/s}$, you have

$$\begin{aligned}\frac{dv}{dt} &= a = -9.81 \text{ m/s}^2 \\ \int_{v_0=10}^v dv &= - \int_0^t 9.81 \, dt \\ [v]_{10}^v &= -[9.81t]_0^t \\ v - 10 &= -9.81t\end{aligned}$$

$$v = 10 - 9.81t \quad (1) \quad \blacktriangleleft$$

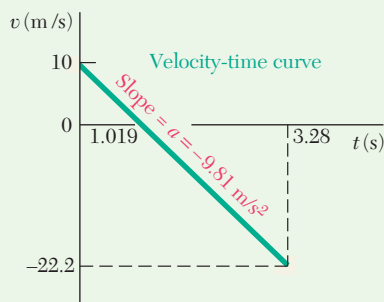


Fig. 2 Velocity of the ball as a function of time.

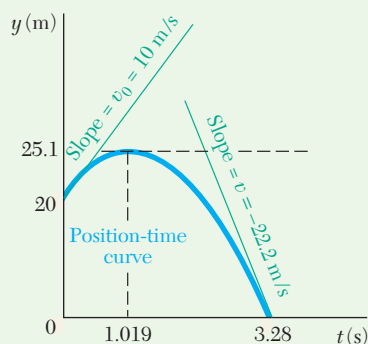


Fig. 3 Height of the ball as a function of time.

Substituting for v in $v = dy/dt$ and noting that when $t = 0$, $y_0 = 20$ m, you have

$$\begin{aligned}\frac{dy}{dt} &= v = 10 - 9.81t \\ \int_{y_0=20}^y dy &= \int_0^t (10 - 9.81t) dt \\ [y]_{20}^y &= [10t - 4.905t^2]_0^t \\ y - 20 &= 10t - 4.905t^2 \\ y &= 20 + 10t - 4.905t^2 \quad (2) \quad \blacktriangleleft\end{aligned}$$

Graphs of these equations are shown in Figs. 2 and 3.

b. Highest Elevation. The ball reaches its highest elevation when $v = 0$. Substituting into Eq. (1), you obtain

$$10 - 9.81t = 0 \quad t = 1.019 \text{ s} \quad \blacktriangleleft$$

Substituting $t = 1.019$ s into Eq. (2), you find

$$y = 20 + 10(1.019) - 4.905(1.019)^2 \quad y = 25.1 \text{ m} \quad \blacktriangleleft$$

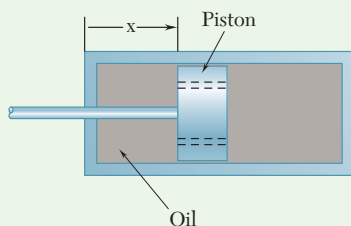
c. Ball Hits the Ground. The ball hits the ground when $y = 0$. Substituting into Eq. (2), you obtain

$$20 + 10t - 4.905t^2 = 0 \quad t = -1.243 \text{ s} \quad \text{and} \quad t = +3.28 \text{ s} \quad \blacktriangleleft$$

Only the root $t = +3.28$ s corresponds to a time after the motion has begun. Carrying this value of t into Eq. (1), you find

$$v = 10 - 9.81(3.28) = -22.2 \text{ m/s} \quad v = 22.2 \text{ m/s} \downarrow \quad \blacktriangleleft$$

REFLECT and THINK: When the acceleration is constant, the velocity changes linearly, and the position is a quadratic function of time. You will see in Sec. 11.2 that the motion in this problem is an example of free fall, where the acceleration in the vertical direction is constant and equal to $-g$.



Sample Problem 11.3

Many mountain bike shocks utilize a piston that travels in an oil-filled cylinder to provide shock absorption; this system is shown schematically. When the front tire goes over a bump, the cylinder is given an initial velocity v_0 . The piston, which is attached to the fork, then moves with respect to the cylinder, and oil is forced through orifices in the piston. This causes the piston to decelerate at a rate proportional to the velocity at $a = -kv$. At time $t = 0$, the position of the piston is $x = 0$. Express (a) the velocity v in terms of t , (b) the position x in terms of t , (c) the velocity v in terms of x . Draw the corresponding motion curves.

(continued)

STRATEGY: Because the acceleration is given as a function of velocity, you need to use either $a = dv/dt$ or $a = v dv/dx$ and then separate variables and integrate. Which one you use depends on what you are asked to find. Since part *a* asks for v in terms of t , use $a = dv/dt$. You can integrate this again using $v = dx/dt$ for part *b*. Since part *c* asked for $v(x)$, you should use $a = v dv/dx$ and then separate the variables and integrate.

MODELING and ANALYSIS: Rotation of the piston is not relevant, so you can model it as a particle undergoing rectilinear motion.

a. v in Terms of t . Substitute $-kv$ for a in the fundamental formula defining acceleration, $a = dv/dt$. You obtain

$$\begin{aligned} -kv &= \frac{dv}{dt} & \frac{dv}{v} &= -k dt & \int_{v_0}^v \frac{dv}{v} &= -k \int_0^t dt \\ \ln \frac{v}{v_0} &= -kt & v &= v_0 e^{-kt} \end{aligned}$$

b. x in Terms of t . Substitute the expression just obtained for v into $v = dx/dt$. You get

$$\begin{aligned} v_0 e^{-kt} &= \frac{dx}{dt} \\ \int_0^x dx &= v_0 \int_0^t e^{-kt} dt \\ x &= -\frac{v_0}{k} [e^{-kt}]_0^t = -\frac{v_0}{k} (e^{-kt} - 1) \\ x &= \frac{v_0}{k} (1 - e^{-kt}) \end{aligned}$$

c. v in Terms of x . Substitute $-kv$ for a in $a = v dv/dx$. You have

$$\begin{aligned} -kv &= v \frac{dv}{dx} \\ dv &= -k dx \\ \int_{v_0}^v dv &= -k \int_0^x dx \\ v - v_0 &= -kx \\ v &= v_0 - kx \end{aligned}$$

The motion curves are shown in Fig. 1.

REFLECT and THINK: You could have solved part *c* by eliminating t from the answers obtained for parts *a* and *b*. You could use this alternative method as a check. From part *a*, you obtain $e^{-kt} = v/v_0$; substituting into the answer of part *b*, you have

$$x = \frac{v_0}{k} (1 - e^{-kt}) = \frac{v_0}{k} \left(1 - \frac{v}{v_0} \right) \quad v = v_0 - kx \quad (\text{checks})$$

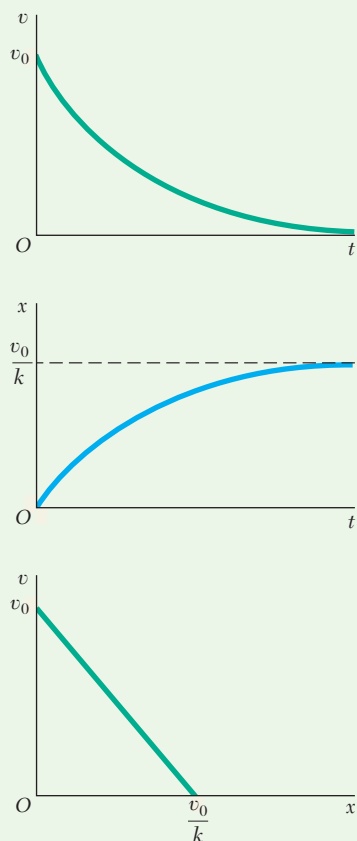
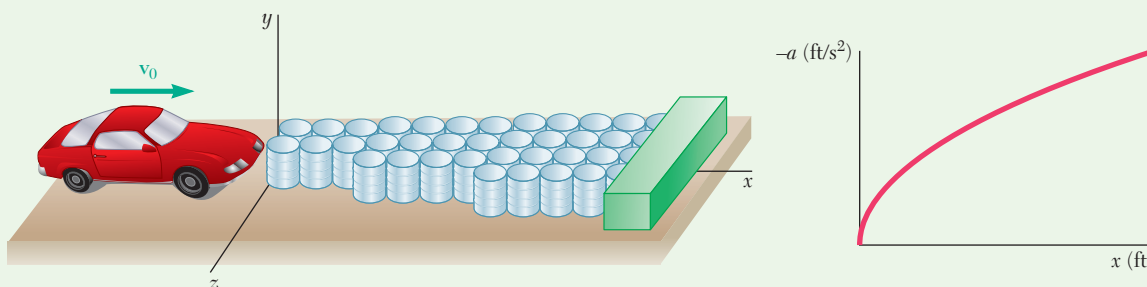


Fig. 1 Motion curves for the piston

Sample Problem 11.4

An uncontrolled automobile traveling at 45 mph strikes a highway crash barrier square on. After initially hitting the barrier, the automobile decelerates at a rate proportional to the distance x the automobile has moved into the barrier; specifically, $a = -60\sqrt{x}$, where a and x are expressed in ft/s^2 and ft , respectively. Determine the distance the automobile will move into the barrier before it comes to rest.



STRATEGY: Since you are given the deceleration as a function of displacement, you should start with the basic kinematic relationship $a = v \, dv/dx$.

MODELING and ANALYSIS: Model the car as a particle. First find the initial speed in ft/s ,

$$v_0 = \left(45 \frac{\text{mi}}{\text{hr}}\right) \left(\frac{1 \text{ hr}}{3600 \text{ s}}\right) \left(\frac{5280 \text{ ft}}{\text{mi}}\right) = 66 \frac{\text{ft}}{\text{s}}$$

Substituting $a = -60\sqrt{x}$ into $a = v \, dv/dx$ gives

$$a = -60\sqrt{x} = \frac{v \, dv}{dx}$$

Separating variables and integrating gives

$$\begin{aligned} v \, dv &= -60\sqrt{x} \, dx \rightarrow \int_{v_0}^0 v \, dv = - \int_0^x 60\sqrt{x} \, dx \\ \frac{1}{2}v^2 - \frac{1}{2}v_0^2 &= -40x^{3/2} \rightarrow x = \left(\frac{1}{80}(v_0^2 - v^2)\right)^{2/3} \end{aligned} \quad (1)$$

Substituting $v = 0$, $v_0 = 45 \text{ ft/s}$ gives

$$d = 14.37 \text{ ft} \quad \blacktriangleleft$$

REFLECT and THINK: A distance of 14 ft seems reasonable for a barrier of this type. If you substitute d into the equation for a , you find a maximum deceleration of about 7 g 's. Note that this problem would have been much harder to solve if you had been asked to find the time for the automobile to stop. In this case, you would need to determine $v(t)$ from Eq. (1). This gives $v = \sqrt{v_0^2 - 80x^{3/2}}$. Using the basic kinematic relationship $v = dx/dt$, you can easily show that

$$\int_0^t dt = \int_0^x \frac{dx}{\sqrt{v_0^2 - 80x^{3/2}}}$$

Unfortunately, there is no closed-form solution to this integral, so you would need to solve it numerically.

SOLVING PROBLEMS ON YOUR OWN

In the problems for this section, you will be asked to determine the **position**, **velocity**, and/or **acceleration** of a particle in **rectilinear motion**. As you read each problem, it is important to identify both the independent variable (typically t or x) and what is required (for example, the need to express v as a function of x). You may find it helpful to start each problem by writing down both the given information and a simple statement of what is to be determined.

1. Determining $v(t)$ and $a(t)$ for a given $x(t)$. As explained in Sec. 11.1A, the first and second derivatives of x with respect to t are equal to the velocity and the acceleration, respectively, of the particle [Eqs. (11.1) and (11.2)]. If the velocity and acceleration have opposite signs, the particle can come to rest and then move in the opposite direction [Sample Prob. 11.1]. Thus, when computing the total distance traveled by a particle, you should first determine if the particle comes to rest during the specified interval of time. Constructing a diagram similar to that of Sample Prob. 11.1, which shows the position and the velocity of the particle at each critical instant ($v = v_{\max}$, $v = 0$, etc.), will help you to visualize the motion.

2. Determining $v(t)$ and $x(t)$ for a given $a(t)$. We discussed the solution of problems of this type in the first part of Sec. 11.1B. We used the initial conditions, $t = 0$ and $v = v_0$, for the lower limits of the integrals in t and v , but any other known state (for example, $t = t_1$ and $v = v_1$) could be used instead. Also, if the given function $a(t)$ contains an unknown constant (for example, the constant k if $a = kt$), you will first have to determine that constant by substituting a set of known values of t and a in the equation defining $a(t)$.

3. Determining $v(x)$ and $x(t)$ for a given $a(x)$. This is the second case considered in Sec. 11.1B. We again note that the lower limits of integration can be any known state (for example, $x = x_1$ and $v = v_1$). In addition, since $v = v_{\max}$ when $a = 0$, you can determine the positions where the maximum or minimum values of the velocity occur by setting $a(x) = 0$ and solving for x .

4. Determining $v(x)$, $v(t)$, and $x(t)$ for a given $a(v)$. This is the last case treated in Sec. 11.1B; the appropriate solution techniques for problems of this type are illustrated in Sample Probs. 11.3 and 11.4. All of the general comments for the preceding cases once again apply. Note that Sample Prob. 11.3 provides a summary of how and when to use the equations $v = dx/dt$, $a = dv/dt$, and $a = v dv/dx$.

We can summarize these relationships in Table 11.1.

Table 11.1

If....	Kinematic relationship	Integrate
$a = a(t)$	$\frac{dv}{dt} = a(t)$	$\int_{v_0}^v dv = \int_0^t a(t) dt$
$a = a(x)$	$v \frac{dv}{dx} = a(x)$	$\int_{v_0}^v v dv = \int_{x_0}^x a(x) dx$
$a = a(v)$	$\frac{dv}{dt} = a(v)$	$\int_{v_0}^v \frac{dv}{a(v)} = \int_0^t dt$
	$v \frac{dv}{dx} = a(v)$	$\int_{x_0}^x dx = \int_{v_0}^v \frac{v dv}{a(v)}$

Problems[†]

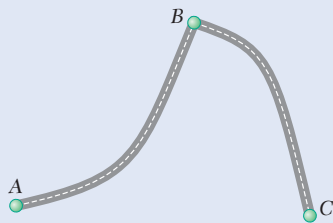


Fig. P11.CQ1

CONCEPT QUESTIONS

11.CQ1 A bus travels the 100 miles between *A* and *B* at 50 mi/h and then another 100 miles between *B* and *C* at 70 mi/h. The average speed of the bus for the entire 200-mile trip is:

- More than 60 mi/h.
- Equal to 60 mi/h.
- Less than 60 mi/h.

11.CQ2 Two cars *A* and *B* race each other down a straight road. The position of each car as a function of time is shown. Which of the following statements are true (more than one answer can be correct)?

- At time t_2 both cars have traveled the same distance.
- At time t_1 both cars have the same speed.
- Both cars have the same speed at some time $t < t_1$.
- Both cars have the same acceleration at some time $t < t_1$.
- Both cars have the same acceleration at some time $t_1 < t < t_2$.

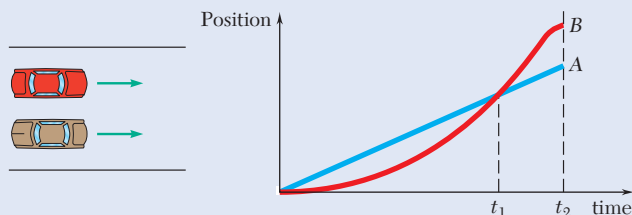


Fig. P11.CQ2

END-OF-SECTION PROBLEMS

11.1 A snowboarder starts from rest at the top of a double black diamond hill. As she rides down the slope, GPS coordinates are used to determine her displacement as a function of time: $x = 0.5t^3 + t^2 + 2t$, where x and t are expressed in feet and seconds, respectively. Determine the position, velocity, and acceleration of the boarder when $t = 5$ seconds.

11.2 The motion of a particle is defined by the relation $x = 2t^3 - 9t^2 + 12t + 10$, where x and t are expressed in feet and seconds, respectively. Determine the time, the position, and the acceleration of the particle when $v = 0$.

11.3 The vertical motion of mass *A* is defined by the relation $x = 10 \sin 2t + 15 \cos 2t + 100$, where x and t are expressed in millimeters and seconds, respectively. Determine (a) the position, velocity, and acceleration of *A* when $t = 1$ s, (b) the maximum velocity and acceleration of *A*.

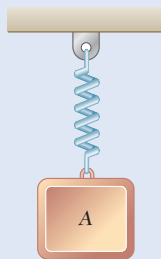


Fig. P11.3

[†]Answers to all problems set in straight type (such as 11.1) are given at the end of the book. Answers to problems with a number set in italic type (such as 11.6) are not given.

- 11.4** A loaded railroad car is rolling at a constant velocity when it couples with a spring and dashpot bumper system. After the coupling, the motion of the car is defined by the relation $x = 60e^{-4.8t} \sin 16t$, where x and t are expressed in millimeters and seconds, respectively. Determine the position, the velocity, and the acceleration of the railroad car when (a) $t = 0$, (b) $t = 0.3$ s.

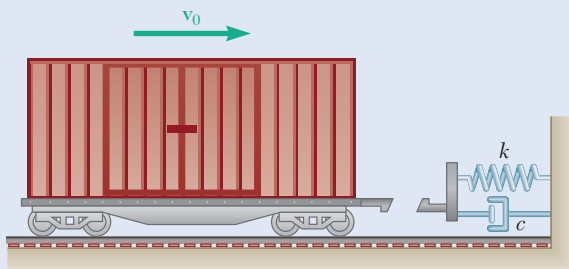


Fig. P11.4

- 11.5** The motion of a particle is defined by the relation $x = 6t^4 - 2t^3 - 12t^2 + 3t + 3$, where x and t are expressed in meters and seconds, respectively. Determine the time, the position, and the velocity when $a = 0$.
- 11.6** The motion of a particle is defined by the relation $x = t^3 - 9t^2 + 24t - 8$, where x and t are expressed in inches and seconds, respectively. Determine (a) when the velocity is zero, (b) the position and the total distance traveled when the acceleration is zero.
- 11.7** A girl operates a radio-controlled model car in a vacant parking lot. The girl's position is at the origin of the xy coordinate axes, and the surface of the parking lot lies in the x - y plane. She drives the car in a straight line so that the x coordinate is defined by the relation $x(t) = 0.5t^3 - 3t^2 + 3t + 2$, where x and t are expressed in meters and seconds, respectively. Determine (a) when the velocity is zero, (b) the position and total distance travelled when the acceleration is zero.

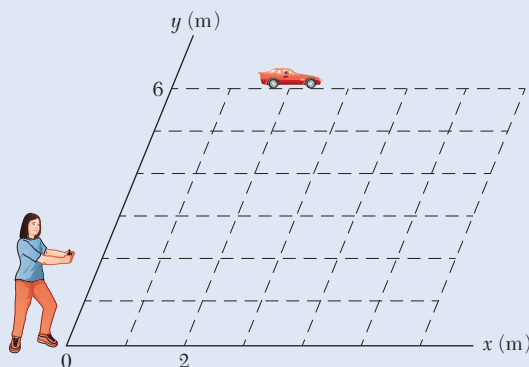


Fig. P11.7

- 11.8** The motion of a particle is defined by the relation $x = t^2 - (t - 2)^3$, where x and t are expressed in feet and seconds, respectively. Determine (a) the two positions at which the velocity is zero (b) the total distance traveled by the particle from $t = 0$ to $t = 4$ s.

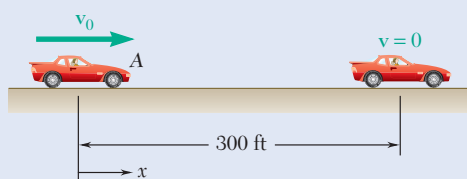


Fig. P11.9

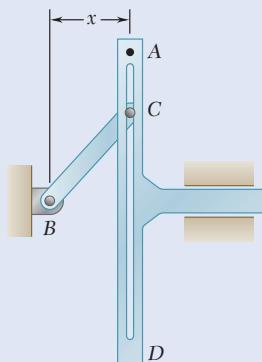


Fig. P11.13 and P11.14

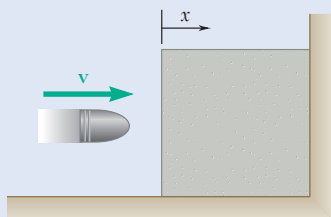


Fig. P11.16



Fig. P11.15

- 11.9** The brakes of a car are applied, causing it to slow down at a rate of 10 ft/s^2 . Knowing that the car stops in 300 ft, determine (a) how fast the car was traveling immediately before the brakes were applied, (b) the time required for the car to stop.
- 11.10** The acceleration of a particle is defined by the relation $a = 3e^{-0.2t}$, where a and t are expressed in ft/s^2 and seconds, respectively. Knowing that $x = 0$ and $v = 0$ at $t = 0$, determine the velocity and position of the particle when $t = 0.5 \text{ s}$.
- 11.11** The acceleration of a particle is directly proportional to the square of the time t . When $t = 0$, the particle is at $x = 24 \text{ m}$. Knowing that at $t = 6 \text{ s}$, $x = 96 \text{ m}$ and $v = 18 \text{ m/s}$, express x and v in terms of t .
- 11.12** The acceleration of a particle is defined by the relation $a = kt^2$. (a) Knowing that $v = -8 \text{ m/s}$ when $t = 0$ and that $v = +8 \text{ m/s}$ when $t = 2 \text{ s}$, determine the constant k . (b) Write the equations of motion, knowing also that $x = 0$ when $t = 2 \text{ s}$.
- 11.13** A Scotch yoke is a mechanism that transforms the circular motion of a crank into the reciprocating motion of a shaft (or vice versa). It has been used in a number of different internal combustion engines and in control valves. In the Scotch yoke shown, the acceleration of point A is defined by the relation $a = -1.8 \sin kt$, where a and t are expressed in m/s^2 and seconds, respectively, and $k = 3 \text{ rad/s}$. Knowing that $x = 0$ and $v = 0.6 \text{ m/s}$ when $t = 0$, determine the velocity and position of point A when $t = 0.5 \text{ s}$.
- 11.14** For the Scotch yoke mechanism shown, the acceleration of point A is defined by the relation $a = -1.08 \sin kt - 1.44 \cos kt$, where a and t are expressed in m/s^2 and seconds, respectively, and $k = 3 \text{ rad/s}$. Knowing that $x = 0.16 \text{ m}$ and $v = 0.36 \text{ m/s}$ when $t = 0$, determine the velocity and position of point A when $t = 0.5 \text{ s}$.
- 11.15** A piece of electronic equipment that is surrounded by packing material is dropped so that it hits the ground with a speed of 4 m/s . After contact the equipment experiences an acceleration of $a = -kx$, where k is a constant and x is the compression of the packing material. If the packing material experiences a maximum compression of 20 mm , determine the maximum acceleration of the equipment.

- 11.16** A projectile enters a resisting medium at $x = 0$ with an initial velocity $v_0 = 900 \text{ ft/s}$ and travels 4 in. before coming to rest. Assuming that the velocity of the projectile is defined by the relation $v = v_0 - kx$, where v is expressed in ft/s and x is in feet, determine (a) the initial acceleration of the projectile, (b) the time required for the projectile to penetrate 3.9 in. into the resisting medium.

- 11.17** The acceleration of a particle is defined by the relation $a = -k/x$. It has been experimentally determined that $v = 15$ ft/s when $x = 0.6$ ft and that $v = 9$ ft/s when $x = 1.2$ ft. Determine (a) the velocity of the particle when $x = 1.5$ ft, (b) the position of the particle at which its velocity is zero.

- 11.18** A brass (nonmagnetic) block A and a steel magnet B are in equilibrium in a brass tube under the magnetic repelling force of another steel magnet C located at a distance $x = 0.004$ m from B . The force is inversely proportional to the square of the distance between B and C . If block A is suddenly removed, the acceleration of block B is $a = -9.81 + k/x^2$, where a and x are expressed in m/s^2 and meters, respectively, and $k = 4 \times 10^{-4} \text{ m}^3/\text{s}^2$. Determine the maximum velocity and acceleration of B .

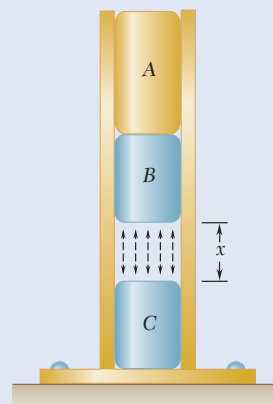


Fig. P11.18

- 11.19** Based on experimental observations, the acceleration of a particle is defined by the relation $a = -(0.1 + \sin x/b)$, where a and x are expressed in m/s^2 and meters, respectively. Knowing that $b = 0.8$ m and that $v = 1$ m/s when $x = 0$, determine (a) the velocity of the particle when $x = -1$ m, (b) the position where the velocity is maximum, (c) the maximum velocity.

- 11.20** A spring AB is attached to a support at A and to a collar. The unstretched length of the spring is l . Knowing that the collar is released from rest at $x = x_0$ and has an acceleration defined by the relation $a = -100(x - lx/\sqrt{l^2 + x^2})$, determine the velocity of the collar as it passes through point C .

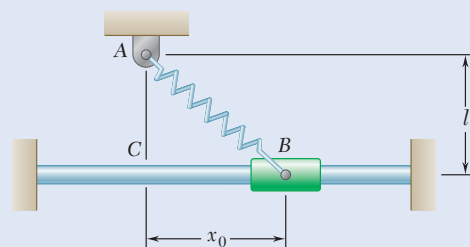


Fig. P11.20

- 11.21** The acceleration of a particle is defined by the relation $a = k(1 - e^{-x})$, where k is a constant. Knowing that the velocity of the particle is $v = +9$ m/s when $x = -3$ m and that the particle comes to rest at the origin, determine (a) the value of k , (b) the velocity of the particle when $x = -2$ m.

- 11.22** Starting from $x = 0$ with no initial velocity, a particle is given an acceleration $a = 0.1\sqrt{v^2 + 16}$, where a and v are expressed in ft/s^2 and ft/s , respectively. Determine (a) the position of the particle when $v = 3$ ft/s, (b) the speed and acceleration of the particle when $x = 4$ ft.

- 11.23** A ball is dropped from a boat so that it strikes the surface of a lake with a speed of 16.5 ft/s. While in the water the ball experiences an acceleration of $a = 10 - 0.8v$, where a and v are expressed in ft/s^2 and ft/s , respectively. Knowing the ball takes 3 s to reach the bottom of the lake, determine (a) the depth of the lake, (b) the speed of the ball when it hits the bottom of the lake.

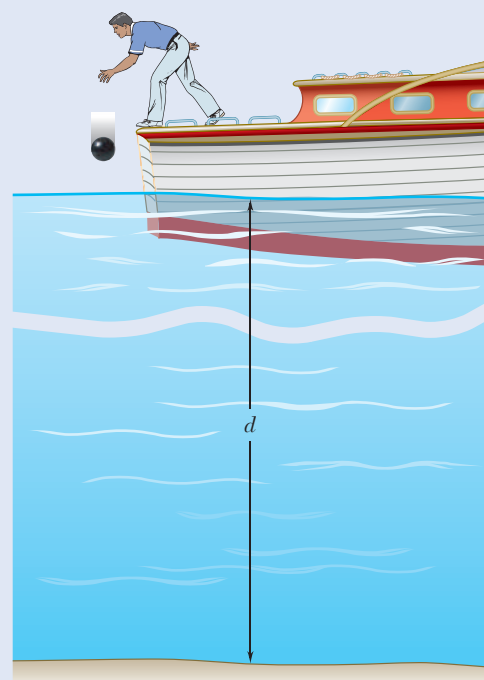


Fig. P11.23

- 11.24** The acceleration of a particle is defined by the relation $a = -k\sqrt{v}$, where k is a constant. Knowing that $x = 0$ and $v = 81$ m/s at $t = 0$ and that $v = 36$ m/s when $x = 18$ m, determine (a) the velocity of the particle when $x = 20$ m, (b) the time required for the particle to come to rest.

- 11.25** The acceleration of a particle is defined by the relation $a = -kv^{2.5}$, where k is a constant. The particle starts at $x = 0$ with a velocity of 16 mm/s, and when $x = 6$ mm, the velocity is observed to be 4 mm/s. Determine (a) the velocity of the particle when $x = 5$ mm, (b) the time at which the velocity of the particle is 9 mm/s.



Fig. P11.26

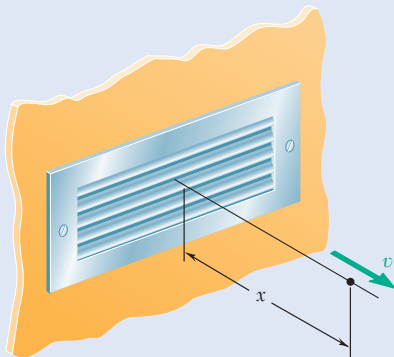


Fig. P11.27

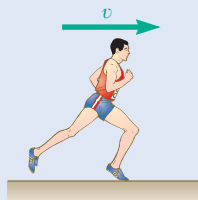


Fig. P11.28

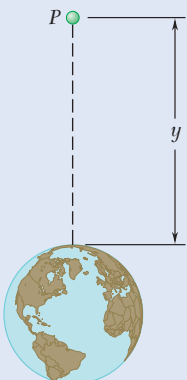


Fig. P11.29

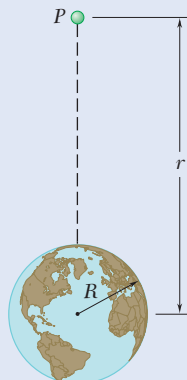


Fig. P11.30

11.26 A human-powered vehicle (HPV) team wants to model the acceleration during the 260-m sprint race (the first 60 m is called a flying start) using $a = A - Cv^2$, where a is acceleration in m/s^2 and v is the velocity in m/s . From wind tunnel testing, they found that $C = 0.0012 \text{ m}^{-1}$. Knowing that the cyclist is going 100 km/h at the 260-meter mark, what is the value of A ?

11.27 Experimental data indicate that in a region downstream of a given louvered supply vent the velocity of the emitted air is defined by $v = 0.18v_0/x$, where v and x are expressed in m/s and meters, respectively, and v_0 is the initial discharge velocity of the air. For $v_0 = 3.6 \text{ m/s}$, determine (a) the acceleration of the air at $x = 2 \text{ m}$, (b) the time required for the air to flow from $x = 1$ to $x = 3 \text{ m}$.

11.28 Based on observations, the speed of a jogger can be approximated by the relation $v = 7.5(1 - 0.04x)^{0.3}$, where v and x are expressed in mi/h and miles, respectively. Knowing that $x = 0$ at $t = 0$, determine (a) the distance the jogger has run when $t = 1 \text{ h}$, (b) the jogger's acceleration in ft/s^2 at $t = 0$, (c) the time required for the jogger to run 6 mi.

11.29 The acceleration due to gravity at an altitude y above the surface of the earth can be expressed as

$$a = \frac{-32.2}{[1 + (y/20.9 \times 10^6)]^2}$$

where a and y are expressed in ft/s^2 and feet, respectively. Using this expression, compute the height reached by a projectile fired vertically upward from the surface of the earth if its initial velocity is (a) 1800 ft/s , (b) 3000 ft/s , (c) 36,700 ft/s .

11.30 The acceleration due to gravity of a particle falling toward the earth is $a = -gR^2/r^2$, where r is the distance from the center of the earth to the particle, R is the radius of the earth, and g is the acceleration due to gravity at the surface of the earth. If $R = 3960 \text{ mi}$, calculate the *escape velocity*, that is, the minimum velocity with which a particle must be projected vertically upward from the surface of the earth if it is not to return to the earth. (Hint: $v = 0$ for $r = \infty$.)

11.31 The velocity of a particle is $v = v_0[1 - \sin(\pi t/T)]$. Knowing that the particle starts from the origin with an initial velocity v_0 , determine (a) its position and its acceleration at $t = 3T$, (b) its average velocity during the interval $t = 0$ to $t = T$.

11.32 An eccentric circular cam, which serves a similar function as the Scotch yoke mechanism in Problem 11.13, is used in conjunction with a flat face follower to control motion in pumps and in steam engine valves. Knowing that the eccentricity is denoted by e , the maximum range of the displacement of the follower is $d_{\max}$ and the maximum velocity of the follower is $v_{\max}$, determine the displacement, velocity, and acceleration of the follower.

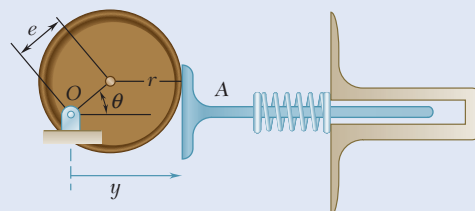


Fig. P11.32

11.2 SPECIAL CASES AND RELATIVE MOTION

In this section, we derive the equations that describe uniform rectilinear motion and uniformly accelerated rectilinear motion. We also introduce the concept of relative motion, which is of fundamental importance whenever we consider the motion of more than one particle at the same time.

11.2A Uniform Rectilinear Motion

Uniform rectilinear motion is a type of straight-line motion that is frequently encountered in practical applications. In this motion, the acceleration a of the particle is zero for every value of t . The velocity v is therefore constant, and Eq. (11.1) becomes

$$\frac{dx}{dt} = v = \text{constant}$$

We can obtain the position coordinate x by integrating this equation. Denoting the initial value of x by x_0 , we have

Distance in uniform rectilinear motion

$$\begin{aligned} \int_{x_0}^x dx &= v \int_0^t dt \\ x - x_0 &= vt \\ x &= x_0 + vt \end{aligned} \quad (11.5)$$

This equation can be used *only if the velocity of the particle is known to be constant*. For example, this would be true for an airplane in steady flight or a car cruising along a highway at a constant speed.

11.2B Uniformly Accelerated Rectilinear Motion

Uniformly accelerated rectilinear motion is another common type of motion. In this case, the acceleration a of the particle is constant, and Eq. (11.2) becomes

$$\frac{dv}{dt} = a = \text{constant}$$

We obtain the velocity v of the particle by integrating this equation as

$$\begin{aligned} \int_{v_0}^v dv &= a \int_0^t dt \\ v - v_0 &= at \\ v &= v_0 + at \end{aligned} \quad (11.6)$$

where v_0 is the initial velocity. Substituting for v in Eq. (11.1), we have

$$\frac{dx}{dt} = v_0 + at$$

Denoting by x_0 the initial value of x and integrating, we have

$$\begin{aligned}\int_{x_0}^x dx &= \int_0^t (v_0 + at) dt \\ x - x_0 &= v_0 t + \frac{1}{2}at^2 \\ x &= x_0 + v_0 t + \frac{1}{2}at^2\end{aligned}\quad (11.7)$$

We can also use Eq. (11.4) and write

$$\begin{aligned}v \frac{dv}{dx} &= a = \text{constant} \\ v dv &= a dx\end{aligned}$$

Integrating both sides, we obtain

$$\begin{aligned}\int_{v_0}^v v dv &= a \int_{x_0}^x dx \\ \frac{1}{2}(v^2 - v_0^2) &= a(x - x_0) \\ v^2 &= v_0^2 + 2a(x - x_0)\end{aligned}\quad (11.8)$$

The three equations we have derived provide useful relations among position, velocity, and time in the case of constant acceleration, once you have provided appropriate values for a , v_0 , and x_0 . You first need to define the origin O of the x axis and choose a positive direction along the axis; this direction determines the signs of a , v_0 , and x_0 . Equation (11.6) relates v and t and should be used when the value of v corresponding to a given value of t is desired, or inversely. Equation (11.7) relates x and t ; Eq. (11.8) relates v and x . An important application of uniformly accelerated motion is the motion of a body in **free fall**. The acceleration of a body in free fall (usually denoted by g) is equal to 9.81 m/s^2 or 32.2 ft/s^2 (we ignore air resistance in this case).

It is important to keep in mind that the three equations can be used *only when the acceleration of the particle is known to be constant*. If the acceleration of the particle is variable, you need to determine its motion from the fundamental Eqs. (11.1) through (11.4) according to the methods outlined in Sec. 11.1B.

11.2C Motion of Several Particles

When several particles move independently along the same line, you can write independent equations of motion for each particle. Whenever possible, you should record time from the same initial instant for all particles and measure displacements from the same origin and in the same direction. In other words, use a single clock and a single measuring tape.

Relative Motion of Two Particles. Consider two particles A and B moving along the same straight line (Fig. 11.7). If we measure the position coordinates x_A and x_B from the same origin, the difference $x_B - x_A$ defines the **relative position coordinate of B with respect to A** , which is denoted by $x_{B/A}$. We have

**Relative position
of two particles**

$$x_{B/A} = x_B - x_A \quad \text{OR} \quad x_B = x_A + x_{B/A} \quad (11.9)$$

Regardless of the positions of A and B with respect to the origin, a positive sign for $x_{B/A}$ means that B is to the right of A , and a negative sign means that B is to the left of A .

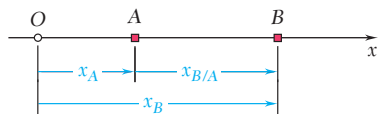


Fig. 11.7 Two particles A and B in motion along the same straight line.

The rate of change of $x_{B/A}$ is known as the **relative velocity of B with respect to A** and is denoted by $v_{B/A}$. Differentiating Eq. (11.9), we obtain

Relative velocity of two particles

$$v_{B/A} = v_B - v_A \quad \text{or} \quad v_B = v_A + v_{B/A} \quad (11.10)$$

A positive sign for $v_{B/A}$ means that B is *observed from A* to move in the positive direction; a negative sign means that it is observed to move in the negative direction.

The rate of change of $v_{B/A}$ is known as the **relative acceleration of B with respect to A** and is denoted by $a_{B/A}$. Differentiating Eq. (11.10), we obtain[†]

Relative acceleration of two particles

$$a_{B/A} = a_B - a_A \quad \text{or} \quad a_B = a_A + a_{B/A} \quad (11.11)$$

Dependent Motion of Particles. Sometimes, the position of a particle depends upon the position of another particle or of several other particles. These motions are called **dependent**. For example, the position of block B in Fig. 11.8 depends upon the position of block A . Since the rope $ACDEFG$ is of constant length, and since the lengths of the portions of rope CD and EF wrapped around the pulleys remain constant, it follows that the sum of the lengths of the segments AC , DE , and FG is constant. Observing that the length of the segment AC differs from x_A only by a constant and that, similarly, the lengths of the segments DE and FG differ from x_B only by a constant, we have

$$x_A + 2x_B = \text{constant}$$

Since only one of the two coordinates x_A and x_B can be chosen arbitrarily, we say that the system shown in Fig. 11.8 has **one degree of freedom**. From the relation between the position coordinates x_A and x_B , it follows that if x_A is given an increment Δx_A —that is, if block A is lowered by an amount Δx_A —the coordinate x_B receives an increment $\Delta x_B = -\frac{1}{2}\Delta x_A$. In other words, block B rises by half the same amount. You can check this directly from Fig. 11.8.

In the case of the three blocks of Fig. 11.9, we can again observe that the length of the rope that passes over the pulleys is constant. Thus, the following relation must be satisfied by the position coordinates of the three blocks:

$$2x_A + 2x_B + x_C = \text{constant}$$

Since two of the coordinates can be chosen arbitrarily, we say that the system shown in Fig. 11.9 has **two degrees of freedom**.

When the relation existing between the position coordinates of several particles is *linear*, a similar relation holds between the velocities and between the accelerations of the particles. In the case of the blocks of Fig. 11.9, for instance, we can differentiate the position equation twice and obtain

$$\begin{aligned} 2\frac{dx_A}{dt} + 2\frac{dx_B}{dt} + \frac{dx_C}{dt} &= 0 & \text{or} & & 2v_A + 2v_B + v_C &= 0 \\ 2\frac{dv_A}{dt} + 2\frac{dv_B}{dt} + \frac{dv_C}{dt} &= 0 & \text{or} & & 2a_A + 2a_B + a_C &= 0 \end{aligned}$$

[†]Note that the product of the subscripts A and B/A used in the right-hand sides of Eqs. (11.9), (11.10), and (11.11) is equal to the subscript B that appears in the left-hand sides. This may help you remember the correct order of subscripts in various situations.



Photo 11.2 Multiple cables and pulleys are used by this shipyard crane.

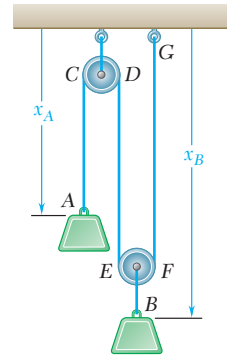


Fig. 11.8 A system of blocks and pulleys with one degree of freedom.

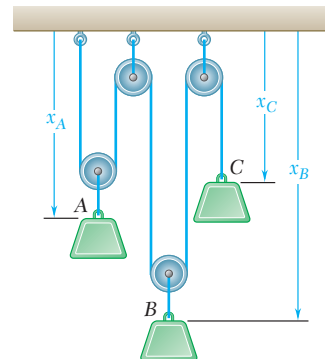


Fig. 11.9 A system of blocks and pulleys with two degrees of freedom.

Sample Problem 11.5

In an elevator shaft, a ball is thrown vertically upward with an initial velocity of 18 m/s from a height of 12 m above ground. At the same instant, an open-platform elevator passes the 5-m level, moving upward with a constant velocity of 2 m/s. Determine (a) when and where the ball hits the elevator (b) the relative velocity of the ball with respect to the elevator when the ball hits the elevator.

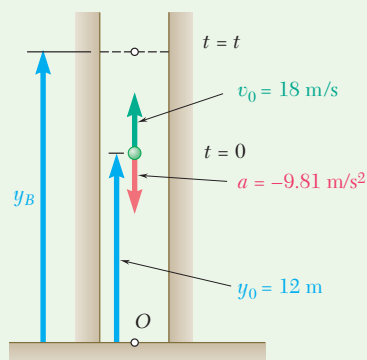


Fig. 1 Acceleration, initial velocity, and initial position of the ball.

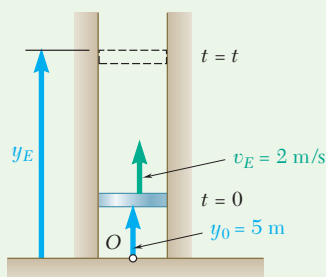


Fig. 2 Initial velocity and initial position of the elevator.

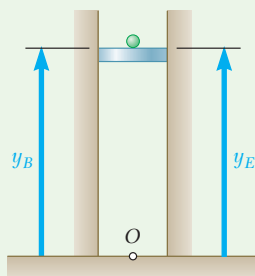


Fig. 3 Position of ball and elevator at time t .

STRATEGY: The ball has a constant acceleration, so its motion is *uniformly accelerated*. The elevator has a constant velocity, so its motion is *uniform*. You can write equations to describe each motion and then set the position coordinates equal to each other to find when the particles meet. The relative velocity is determined from the calculated motion of each particle.

MODELING and ANALYSIS:

Motion of Ball. Place the origin O of the y axis at ground level and choose its positive direction upward (Fig. 1). Then the initial position of the ball is $y_0 = +12$ m, its initial velocity is $v_0 = +18$ m/s, and its acceleration is $a = -9.81$ m/s². Substituting these values in the equations for uniformly accelerated motion, you get

$$v_B = v_0 + at \qquad v_B = 18 - 9.81t \qquad (1)$$

$$y_B = y_0 + v_0 t + \frac{1}{2}at^2 \qquad y_B = 12 + 18t - 4.905t^2 \qquad (2)$$

Motion of Elevator. Again place the origin O at ground level and choose the positive direction upward (Fig. 2). Noting that $y_0 = +5$ m, you have

$$v_E = +2 \text{ m/s} \qquad (3)$$

$$y_E = y_0 + v_E t \qquad y_E = 5 + 2t \qquad (4)$$

Ball Hits Elevator. First note that you used the same time t and the same origin O in writing the equations of motion for both the ball and the elevator. From Fig. 3, when the ball hits the elevator,

$$y_E = y_B \qquad (5)$$

Substituting for y_E and y_B from Eqs. (2) and (4) into Eq. (5), you have

$$5 + 2t = 12 + 18t - 4.905t^2$$

$$t = -0.39 \text{ s} \qquad \text{and} \qquad t = 3.65 \text{ s} \quad \blacktriangleleft$$

Only the root $t = 3.65$ s corresponds to a time after the motion has begun. Substituting this value into Eq. (4), you obtain

$$y_E = 5 + 2(3.65) = 12.30 \text{ m}$$

Elevation from ground = 12.30 m ◀

Relative Velocity. The relative velocity of the ball with respect to the elevator is

$$v_{B/E} = v_B - v_E = (18 - 9.81t) - 2 = 16 - 9.81t$$

When the ball hits the elevator at time $t = 3.65$ s, you have

$$v_{B/E} = 16 - 9.81(3.65) \quad v_{B/E} = -19.81 \text{ m/s} \quad \blacktriangleleft$$

The negative sign means that if you are riding on the elevator, it will appear as if the ball is moving downward.

REFLECT and THINK: The key insight is that, when two particles collide, their position coordinates must be equal. Also, although you can use the basic kinematic relationships in this problem, you may find it easier to use the equations relating a , v , x , and t when the acceleration is constant or zero.

Sample Problem 11.6

Car A is travelling at a constant 90 mi/h when she passes a parked police officer B, who gives chase when the car passes her. The officer accelerates at a constant rate until she reaches the speed of 105 mi/h. Thereafter, her speed remains constant. The police officer catches the car 3 mi from her starting point. Determine the initial acceleration of the police officer.

STRATEGY: One car is traveling at a constant speed and the other has a constant acceleration, so you can start with the algebraic relationships found in Sec. 11.2 rather than separating and integrating the basic kinematic relationships.

MODELING and ANALYSIS: A clearly labeled picture will help you understand the problem better (Fig. 1). The position, x , is defined from the point the car passes the officer.

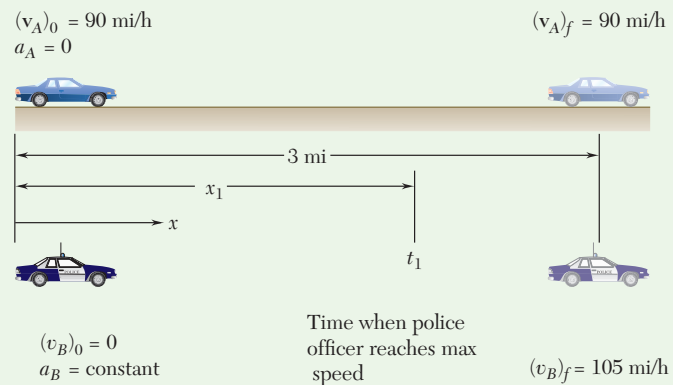


Fig. 1 Velocities and accelerations of the cars at various times.

(continued)

Unit Conversions. First you should convert everything to units of feet and seconds. Use the subscript A for the car and B for the officer

$$v_A = \left(90 \frac{\text{mi}}{\text{hr}}\right) \left(\frac{1 \text{ hr}}{3600 \text{ s}}\right) \left(\frac{5280 \text{ ft}}{\text{mi}}\right) = 132 \frac{\text{ft}}{\text{s}}$$

$$v_B = \left(105 \frac{\text{mi}}{\text{hr}}\right) \left(\frac{1 \text{ hr}}{3600 \text{ s}}\right) \left(\frac{5280 \text{ ft}}{\text{mi}}\right) = 154 \frac{\text{ft}}{\text{s}}$$

Motion of the Speeding Car A. Since the car has a constant speed,

$$x_A = v_A t = 132 t \quad (1)$$

Motion of the Officer B. The officer has a constant acceleration until she reaches a final speed of 105 mph. This time is labeled t_1 in Fig. 1. Therefore, from time $0 < t < t_1$, the officer has a velocity of

$$v_B = a_B t \quad \text{for } 0 < t < t_1$$

or at time $t = t_1$, it is

$$154 = a_B t_1 \quad (2)$$

The distance the officer travels is going to be the distance from 0 to t_1 and then from t_1 to t_f . Hence,

$$x_B = \frac{1}{2} a_B t_1^2 + v_B (t - t_1) \quad \text{for } t > t_1 \quad (3)$$

The officer catches the speeder when $x_A = x_B = 3 \text{ mi} = 15,840 \text{ ft}$. From Eq. (1), you can solve for the time $t_f = (15,840 \text{ ft}) / (132 \text{ ft/s}) = 120 \text{ s}$. Therefore, you have two equations: Eq. (2) and

$$15,840 = \frac{1}{2} a_B t_1^2 + 154(120 - t_1) \quad (4)$$

Substituting Eq. (2) into Eq. (4) allows you to solve for t_1 :

$$t_1 = 34.39 \text{ s}$$

Substituting this into Eq. (2) gives

$$a_B = 4.49 \text{ ft/s}^2 \quad \blacktriangleleft$$

REFLECT and THINK: It is important to use the same origin for the position of both vehicles. The time to accelerate from 0 to 105 mph seems reasonable, although it is perhaps longer than you would expect. A high-performance sports car can go from 0 to 60 mph in less than 5 seconds. It is very likely that the officer could have accelerated to 105 mph in less time if she had wanted to, but perhaps she had to consider the safety of other motorists.

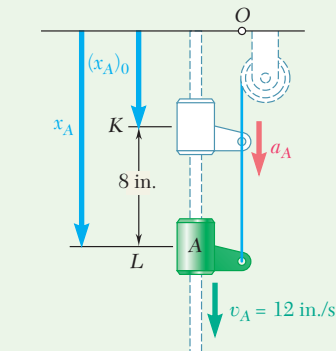
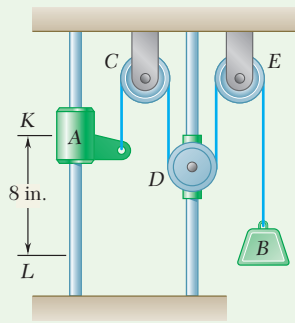


Fig. 1 Position, velocity, and acceleration of collar A.

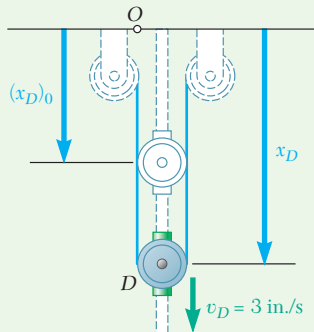


Fig. 2 Position and velocity of pulley D.

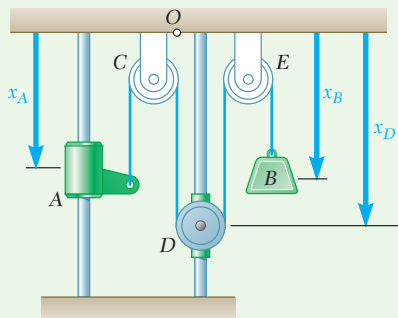


Fig. 3 Position of A, B, and D.

Sample Problem 11.7

Collar A and block B are connected by a cable passing over three pulleys C, D, and E as shown. Pulleys C and E are fixed, but D is attached to a collar that is pulled downward with a constant velocity of 3 in./s. At $t = 0$, collar A starts moving downward from position K with a constant acceleration and no initial velocity. Knowing that the velocity of collar A is 12 in./s as it passes through point L, determine the change in elevation, the velocity, and the acceleration of block B when collar A passes through L.

STRATEGY: You have multiple objects connected by cables, so this is a problem in *dependent motion*. Use the given data to write a single equation relating the changes in position coordinates of collar A, pulley D, and block B. Based on the given information, you will also need to use the algebraic relationships we found for uniformly accelerated motion.

MODELING and ANALYSIS:

Motion of Collar A. Place the origin O at the upper horizontal surface and choose the positive direction downward. Then when $t = 0$, collar A is at position K and $(v_A)_0 = 0$ (Fig. 1). Since $v_A = 12$ in./s and $x_A - (x_A)_0 = 8$ in. when the collar passes through L, you have

$$v_A^2 = (v_A)_0^2 + 2a_A[x_A - (x_A)_0] \quad (12)^2 = 0 + 2a_A(8) \\ a_A = 9 \text{ in./s}^2$$

To find the time at which collar A reaches point L, use the equation for velocity as a function of time with uniform acceleration. Thus,

$$v_A = (v_A)_0 + a_A t \quad 12 = 0 + 9t \quad t = 1.333 \text{ s}$$

Motion of Pulley D. Since the positive direction is downward, you have (Fig. 2)

$$a_D = 0 \quad v_D = 3 \text{ in./s} \quad x_D = (x_D)_0 + v_D t = (x_D)_0 + 3t$$

When collar A reaches L at $t = 1.333$ s, the position of pulley D is

$$x_D = (x_D)_0 + 3(1.333) = (x_D)_0 + 4$$

Thus,

$$x_D - (x_D)_0 = 4 \text{ in.}$$

Motion of Block B. Note that the total length of cable ACDEB differs from the quantity $(x_A + 2x_D + x_B)$ only by a constant. Since the cable length is constant during the motion, this quantity must also remain constant. Thus, considering the times $t = 0$ and $t = 1.333$ s, you can write

$$x_A + 2x_D + x_B = (x_A)_0 + 2(x_D)_0 + (x_B)_0 \quad (1)$$

$$[x_A - (x_A)_0] + 2[x_D - (x_D)_0] + [x_B - (x_B)_0] = 0 \quad (2)$$

But you know that $x_A - (x_A)_0 = 8$ in. and $x_D - (x_D)_0 = 4$ in. Substituting these values in Eq. (2), you find

$$8 + 2(4) + [x_B - (x_B)_0] = 0 \quad x_B - (x_B)_0 = -16 \text{ in.}$$

Thus,

Change in elevation of B = 16 in. ↑ ◀
(continued)

Differentiating Eq. (1) twice, you obtain equations relating the velocities and the accelerations of A , B , and D . Substituting for the velocities and accelerations of A and D at $t = 1.333$ s, you have

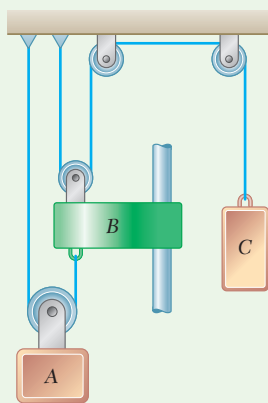
$$v_A + 2v_D + v_B = 0: \quad 12 + 2(3) + v_B = 0$$

$$v_B = -18 \text{ in./s} \quad v_B = 18 \text{ in./s} \uparrow \quad \blacktriangleleft$$

$$a_A + 2a_D + a_B = 0: \quad 9 + 2(0) + a_B = 0$$

$$a_B = -9 \text{ in./s}^2 \quad a_B = 9 \text{ in./s}^2 \uparrow \quad \blacktriangleleft$$

REFLECT and THINK: In this case, the relationship we needed was not between position coordinates, but between changes in position coordinates at two different times. The key step is to clearly define your position vectors. This is a two-degree-of-freedom system, because two coordinates are required to completely describe it.



Sample Problem 11.8

Block C starts from rest and moves down with a constant acceleration. Knowing that after block A has moved 1.5 ft its velocity is 0.6 ft/s, determine (a) the acceleration of A and C , (b) the change in velocity and the change in position of block B after 2.5 seconds.

STRATEGY: Since you have blocks connected by cables, this is a dependent-motion problem. You should define coordinates for each mass and write constraint equations for both cables.

MODELING and ANALYSIS: Define position vectors as shown in Fig. 1, where positive is defined to be down.

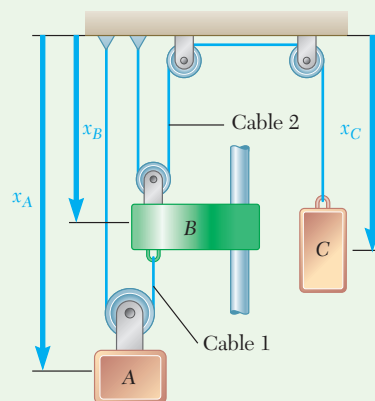


Fig. 1 Position of A , B , and C .

Constraint Equations. Assuming the cables are inextensible, you can write the lengths in terms of the defined coordinates and then differentiate.

Cable 1: $x_A + (x_A - x_B) = \text{constant}$

Differentiating this, you find

$$2v_A = v_B \quad \text{and} \quad 2a_A = a_B \quad (1)$$

Cable 2: $2x_B + x_C = \text{constant}$

Differentiating this, you find

$$v_C = -2v_B \quad \text{and} \quad a_C = -2a_B \quad (2)$$

Substituting Eq. (1) into Eq. (2) gives

$$v_C = -4v_A \quad \text{and} \quad a_C = -4a_A \quad (3)$$

Motion of A. You can use the constant-acceleration equations for block A:, as

$$v_A^2 - v_{A_0}^2 = 2a_A[x_A - (x_A)_0] \quad \text{or} \quad a_A = \frac{v_A^2 - (v_A)_0^2}{2[x_A - (x_A)_0]} \quad (4)$$

a. Acceleration of A and C. You know v_C and a_C are down, so from Eq. (3), you also know v_A and a_A are up. Substituting the given values into Eq. (4), you find

$$a_A = \frac{(0.6 \text{ ft/s})^2 - 0}{2(-1.5 \text{ ft})} = -0.12 \text{ ft/s}^2 \quad \mathbf{a_A = 0.120 \text{ ft/s}^2 \uparrow} \quad \blacktriangleleft$$

Substituting this value into $a_C = -4a_A$, you obtain

$$\mathbf{a_C = 0.480 \text{ ft/s}^2 \downarrow} \quad \blacktriangleleft$$

b. Velocity and change in position of B after 2.5 s. Substituting a_A in $a_B = 2a_A$ gives

$$a_B = 2(-0.12 \text{ ft/s}^2) = -0.24 \text{ ft/s}^2$$

You can use the equations of constant acceleration to find

$$\Delta v_B = a_B t = (-0.24 \text{ ft/s}^2)(2.5 \text{ s}) = -0.600 \text{ ft/s} \quad \Delta v_B = 0.600 \text{ ft/s} \uparrow \quad \blacktriangleleft$$

$$\Delta x_B = \frac{1}{2} a_B t^2 = \frac{1}{2} (-0.24 \text{ ft/s}^2)(2.5 \text{ s})^2 = -0.750 \text{ ft} \quad \Delta x_B = 0.750 \text{ ft} \uparrow \quad \blacktriangleleft$$

REFLECT and THINK: One of the keys to solving this problem is recognizing that since there are two cables, you need to write two constraint equations. The directions of the answers also make sense. If block C is accelerating downward, you would expect A and B to accelerate upward.

SOLVING PROBLEMS ON YOUR OWN

In this section, we derived the equations that describe **uniform rectilinear motion** (constant velocity) and **uniformly accelerated rectilinear motion** (constant acceleration). We also introduced the concept of **relative motion**. We can apply the equations for relative motion [Eqs. (11.9) through (11.11)] to the independent or dependent motions of any two particles moving along the same straight line.

A. Independent motion of one or more particles. Organize the solution of problems of this type as follows.

1. Begin your solution by listing the given information, sketching the system, and selecting the origin and the positive direction of the coordinate axis [Sample Prob. 11.5]. It is always advantageous to have a visual representation of problems of this type.

2. Write the equations that describe the motions of the various particles as well as those that describe how these motions are related [Eq. (5) of Sample Prob. 11.5].

3. Define the initial conditions, i.e., specify the state of the system corresponding to $t = 0$. This is especially important if the motions of the particles begin at different times. In such cases, either of two approaches can be used.

a. Let $t = 0$ be the time when the last particle begins to move. You must then determine the initial position x_0 and the initial velocity v_0 of each of the other particles.

b. Let $t = 0$ be the time when the first particle begins to move. You must then, in each of the equations describing the motion of another particle, replace t with $t - t_0$, where t_0 is the time at which that specific particle begins to move. It is important to recognize that the equations obtained in this way are valid only for $t \geq t_0$.

B. Dependent motion of two or more particles. In problems of this type, the particles of the system are connected to each other, typically by ropes or cables. The method of solution of these problems is similar to that of the preceding group of problems, except that it is now necessary to describe the *physical connections* between the particles. In the following problems, the connection is provided by one or more cables. For each cable, you will have to write equations similar to the last three equations of Sec. 11.2C. We suggest that you use the following procedure.

1. Draw a sketch of the system and select a coordinate system, indicating clearly a positive sense for each of the coordinate axes. For example, in Sample Probs. 11.7 and 11.8, we measured lengths downward from the upper horizontal support. It thus follows that those displacements, velocities, and accelerations that have positive values are directed downward.

2. Write the equation describing the constraint imposed by each cable on the motion of the particles involved. Differentiating this equation twice, you obtain the corresponding relations among velocities and accelerations.

3. If several directions of motion are involved, you must select a coordinate axis and a positive sense for each of these directions. You should also try to locate the origins of your coordinate axes so that the equations of constraints are as simple as possible. For example, in Sample Prob. 11.7, it is easier to define the various coordinates by measuring them downward from the upper support than by measuring them upward from the bottom support.

Finally, keep in mind that the method of analysis described in this section and the corresponding equations can be used only for particles moving with *uniform* or *uniformly accelerated rectilinear motion*.

Problems

- 11.33** An airplane begins its take-off run at A with zero velocity and a constant acceleration a . Knowing that it becomes airborne 30 s later at B and that the distance AB is 900 m, determine (a) the acceleration a (b) the take-off velocity v_B .

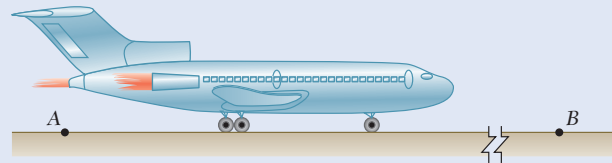


Fig. P11.33

- 11.34** A motorist is traveling at 54 km/h when she observes that a traffic light 240 m ahead of her turns red. The traffic light is timed to stay red for 24 s. If the motorist wishes to pass the light without stopping just as it turns green again, determine (a) the required uniform deceleration of the car, (b) the speed of the car as it passes the light.

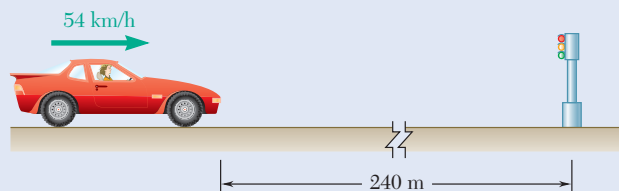


Fig. P11.34

- 11.35** Steep safety ramps are built beside mountain highways to enable vehicles with defective brakes to stop safely. A truck enters a 750-ft ramp at a high speed v_0 and travels 540 ft in 6 s at constant deceleration before its speed is reduced to $v_0/2$. Assuming the same constant deceleration, determine (a) the additional time required for the truck to stop (b) the additional distance traveled by the truck.

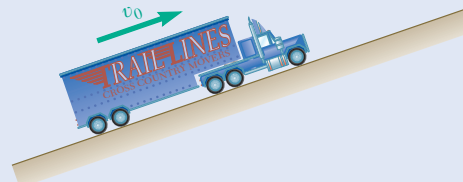


Fig. P11.35

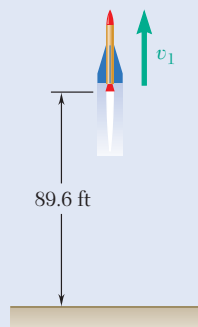


Fig. P11.36

- 11.36** A group of students launches a model rocket in the vertical direction. Based on tracking data, they determine that the altitude of the rocket was 89.6 ft at the end of the powered portion of the flight and that the rocket landed 16 s later. Knowing that the descent parachute failed to deploy so that the rocket fell freely to the ground after reaching its maximum altitude and assuming that $g = 32.2 \text{ ft/s}^2$, determine (a) the speed v_1 of the rocket at the end of powered flight, (b) the maximum altitude reached by the rocket.

- 11.37** A small package is released from rest at A and moves along the skate wheel conveyor $ABCD$. The package has a uniform acceleration of 4.8 m/s^2 as it moves down sections AB and CD , and its velocity is constant between B and C . If the velocity of the package at D is 7.2 m/s , determine (a) the distance d between C and D , (b) the time required for the package to reach D .

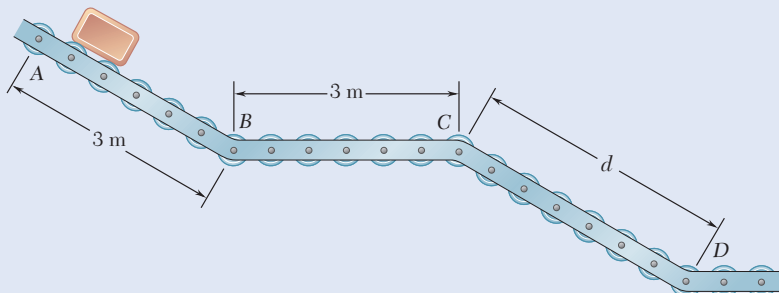


Fig. P11.37

- 11.38** A sprinter in a 100-m race accelerates uniformly for the first 35 m and then runs with constant velocity. If the sprinter's time for the first 35 m is 5.4 s, determine (a) his acceleration, (b) his final velocity, (c) his time for the race.

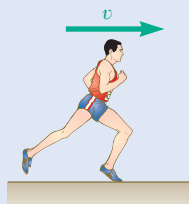


Fig. P11.38

- 11.39** Automobile A starts from O and accelerates at the constant rate of 0.75 m/s^2 . A short time later it is passed by bus B which is traveling in the opposite direction at a constant speed of 6 m/s . Knowing that bus B passes point O 20 s after automobile A started from there, determine when and where the vehicles passed each other.

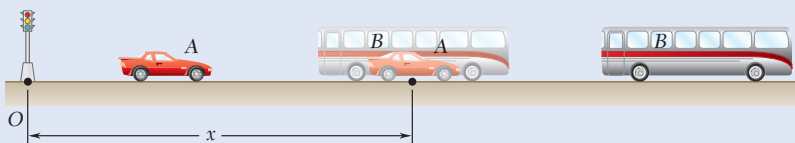


Fig. P11.39

- 11.40** In a boat race, boat A is leading boat B by 50 m and both boats are traveling at a constant speed of 180 km/h . At $t = 0$, the boats accelerate at constant rates. Knowing that when B passes A , $t = 8 \text{ s}$ and $v_A = 225 \text{ km/h}$, determine (a) the acceleration of A , (b) the acceleration of B .

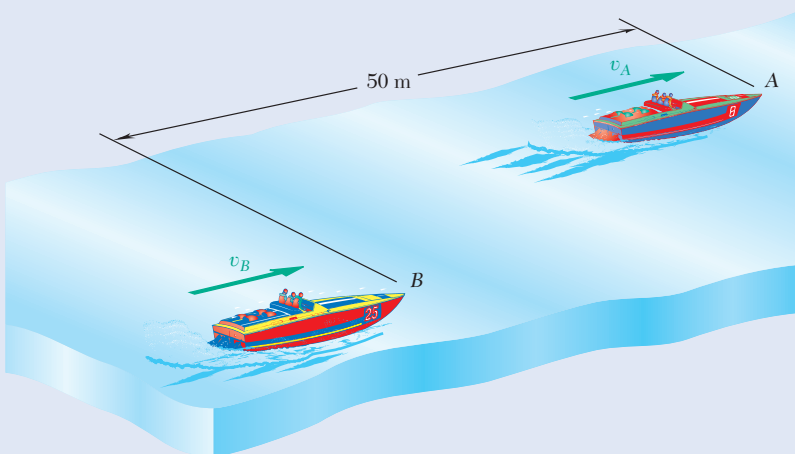


Fig. P11.40

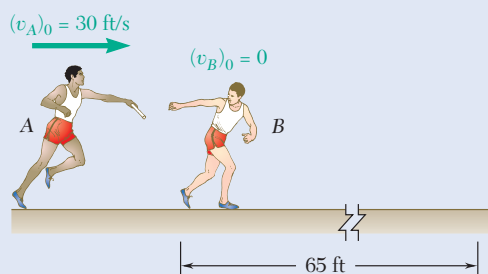


Fig. P11.41

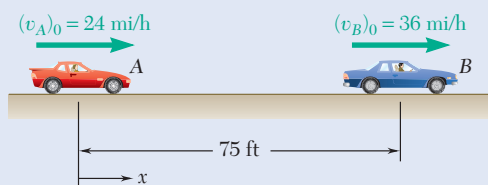


Fig. P11.42

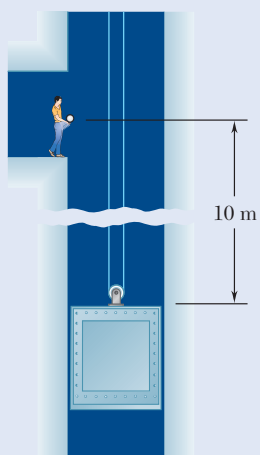


Fig. P11.44

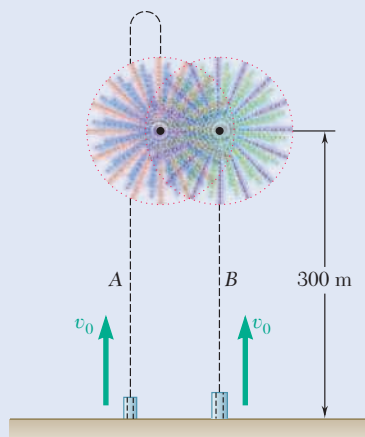


Fig. P11.45

11.41 As relay runner *A* enters the 65-ft-long exchange zone with a speed of 30 ft/s, he begins to slow down. He hands the baton to runner *B* 2.5 s later as they leave the exchange zone with the same velocity. Determine (a) the uniform acceleration of each of the runners, (b) when runner *B* should begin to run.

11.42 Automobiles *A* and *B* are traveling in adjacent highway lanes and at $t = 0$ have the positions and speeds shown. Knowing that automobile *A* has a constant acceleration of 1.8 ft/s^2 and that *B* has a constant deceleration of 1.2 ft/s^2 , determine (a) when and where *A* will overtake *B*, (b) the speed of each automobile at that time.

11.43 Two automobiles *A* and *B* are approaching each other in adjacent highway lanes. At $t = 0$, *A* and *B* are 3200 ft apart, their speeds are $v_A = 65 \text{ mi/h}$ and $v_B = 40 \text{ mi/h}$, and they are at points *P* and *Q*, respectively. Knowing that *A* passes point *Q* 40 s after *B* was there and that *B* passes point *P* 42 s after *A* was there, determine (a) the uniform accelerations of *A* and *B*, (b) when the vehicles pass each other, (c) the speed of *B* at that time.

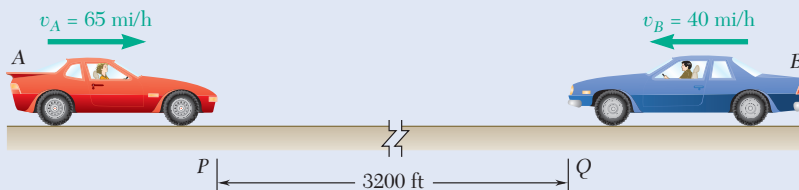


Fig. P11.43

11.44 An elevator is moving upward at a constant speed of 4 m/s. A man standing 10 m above the top of the elevator throws a ball upward with a speed of 3 m/s. Determine (a) when the ball will hit the elevator, (b) where the ball will hit the elevator with respect to the location of the man.

11.45 Two rockets are launched at a fireworks display. Rocket *A* is launched with an initial velocity $v_0 = 100 \text{ m/s}$ and rocket *B* is launched t_1 seconds later with the same initial velocity. The two rockets are timed to explode simultaneously at a height of 300 m as *A* is falling and *B* is rising. Assuming a constant acceleration $g = 9.81 \text{ m/s}^2$, determine (a) the time t_1 , (b) the velocity of *B* relative to *A* at the time of the explosion.

11.46 Car *A* is parked along the northbound lane of a highway, and car *B* is traveling in the southbound lane at a constant speed of 60 mi/h. At $t = 0$, *A* starts and accelerates at a constant rate a_A , while at $t = 5 \text{ s}$, *B* begins to slow down with a constant deceleration of magnitude $a_A/6$. Knowing that when the cars pass each other $x = 294 \text{ ft}$ and $v_A = v_B$, determine (a) the acceleration a_A , (b) when the vehicles pass each other, (c) the distance d between the vehicles at $t = 0$.

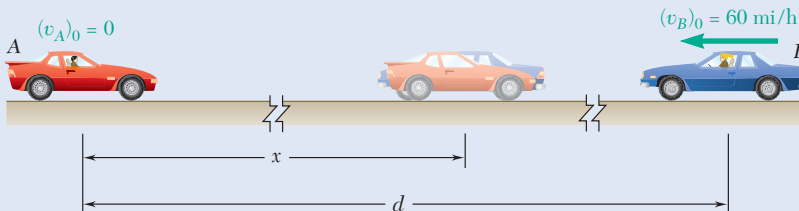


Fig. P11.46

- 11.47** The elevator E shown in the figure moves downward with a constant velocity of 4 m/s. Determine (a) the velocity of the cable C , (b) the velocity of the counterweight W , (c) the relative velocity of the cable C with respect to the elevator, (d) the relative velocity of the counterweight W with respect to the elevator.
- 11.48** The elevator E shown starts from rest and moves upward with a constant acceleration. If the counterweight W moves through 30 ft in 5 s, determine (a) the acceleration of the elevator and the cable C , (b) the velocity of the elevator after 5 s.
- 11.49** An athlete pulls handle A to the left with a constant velocity of 0.5 m/s. Determine (a) the velocity of the weight B , (b) the relative velocity of weight B with respect to the handle A .

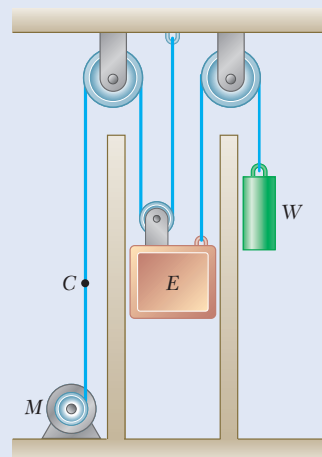


Fig. P11.47 and P11.48

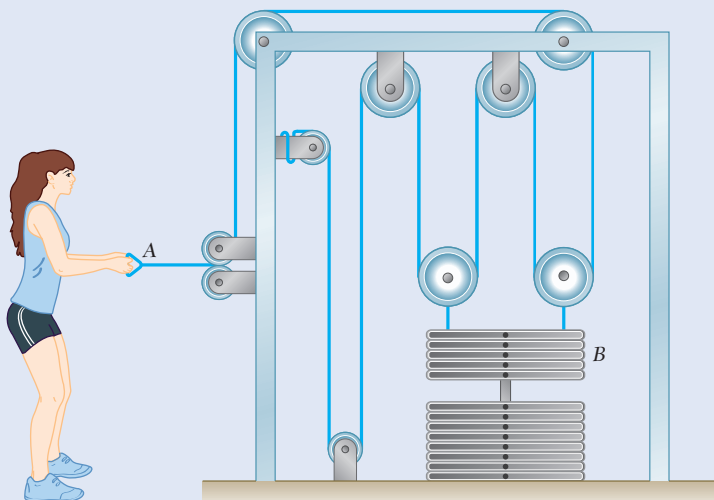


Fig. P11.49

- 11.50** An athlete pulls handle A to the left with a constant acceleration. Knowing that after the weight B has been lifted 4 in. its velocity is 2 ft/s, determine (a) the accelerations of handle A and weight B , (b) the velocity and change in position of handle A after 0.5 sec.

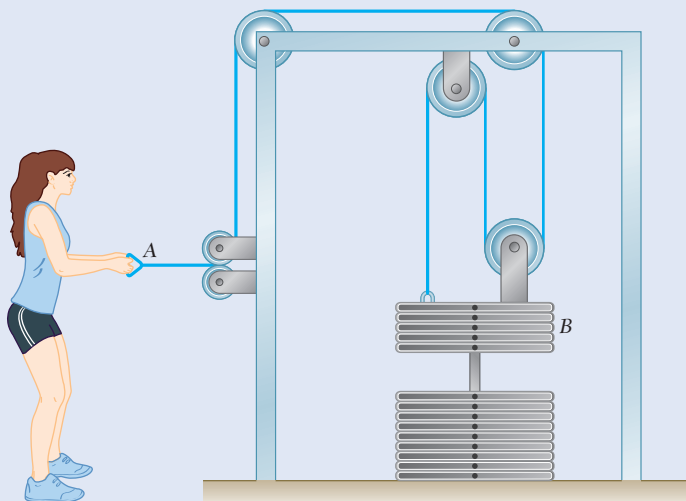


Fig. P11.50

- 11.51** Slider block B moves to the right with a constant velocity of 300 mm/s. Determine (a) the velocity of slider block A , (b) the velocity of portion C of the cable, (c) the velocity of portion D of the cable, (d) the relative velocity of portion C of the cable with respect to slider block A .

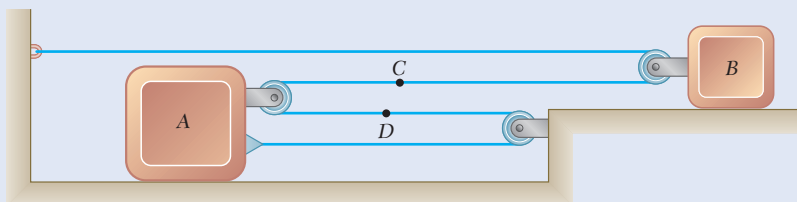


Fig. P11.51 and P11.52

- 11.52** At the instant shown, slider block B is moving with a constant acceleration, and its speed is 150 mm/s. Knowing that after slider block A has moved 240 mm to the right its velocity is 60 mm/s, determine (a) the accelerations of A and B , (b) the acceleration of portion D of the cable, (c) the velocity and the change in position of slider block B after 4 s.

- 11.53** A farmer lifts his hay bales into the top loft of his barn by walking his horse forward with a constant velocity of 1 ft/s. Determine the velocity and acceleration of the hay bale when the horse is 10 ft away from the barn.

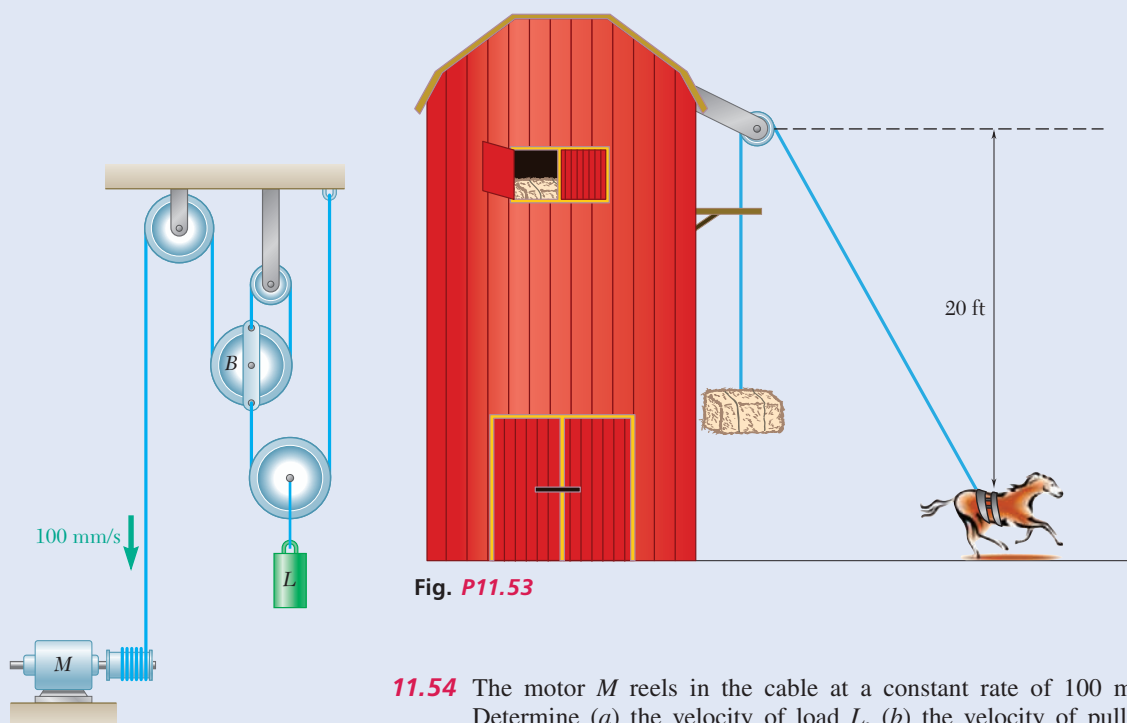


Fig. P11.53

- 11.54** The motor M reels in the cable at a constant rate of 100 mm/s. Determine (a) the velocity of load L , (b) the velocity of pulley B with respect to load L .

Fig. P11.54

- 11.55** Collar *A* starts from rest at $t = 0$ and moves upward with a constant acceleration of 3.6 in./s^2 . Knowing that collar *B* moves downward with a constant velocity of 18 in./s , determine (a) the time at which the velocity of block *C* is zero, (b) the corresponding position of block *C*.
- 11.56** Block *A* starts from rest at $t = 0$ and moves downward with a constant acceleration of 6 in./s^2 . Knowing that block *B* moves up with a constant velocity of 3 in./s , determine (a) the time when the velocity of block *C* is zero, (b) the corresponding position of block *C*.

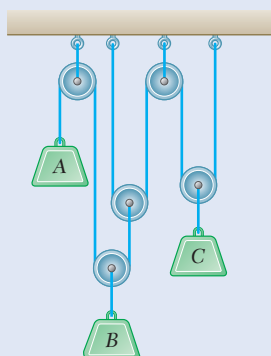


Fig. P11.56

- 11.57** Block *B* starts from rest, block *A* moves with a constant acceleration, and slider block *C* moves to the right with a constant acceleration of 75 mm/s^2 . Knowing that at $t = 2 \text{ s}$ the velocities of *B* and *C* are 480 mm/s downward and 280 mm/s to the right, respectively, determine (a) the accelerations of *A* and *B*, (b) the initial velocities of *A* and *C*, (c) the change in position of slider block *C* after 3 s .
- 11.58** Block *B* moves downward with a constant velocity of 20 mm/s . At $t = 0$, block *A* is moving upward with a constant acceleration, and its velocity is 30 mm/s . Knowing that at $t = 3 \text{ s}$ slider block *C* has moved 57 mm to the right, determine (a) the velocity of slider block *C* at $t = 0$, (b) the accelerations of *A* and *C*, (c) the change in position of block *A* after 5 s .
- 11.59** The system shown starts from rest, and each component moves with a constant acceleration. If the relative acceleration of block *C* with respect to collar *B* is 60 mm/s^2 upward and the relative acceleration of block *D* with respect to block *A* is 110 mm/s^2 downward, determine (a) the velocity of block *C* after 3 s , (b) the change in position of block *D* after 5 s .
- *11.60** The system shown starts from rest, and the length of the upper cord is adjusted so that *A*, *B*, and *C* are initially at the same level. Each component moves with a constant acceleration, and after 2 s the relative change in position of block *C* with respect to block *A* is 280 mm upward. Knowing that when the relative velocity of collar *B* with respect to block *A* is 80 mm/s downward, the displacements of *A* and *B* are 160 mm downward and 320 mm downward, respectively, determine (a) the accelerations of *A* and *B* if $a_B > 10 \text{ mm/s}^2$, (b) the change in position of block *D* when the velocity of block *C* is 600 mm/s upward.

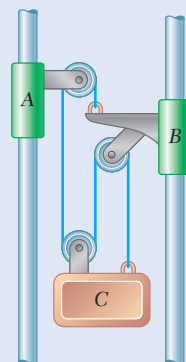


Fig. P11.55

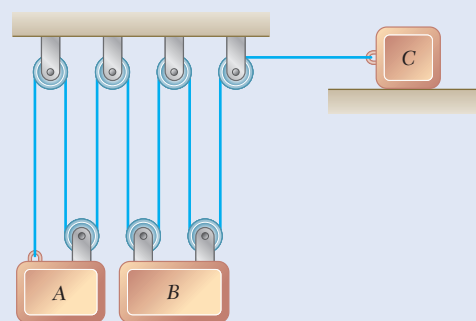


Fig. P11.57 and P11.58

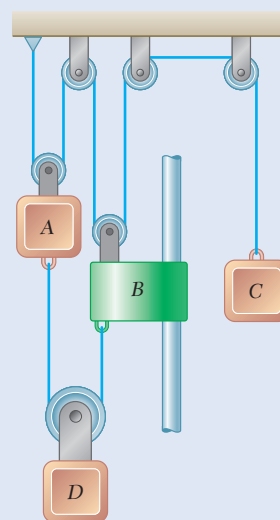


Fig. P11.59 and P11.60

*11.3 GRAPHICAL SOLUTIONS

In analyzing problems in rectilinear motion, it is often useful to draw graphs of position, velocity, or acceleration versus time. Sometimes these graphs can provide insight into the situation by indicating when quantities increase, decrease, or stay the same. In other cases, the graphs can provide numerical solutions when analytical methods are not available. In many experimental situations, data are collected as a function of time, and the methods of this section are very useful for the analysis.

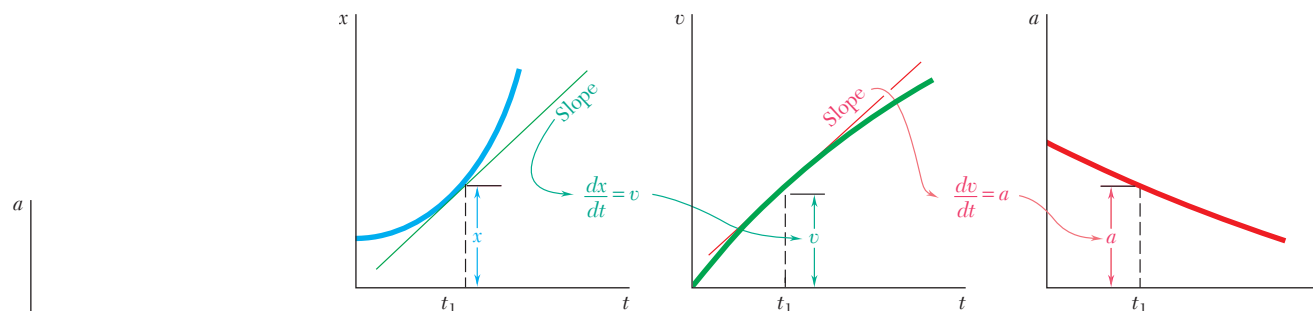


Fig. 11.10 The slope of an x - t curve at time t_1 equals the velocity v at that time; the slope of the v - t curve at time t_1 equals the acceleration a at that time.

We observed in Sec. 11.1 that the fundamental formulas

$$v = \frac{dx}{dt} \quad \text{and} \quad a = \frac{dv}{dt}$$

have a geometrical significance. The first formula says that the velocity at any instant is equal to the slope of the x - t curve at that instant (Fig. 11.10). The second formula states that the acceleration is equal to the slope of the v - t curve. We can use these two properties to determine graphically the v - t and a - t curves of a motion when the x - t curve is known.

Integrating the two fundamental formulas from a time t_1 to a time t_2 , we have

$$x_2 - x_1 = \int_{t_1}^{t_2} v dt \quad \text{and} \quad v_2 - v_1 = \int_{t_1}^{t_2} a dt \quad (11.12)$$

The first formula says that the area measured under the v - t curve from t_1 to t_2 is equal to the change in x during that time interval (Fig. 11.11). Similarly, the second formula states that the area measured under the a - t curve from t_1 to t_2 is equal to the change in v during that time interval. We can use these two properties to determine graphically the x - t curve of a motion when its v - t curve or its a - t curve is known (see Sample Prob. 11.9).

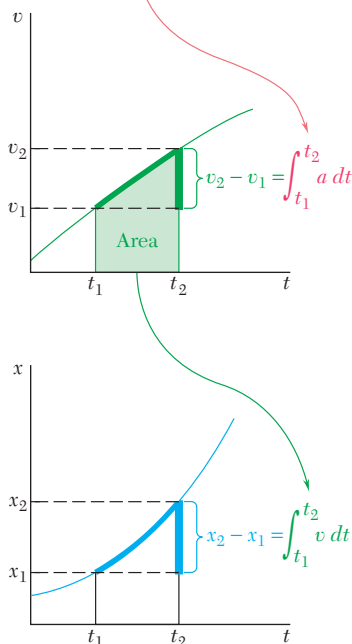
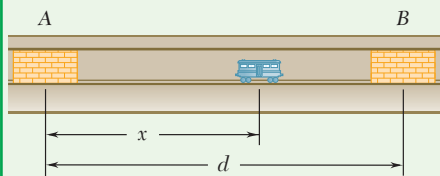


Fig. 11.11 The area under an a - t curve equals the change in velocity during that time interval; the area under the v - t curve equals the change in position during that time interval.

Graphical solutions are particularly useful when the motion considered is defined from experimental data and when x , v , and a are not analytical functions of t . They also can be used to advantage when the motion consists of distinct parts and when its analysis requires writing a different equation for each of its parts. When using a graphical solution, however, be careful to note that (1) the area under the v - t curve measures the *change in x* —not x itself—and similarly, that the area under the a - t curve measures the change in v ; (2) an area above the t axis corresponds to an *increase* in x or v , whereas an area located below the t axis measures a *decrease* in x or v .

In drawing motion curves, it is useful to remember that, if the velocity is constant, it is represented by a horizontal straight line; the position coordinate x is then a linear function of t and is represented by an oblique straight line. If the acceleration is constant and different from zero, it is represented by a horizontal straight line; v is then a linear function of t and is represented by an oblique straight line, and x is a second-degree polynomial in t and is represented by a parabola. If the acceleration is a linear function of t , the velocity and the position coordinate are equal, respectively, to second-degree and third-degree polynomials; a is then represented by an oblique straight line, v by a parabola, and x by a cubic. In general, if the acceleration is a polynomial of degree n in t , the velocity is a polynomial of degree $n + 1$, and the position coordinate is a polynomial of degree $n + 2$. These polynomials are represented by motion curves of a corresponding degree.

Sample Problem 11.9



A subway car leaves station A; it gains speed at the rate of 4 ft/s^2 for 6 s and then at the rate of 6 ft/s^2 until it has reached the speed of 48 ft/s . The car maintains the same speed until it approaches station B; then the driver applies the brakes, giving the car a constant deceleration and bringing it to a stop in 6 s. The total running time from A to B is 40 s. Draw the a - t , v - t , and x - t curves, and determine the distance between stations A and B.

STRATEGY: You are given acceleration data, so first draw the graph of a versus t . You can calculate areas under the curve to determine the v - t curve and calculate areas under the v - t curve to determine the x - t curve.

MODELING and ANALYSIS: You can model the subway car as a particle without drag.

Acceleration–Time Curve. Since the acceleration is either constant or zero, the a - t curve consists of horizontal straight-line segments. Determine the values of t_2 and a_4 as

$$\begin{aligned} 0 < t < 6: & \quad \text{Change in } v = \text{area under } a\text{-}t \text{ curve} \\ & \quad v_6 - 0 = (6 \text{ s})(4 \text{ ft/s}^2) = 24 \text{ ft/s} \\ 6 < t < t_2: & \quad \text{Since the velocity increases from 24 to 48 ft/s,} \\ & \quad \text{Change in } v = \text{area under } a\text{-}t \text{ curve} \\ & \quad 48 \text{ ft/s} - 24 \text{ ft/s} = (t_2 - 6)(6 \text{ ft/s}^2) \quad t_2 = 10 \text{ s} \\ t_2 < t < 34: & \quad \text{Since the velocity is constant, the acceleration is zero.} \\ 34 < t < 40: & \quad \text{Change in } v = \text{area under } a\text{-}t \text{ curve} \\ & \quad 0 - 48 \text{ ft/s} = (6 \text{ s})a_4 \quad a_4 = -8 \text{ ft/s}^2 \end{aligned}$$

The acceleration is negative, so the corresponding area is below the t axis; this area represents a decrease in velocity (Fig. 1).

Velocity–Time Curve. Since the acceleration is either constant or zero, the v - t curve consists of straight-line segments connecting the points determined previously (Fig. 2).

$$\begin{aligned} \text{Change in } x &= \text{area under } v\text{-}t \text{ curve} \\ 0 < t < 6: & \quad x_6 - 0 = \frac{1}{2}(6)(24) = 72 \text{ ft} \\ 6 < t < 10: & \quad x_{10} - x_6 = \frac{1}{2}(4)(24 + 48) = 144 \text{ ft} \\ 10 < t < 34: & \quad x_{34} - x_{10} = (24)(48) = 1152 \text{ ft} \\ 34 < t < 40: & \quad x_{40} - x_{34} = \frac{1}{2}(6)(48) = 144 \text{ ft} \end{aligned}$$

Adding the changes in x gives you the distance from A to B:

$$d = x_{40} - 0 = 1512 \text{ ft}$$

$$d = 1512 \text{ ft} \quad \blacktriangleleft$$

Position–Time Curve. The points determined previously should be joined by three parabolic arcs and one straight-line segment (Fig. 3). In constructing the x - t curve, keep in mind that for any value of t , the slope of the tangent to the x - t curve is equal to the value of v at that instant.

REFLECT and THINK: This problem also could have been solved using the uniform motion equations for each interval of time that has a different acceleration, but it would have been much more difficult and time consuming. For a real subway car, the acceleration does not instantaneously change from one value to another.

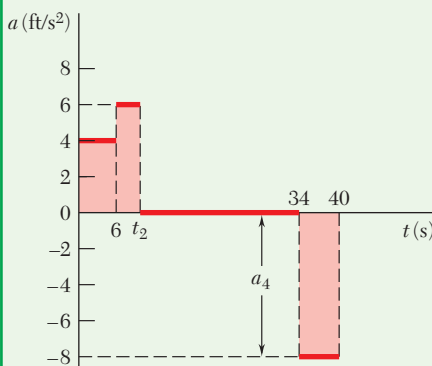


Fig. 1 Acceleration of the subway car as a function of time.

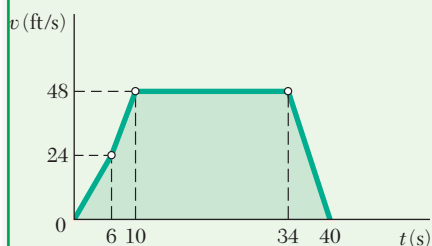


Fig. 2 Velocity of the subway car as a function of time.

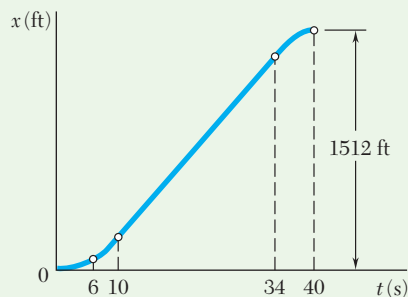


Fig. 3 Position of the subway car as a function of time.

SOLVING PROBLEMS ON YOUR OWN

In this section, we reviewed and developed several **graphical techniques** for the solution of problems involving rectilinear motion. These techniques can be used to solve problems directly or to complement analytical methods of solution by providing a visual description, and thus a better understanding, of the motion of a given body. We suggest that you sketch one or more motion curves for several of the problems in this section, even if these problems are not part of your homework assignment.

1. Drawing x - t , v - t , and a - t curves and applying graphical methods. We described the following properties in Sec. 11.3, and they should be kept in mind as you use a graphical method of solution.

a. The slopes of the x - t and v - t curves at a time t_1 are equal to the velocity and the acceleration at time t_1 , respectively.

b. The areas under the a - t and v - t curves between the times t_1 and t_2 are equal to the change Δv in the velocity and to the change Δx in the position coordinate, respectively, during that time interval.

c. If you know one of the motion curves, the fundamental properties we have summarized in paragraphs *a* and *b* will enable you to construct the other two curves. However, when using the properties of paragraph *b*, you must know the velocity and the position coordinate at time t_1 in order to determine the velocity and the position coordinate at time t_2 . Thus, in Sample Prob. 11.9, knowing that the initial value of the velocity was zero allowed us to find the velocity at $t = 6$ s: $v_6 = v_0 + \Delta v = 0 + 24 \text{ ft/s} = 24 \text{ ft/s}$.

If you have studied the shear and bending-moment diagrams for a beam previously, you should recognize the analogy between the three motion curves and the three diagrams representing, respectively, the distributed load, the shear, and the bending moment in the beam. Thus, any techniques that you have learned regarding the construction of these diagrams can be applied when drawing the motion curves.

2. Using approximate methods. When the a - t and v - t curves are not represented by analytical functions or when they are based on experimental data, it is often necessary to use approximate methods to calculate the areas under these curves. In those cases, the given area is approximated by a series of rectangles of width Δt . The smaller the value of Δt , the smaller is the error introduced by the approximation. You can obtain the velocity and the position coordinate from

$$v = v_0 + \sum a_{\text{ave}} \Delta t \quad x = x_0 + \sum v_{\text{ave}} \Delta t$$

where a_{ave} and v_{ave} are the heights of an acceleration rectangle and a velocity rectangle, respectively.

Problems

- 11.61** A particle moves in a straight line with a constant acceleration of -4 ft/s^2 for 6 s, zero acceleration for the next 4 s, and a constant acceleration of $+4 \text{ ft/s}^2$ for the next 4 s. Knowing that the particle starts from the origin and that its velocity is -8 ft/s during the zero acceleration time interval, (a) construct the v - t and x - t curves for $0 \leq t \leq 14 \text{ s}$, (b) determine the position and the velocity of the particle and the total distance traveled when $t = 14 \text{ s}$.

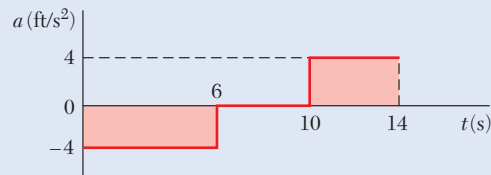


Fig. P11.61 and P11.62

- 11.62** A particle moves in a straight line with a constant acceleration of -4 ft/s^2 for 6 s, zero acceleration for the next 4 s, and a constant acceleration of $+4 \text{ ft/s}^2$ for the next 4 s. Knowing that the particle starts from the origin with $v_0 = 16 \text{ ft/s}$, (a) construct the v - t and x - t curves for $0 \leq t \leq 14 \text{ s}$, (b) determine the amount of time during which the particle is further than 16 ft from the origin.
- 11.63** A particle moves in a straight line with the velocity shown in the figure. Knowing that $x = -540 \text{ m}$ at $t = 0$, (a) construct the a - t and x - t curves for $0 < t < 50 \text{ s}$, and determine (b) the total distance traveled by the particle when $t = 50 \text{ s}$, (c) the two times at which $x = 0$.

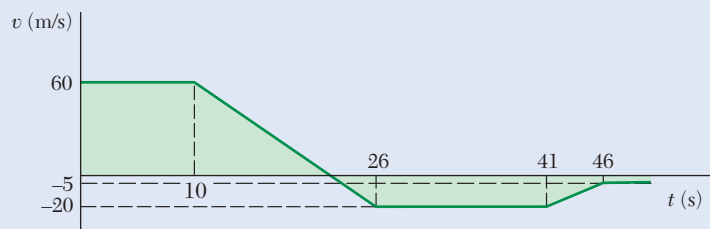


Fig. P11.63 and P11.64

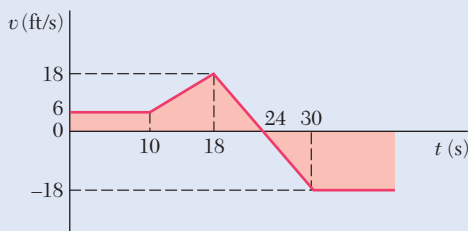


Fig. P11.65

- 11.64** A particle moves in a straight line with the velocity shown in the figure. Knowing that $x = -540 \text{ m}$ at $t = 0$, (a) construct the a - t and x - t curves for $0 < t < 50 \text{ s}$, and determine (b) the maximum value of the position coordinate of the particle, (c) the values of t for which the particle is at $x = 100 \text{ m}$.
- 11.65** A particle moves in a straight line with the velocity shown in the figure. Knowing that $x = -48 \text{ ft}$ at $t = 0$, draw the a - t and x - t curves for $0 < t < 40 \text{ s}$ and determine (a) the maximum value of the position coordinate of the particle, (b) the values of t for which the particle is at a distance of 108 ft from the origin.

- 11.66** A parachutist is in free fall at a rate of 200 km/h when he opens his parachute at an altitude of 600 m. Following a rapid and constant deceleration, he then descends at a constant rate of 50 km/h from 586 m to 30 m, where he maneuvers the parachute into the wind to further slow his descent. Knowing that the parachutist lands with a negligible downward velocity, determine (a) the time required for the parachutist to land after opening his parachute, (b) the initial deceleration.

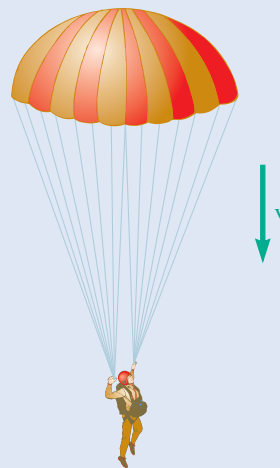


Fig. P11.66

- 11.67** A commuter train traveling at 40 mi/h is 3 mi from a station. The train then decelerates so that its speed is 20 mi/h when it is 0.5 mi from the station. Knowing that the train arrives at the station 7.5 min after beginning to decelerate and assuming constant decelerations, determine (a) the time required for the train to travel the first 2.5 mi, (b) the speed of the train as it arrives at the station, (c) the final constant deceleration of the train.

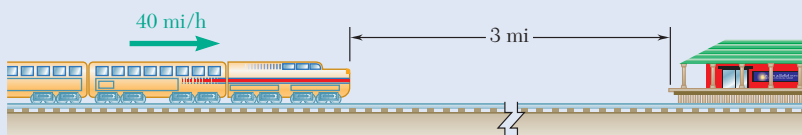


Fig. P11.67

- 11.68** A temperature sensor is attached to slider AB which moves back and forth through 60 in. The maximum velocities of the slider are 12 in./s to the right and 30 in./s to the left. When the slider is moving to the right, it accelerates and decelerates at a constant rate of 6 in./s²; when moving to the left, the slider accelerates and decelerates at a constant rate of 20 in./s². Determine the time required for the slider to complete a full cycle, and construct the v - t and x - t curves of its motion.

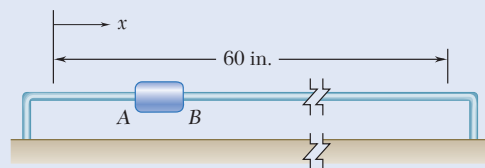


Fig. P11.68

- 11.69** In a water-tank test involving the launching of a small model boat, the model's initial horizontal velocity is 6 m/s and its horizontal acceleration varies linearly from -12 m/s^2 at $t = 0$ to -2 m/s^2 at $t = t_1$ and then remains equal to -2 m/s^2 until $t = 1.4 \text{ s}$. Knowing that $v = 1.8 \text{ m/s}$ when $t = t_1$, determine (a) the value of t_1 , (b) the velocity and the position of the model at $t = 1.4 \text{ s}$.

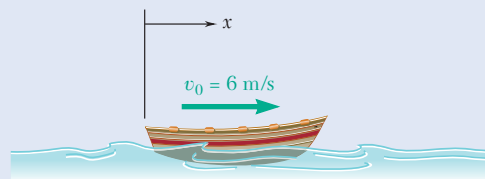


Fig. P11.69

- 11.70** The acceleration record shown was obtained for a small airplane traveling along a straight course. Knowing that $x = 0$ and $v = 60 \text{ m/s}$ when $t = 0$, determine (a) the velocity and position of the plane at $t = 20 \text{ s}$, (b) its average velocity during the interval $6 \text{ s} < t < 14 \text{ s}$.

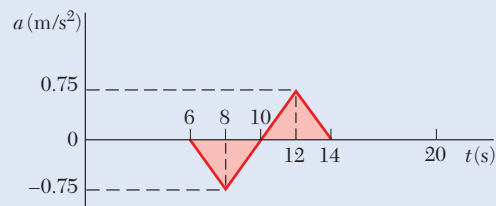


Fig. P11.70

- 11.71** In a 400-m race, runner A reaches her maximum velocity v_A in 4 s with constant acceleration and maintains that velocity until she reaches the halfway point with a split time of 25 s. Runner B reaches her maximum velocity v_B in 5 s with constant acceleration and maintains that velocity until she reaches the halfway point with a split time of 25.2 s. Both runners then run the second half of the race with the same constant deceleration of 0.1 m/s^2 . Determine (a) the race times for both runners, (b) the position of the winner relative to the loser when the winner reaches the finish line.

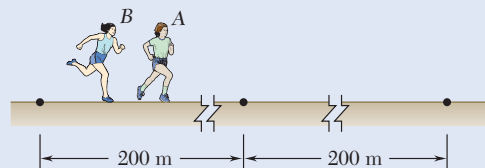


Fig. P11.71

- 11.72** A car and a truck are both traveling at the constant speed of 35 mi/h; the car is 40 ft behind the truck. The driver of the car wants to pass the truck, i.e., he wishes to place his car at *B*, 40 ft in front of the truck, and then resume the speed of 35 mi/h. The maximum acceleration of the car is 5 ft/s^2 and the maximum deceleration obtained by applying the brakes is 20 ft/s^2 . What is the shortest time in which the driver of the car can complete the passing operation if he does not at any time exceed a speed of 50 mi/h? Draw the v - t curve.

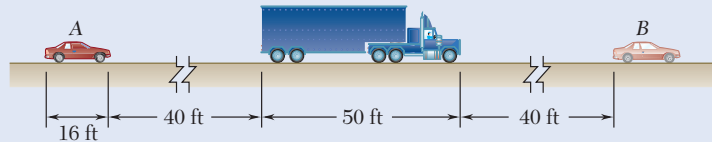


Fig. P11.72

- 11.73** Solve Prob. 11.72, assuming that the driver of the car does not pay any attention to the speed limit while passing and concentrates on reaching position *B* and resuming a speed of 35 mi/h in the shortest possible time. What is the maximum speed reached? Draw the v - t curve.

- 11.74** Car *A* is traveling on a highway at a constant speed $(v_A)_0 = 60 \text{ mi/h}$ and is 380 ft from the entrance of an access ramp when car *B* enters the acceleration lane at that point at a speed $(v_B)_0 = 15 \text{ mi/h}$. Car *B* accelerates uniformly and enters the main traffic lane after traveling 200 ft in 5 s. It then continues to accelerate at the same rate until it reaches a speed of 60 mi/h, which it then maintains. Determine the final distance between the two cars.

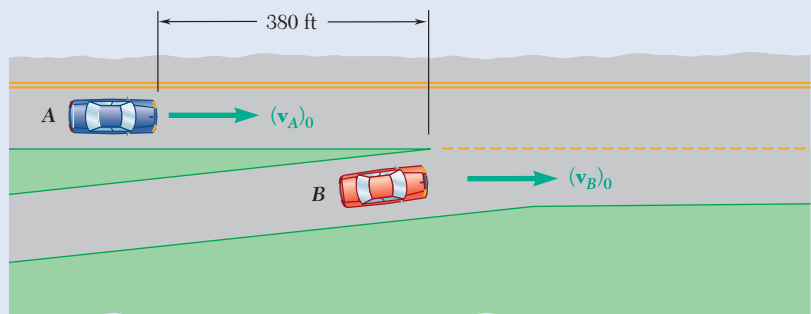


Fig. P11.74

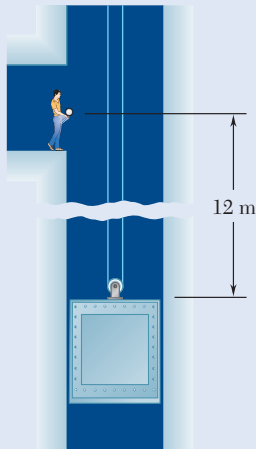


Fig. P11.75

- 11.75** An elevator starts from rest and moves upward, accelerating at a rate of 1.2 m/s^2 until it reaches a speed of 7.8 m/s , which it then maintains. Two seconds after the elevator begins to move, a man standing 12 m above the initial position of the top of the elevator throws a ball upward with an initial velocity of 20 m/s . Determine when the ball will hit the elevator.

- 11.76** Car A is traveling at 40 mi/h when it enters a 30 mi/h speed zone. The driver of car A decelerates at a rate of 16 ft/s^2 until reaching a speed of 30 mi/h, which she then maintains. When car B, which was initially 60 ft behind car A and traveling at a constant speed of 45 mi/h, enters the speed zone, its driver decelerates at a rate of 20 ft/s^2 until reaching a speed of 28 mi/h. Knowing that the driver of car B maintains a speed of 28 mi/h, determine (a) the closest that car B comes to car A, (b) the time at which car A is 70 ft in front of car B.

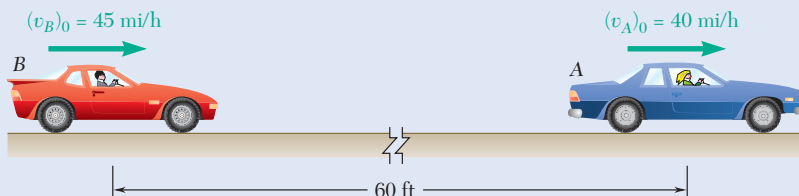


Fig. P11.76

- 11.77** An accelerometer record for the motion of a given part of a mechanism is approximated by an arc of a parabola for 0.2 s and a straight line for the next 0.2 s as shown in the figure. Knowing that $v = 0$ when $t = 0$ and $x = 0.8 \text{ ft}$ when $t = 0.4 \text{ s}$, (a) construct the $v-t$ curve for $0 \leq t \leq 0.4 \text{ s}$, (b) determine the position of the part at $t = 0.3 \text{ s}$ and $t = 0.2 \text{ s}$.

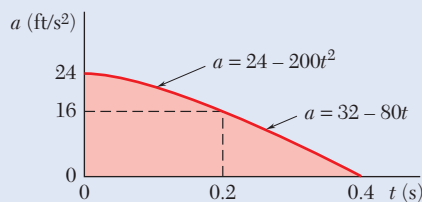


Fig. P11.77

- 11.78** A car is traveling at a constant speed of 54 km/h when its driver sees a child run into the road. The driver applies her brakes until the child returns to the sidewalk and then accelerates to resume her original speed of 54 km/h; the acceleration record of the car is shown in the figure. Assuming $x = 0$ when $t = 0$, determine (a) the time t_1 at which the velocity is again 54 km/h, (b) the position of the car at that time, (c) the average velocity of the car during the interval $1 \text{ s} \leq t \leq t_1$.

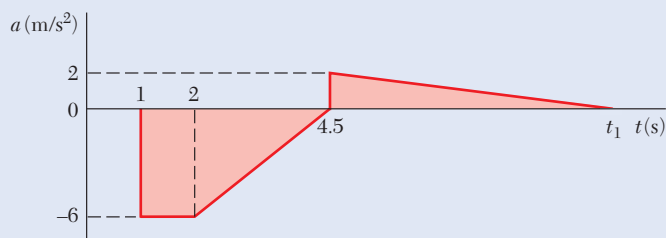


Fig. P11.78

11.79 An airport shuttle train travels between two terminals that are 1.6 mi apart. To maintain passenger comfort, the acceleration of the train is limited to $\pm 4 \text{ ft/s}^2$, and the jerk, or rate of change of acceleration, is limited to $\pm 0.8 \text{ ft/s}^2$ per second. If the shuttle has a maximum speed of 20 mi/h, determine (a) the shortest time for the shuttle to travel between the two terminals, (b) the corresponding average velocity of the shuttle.

11.80 During a manufacturing process, a conveyor belt starts from rest and travels a total of 1.2 ft before temporarily coming to rest. Knowing that the jerk, or rate of change of acceleration, is limited to $\pm 4.8 \text{ ft/s}^2$ per second, determine (a) the shortest time required for the belt to move 1.2 ft, (b) the maximum and average values of the velocity of the belt during that time.

11.81 Two seconds are required to bring the piston rod of an air cylinder to rest; the acceleration record of the piston rod during the 2 s is as shown. Determine by approximate means (a) the initial velocity of the piston rod, (b) the distance traveled by the piston rod as it is brought to rest.

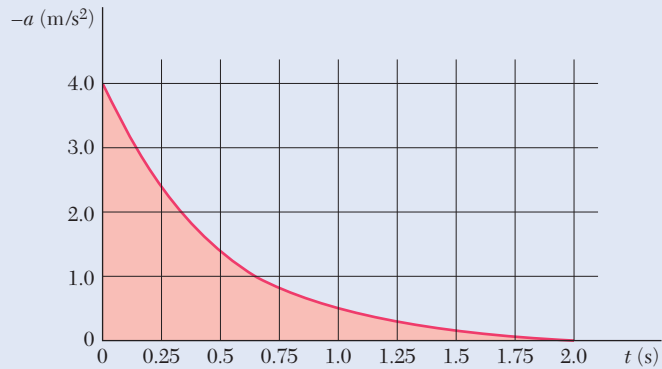


Fig. P11.81

11.82 The acceleration record shown was obtained during the speed trials of a sports car. Knowing that the car starts from rest, determine by approximate means (a) the velocity of the car at $t = 8 \text{ s}$, (b) the distance the car has traveled at $t = 20 \text{ s}$.

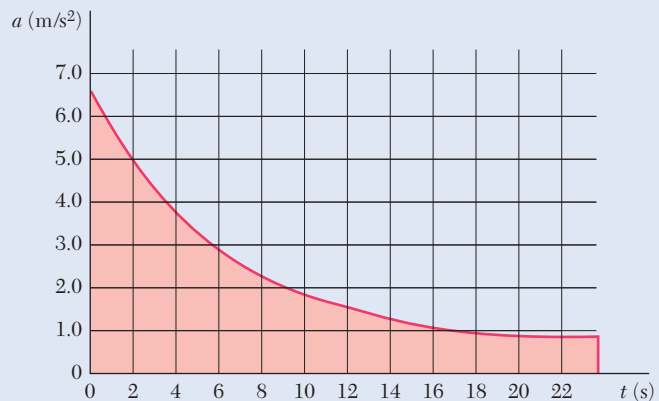


Fig. P11.82

- 11.83** A training airplane has a velocity of 126 ft/s when it lands on an aircraft carrier. As the arresting gear of the carrier brings the airplane to rest, the velocity and the acceleration of the airplane are recorded; the results are shown (solid curve) in the figure. Determine by approximate means (a) the time required for the airplane to come to rest, (b) the distance traveled in that time.

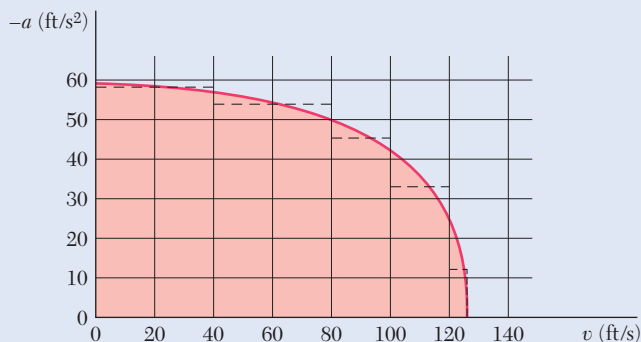


Fig. P11.83

- 11.84** Shown in the figure is a portion of the experimentally determined v - x curve for a shuttle cart. Determine by approximate means the acceleration of the cart when (a) $x = 10$ in., (b) $v = 80$ in./s.

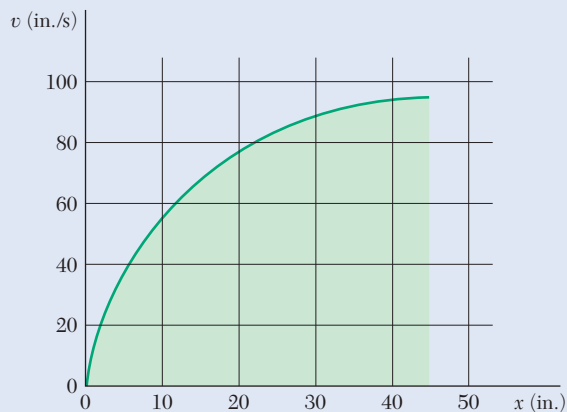


Fig. P11.84

- 11.85** An elevator starts from rest and rises 40 m to its maximum velocity in T s with the acceleration record shown in the figure. Determine (a) the required time T , (b) the maximum velocity, (c) the velocity and position of the elevator at $t = T/2$.

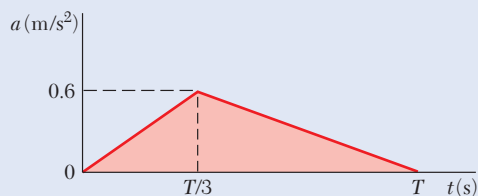


Fig. P11.85

- 11.86** Two road rally checkpoints *A* and *B* are located on the same highway and are 8 mi apart. The speed limits for the first 5 mi and the last 3 mi are 60 mi/h and 35 mi/h, respectively. Drivers must stop at each checkpoint, and the specified time between points *A* and *B* is 10 min 20 s. Knowing that the driver accelerates and decelerates at the same constant rate, determine the magnitude of her acceleration if she travels at the speed limit as much as possible.

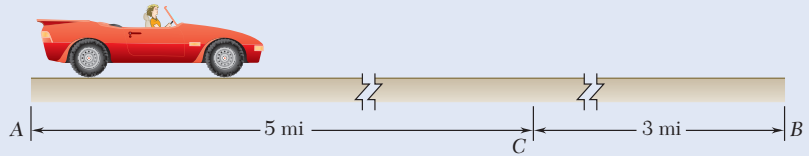


Fig. P11.86

- 11.87** As shown in the figure, from $t = 0$ to $t = 4$ s, the acceleration of a given particle is represented by a parabola. Knowing that $x = 0$ and $v = 8$ m/s when $t = 0$, (a) construct the v - t and x - t curves for $0 < t < 4$ s, (b) determine the position of the particle at $t = 3$ s. (Hint: Use table inside the front cover.)

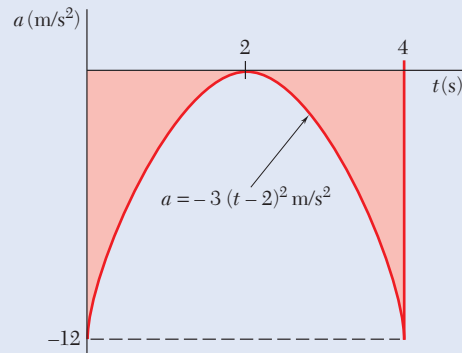


Fig. P11.87

- 11.88** A particle moves in a straight line with the acceleration shown in the figure. Knowing that the particle starts from the origin with $v_0 = -2$ m/s, (a) construct the v - t and x - t curves for $0 < t < 18$ s, (b) determine the position and the velocity of the particle and the total distance traveled when $t = 18$ s.

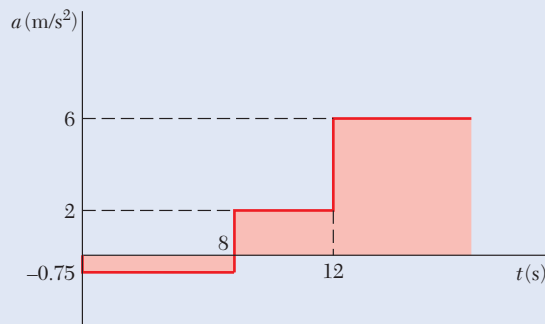


Fig. P11.88

11.4 CURVILINEAR MOTION OF PARTICLES

When a particle moves along a curve other than a straight line, we say that the particle is in **curvilinear motion**. We can use position, velocity, and acceleration to describe the motion, but now we must treat these quantities as vectors because they can have directions in two or three dimensions.

11.4A Position, Velocity, and Acceleration Vectors

To define the position P occupied by a particle in curvilinear motion at a given time t , we select a fixed reference system, such as the x, y, z axes shown in Fig. 11.12a, and draw the vector $\mathbf{r}$ joining the origin O and point P . The vector $\mathbf{r}$ is characterized by its magnitude r and its direction with respect to the reference axes, so it completely defines the position of the particle with respect to those axes. We refer to vector $\mathbf{r}$ as the **position vector** of the particle at time t .

Consider now the vector $\mathbf{r}'$ defining the position P' occupied by the same particle at a later time $t + \Delta t$. The vector $\Delta \mathbf{r}$ joining P and P' represents the change in the position vector during the time interval Δt and is called the **displacement vector**. We can check this directly from Fig. 11.12a, where we obtain the vector $\mathbf{r}'$ by adding the vectors $\mathbf{r}$ and $\Delta \mathbf{r}$ according to the triangle rule. Note that $\Delta \mathbf{r}$ represents a change in *direction* as well as a change in *magnitude* of the position vector $\mathbf{r}$.

We define the **average velocity** of the particle over the time interval Δt as the quotient of $\Delta \mathbf{r}$ and Δt . Since $\Delta \mathbf{r}$ is a vector and Δt is a scalar, the quotient $\Delta \mathbf{r}/\Delta t$ is a vector attached at P with the same direction as $\Delta \mathbf{r}$ and a magnitude equal to the magnitude of $\Delta \mathbf{r}$ divided by Δt (Fig. 11.12b).

We obtain the **instantaneous velocity** of the particle at time t by taking the limit as the time interval Δt approaches zero. The instantaneous velocity is thus represented by the vector

$$\mathbf{v} = \lim_{\Delta t \rightarrow 0} \frac{\Delta \mathbf{r}}{\Delta t} \quad (11.13)$$

As Δt and $\Delta \mathbf{r}$ become shorter, the points P and P' get closer together. Thus, the vector $\mathbf{v}$ obtained in the limit must be tangent to the path of the particle (Fig. 11.12c).

Because the position vector $\mathbf{r}$ depends upon the time t , we can refer to it as a **vector function** of the scalar variable t and denote it by $\mathbf{r}(t)$. Extending the concept of the derivative of a scalar function introduced in elementary calculus, we refer to the limit of the quotient $\Delta \mathbf{r}/\Delta t$ as the **derivative** of the vector function $\mathbf{r}(t)$. We have

Velocity vector

$$\mathbf{v} = \frac{d\mathbf{r}}{dt} \quad (11.14)$$

The magnitude v of the vector $\mathbf{v}$ is called the **speed** of the particle. We can obtain the speed by substituting the magnitude of this vector,

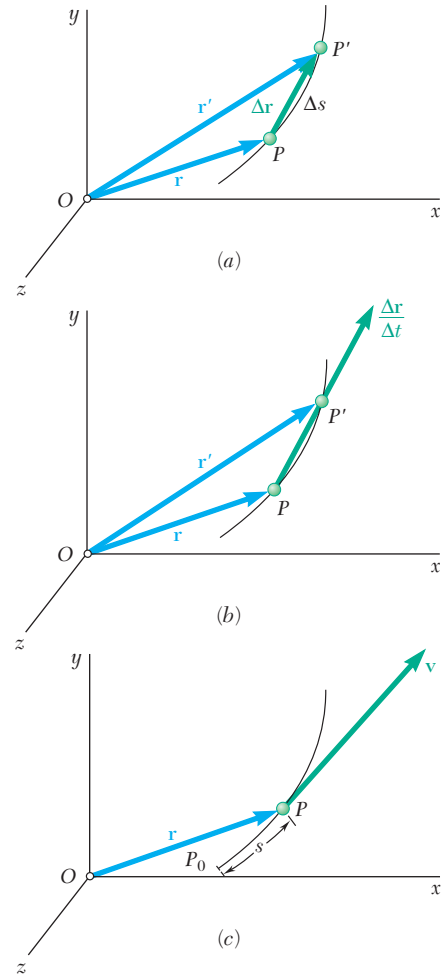


Fig. 11.12 (a) Position vectors for a particle moving along a curve from P to P' ; (b) the average velocity vector is the quotient of the change in position to the elapsed time interval; (c) the instantaneous velocity vector is tangent to the particle's path.

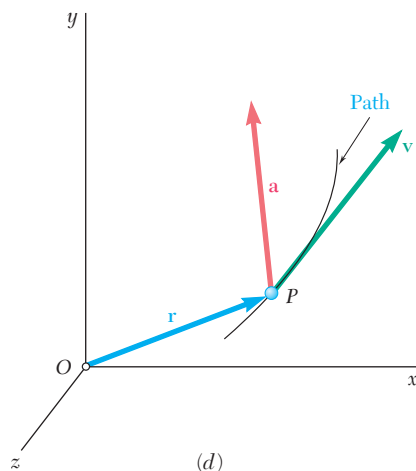
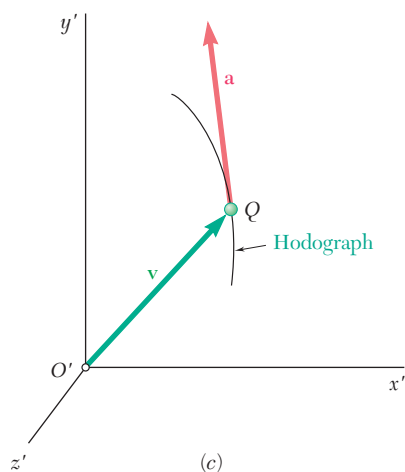
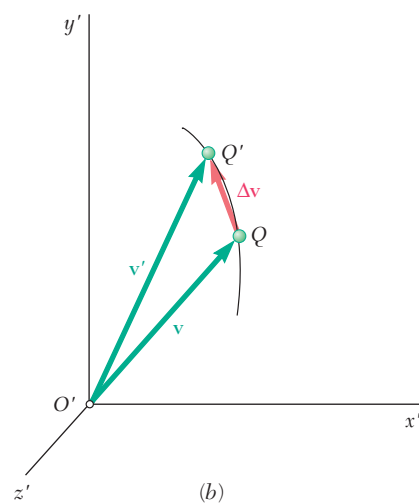
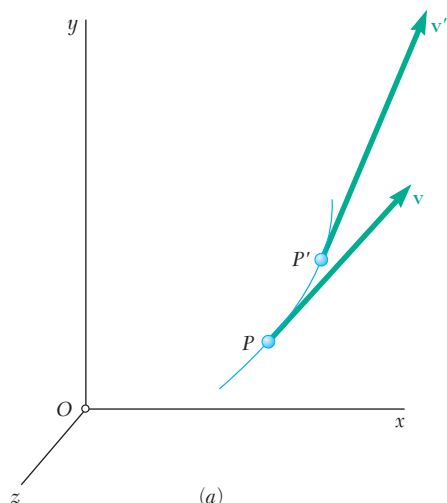


Fig. 11.13 (a) Velocities $\mathbf{v}$ and $\mathbf{v}'$ of a particle at two different times; (b) the vector change in the particle's velocity during the time interval; (c) the instantaneous acceleration vector is tangent to the hodograph; (d) in general, the acceleration vector is not tangent to the particle's path.

which is represented by the straight-line segment PP' , for the vector $\Delta \mathbf{r}$ in formula (11.13). However, the length of segment PP' approaches the length Δs of arc PP' as Δt decreases (Fig. 11.12a). Therefore, we can write

$$v = \lim_{\Delta t \rightarrow 0} \frac{PP'}{\Delta t} = \lim_{\Delta t \rightarrow 0} \frac{\Delta s}{\Delta t} \quad v = \frac{ds}{dt} \quad (11.15)$$

Thus, we obtain the speed v by finding the length s of the arc described by the particle and differentiating it with respect to t .

Now let's consider the velocity $\mathbf{v}$ of the particle at time t and its velocity $\mathbf{v}'$ at a later time $t + \Delta t$ (Fig. 11.13a). Let us draw both vectors $\mathbf{v}$ and $\mathbf{v}'$ from the same origin O' (Fig. 11.13b). The vector $\Delta \mathbf{v}$ joining Q and Q' represents the change in the velocity of the particle during the time interval Δt , since we can obtain the vector $\mathbf{v}'$ by adding the vectors $\mathbf{v}$ and $\Delta \mathbf{v}$. Again, note that $\Delta \mathbf{v}$ represents a change in the *direction* of the velocity as well as a change in *speed*. We define the **average acceleration** of the particle over the time interval Δt as the quotient of $\Delta \mathbf{v}$ and Δt . Since $\Delta \mathbf{v}$ is a vector and Δt is a scalar, the quotient $\Delta \mathbf{v}/\Delta t$ is a vector in the same direction as $\Delta \mathbf{v}$.

We obtain the **instantaneous acceleration** of the particle at time t by choosing increasingly smaller values for Δt and $\Delta \mathbf{v}$. The instantaneous acceleration is thus represented by the vector

$$\mathbf{a} = \lim_{\Delta t \rightarrow 0} \frac{\Delta \mathbf{v}}{\Delta t} \quad (11.16)$$

Noting that the velocity $\mathbf{v}$ is a vector function $\mathbf{v}(t)$ of the time t , we can refer to the limit of the quotient $\Delta \mathbf{v}/\Delta t$ as the derivative of $\mathbf{v}$ with respect to t . We have

Acceleration vector

$$\mathbf{a} = \frac{d\mathbf{v}}{dt} \quad (11.17)$$

Observe that the acceleration $\mathbf{a}$ is tangent to the curve described by the tip Q of the vector $\mathbf{v}$ when we draw $\mathbf{v}$ from a fixed origin O' (Fig. 11.13c). However, in general, the acceleration is *not* tangent to the path of the particle (Fig. 11.13d). The curve described by the tip of $\mathbf{v}$ and shown in Fig. 11.13c is called the *hodograph* of the motion.

11.4B Derivatives of Vector Functions

We have just seen that we can represent the velocity $\mathbf{v}$ of a particle in curvilinear motion by the derivative of the vector function $\mathbf{r}(t)$ characterizing the position of the particle. Similarly, we can represent the acceleration $\mathbf{a}$ of the particle by the derivative of the vector function $\mathbf{v}(t)$. Here we give a formal definition of the derivative of a vector function and establish a few rules governing the differentiation of sums and products of vector functions.

Let $\mathbf{P}(u)$ be a vector function of the scalar variable u . By that, we mean that the scalar u completely defines the magnitude and direction of the vector $\mathbf{P}$. If the vector $\mathbf{P}$ is drawn from a fixed origin O and the scalar u is allowed to vary, the tip of $\mathbf{P}$ describes a given curve in space. Consider the vectors $\mathbf{P}$ corresponding, respectively, to the values u and $u + \Delta u$ of the scalar variable (Fig. 11.14a). Let $\Delta \mathbf{P}$ be the vector joining the tips of the two given vectors. Then we have

$$\Delta \mathbf{P} = \mathbf{P}(u + \Delta u) - \mathbf{P}(u)$$

Dividing through by Δu and letting Δu approach zero, we define the derivative of the vector function $\mathbf{P}(u)$ as

$$\frac{d\mathbf{P}}{du} = \lim_{\Delta u \rightarrow 0} \frac{\Delta \mathbf{P}}{\Delta u} = \lim_{\Delta u \rightarrow 0} \frac{\mathbf{P}(u + \Delta u) - \mathbf{P}(u)}{\Delta u} \quad (11.18)$$

As Δu approaches zero, the line of action of $\Delta \mathbf{P}$ becomes tangent to the curve of Fig. 11.14a. Thus, the derivative $d\mathbf{P}/du$ of the vector function $\mathbf{P}(u)$ is *tangent to the curve described by the tip of $\mathbf{P}(u)$* (Fig. 11.14b).

The standard rules for the differentiation of the sums and products of scalar functions extend to vector functions. Consider first the **sum of two vector functions** $\mathbf{P}(u)$ and $\mathbf{Q}(u)$ of the same scalar variable u . According to the definition given in Eq. (11.18), the derivative of the vector $\mathbf{P} + \mathbf{Q}$ is

$$\frac{d(\mathbf{P} + \mathbf{Q})}{du} = \lim_{\Delta u \rightarrow 0} \frac{\Delta(\mathbf{P} + \mathbf{Q})}{\Delta u} = \lim_{\Delta u \rightarrow 0} \left(\frac{\Delta \mathbf{P}}{\Delta u} + \frac{\Delta \mathbf{Q}}{\Delta u} \right)$$

or since the limit of a sum is equal to the sum of the limits of its terms,

$$\frac{d(\mathbf{P} + \mathbf{Q})}{du} = \lim_{\Delta u \rightarrow 0} \frac{\Delta \mathbf{P}}{\Delta u} + \lim_{\Delta u \rightarrow 0} \frac{\Delta \mathbf{Q}}{\Delta u}$$

$$\frac{d(\mathbf{P} + \mathbf{Q})}{du} = \frac{d\mathbf{P}}{du} + \frac{d\mathbf{Q}}{du} \quad (11.19)$$

That is, the derivative of a sum of vector functions equals the sum of the derivative of each function separately.

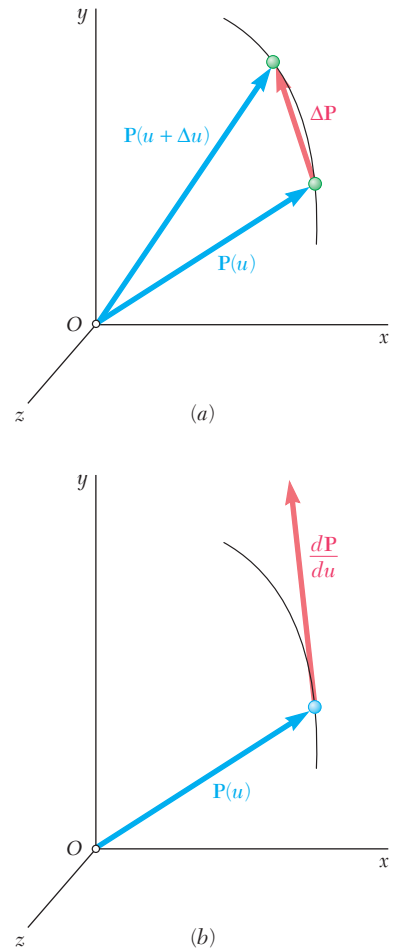


Fig. 11.14 (a) The change in vector function for a particle moving along a curvilinear path; (b) the derivative of the vector function is tangent to the path described by the tip of the function.

We now consider the **product of a scalar function** $f(u)$ **and a vector function** $\mathbf{P}(u)$ of the same scalar variable u . The derivative of the vector $f\mathbf{P}$ is

$$\frac{d(f\mathbf{P})}{du} = \lim_{\Delta u \rightarrow 0} \frac{(f + \Delta f)(\mathbf{P} + \Delta \mathbf{P}) - f\mathbf{P}}{\Delta u} = \lim_{\Delta u \rightarrow 0} \left(\frac{\Delta f}{\Delta u} \mathbf{P} + f \frac{\Delta \mathbf{P}}{\Delta u} \right)$$

or recalling the properties of the limits of sums and products,

$$\frac{d(f\mathbf{P})}{du} = \frac{df}{du} \mathbf{P} + f \frac{d\mathbf{P}}{du} \quad (11.20)$$

In a similar way, we can obtain the derivatives of the **scalar product** and the **vector product** of two vector functions $\mathbf{P}(u)$ and $\mathbf{Q}(u)$. Thus,

$$\frac{d(\mathbf{P} \cdot \mathbf{Q})}{du} = \frac{d\mathbf{P}}{du} \cdot \mathbf{Q} + \mathbf{P} \cdot \frac{d\mathbf{Q}}{du} \quad (11.21)$$

$$\frac{d(\mathbf{P} \times \mathbf{Q})}{du} = \frac{d\mathbf{P}}{du} \times \mathbf{Q} + \mathbf{P} \times \frac{d\mathbf{Q}}{du} \quad (11.22)^\dagger$$

We can use the properties just established to determine the **rectangular components of the derivative of a vector function** $\mathbf{P}(u)$. Resolving $\mathbf{P}$ into components along fixed rectangular axes x , y , and z , we have

$$\mathbf{P} = P_x \mathbf{i} + P_y \mathbf{j} + P_z \mathbf{k} \quad (11.23)$$

where P_x , P_y , and P_z are the rectangular scalar components of the vector $\mathbf{P}$, and $\mathbf{i}$, $\mathbf{j}$, and $\mathbf{k}$ are the unit vectors corresponding, respectively, to the x , y , and z axes (Sec. 2.12 or Appendix A). From Eq. (11.19), the derivative of $\mathbf{P}$ is equal to the sum of the derivatives of the terms in the right-hand side. Since each of these terms is the product of a scalar and a vector function, we should use Eq. (11.20). However, the unit vectors $\mathbf{i}$, $\mathbf{j}$, and $\mathbf{k}$ have a constant magnitude (equal to 1) and fixed directions. Their derivatives are therefore zero, and we obtain

$$\frac{d\mathbf{P}}{du} = \frac{dP_x}{du} \mathbf{i} + \frac{dP_y}{du} \mathbf{j} + \frac{dP_z}{du} \mathbf{k} \quad (11.24)$$

Note that the coefficients of the unit vectors are, by definition, the scalar components of the vector $d\mathbf{P}/du$. We conclude that we can obtain the rectangular scalar components of the derivative $d\mathbf{P}/du$ of the vector function $\mathbf{P}(u)$ by differentiating the corresponding scalar components of $\mathbf{P}$.

Rate of Change of a Vector. When the vector $\mathbf{P}$ is a function of the time t , its derivative $d\mathbf{P}/dt$ represents the **rate of change** of $\mathbf{P}$ with respect to the frame $Oxyz$. Resolving $\mathbf{P}$ into rectangular components and using Eq. (11.24), we have

$$\frac{d\mathbf{P}}{dt} = \frac{dP_x}{dt} \mathbf{i} + \frac{dP_y}{dt} \mathbf{j} + \frac{dP_z}{dt} \mathbf{k}$$

[†]Since the vector product is not commutative (see Sec. 3.4), the order of the factors in Eq. (11.22) must be maintained.

Alternatively, using dots to indicate differentiation with respect to t gives

$$\dot{\mathbf{P}} = \dot{P}_x \mathbf{i} + \dot{P}_y \mathbf{j} + \dot{P}_z \mathbf{k} \quad (11.24')$$

As you will see in Sec. 15.5, the rate of change of a vector as observed from a *moving frame of reference* is, in general, different from its rate of change as observed from a fixed frame of reference. However, if the moving frame $O'x'y'z'$ is in *translation*, i.e., if its axes remain parallel to the corresponding axes of the fixed frame $Oxyz$ (Fig. 11.15), we can use the same unit vectors $\mathbf{i}$, $\mathbf{j}$, and $\mathbf{k}$ in both frames, and at any given instant, the vector $\mathbf{P}$ has the same components P_x , P_y , and P_z in both frames. It follows from Eq. (11.24') that the rate of change $\dot{\mathbf{P}}$ is the same with respect to the frames $Oxyz$ and $O'x'y'z'$. Therefore,

The rate of change of a vector is the same with respect to a fixed frame and with respect to a frame in translation.

This property will greatly simplify our work, since we will be concerned mainly with frames in translation.

11.4C Rectangular Components of Velocity and Acceleration

Suppose the position of a particle P is defined at any instant by its rectangular coordinates x , y , and z . In this case, it is often convenient to resolve the velocity $\mathbf{v}$ and the acceleration $\mathbf{a}$ of the particle into rectangular components (Fig. 11.16).

To resolve the position vector $\mathbf{r}$ of the particle into rectangular components, we write

$$\mathbf{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k} \quad (11.25)$$

Here the coordinates x , y , and z are functions of t . Differentiating twice, we obtain

Velocity and acceleration in rectangular components

$$\mathbf{v} = \frac{d\mathbf{r}}{dt} = \dot{x}\mathbf{i} + \dot{y}\mathbf{j} + \dot{z}\mathbf{k} \quad (11.26)$$

$$\mathbf{a} = \frac{d\mathbf{v}}{dt} = \ddot{x}\mathbf{i} + \ddot{y}\mathbf{j} + \ddot{z}\mathbf{k} \quad (11.27)$$

where $\dot{x}$, $\dot{y}$, and $\dot{z}$ and $\ddot{x}$, $\ddot{y}$, and $\ddot{z}$ represent, respectively, the first and second derivatives of x , y , and z with respect to t . It follows from Eqs. (11.26) and (11.27) that the scalar components of the velocity and acceleration are

$$v_x = \dot{x} \quad v_y = \dot{y} \quad v_z = \dot{z} \quad (11.28)$$

$$a_x = \ddot{x} \quad a_y = \ddot{y} \quad a_z = \ddot{z} \quad (11.29)$$

A positive value for v_x indicates that the vector component $\mathbf{v}_x$ is directed to the right, and a negative value indicates that it is directed to the left. The sense of each of the other vector components is determined in a similar way from the sign of the corresponding scalar component. If desired, we can obtain the magnitudes and directions of the velocity and acceleration from their scalar components using the methods of Secs. 2.2A and 2.4A (or Appendix A).

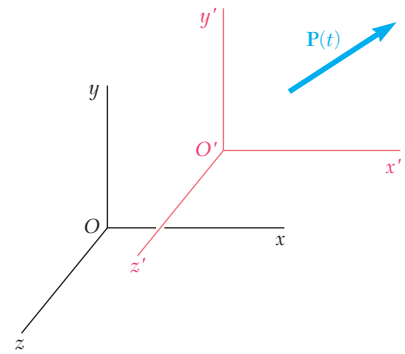


Fig. 11.15 The rate of change of a vector is the same with respect to a fixed frame of reference and with respect to a frame in translation.

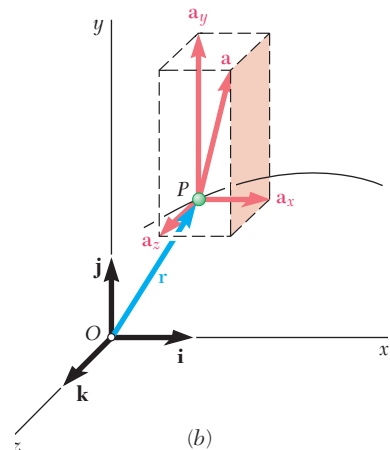
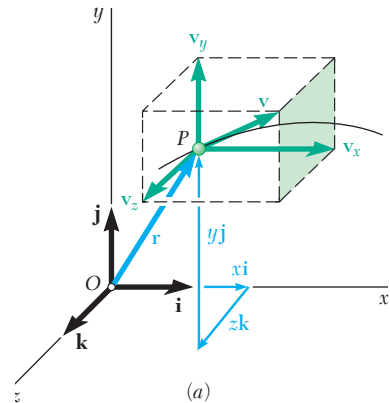
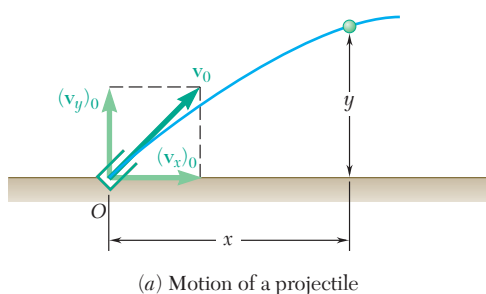


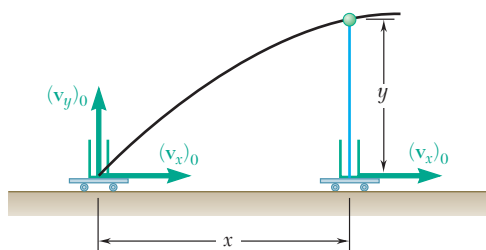
Fig. 11.16 (a) Rectangular components of position and velocity for a particle P ; (b) rectangular components of acceleration for particle P .



Photo 11.3 The motion of this snowboarder in the air is a parabola, assuming we can neglect air resistance.



(a) Motion of a projectile



(b) Equivalent rectilinear motions

Fig. 11.17 The motion of a projectile (a) consists of uniform horizontal motion and uniformly accelerated vertical motion and (b) is equivalent to two independent rectilinear motions.

The use of rectangular components to describe the position, velocity, and acceleration of a particle is particularly effective when the component a_x of the acceleration depends only upon t , x , and/or v_x , and similarly when a_y depends only upon t , y , and/or v_y , and when a_z depends upon t , z , and/or v_z . In this case, we can integrate Equations (11.28) and (11.29) independently. In other words, the motion of the particle in the x direction, its motion in the y direction, and its motion in the z direction can be treated separately.

In the case of the **motion of a projectile**, we can show (see Sec. 12.1D) that the components of the acceleration are

$$a_x = \ddot{x} = 0 \quad a_y = \ddot{y} = -g \quad a_z = \ddot{z} = 0$$

if the resistance of the air is neglected. Denoting the coordinates of a gun by x_0 , y_0 , and z_0 and the components of the initial velocity $\mathbf{v}_0$ of the projectile by $(v_x)_0$, $(v_y)_0$, and $(v_z)_0$, we can integrate twice in t and obtain

$$\begin{aligned} v_x = \dot{x} &= (v_x)_0 & v_y = \dot{y} &= (v_y)_0 - gt & v_z = \dot{z} &= (v_z)_0 \\ x &= x_0 + (v_x)_0 t & y &= y_0 + (v_y)_0 t - \frac{1}{2}gt^2 & z &= z_0 + (v_z)_0 t \end{aligned}$$

If the projectile is fired in the xy plane from the origin O , we have $x_0 = y_0 = z_0 = 0$ and $(v_z)_0 = 0$, so the equations of motion reduce to

$$\begin{aligned} v_x &= (v_x)_0 & v_y &= (v_y)_0 - gt & v_z &= 0 \\ x &= (v_x)_0 t & y &= (v_y)_0 t - \frac{1}{2}gt^2 & z &= 0 \end{aligned}$$

These equations show that the projectile remains in the xy plane, that its motion in the horizontal direction is uniform, and that its motion in the vertical direction is uniformly accelerated. Thus, we can replace the motion of a projectile by two independent rectilinear motions, which are easily visualized if we assume that the projectile is fired vertically with an initial velocity $(v_y)_0$ from a platform moving with a constant horizontal velocity $(v_x)_0$ (Fig. 11.17). The coordinate x of the projectile is equal at any instant to the distance traveled by the platform, and we can compute its coordinate y as if the projectile were moving along a vertical line. Additionally, because the $(v_x)_0$ values are the same, the projectile will land on the platform regardless of the value of $(v_y)_0$.

Note that the equations defining the coordinates x and y of a projectile at any instant are the parametric equations of a parabola. Thus, the trajectory of a projectile is *parabolic*. This result, however, ceases to be valid if we take into account the resistance of the air or the variation with altitude of the acceleration due to gravity.

11.4D Motion Relative to a Frame in Translation

We have just seen how to describe the motion of a particle by using a single frame of reference. In most cases, this frame was attached to the earth and was considered to be fixed. Now we want to analyze situations in which it is convenient to use several frames of reference simultaneously. If one of the frames is attached to the earth, it is called a **fixed frame of reference**, and the other frames are referred to as **moving frames of reference**. You should recognize, however, that the selection of a fixed frame of reference is purely arbitrary. Any frame can be designated as “fixed”; all other frames not rigidly attached to this frame are then described as “moving.”

Consider two particles A and B moving in space (Fig. 11.18). The vectors $\mathbf{r}_A$ and $\mathbf{r}_B$ define their positions at any given instant with respect to the fixed frame of reference $Oxyz$. Consider now a system of axes x' , y' , and z' centered at A and parallel to the x , y , and z axes. Suppose that, while the origin of these axes moves, their orientation remains the same; then the frame of reference $Ax'y'z'$ is in *translation* with respect to $Oxyz$. The vector $\mathbf{r}_{B/A}$ joining A and B defines **the position of B relative to the moving frame $Ax'y'z'$** (or for short, **the position of B relative to A**).

Figure 11.18 shows that the position vector $\mathbf{r}_B$ of particle B is the sum of the position vector $\mathbf{r}_A$ of particle A and of the position vector $\mathbf{r}_{B/A}$ of B relative to A ; that is,

Relative position

$$\mathbf{r}_B = \mathbf{r}_A + \mathbf{r}_{B/A} \quad (11.30)$$

Differentiating Eq. (11.30) with respect to t within the fixed frame of reference, and using dots to indicate time derivatives, we have

$$\dot{\mathbf{r}}_B = \dot{\mathbf{r}}_A + \dot{\mathbf{r}}_{B/A} \quad (11.31)$$

The derivatives $\dot{\mathbf{r}}_A$ and $\dot{\mathbf{r}}_B$ represent, respectively, the velocities $\mathbf{v}_A$ and $\mathbf{v}_B$ of the particles A and B . Since $Ax'y'z'$ is in translation, the derivative $\dot{\mathbf{r}}_{B/A}$ represents the rate of change of $\mathbf{r}_{B/A}$ with respect to the frame $Ax'y'z'$ as well as with respect to the fixed frame (Sec. 11.4B). This derivative, therefore, defines **the velocity $\mathbf{v}_{B/A}$ of B relative to the frame $Ax'y'z'$** (or for short, **the velocity $\mathbf{v}_{B/A}$ of B relative to A**). We have

Relative velocity

$$\mathbf{v}_B = \mathbf{v}_A + \mathbf{v}_{B/A} \quad (11.32)$$

Differentiating Eq. (11.32) with respect to t , and using the derivative $\dot{\mathbf{v}}_{B/A}$ to define **the acceleration $\mathbf{a}_{B/A}$ of B relative to the frame $Ax'y'z'$** (or for short, **the acceleration $\mathbf{a}_{B/A}$ of B relative to A**), we obtain

Relative acceleration

$$\mathbf{a}_B = \mathbf{a}_A + \mathbf{a}_{B/A} \quad (11.33)$$

We refer to the motion of B with respect to the fixed frame $Oxyz$ as the **absolute motion of B** . The equations derived in this section show that **we can obtain the absolute motion of B by combining the motion of A and the relative motion of B with respect to the moving frame attached to A** . Equation (11.32), for example, expresses that the absolute velocity $\mathbf{v}_B$ of particle B can be obtained by vectorially adding the velocity of A and the velocity of B relative to the frame $Ax'y'z'$. Equation (11.33) expresses a similar property in terms of the accelerations. (Note that the product of the subscripts A and B/A used in the right-hand sides of Eqs. (11.30) through (11.33) is equal to the subscript B used in their left-hand sides.) Keep in mind, however, that the frame $Ax'y'z'$ is in *translation*; that is, while it moves with A , it maintains the same orientation. As you will see later (Sec. 15.7), you must use different relations in the case of a rotating frame of reference.

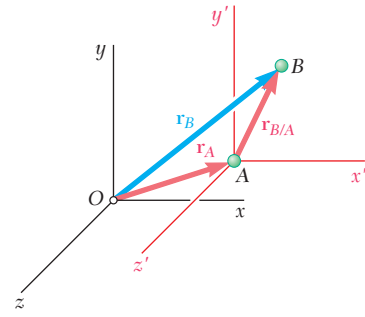


Fig. 11.18 The vector $\mathbf{r}_{B/A}$ defines the position of B with respect to moving frame A .



Photo 11.4 The pilot of a helicopter landing on a moving carrier must take into account the relative motion of the ship.

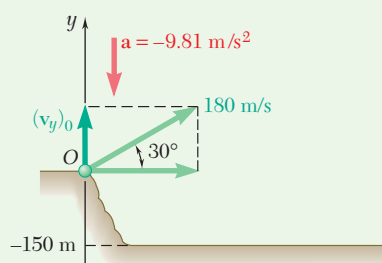
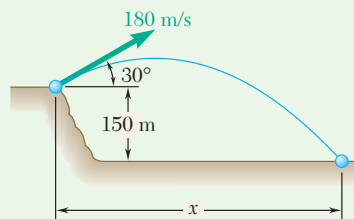


Fig. 1 Acceleration and initial velocity of the projectile in the y -direction.

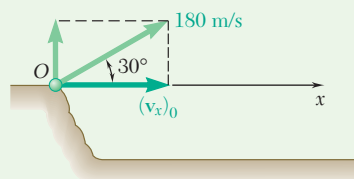


Fig. 2 Initial velocity of the projectile in the x -direction.

Sample Problem 11.10

A projectile is fired from the edge of a 150-m cliff with an initial velocity of 180 m/s at an angle of 30° with the horizontal. Neglecting air resistance, find (a) the horizontal distance from the gun to the point where the projectile strikes the ground, (b) the greatest elevation above the ground reached by the projectile.

STRATEGY: This is a projectile motion problem, so you can consider the vertical and horizontal motions separately. First determine the equations governing each direction, and then use them to find the distances.

MODELING and ANALYSIS: Model the projectile as a particle and neglect the effects of air resistance. The vertical motion has a constant acceleration. Choosing the positive sense of the y axis upward and placing the origin O at the gun (Fig. 1), you have

$$(v_y)_0 = (180 \text{ m/s}) \sin 30^\circ = +90 \text{ m/s}$$

$$a = -9.81 \text{ m/s}^2$$

Substitute these values into the equations for motion with constant acceleration. Thus,

$$v_y = (v_y)_0 + at \quad v_y = 90 - 9.81t \quad (1)$$

$$y = (v_y)_0 t + \frac{1}{2}at^2 \quad y = 90t - 4.90t^2 \quad (2)$$

$$v_y^2 = (v_y)_0^2 + 2ay \quad v_y^2 = 8100 - 19.62y \quad (3)$$

The horizontal motion has zero acceleration. Choose the positive sense of the x axis to the right (Fig. 2), which gives you

$$(v_x)_0 = (180 \text{ m/s}) \cos 30^\circ = +155.9 \text{ m/s}$$

Substituting into the equation for constant acceleration, you obtain

$$x = (v_x)_0 t \quad x = 155.9t \quad (4)$$

a. Horizontal Distance. When the projectile strikes the ground,

$$y = -150 \text{ m}$$

Substituting this value into Eq. (2) for the vertical motion, you have

$$-150 = 90t - 4.90t^2 \quad t^2 - 18.37t - 30.6 = 0 \quad t = 19.91 \text{ s}$$

Substituting $t = 19.91 \text{ s}$ into Eq. (4) for the horizontal motion, you obtain

$$x = 155.9(19.91) \quad x = 3100 \text{ m} \quad \blacktriangleleft$$

b. Greatest Elevation. When the projectile reaches its greatest elevation, $v_y = 0$; substituting this value into Eq. (3) for the vertical motion, you have

$$0 = 8100 - 19.62y \quad y = 413 \text{ m}$$

$$\text{Greatest elevation above ground} = 150 \text{ m} + 413 \text{ m} = 563 \text{ m} \quad \blacktriangleleft$$

REFLECT and THINK: Because there is no air resistance, you can treat the vertical and horizontal motions separately and can immediately write down the algebraic equations of motion. If you did want to include air resistance, you must know the acceleration as a function of the speed (you will see how to derive this in Chapter 12), and then you need to use the basic kinematic relationships, separate variables, and integrate.

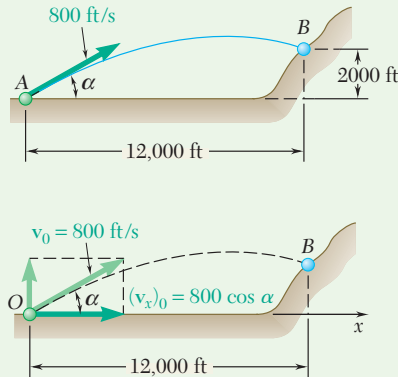


Fig. 1 Initial velocity of the projectile in the x -direction.

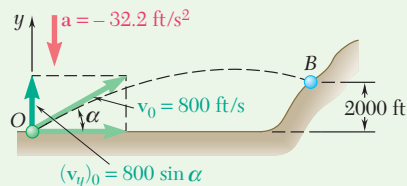


Fig. 2 Acceleration and initial velocity of the projectile in the y -direction.

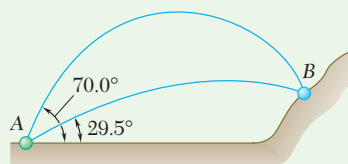


Fig. 3 Firing angles that will hit target B .

Sample Problem 11.11

A projectile is fired with an initial velocity of 800 ft/s at a target B located 2000 ft above the gun A and at a horizontal distance of 12,000 ft. Neglecting air resistance, determine the value of the firing angle α needed to hit the target.

STRATEGY: This is a projectile motion problem, so you can consider the vertical and horizontal motions separately. First determine the equations governing the motion in each direction, and then use them to find the firing angle.

MODELING and ANALYSIS:

Horizontal Motion. Place the origin of the coordinate axes at the gun (Fig. 1). Then

$$(v_x)_0 = 800 \cos \alpha$$

Substituting into the equation of uniform horizontal motion, you obtain

$$x = (v_x)_0 t \quad x = (800 \cos \alpha) t$$

Obtain the time required for the projectile to move through a horizontal distance of 12,000 ft by setting x equal to 12,000 ft.

$$12,000 = (800 \cos \alpha) t$$

$$t = \frac{12,000}{800 \cos \alpha} = \frac{15}{\cos \alpha}$$

Vertical Motion. Again, place the origin at the gun (Fig. 2).

$$(v_y)_0 = 800 \sin \alpha \quad a = -32.2 \text{ ft/s}^2$$

Substituting into the equation for constant acceleration in the vertical direction, you obtain

$$y = (v_y)_0 t + \frac{1}{2} a t^2 \quad y = (800 \sin \alpha) t - 16.1 t^2$$

Projectile Hits Target. When $x = 12,000$ ft, you want $y = 2000$ ft. Substituting for y and setting t equal to the value found previously, you have

$$2000 = 800 \sin \alpha \frac{15}{\cos \alpha} - 16.1 \left(\frac{15}{\cos \alpha} \right)^2 \quad (1)$$

Since $1/\cos^2 \alpha = \sec^2 \alpha = 1 + \tan^2 \alpha$, you have

$$2000 = 800(15) \tan \alpha - 16.1(15^2)(1 + \tan^2 \alpha)$$

$$3622 \tan^2 \alpha - 12,000 \tan \alpha + 5622 = 0$$

Solving this quadratic equation for $\tan \alpha$ gives you

$$\tan \alpha = 0.565 \quad \text{and} \quad \tan \alpha = 2.75$$

$$\alpha = 29.5^\circ \quad \text{and} \quad \alpha = 70.0^\circ \quad \blacktriangleleft$$

The target will be hit if either of these two firing angles is used (Fig. 3).

REFLECT and THINK: It is a well-known characteristic of projectile motion that you can hit the same target by using either of two firing angles. We used trigonometry to write the equation in terms of $\tan \alpha$, but most calculators or computer programs like Maple, Matlab, or Mathematica also can be used to solve (1) for α . You must be careful when using these tools, however, to make sure that you find both angles.

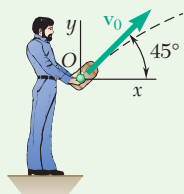
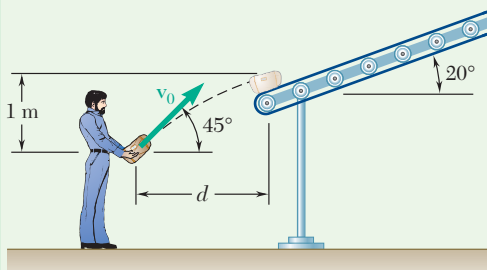


Fig. 1 Initial velocity of the package.

Sample Problem 11.12

A conveyor belt at an angle of 20° with the horizontal is used to transfer small packages to other parts of an industrial plant. A worker tosses a package with an initial velocity v_0 at an angle of 45° so that its velocity is parallel to the belt as it lands 1 m above the release point. Determine (a) the magnitude of v_0 , (b) the horizontal distance d .

STRATEGY: This is a projectile motion problem, so you can consider the vertical and the horizontal motions separately. First determine the equations governing the motion in each direction, then use them to determine the unknown quantities.

MODELING and ANALYSIS:

Horizontal Motion. Placing the axes of your origin at the location where the package leaves the workers hands (Fig. 1), you can write

$$\text{Horizontal: } v_x = v_0 \cos 45^\circ \quad \text{and} \quad x = (v_0 \cos 45^\circ)t$$

$$\text{Vertical: } v_y = v_0 \sin 45^\circ - gt \quad \text{and} \quad y = (v_0 \sin 45^\circ)t - \frac{1}{2}gt^2$$

Landing on the Belt. The problem statement indicates that when the package lands on the belt, its velocity vector will be in the same direction as the belt is moving. If this happens when $t = t_1$, you can write

$$\frac{v_y}{v_x} = \tan 20^\circ = \frac{v_0 \sin 45^\circ - gt_1}{v_0 \cos 45^\circ} = 1 - \frac{gt_1}{v_0 \cos 45^\circ} \quad (1)$$

This equation has two unknown quantities: t_1 and v_0 . Therefore, you need more equations. Substituting $t = t_1$ into the remaining projectile motion equations gives

$$d = (v_0 \cos 45^\circ)t \quad (2)$$

$$1 \text{ m} = (v_0 \sin 45^\circ)t_1 - \frac{1}{2}gt_1^2 \quad (3)$$

You now have three equations (1), (2), and (3) and three unknowns t_1 , v_0 , and d . Using $g = 9.81 \text{ m/s}^2$ and solving these three equations give $t_1 = 0.3083 \text{ s}$ and

$$v_0 = 6.73 \text{ m/s} \quad \blacktriangleleft$$

$$d = 1.466 \text{ m} \quad \blacktriangleleft$$

REFLECT and THINK: All of these projectile problems are similar. You write down the governing equations for motion in the horizontal and vertical directions and then use additional information in the problem statement to solve the problem. In this case, the distance is just less than 1.5 meters, which is a reasonable distance for a worker to toss a package.

Sample Problem 11.13

Airplane *B*, which is travelling at a constant 560 km/h, is pursuing airplane *A*, which is travelling northeast at a constant 800 km/hr. At time $t = 0$, airplane *A* is 640 km east of airplane *B*. Determine (a) the direction of the course airplane *B* should follow (measured from the east) to intercept plane *A*, (b) the rate at which the distance between the airplanes is decreasing, (c) how long it takes for airplane *B* to catch airplane *A*.

STRATEGY: To find when *B* intercepts *A*, you just need to find out when the two planes are at the same location. The rate at which the distance is decreasing is the magnitude of $\mathbf{v}_{B/A}$, so you can use the relative velocity equation.

MODELING and ANALYSIS: Choose x to be east, y to be north, and place the origin of your coordinate system at *B* (Fig. 1).

Positions of the Planes: You know that each plane has a constant speed, so you can write a position vector for each plane. Thus,

$$\mathbf{r}_A = [(v_A \cos 45^\circ)t + 640 \text{ km}]\mathbf{i} + [(v_A \sin 45^\circ)t]\mathbf{j} \quad (1)$$

$$\mathbf{r}_B = [(v_B \cos \theta)t]\mathbf{i} + [(v_B \sin \theta)t]\mathbf{j} \quad (2)$$

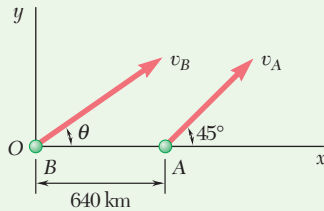


Fig. 1 Initial velocity of airplanes *A* and *B*.

a. Direction of *B*. Plane *B* will catch up when they are at the same location, that is, $\mathbf{r}_A = \mathbf{r}_B$. You can equate components in the $\mathbf{j}$ direction to find

$$v_A \sin 45^\circ t_1 = v_B \sin \theta t_1$$

After you substitute in values,

$$\sin \theta = \frac{(v_A \sin 45^\circ)t_1}{v_B t_1} = \frac{(560 \text{ km/hr})\sin 45^\circ}{800 \text{ km/hr}} = 0.4950$$

$$\theta = \sin^{-1} 0.4950 = 29.67^\circ \quad \theta = 29.7^\circ \quad \blacktriangleleft$$

b. Rate. The rate at which the distance is decreasing is the magnitude of $\mathbf{v}_{B/A}$, so

$$\begin{aligned} \mathbf{v}_{B/A} &= \mathbf{v}_B - \mathbf{v}_A = (v_B \cos \theta \mathbf{i} + v_B \sin \theta \mathbf{j}) - (v_A \cos 45^\circ \mathbf{i} + v_A \sin 45^\circ \mathbf{j}) \\ &= [(800 \text{ km/h})\cos 29.668^\circ - (560 \text{ km/h})\cos 45^\circ]\mathbf{i} \\ &\quad + [(800 \text{ km/h})\sin 29.668^\circ - (560 \text{ km/h})\sin 45^\circ]\mathbf{j} \\ &= 299.15 \text{ km/h } \mathbf{i} \quad |\mathbf{v}_{B/A}| = 299 \text{ km/h } \quad \blacktriangleleft \end{aligned}$$

c. Time for *B* to catch up with *A*. To find the time, you equate the $\mathbf{i}$ components of each position vector, giving

$$(v_A \cos 45^\circ)t_1 + 640 \text{ km} = (v_B \cos \theta)t_1$$

Solve this for t_1 . Thus,

$$\begin{aligned} t_1 &= \frac{640 \text{ km}}{v_B \cos \theta - v_A \cos 45^\circ} \\ &= \frac{640 \text{ km}}{(800 \text{ km/h})\cos 29.67^\circ - (560 \text{ km/h})\cos 45^\circ} = 2.139 \text{ h} \\ t_1 &= 2.14 \text{ h } \quad \blacktriangleleft \end{aligned}$$

REFLECT and THINK: The relative velocity is only in the horizontal (eastern) direction. This makes sense, because the vertical (northern) components have to be equal in order for the two planes to intersect.

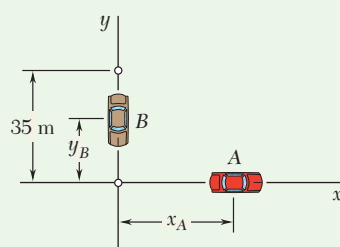
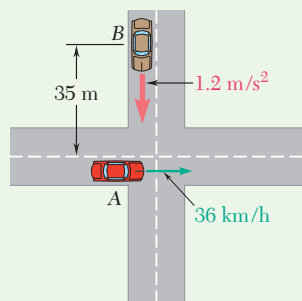


Fig. 1 Initial positions of car A and B.

Sample Problem 11.14

Automobile A is traveling east at the constant speed of 36 km/h. As automobile A crosses the intersection shown, automobile B starts from rest 35 m north of the intersection and moves south with a constant acceleration of 1.2 m/s^2 . Determine the position, velocity, and acceleration of B relative to A 5 s after A crosses the intersection.

STRATEGY: This is a relative motion problem. Determine the motion of each vehicle independently, and then use the definition of relative motion to determine the desired quantities.

MODELING and ANALYSIS:

Motion of Automobile A. Choose x and y axes with the origin at the intersection of the two streets and with positive senses directed east and north, respectively. First express the speed in m/s, as

$$v_A = \left(36 \frac{\text{km}}{\text{h}}\right) \left(\frac{1000 \text{ m}}{1 \text{ km}}\right) \left(\frac{1 \text{ h}}{3600 \text{ s}}\right) = 10 \text{ m/s}$$

The motion of A is uniform, so for any time t

$$\begin{aligned} a_A &= 0 \\ v_A &= +10 \text{ m/s} \\ x_A &= (x_A)_0 + v_A t = 0 + 10t \end{aligned}$$

For $t = 5 \text{ s}$, you have (Fig. 1)

$$\begin{aligned} a_A &= 0 & \mathbf{a}_A &= 0 \\ v_A &= +10 \text{ m/s} & \mathbf{v}_A &= 10 \text{ m/s} \rightarrow \\ x_A &= +(10 \text{ m/s})(5 \text{ s}) = +50 \text{ m} & \mathbf{r}_A &= 50 \text{ m} \rightarrow \end{aligned}$$

Motion of Automobile B. The motion of B is uniformly accelerated, so

$$\begin{aligned} a_B &= -1.2 \text{ m/s}^2 \\ v_B &= (v_B)_0 + at = 0 - 1.2t \\ y_B &= (y_B)_0 + (v_B)_0 t + \frac{1}{2} a_B t^2 = 35 + 0 - \frac{1}{2} (1.2) t^2 \end{aligned}$$

For $t = 5 \text{ s}$, you have (Fig. 1)

$$\begin{aligned} a_B &= -1.2 \text{ m/s}^2 & \mathbf{a}_B &= 1.2 \text{ m/s}^2 \downarrow \\ v_B &= -(1.2 \text{ m/s}^2)(5 \text{ s}) = -6 \text{ m/s} & \mathbf{v}_B &= 6 \text{ m/s} \downarrow \\ y_B &= 35 - \frac{1}{2} (1.2 \text{ m/s}^2)(5 \text{ s})^2 = +20 \text{ m} & \mathbf{r}_B &= 20 \text{ m} \uparrow \end{aligned}$$

Motion of B Relative to A. Draw the triangle corresponding to the vector equation $\mathbf{r}_B = \mathbf{r}_A + \mathbf{r}_{B/A}$ (Fig. 2) and obtain the magnitude and direction of the position vector of B relative to A.

$$r_{B/A} = 53.9 \text{ m} \quad \alpha = 21.8^\circ \quad \mathbf{r}_{B/A} = 53.9 \text{ m} \nearrow 21.8^\circ$$

Proceeding in a similar fashion (Fig. 2), find the velocity and acceleration of B relative to A. Hence,

$$\begin{aligned} \mathbf{v}_B &= \mathbf{v}_A + \mathbf{v}_{B/A} \\ v_{B/A} &= 11.66 \text{ m/s} \quad \beta = 31.0^\circ \quad \mathbf{v}_{B/A} = 11.66 \text{ m/s} \nearrow 31.0^\circ \\ \mathbf{a}_B &= \mathbf{a}_A + \mathbf{a}_{B/A} \\ a_{B/A} &= 1.2 \text{ m/s}^2 \downarrow \end{aligned}$$

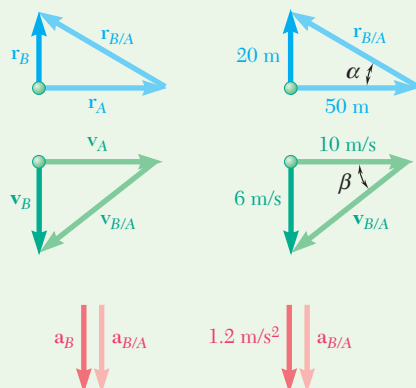


Fig. 2 Vector triangles for position, velocity, and acceleration.

REFLECT and THINK: Note that the relative position and velocity of B relative to A change with time; the values given here are only for the moment $t = 5$ s. Rather than drawing triangles, you could have also used vector algebra. When the vectors are at right angles, as in this problem, drawing vector triangles is usually easiest.

Sample Problem 11.15

Knowing that at the instant shown cylinder/ramp A has a velocity of 8 in./s directed down, determine the velocity of block B .

STRATEGY: You have objects connected by cables, so this is a dependent-motion problem. You should define coordinates for each block-object and write a constraint equation for the cable. You will also need to use relative motion, since B slides on A .

MODELING and ANALYSIS: Define position vectors, as shown in Fig. 1.

Constraint Equations. Assuming the cable is inextensible, you can write the length in terms of the coordinates and then differentiate.

The constraint equation for the cable is

$$x_A + 2x_{B/A} = \text{constant}$$

Differentiating this gives

$$v_A = -2v_{B/A} \quad (1)$$

Substituting for v_A gives $v_{B/A} = -4$ in./s or 4 in./s up the incline.

Dependent Motion: You know that the direction of $v_{B/A}$ is directed up the incline. Therefore, the relative motion equation relating the velocities of blocks A and B is $\mathbf{v}_B = \mathbf{v}_A + \mathbf{v}_{B/A}$. You could either draw a vector triangle or use vector algebra. Let's use vector algebra. Using the coordinate system shown in Fig. 2 and substituting in the magnitudes gives

$$(v_B)_x \mathbf{i} + (v_B)_y \mathbf{j} = (-8 \text{ in./s})\mathbf{j} + (-4 \text{ in./s}) \sin 50^\circ \mathbf{i} + (4 \text{ in./s}) \cos 50^\circ \mathbf{j}$$

Equating components gives

$$\mathbf{i}: (v_B)_x = -(4 \text{ in./s}) \sin 50^\circ \quad \rightarrow v_{B_x} = -3.064 \text{ in./s}$$

$$\mathbf{j}: (v_B)_y = (-8 \text{ in./s}) + (4 \text{ in./s}) \cos 50^\circ \quad \rightarrow v_{B_y} = -5.429 \text{ in./s}$$

Finding the magnitude and direction gives

$$\mathbf{v}_B = 6.23 \text{ in./s} \nearrow 60.6^\circ$$

REFLECT and THINK: Rather than using vector algebra, you could have also drawn a vector triangle, as shown in Fig. 3. To use this vector triangle, you need to use the law of cosines and the law of sines. Looking at the mechanism, block B should move up the incline if block A moves downward; our mathematical result is consistent with this. It is also interesting to note that, even though B moves up the incline relative to A , block B is actually moving down and to the left, as shown in the calculation here. This occurs because block A is also moving down.

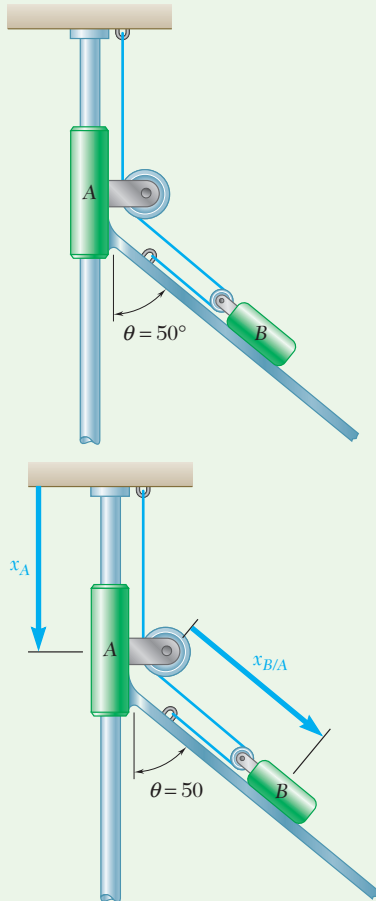


Fig. 1 Position vectors to A and B .

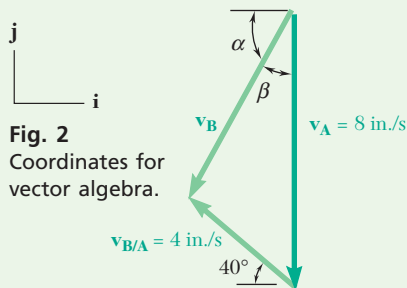


Fig. 2 Coordinates for vector algebra.

Fig. 3 Vector triangle for velocity of blocks A and B .

SOLVING PROBLEMS ON YOUR OWN

In the problems for this section, you will analyze the **curvilinear motion** of a particle. The physical interpretations of velocity and acceleration are the same as in the first sections of the chapter, but you should remember that these quantities are vectors. In addition, recall from your experience with vectors in statics that it is often advantageous to express position vectors, velocities, and accelerations in terms of their rectangular scalar components [Eqs. (11.25) through (11.27)].

A. Analyzing the motion of a projectile. Many of the following problems deal with the two-dimensional motion of a projectile where we can neglect the resistance of the air. In Sec. 11.4C, we developed the equations that describe this type of motion, and we observed that the horizontal component of the velocity remains constant (uniform motion), while the vertical component of the acceleration is constant (uniformly accelerated motion). We are able to consider the horizontal and the vertical motions of the particle separately. Assuming that the projectile is fired from the origin, we can write the two equations as

$$x = (v_x)_0 t \quad y = (v_y)_0 t - \frac{1}{2}gt^2$$

1. If you know the initial velocity and firing angle, you can obtain the value of y corresponding to any given value of x (or the value of x for any value of y) by solving one of the previous equations for t and substituting for t into the other equation [Sample Prob. 11.10].

2. If you know the initial velocity and the coordinates of a point of the trajectory and you wish to determine the firing angle α , begin your solution by expressing the components $(v_x)_0$ and $(v_y)_0$ of the initial velocity as functions of α . Then substitute these expressions and the known values of x and y into the previous equations. Finally, solve the first equation for t and substitute that value of t into the second equation to obtain a trigonometric equation in α , which you can solve for that unknown [Sample Prob. 11.11].

B. Solving translational two-dimensional relative-motion problems. You saw in Sec. 11.4D that you can obtain the absolute motion of a particle B by combining the motion of a particle A and the **relative motion** of B with respect to a frame attached to A that is in *translation* [Sample Probs. 11.12 and 11.13]. You can then express the velocity and acceleration of B as shown in Eqs. (11.32) and (11.33), respectively.

1. To visualize the relative motion of B with respect to A , imagine that you are attached to particle A as you observe the motion of particle B . For example, to a passenger in automobile A of Sample Prob. 11.14, automobile B appears to be heading in a southwesterly direction (*south* should be obvious; *west* is due to the fact that automobile A is moving to the east—automobile B then appears to travel to the west). Note that this conclusion is consistent with the direction of $\mathbf{v}_{B/A}$.

2. To solve a relative-motion problem, first write the vector equations (11.30), (11.32), and (11.33), which relate the motions of particles A and B . You may then use either of the following methods.

a. Construct the corresponding vector triangles and solve them for the desired position vector, velocity, and acceleration [Sample Prob. 11.14].

b. Express all vectors in terms of their rectangular components and solve the resulting two independent sets of scalar equations [Sample Prob. 11.15]. If you choose this approach, be sure to select the same positive direction for the displacement, velocity, and acceleration of each particle.

Problems

CONCEPT QUESTIONS

11.CQ3 Two model rockets are fired simultaneously from a ledge and follow the trajectories shown. Neglecting air resistance, which of the rockets will hit the ground first?

- A.
- B.
- They hit at the same time.
- The answer depends on h .

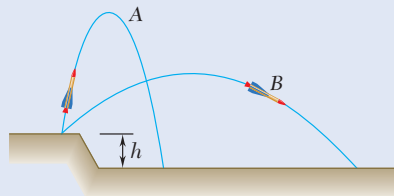


Fig. P6.CQ3

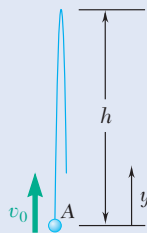


Fig. P6.CQ4

11.CQ4 Ball A is thrown straight up. Which of the following statements about the ball are true at the highest point in its path?

- The velocity and acceleration are both zero.
- The velocity is zero, but the acceleration is not zero.
- The velocity is not zero, but the acceleration is zero.
- Neither the velocity nor the acceleration is zero.

11.CQ5 Ball A is thrown straight up with an initial speed v_0 and reaches a maximum elevation h before falling back down. When A reaches its maximum elevation, a second ball is thrown straight upward with the same initial speed v_0 . At what height, y , will the balls cross paths?

- $y = h$
- $y > h/2$
- $y = h/2$
- $y < h/2$
- $y = 0$

11.CQ6 Two cars are approaching an intersection at constant speeds as shown. What velocity will car B appear to have to an observer in car A?

- $\rightarrow$
- $\searrow$
- $\nwarrow$
- $\nearrow$
- $\swarrow$

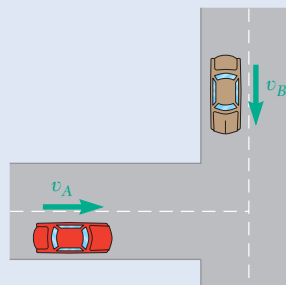


Fig. P6.CQ6

- 11.CQ7** Blocks *A* and *B* are released from rest in the positions shown. Neglecting friction between all surfaces, which figure best indicates the direction α of the acceleration of block *B*?

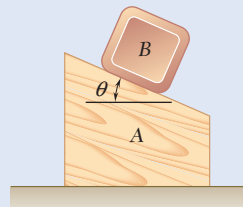
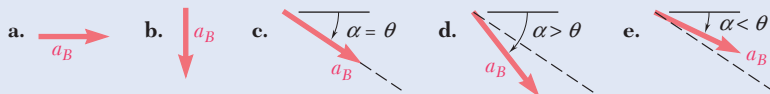


Fig. P6.CQ7

END-OF-SECTION PROBLEMS

- 11.89** A ball is thrown so that the motion is defined by the equations $x = 5t$ and $y = 2 + 6t - 4.9t^2$, where x and y are expressed in meters and t is expressed in seconds. Determine (a) the velocity at $t = 1$ s, (b) the horizontal distance the ball travels before hitting the ground.

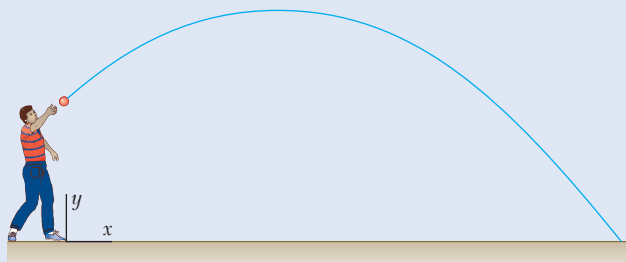


Fig. P11.89

- 11.90** The motion of a vibrating particle is defined by the position vector $\mathbf{r} = 10(1 - e^{-3t})\mathbf{i} + (4e^{-2t} \sin 15t)\mathbf{j}$, where $\mathbf{r}$ and t are expressed in millimeters and seconds, respectively. Determine the velocity and acceleration when (a) $t = 0$, (b) $t = 0.5$ s.

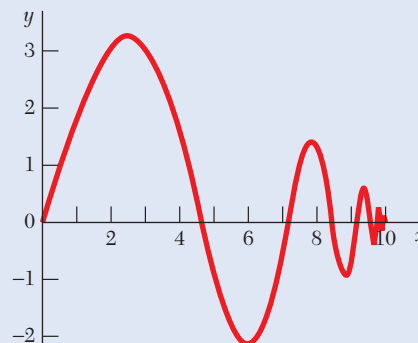


Fig. P11.90

- 11.91** The motion of a vibrating particle is defined by the position vector $\mathbf{r} = (4 \sin \pi t)\mathbf{i} - (\cos 2\pi t)\mathbf{j}$, where r is expressed in inches and t in seconds. (a) Determine the velocity and acceleration when $t = 1$ s. (b) Show that the path of the particle is parabolic.

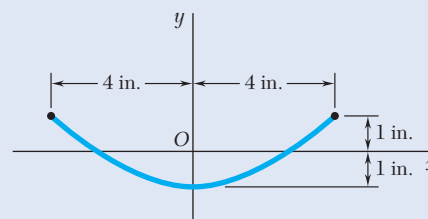


Fig. P11.91

- 11.92** The motion of a particle is defined by the equations $x = 10t - 5 \sin t$ and $y = 10 - 5 \cos t$, where x and y are expressed in feet and t is expressed in seconds. Sketch the path of the particle for the time interval $0 \leq t \leq 2\pi$, and determine (a) the magnitudes of the smallest and largest velocities reached by the particle, (b) the corresponding times, positions, and directions of the velocities.

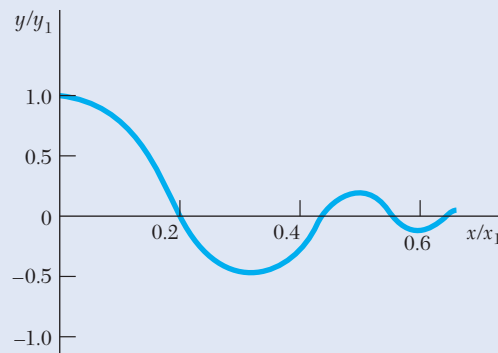


Fig. P11.92

- 11.93** The damped motion of a vibrating particle is defined by the position vector $\mathbf{r} = x_1[1 - 1/(t + 1)]\mathbf{i} + (y_1 e^{-\pi t/2} \cos 2\pi t)\mathbf{j}$, where t is expressed in seconds. For $x_1 = 30$ mm and $y_1 = 20$ mm, determine the position, the velocity, and the acceleration of the particle when (a) $t = 0$, (b) $t = 1.5$ s.

- 11.94** A girl operates a radio-controlled model car in a vacant parking lot. The girl's position is at the origin of the xy coordinate axes, and the surface of the parking lot lies in the x - y plane. The motion of the car is defined by the position vector $\mathbf{r} = (2 + 2t^2)\mathbf{i} + (6 + t^3)\mathbf{j}$ where $\mathbf{r}$ and t are expressed in meters and seconds, respectively. Determine (a) the distance between the car and the girl when $t = 2$ s, (b) the distance the car traveled in the interval from $t = 0$ to $t = 2$ s, (c) the speed and direction of the car's velocity at $t = 2$ s, (d) the magnitude of the car's acceleration at $t = 2$ s.

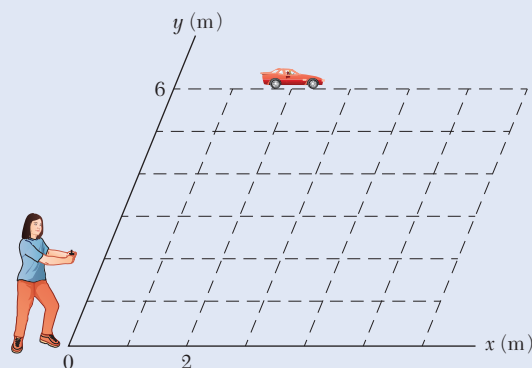


Fig. P11.94

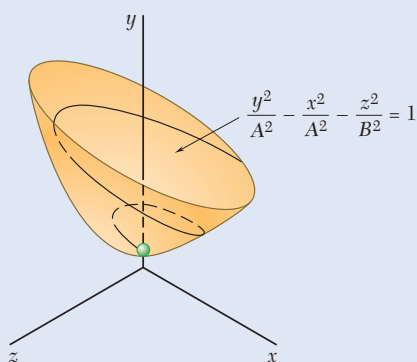


Fig. P11.96

- 11.95** The three-dimensional motion of a particle is defined by the position vector $\mathbf{r} = (Rt \cos \omega_n t)\mathbf{i} + ct\mathbf{j} + (Rt \sin \omega_n t)\mathbf{k}$. Determine the magnitudes of the velocity and acceleration of the particle. (The space curve described by the particle is a conic helix.)
- *11.96** The three-dimensional motion of a particle is defined by the position vector $\mathbf{r} = (At \cos t)\mathbf{i} + (A\sqrt{t^2 + 1})\mathbf{j} + (Bt \sin t)\mathbf{k}$, where r and t are expressed in feet and seconds, respectively. Show that the curve described by the particle lies on the hyperboloid $(y/A)^2 - (x/A)^2 - (z/B)^2 = 1$. For $A = 3$ and $B = 1$, determine (a) the magnitudes of the velocity and acceleration when $t = 0$, (b) the smallest nonzero value of t for which the position vector and the velocity are perpendicular to each other.

- 11.97** An airplane used to drop water on brushfires is flying horizontally in a straight line at 180 mi/h at an altitude of 300 ft. Determine the distance d at which the pilot should release the water so that it will hit the fire at B .

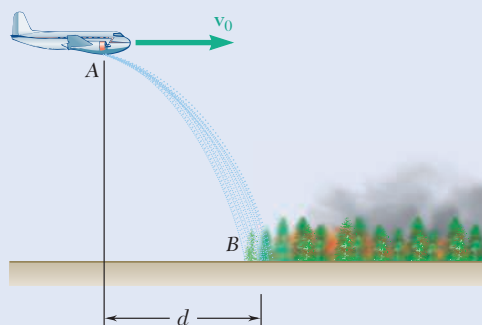


Fig. P11.97

- 11.98** A ski jumper starts with a horizontal take-off velocity of 25 m/s and lands on a straight landing hill inclined at 30° . Determine (a) the time between take-off and landing, (b) the length d of the jump, (c) the maximum vertical distance between the jumper and the landing hill.

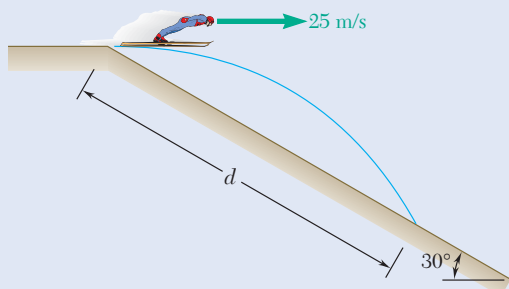


Fig. P11.98

- 11.99** A baseball pitching machine “throws” baseballs with a horizontal velocity v_0 . Knowing that height h varies between 788 mm and 1068 mm, determine (a) the range of values of v_0 , (b) the values of α corresponding to $h = 788$ mm and $h = 1068$ mm.

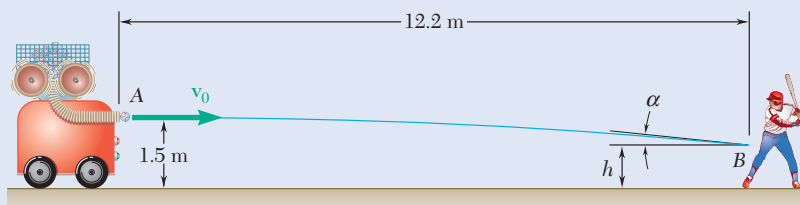


Fig. P11.99

- 11.100** While delivering newspapers, a girl throws a newspaper with a horizontal velocity v_0 . Determine the range of values of v_0 if the newspaper is to land between points B and C.

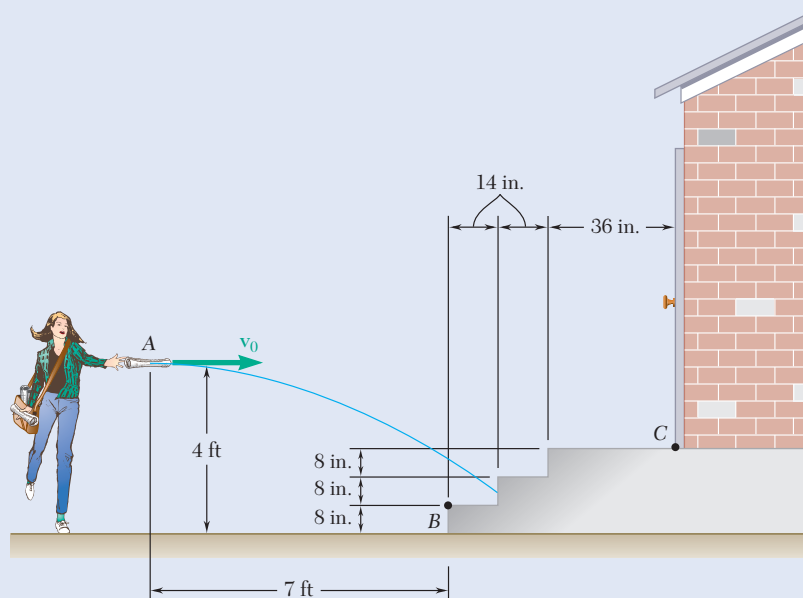


Fig. P11.100

- 11.101** Water flows from a drain spout with an initial velocity of 2.5 ft/s at an angle of 15° with the horizontal. Determine the range of values of the distance d for which the water will enter the trough BC .

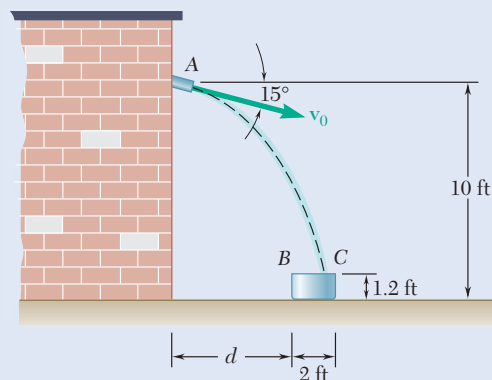


Fig. P11.101

- 11.102** In slow pitch softball, the underhand pitch must reach a maximum height of between 1.8 m and 3.7 m above the ground. A pitch is made with an initial velocity v_0 with a magnitude of 13 m/s at an angle of 33° with the horizontal. Determine (a) if the pitch meets the maximum height requirement, (b) the height of the ball as it reaches the batter.

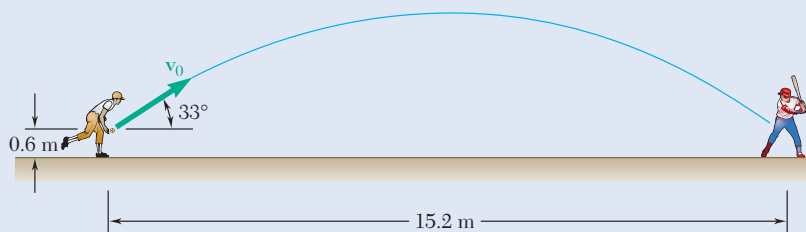


Fig. P11.102

- 11.103** A volleyball player serves the ball with an initial velocity v_0 of magnitude 13.40 m/s at an angle of 20° with the horizontal. Determine (a) if the ball will clear the top of the net, (b) how far from the net the ball will land.

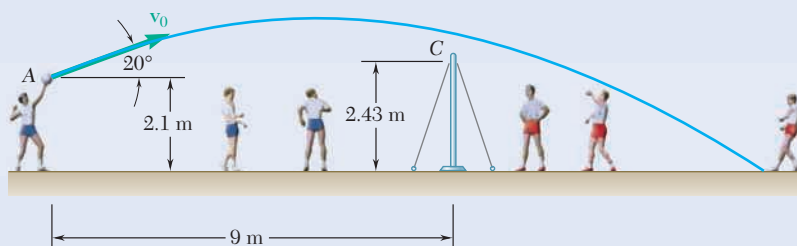


Fig. P11.103

- 11.104** A golfer hits a golf ball with an initial velocity of 160 ft/s at an angle of 25° with the horizontal. Knowing that the fairway slopes downward at an average angle of 5° , determine the distance d between the golfer and point B where the ball first lands.

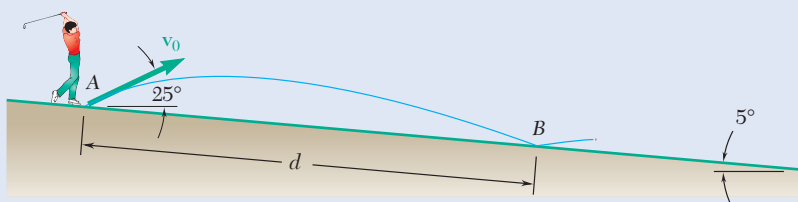


Fig. P11.104

- 11.105** A homeowner uses a snowblower to clear his driveway. Knowing that the snow is discharged at an average angle of 40° with the horizontal, determine the initial velocity v_0 of the snow.

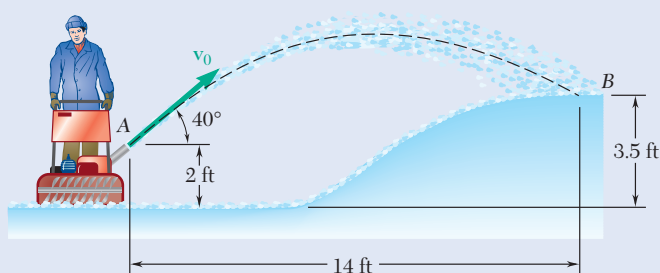


Fig. P11.105

- 11.106** At halftime of a football game souvenir balls are thrown to the spectators with a velocity v_0 . Determine the range of values of v_0 if the balls are to land between points B and C .

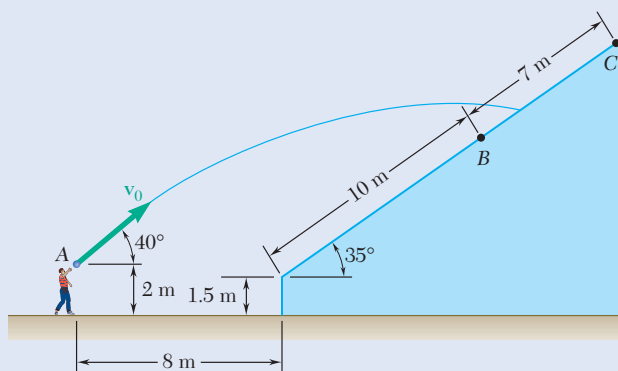


Fig. P11.106

- 11.107** A basketball player shoots when she is 16 ft from the backboard. Knowing that the ball has an initial velocity v_0 at an angle of 30° with the horizontal, determine the value of v_0 when d is equal to (a) 9 in., (b) 17 in.

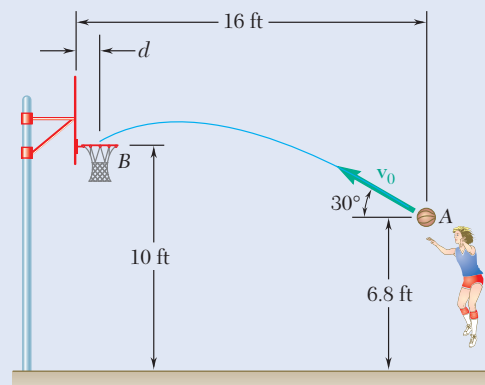


Fig. P11.107

- 11.108** A tennis player serves the ball at a height $h = 2.5$ m with an initial velocity of v_0 at an angle of 5° with the horizontal. Determine the range of v_0 for which the ball will land in the service area that extends to 6.4 m beyond the net.

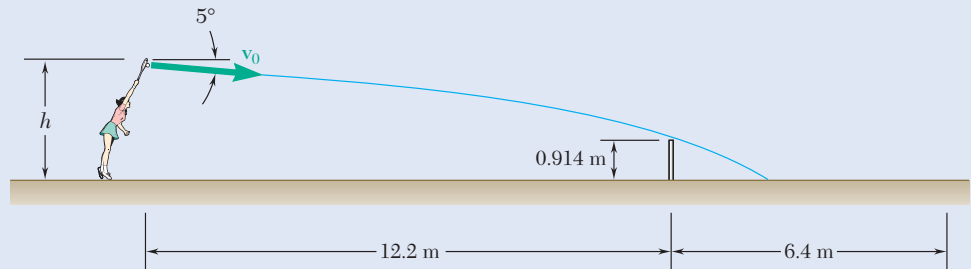


Fig. P11.108

- 11.109** The nozzle at A discharges cooling water with an initial velocity v_0 at an angle of 6° with the horizontal onto a grinding wheel 350 mm in diameter. Determine the range of values of the initial velocity for which the water will land on the grinding wheel between points B and C.

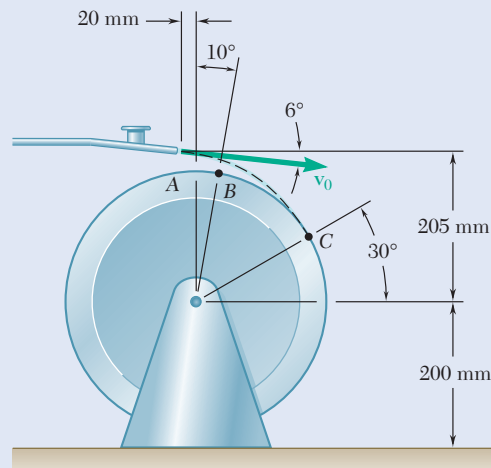


Fig. P11.109

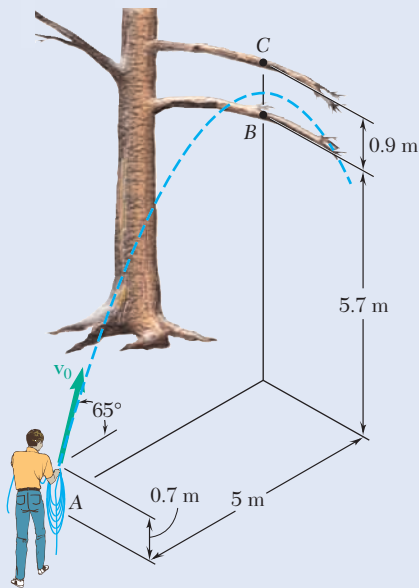


Fig. P11.110

- 11.110** While holding one of its ends, a worker lobs a coil of rope over the lowest limb of a tree. If he throws the rope with an initial velocity v_0 at an angle of 65° with the horizontal, determine the range of values of v_0 for which the rope will go over only the lowest limb.

- 11.111** The pitcher in a softball game throws a ball with an initial velocity v_0 of 72 km/h at an angle α with the horizontal. If the height of the ball at point B is 0.68 m, determine (a) the angle α , (b) the angle θ that the velocity of the ball at point B forms with the horizontal.

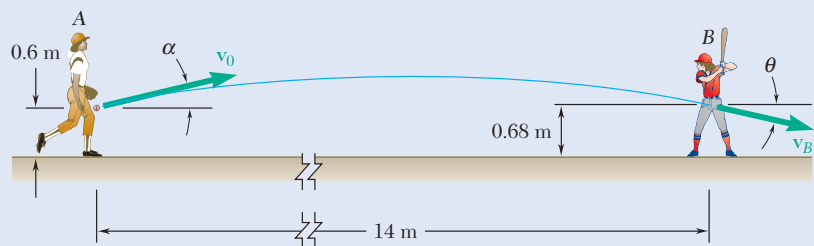


Fig. P11.111

11.112 A model rocket is launched from point A with an initial velocity v_0 of 75 m/s. If the rocket's descent parachute does not deploy and the rocket lands a distance $d = 100$ m from A , determine (a) the angle α that v_0 forms with the vertical, (b) the maximum height above point A reached by the rocket, (c) the duration of the flight.

11.113 The initial velocity v_0 of a hockey puck is 105 mi/h. Determine (a) the largest value (less than 45°) of the angle α for which the puck will enter the net, (b) the corresponding time required for the puck to reach the net.

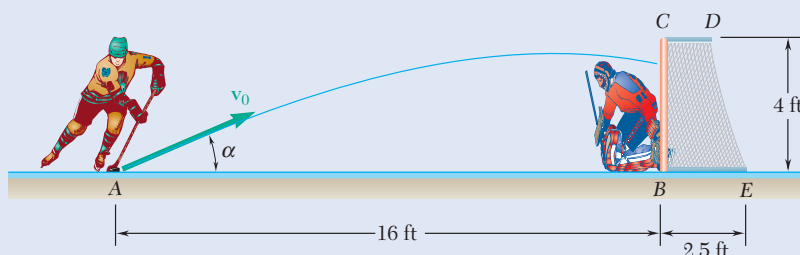


Fig. P11.113

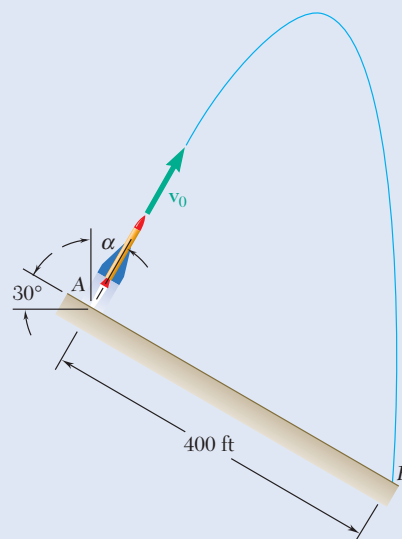


Fig. P11.112

11.114 A worker uses high-pressure water to clean the inside of a long drainpipe. If the water is discharged with an initial velocity v_0 of 11.5 m/s, determine (a) the distance d to the farthest point B on the top of the pipe that the worker can wash from his position at A , (b) the corresponding angle α .

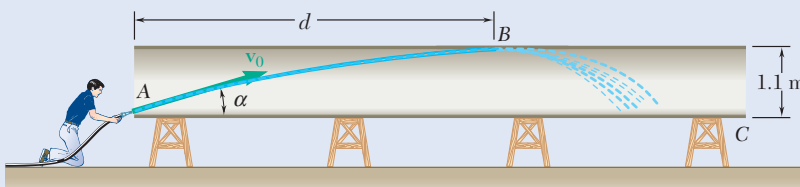


Fig. P11.114

11.115 An oscillating garden sprinkler which discharges water with an initial velocity v_0 of 8 m/s is used to water a vegetable garden. Determine the distance d to the farthest point B that will be watered and the corresponding angle α when (a) the vegetables are just beginning to grow, (b) the height h of the corn is 1.8 m.

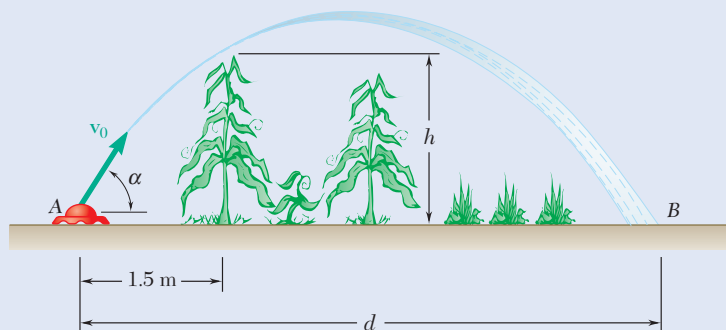


Fig. P11.115

- *11.116** A nozzle at A discharges water with an initial velocity of 36 ft/s at an angle α with the horizontal. Determine (a) the distance d to the farthest point B on the roof that the water can reach, (b) the corresponding angle α . Check that the stream will clear the edge of the roof.

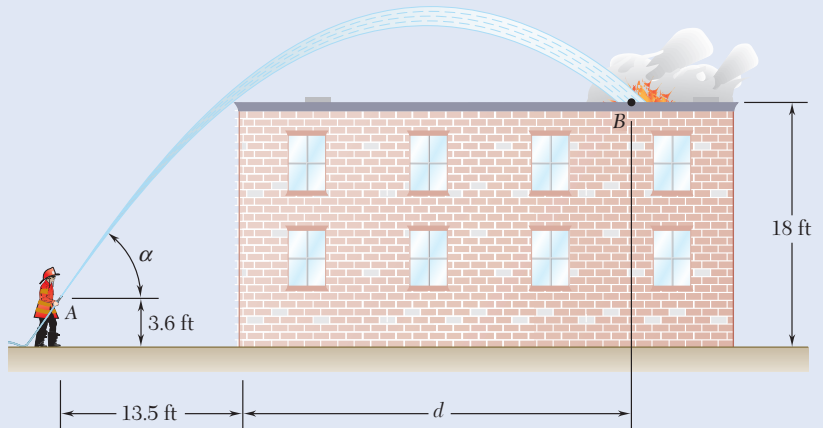


Fig. P11.116

- 11.117** The velocities of skiers A and B are as shown. Determine the velocity of A with respect to B .

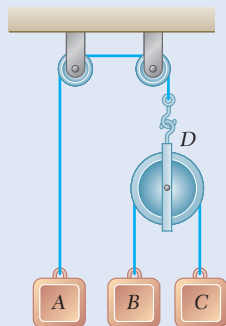


Fig. P11.118

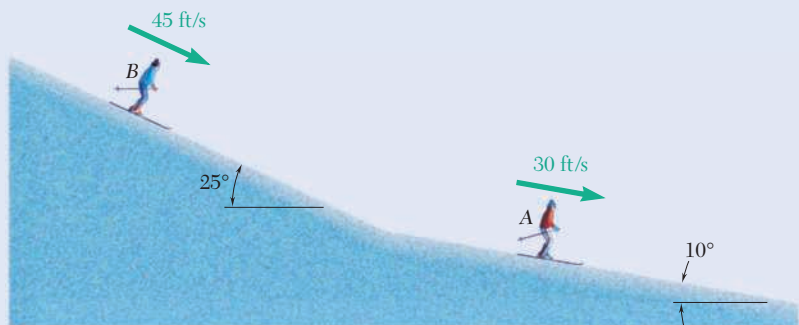


Fig. P11.117

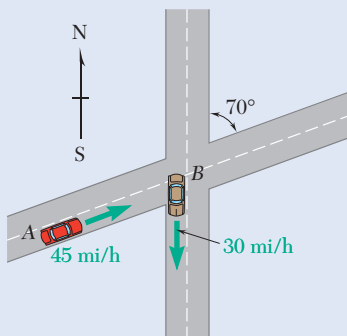


Fig. P11.119

- 11.118** The three blocks shown move with constant velocities. Find the velocity of each block, knowing that the relative velocity of A with respect to C is 300 mm/s upward and that the relative velocity of B with respect to A is 200 mm/s downward.

- 11.119** Three seconds after automobile B passes through the intersection shown, automobile A passes through the same intersection. Knowing that the speed of each automobile is constant, determine (a) the relative velocity of B with respect to A , (b) the change in position of B with respect to A during a 4-s interval, (c) the distance between the two automobiles 2 s after A has passed through the intersection.

11.120 Shore-based radar indicates that a ferry leaves its slip with a velocity $\mathbf{v} = 18 \text{ km/h} \nearrow 70^\circ$, while instruments aboard the ferry indicate a speed of 18.4 km/h and a heading of 30° west of south relative to the river. Determine the velocity of the river.

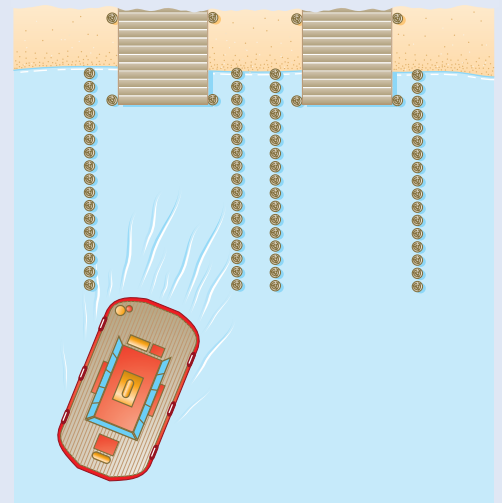


Fig. P11.120

11.121 Airplanes A and B are flying at the same altitude and are tracking the eye of hurricane C . The relative velocity of C with respect to A is $\mathbf{v}_{C/A} = 350 \text{ km/h} \nearrow 75^\circ$, and the relative velocity of C with respect to B is $\mathbf{v}_{C/B} = 400 \text{ km/h} \swarrow 40^\circ$. Determine (a) the relative velocity of B with respect to A , (b) the velocity of A if ground-based radar indicates that the hurricane is moving at a speed of 30 km/h due north, (c) the change in position of C with respect to B during a 15-min interval.

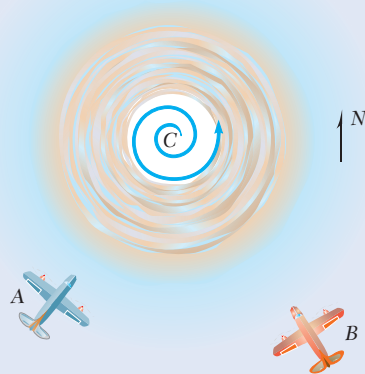


Fig. P11.121

11.122 Instruments in an airplane which is in level flight indicate that the velocity relative to the air (airspeed) is 120 km/h and the direction of the relative velocity vector (heading) is 70° east of north. Instruments on the ground indicate that the velocity of the airplane (ground speed) is 110 km/h and the direction of flight (course) is 60° east of north. Determine the wind speed and direction.

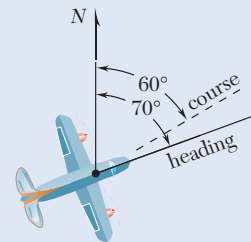


Fig. P11.122

11.123 Knowing that at the instant shown block B has a velocity of 2 ft/s to the right and an acceleration of 3 ft/s^2 to the left, determine (a) the velocity of block A , (b) the acceleration of block A .

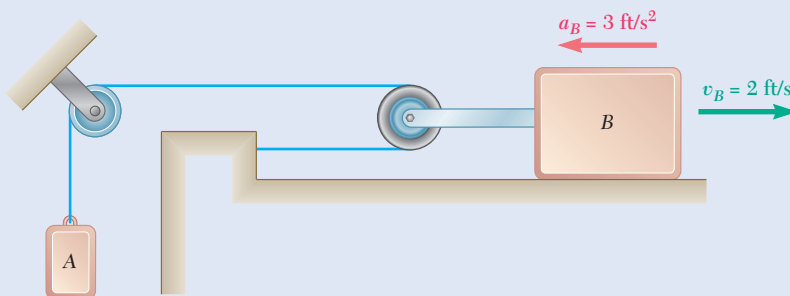


Fig. P11.123

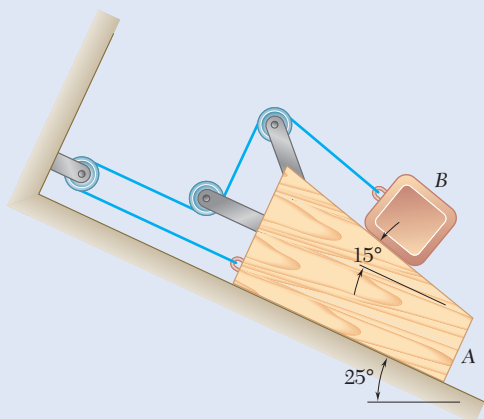


Fig. P11.124

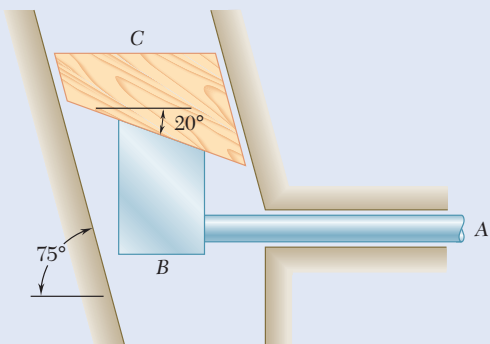


Fig. P11.126

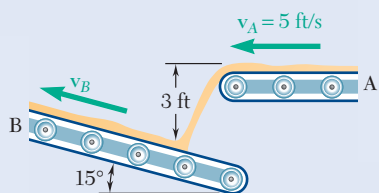


Fig. P11.127

11.124 Knowing that at the instant shown block A has a velocity of 8 in./s and an acceleration of 6 in./s^2 both directed down the incline, determine (a) the velocity of block B , (b) the acceleration of block B .

11.125 A boat is moving to the right with a constant deceleration of 0.3 m/s^2 when a boy standing on the deck D throws a ball with an initial velocity relative to the deck which is vertical. The ball rises to a maximum height of 8 m above the release point and the boy must step forward a distance d to catch it at the same height as the release point. Determine (a) the distance d , (b) the relative velocity of the ball with respect to the deck when the ball is caught.

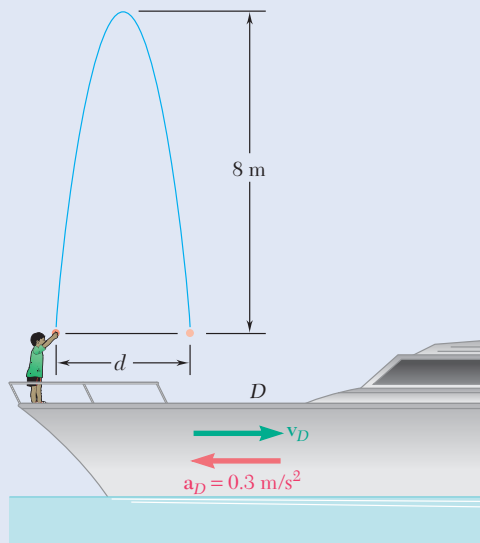


Fig. P11.125

11.126 The assembly of rod A and wedge B starts from rest and moves to the right with a constant acceleration of 2 mm/s^2 . Determine (a) the acceleration of wedge C , (b) the velocity of wedge C when $t = 10 \text{ s}$.

11.127 Determine the required velocity of the belt B if the relative velocity with which the sand hits belt B is to be (a) vertical, (b) as small as possible.

11.128 Conveyor belt A , which forms a 20° angle with the horizontal, moves at a constant speed of 4 ft/s and is used to load an airplane. Knowing that a worker tosses duffel bag B with an initial velocity of 2.5 ft/s at an angle of 30° with the horizontal, determine the velocity of the bag relative to the belt as it lands on the belt.

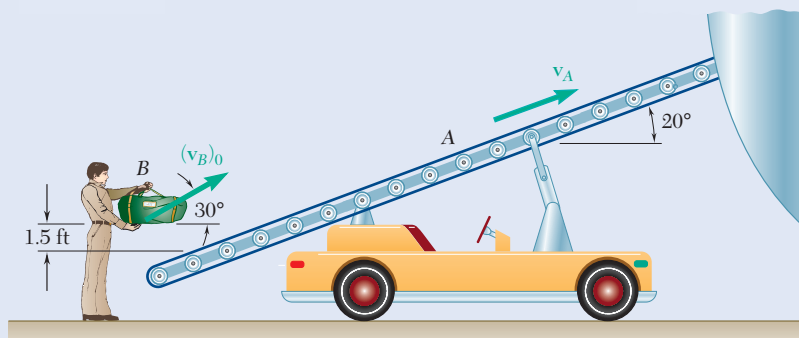
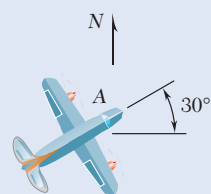


Fig. P11.128

11.129 During a rainstorm the paths of the raindrops appear to form an angle of 30° with the vertical and to be directed to the left when observed from a side window of a train moving at a speed of 15 km/h. A short time later, after the speed of the train has increased to 24 km/h, the angle between the vertical and the paths of the drops appears to be 45° . If the train were stopped, at what angle and with what velocity would the drops be observed to fall?



11.130 Instruments in airplane *A* indicate that, with respect to the air, the plane is headed 30° north of east with an air speed of 300 mi/h. At the same time, radar on ship *B* indicates that the relative velocity of the plane with respect to the ship is 280 mi/h in the direction 33° north of east. Knowing that the ship is steaming due south at 12 mi/h, determine (a) the velocity of the airplane, (b) the wind speed and direction.

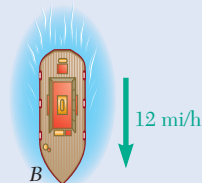


Fig. P11.130

11.131 When a small boat travels north at 5 km/h, a flag mounted on its stern forms an angle $\theta = 50^\circ$ with the centerline of the boat as shown. A short time later, when the boat travels east at 20 km/h, angle θ is again 50° . Determine the speed and the direction of the wind.

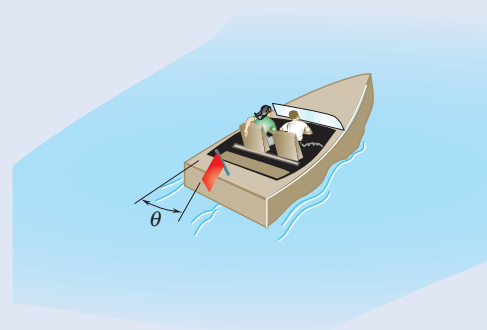


Fig. P11.131

11.132 As part of a department store display, a model train *D* runs on a slight incline between the store's up and down escalators. When the train and shoppers pass point *A*, the train appears to a shopper on the up escalator *B* to move downward at an angle of 22° with the horizontal, and to a shopper on the down escalator *C* to move upward at an angle of 23° with the horizontal and to travel to the left. Knowing that the speed of the escalators is 3 ft/s, determine the speed and the direction of the train.

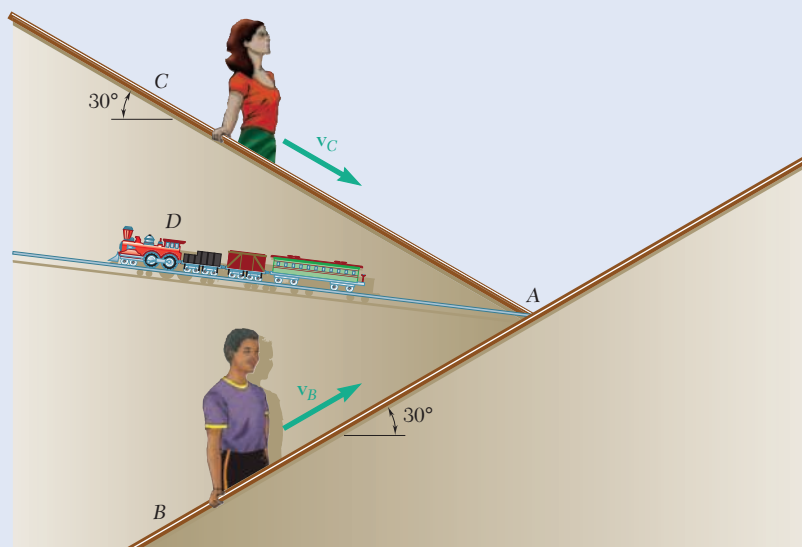


Fig. P11.132

11.5 NON-RECTANGULAR COMPONENTS

Sometimes it is useful to analyze the motion of a particle in a coordinate system that is not rectangular. In this section, we introduce two common and important systems. The first system is based on the path of the particle; the second system is based on the radial distance and angular displacement of the particle.

11.5A Tangential and Normal Components

We saw in Sec. 11.4 that the velocity of a particle is a vector tangent to the path of the particle, but in general, the acceleration is not tangent to the path. It is sometimes convenient to resolve the acceleration into components directed, respectively, along the tangent and the normal to the path of the particle. We will refer to this reference frame as tangential and normal coordinates, which are sometimes called path coordinates.

Planar Motion of a Particle. First we consider a particle that moves along a curve contained in a plane. Let P be the position of the particle at a given instant. We attach at P a unit vector $\mathbf{e}_t$ tangent to the path of the particle and pointing in the direction of motion (Fig. 11.19a). Let $\mathbf{e}'_t$ be the unit vector corresponding to the position P' of the particle at a later instant. Drawing both vectors from the same origin O' , we define the vector $\Delta\mathbf{e}_t = \mathbf{e}'_t - \mathbf{e}_t$ (Fig. 11.19b). Since $\mathbf{e}_t$ and $\mathbf{e}'_t$ are of unit length, their tips lie on a circle with a radius of 1. Denote the angle formed by $\mathbf{e}_t$ and $\mathbf{e}'_t$ by $\Delta\theta$. Then the magnitude of $\Delta\mathbf{e}_t$ is $2 \sin(\Delta\theta/2)$. Considering now the vector $\Delta\mathbf{e}_t/\Delta\theta$, we note that, as $\Delta\theta$ approaches zero, this vector becomes tangent to the unit circle of Fig. 11.19b, i.e., perpendicular to $\mathbf{e}_t$, and that its magnitude approaches

$$\lim_{\Delta\theta \rightarrow 0} \frac{2 \sin(\Delta\theta/2)}{\Delta\theta} = \lim_{\Delta\theta \rightarrow 0} \frac{\sin(\Delta\theta/2)}{\Delta\theta/2} = 1$$

Thus, the vector obtained in the limit is a unit vector along the normal to the path of the particle in the direction toward which $\mathbf{e}_t$ turns. Denoting this vector by $\mathbf{e}_n$, we have

$$\begin{aligned} \mathbf{e}_n &= \lim_{\Delta\theta \rightarrow 0} \frac{\Delta\mathbf{e}_t}{\Delta\theta} \\ \mathbf{e}_n &= \frac{d\mathbf{e}_t}{d\theta} \end{aligned} \quad (11.34)$$

Now, since the velocity $\mathbf{v}$ of the particle is tangent to the path, we can express it as the product of the scalar v and the unit vector $\mathbf{e}_t$. We have

$$\mathbf{v} = v\mathbf{e}_t \quad (11.35)$$

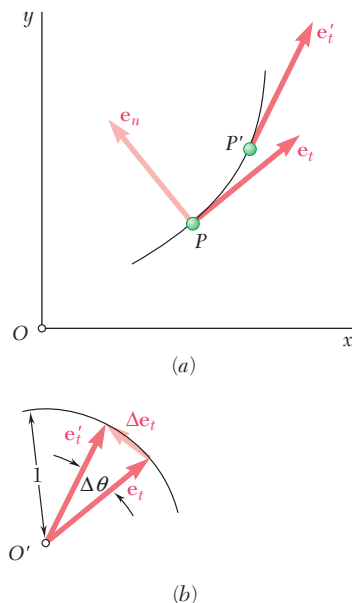


Fig. 11.19 (a) Unit tangent vectors for two positions of particle P ; (b) the angle between the unit tangent vectors and their difference $\Delta\mathbf{e}_t$.

To obtain the acceleration of the particle, we differentiate Eq. (11.35) with respect to t . Applying the rule for the differentiation of the product of a scalar and a vector function (Sec. 11.4B), we have

$$\mathbf{a} = \frac{dv}{dt} \mathbf{e}_t + v \frac{d\mathbf{e}_t}{dt} \quad (11.36)$$

However,

$$\frac{d\mathbf{e}_t}{dt} = \frac{d\mathbf{e}_t}{d\theta} \frac{d\theta}{ds} \frac{ds}{dt}$$

Recall from Eq. (11.15) that $ds/dt = v$, from Eq. (11.34) that $d\mathbf{e}_t/d\theta = \mathbf{e}_n$, and from elementary calculus that $d\theta/ds$ is equal to $1/\rho$, where ρ is the radius of curvature of the path at P (Fig. 11.20). Then we have

$$\frac{d\mathbf{e}_t}{dt} = \frac{v}{\rho} \mathbf{e}_n \quad (11.37)$$

Substituting into Eq. (11.36), we obtain

Acceleration in normal and tangential components

$$\mathbf{a} = \frac{dv}{dt} \mathbf{e}_t + \frac{v^2}{\rho} \mathbf{e}_n \quad (11.38)$$

Thus, the scalar components of the acceleration are

$$a_t = \frac{dv}{dt} \quad a_n = \frac{v^2}{\rho} \quad (11.39)$$

These relations state that the **tangential component** of the acceleration is equal to the **rate of change of the speed of the particle**, whereas the **normal component** is equal to the **square of the speed divided by the radius of curvature of the path at P** . For a given speed, the normal acceleration increases as the radius of curvature decreases. If the particle travels in a straight line, then ρ is infinite, and the normal acceleration is zero. If the speed of the particle increases, a_t is positive, and the vector component $\mathbf{a}_t$ points in the direction of motion. If the speed of the particle decreases, a_t is negative, and $\mathbf{a}_t$ points against the direction of motion. The vector component $\mathbf{a}_n$, on the other hand, **is always directed toward the center of curvature C of the path** (Fig. 11.21).

We conclude from this discussion that the tangential component of the acceleration reflects a change in the speed of the particle, whereas its normal component reflects a change in the direction of motion of the particle. The acceleration of a particle is zero only if both of its components are zero. Thus, the acceleration of a particle moving with constant speed along a curve is not zero unless the particle happens to pass through a point of inflection of the curve (where the radius of curvature is infinite) or unless the curve is a straight line.

The fact that the normal component of acceleration depends upon the radius of curvature of the particle's path is taken into account in the design of structures or mechanisms as widely different as airplane wings, railroad tracks, and cams. In order to avoid sudden changes in the acceleration of the air particles flowing past a wing, wing profiles are designed without any sudden change in curvature. Similar care is taken in designing

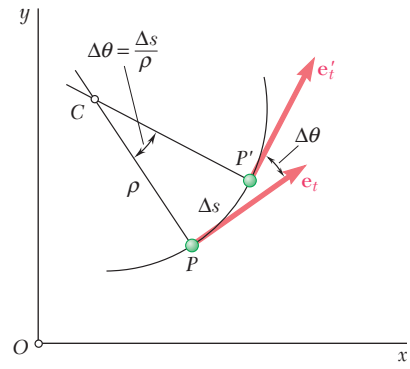


Fig. 11.20 Relationship among $\Delta\theta$, Δs , and ρ . Recall that for a circle, the arc length is equal to the radius multiplied by the angle.

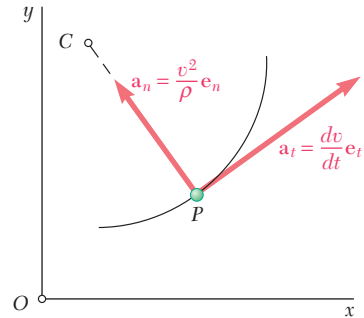


Fig. 11.21 Acceleration components in normal and tangential coordinates; the normal component always points toward the center of curvature of the path.



Photo 11.5 The passengers in a train traveling around a curve experience a normal acceleration toward the center of curvature of the path.

railroad curves to avoid sudden changes in the acceleration of the cars (which would be hard on the equipment and unpleasant for the passengers). A straight section of track, for instance, is never directly followed by a circular section. Special transition sections are used to help pass smoothly from the infinite radius of curvature of the straight section to the finite radius of the circular track. Likewise, in the design of high-speed cams (that can be used to transform rotary motion into translational motion), abrupt changes in acceleration are avoided by using transition curves that produce a continuous change in acceleration.

Motion of a Particle in Space. The relations in Eqs. (11.38) and (11.39) still hold in the case of a particle moving along a space curve. However, since an infinite number of straight lines are perpendicular to the tangent at a given point P of a space curve, it is necessary to define more precisely the direction of the unit vector $\mathbf{e}_n$.

Let us consider again the unit vectors $\mathbf{e}_t$ and $\mathbf{e}'_t$ tangent to the path of the particle at two neighboring points P and P' (Fig. 11.22a). Again the vector $\Delta\mathbf{e}_t$ represents the difference between $\mathbf{e}_t$ and $\mathbf{e}'_t$ (Fig. 11.22b). Let us now imagine a plane through P (Fig. 11.22c) parallel to the plane defined by the vectors $\mathbf{e}_t$, $\mathbf{e}'_t$, and $\Delta\mathbf{e}_t$ (Fig. 11.22b). This plane contains the tangent to the curve at P and is parallel to the tangent at P' . If we let P' approach P , we obtain in the limit the plane that fits the curve most closely in the neighborhood of P . This plane is called the **osculating plane** at P (from the Latin *osculari*, to kiss). It follows from this definition that the osculating plane contains the unit vector $\mathbf{e}_n$, since this vector represents the limit of the vector $\Delta\mathbf{e}_t/\Delta\theta$. The normal defined by $\mathbf{e}_n$ is thus contained in the osculating plane; it is called the **principal normal** at P . The unit vector $\mathbf{e}_b = \mathbf{e}_t \times \mathbf{e}_n$ that completes the right-handed triad $\mathbf{e}_t$, $\mathbf{e}_n$, and $\mathbf{e}_b$ (Fig. 11.22c) defines the **binormal** at P . The binormal is thus perpendicular to the osculating plane. We conclude that the acceleration of the particle at P can be resolved into two components: one along the tangent and the other along the principal normal at P , as indicated in Eq. (11.38). Note that the acceleration has no component along the binormal.

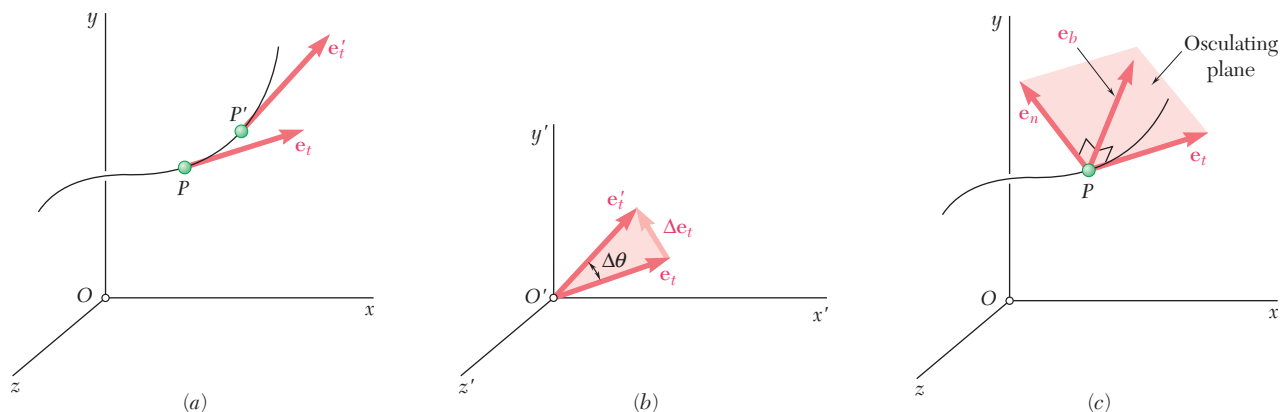


Fig. 11.22 (a) Unit tangent vectors for a particle moving in space; (b) the plane defined by the unit vectors and the vector difference $\Delta\mathbf{e}_t$; (c) the osculating plane contains the unit tangent and principal normal vectors and is perpendicular to the unit binormal vector.

11.5B Radial and Transverse Components

In some situations in planar motion, the position of particle P is defined by its polar coordinates r and θ (Fig. 11.23a). It is then convenient to resolve the velocity and acceleration of the particle into components parallel and perpendicular to the radial line OP . These components are called **radial and transverse components**.

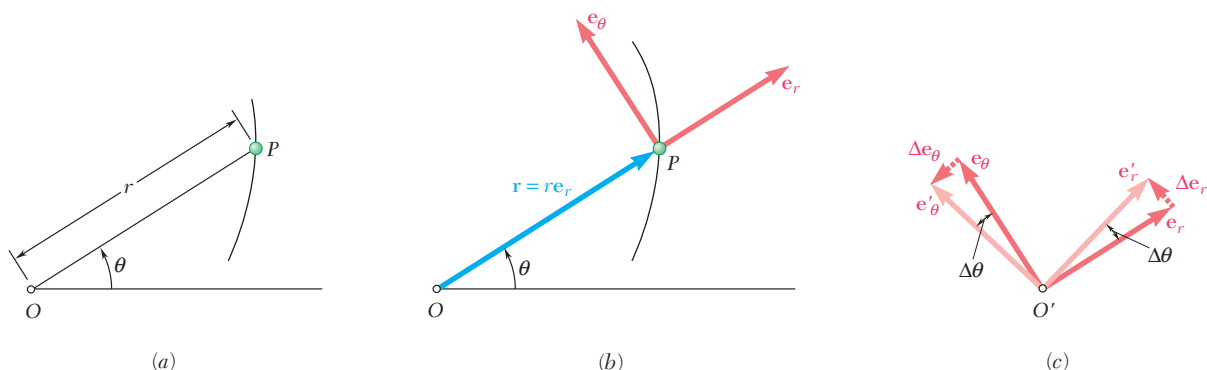


Fig. 11.23 (a) Polar coordinates r and θ of a particle at P ; (b) radial and transverse unit vectors; (c) changes of the radial and transverse unit vectors resulting from a change in angle $\Delta\theta$.

We attach two unit vectors, $\mathbf{e}_r$ and $\mathbf{e}_\theta$, at P (Fig. 11.23b). The vector $\mathbf{e}_r$ is directed along OP and the vector $\mathbf{e}_\theta$ is obtained by rotating $\mathbf{e}_r$ through 90° counterclockwise. The unit vector $\mathbf{e}_r$ defines the **radial** direction, i.e., the direction in which P would move if r were increased and θ were kept constant. The unit vector $\mathbf{e}_\theta$ defines the **transverse** direction, i.e., the direction in which P would move if θ were increased and r were kept constant. A derivation similar to the one we used in the preceding section to determine the unit vector $\mathbf{e}_t$ leads to the relations

$$\frac{d\mathbf{e}_r}{d\theta} = \mathbf{e}_\theta \quad \frac{d\mathbf{e}_\theta}{d\theta} = -\mathbf{e}_r \quad (11.40)$$

Here $-\mathbf{e}_r$ denotes a unit vector with a sense opposite to that of $\mathbf{e}_r$ (Fig. 11.23c). Using the chain rule of differentiation, we express the time derivatives of the unit vectors $\mathbf{e}_r$ and $\mathbf{e}_\theta$ as

$$\frac{d\mathbf{e}_r}{dt} = \frac{d\mathbf{e}_r}{d\theta} \frac{d\theta}{dt} = \mathbf{e}_\theta \frac{d\theta}{dt} \quad \frac{d\mathbf{e}_\theta}{dt} = \frac{d\mathbf{e}_\theta}{d\theta} \frac{d\theta}{dt} = -\mathbf{e}_r \frac{d\theta}{dt}$$

or using dots to indicate differentiation with respect to t as

$$\dot{\mathbf{e}}_r = \dot{\theta} \mathbf{e}_\theta \quad \dot{\mathbf{e}}_\theta = -\dot{\theta} \mathbf{e}_r \quad (11.41)$$

To obtain the velocity $\mathbf{v}$ of particle P , we express the position vector $\mathbf{r}$ of P as the product of the scalar r and the unit vector $\mathbf{e}_r$ and then differentiate with respect to t for

$$\mathbf{v} = \frac{d}{dt}(r\mathbf{e}_r) = r\dot{\mathbf{e}}_r + \dot{r}\mathbf{e}_r$$



Photo 11.6 The foot pedals on an elliptical trainer undergo curvilinear motion.

Using the first of the relations of Eq. (11.41), we can rewrite this as

Velocity in radial and transverse components

$$\mathbf{v} = \dot{r}\mathbf{e}_r + r\dot{\theta}\mathbf{e}_\theta \quad (11.42)$$

Differentiating again with respect to t to obtain the acceleration, we have

$$\mathbf{a} = \frac{d\mathbf{v}}{dt} = \ddot{r}\mathbf{e}_r + \dot{r}\dot{\mathbf{e}}_r + \dot{r}\dot{\theta}\mathbf{e}_\theta + r\ddot{\theta}\mathbf{e}_\theta + r\dot{\theta}\dot{\mathbf{e}}_\theta$$

Substituting for $\dot{\mathbf{e}}_r$ and $\dot{\mathbf{e}}_\theta$ from Eq. (11.41) and factoring $\mathbf{e}_r$ and $\mathbf{e}_\theta$, we obtain

Acceleration in radial and transverse components

$$\mathbf{a} = (\ddot{r} - r\dot{\theta}^2)\mathbf{e}_r + (r\ddot{\theta} + 2\dot{r}\dot{\theta})\mathbf{e}_\theta \quad (11.43)$$

The scalar components of the velocity and the acceleration in the radial and transverse directions are

$$v_r = \dot{r} \quad v_\theta = r\dot{\theta} \quad (11.44)$$

$$a_r = \ddot{r} - r\dot{\theta}^2 \quad a_\theta = r\ddot{\theta} + 2\dot{r}\dot{\theta} \quad (11.45)$$

It is important to note that a_r is *not* equal to the time derivative of v_r and that a_θ is *not* equal to the time derivative of v_θ .

In the case of a particle moving along a circle with a center O , we have $r = \text{constant}$ and $\dot{r} = \ddot{r} = 0$, so the formulas (11.42) and (11.43) reduce, respectively, to

$$\mathbf{v} = r\dot{\theta}\mathbf{e}_\theta \quad \mathbf{a} = -r\dot{\theta}^2\mathbf{e}_r + r\ddot{\theta}\mathbf{e}_\theta \quad (11.46)$$

Compare this to using tangential and normal coordinates for a particle in a circular path. In this case, the radius of curvature ρ is equal to the radius of the circle r , and we have $\mathbf{v} = v\mathbf{e}_t$ and $\mathbf{a} = \dot{v}\mathbf{e}_t + (v^2/r)\mathbf{e}_n$. Note that $\mathbf{e}_r$ and $\mathbf{e}_n$ point in opposite directions ($\mathbf{e}_n$ inward and $\mathbf{e}_r$ outward).

Extension to the Motion of a Particle in Space: Cylindrical Coordinates.

Sometimes it is convenient to define the position of a particle P in space by its cylindrical coordinates R , θ , and z (Fig. 11.24a). We can then use the unit vectors $\mathbf{e}_R$, $\mathbf{e}_\theta$, and $\mathbf{k}$ shown in Fig. 11.24b. Resolving the position vector $\mathbf{r}$ of particle P into components along the unit vectors, we have

$$\mathbf{r} = R\mathbf{e}_R + z\mathbf{k} \quad (11.47)$$

Observe that $\mathbf{e}_R$ and $\mathbf{e}_\theta$ define the radial and transverse directions in the horizontal xy plane, respectively, and that the vector $\mathbf{k}$, which defines the **axial** direction, is constant in direction as well as in magnitude. Then we can verify that

$$\mathbf{v} = \frac{d\mathbf{r}}{dt} = \dot{R}\mathbf{e}_R + R\dot{\theta}\mathbf{e}_\theta + \dot{z}\mathbf{k} \quad (11.48)$$

$$\mathbf{a} = \frac{d\mathbf{v}}{dt} = (\ddot{R} - R\dot{\theta}^2)\mathbf{e}_R + (R\ddot{\theta} + 2\dot{R}\dot{\theta})\mathbf{e}_\theta + \ddot{z}\mathbf{k} \quad (11.49)$$

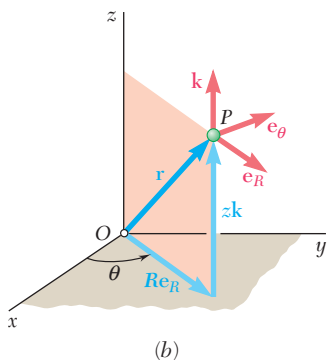
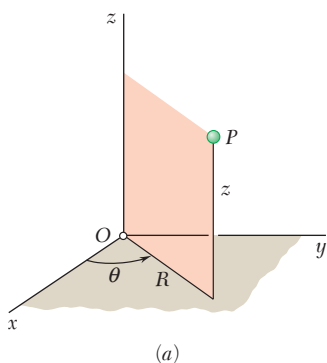
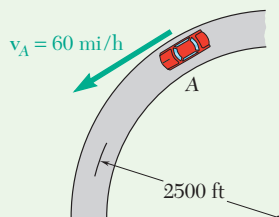


Fig. 11.24 (a) Cylindrical coordinates R , θ , and z ; (b) unit vectors in cylindrical coordinates for a particle in space.



Sample Problem 11.16

A motorist is traveling on a curved section of highway with a radius of 2500 ft at a speed of 60 mi/h. The motorist suddenly applies the brakes, causing the automobile to slow down at a constant rate. If the speed has been reduced to 45 mi/h after 8 s, determine the acceleration of the automobile immediately after the brakes have been applied.

STRATEGY: You know the path of the motion, and that the forward speed of the vehicle defines the direction of $\mathbf{e}_t$. Therefore, you can use tangential and normal components.

MODELING and ANALYSIS:

Tangential Component of Acceleration. First express the speeds in ft/s.

$$60 \text{ mi/h} = \left(60 \frac{\text{mi}}{\text{h}}\right) \left(\frac{5280 \text{ ft}}{1 \text{ mi}}\right) \left(\frac{1 \text{ h}}{3600 \text{ s}}\right) = 88 \text{ ft/s}$$

$$45 \text{ mi/h} = 66 \text{ ft/s}$$

Since the automobile slows down at a constant rate, you have the tangential acceleration of

$$a_t = \text{average } a_t = \frac{\Delta v}{\Delta t} = \frac{66 \text{ ft/s} - 88 \text{ ft/s}}{8 \text{ s}} = -2.75 \text{ ft/s}^2$$

Normal Component of Acceleration. Immediately after the brakes have been applied, the speed is still 88 ft/s. Therefore, you have

$$a_n = \frac{v^2}{\rho} = \frac{(88 \text{ ft/s})^2}{2500 \text{ ft}} = 3.10 \text{ ft/s}^2$$

Magnitude and Direction of Acceleration. The magnitude and direction of the resultant $\mathbf{a}$ of the components $\mathbf{a}_n$ and $\mathbf{a}_t$ are (Fig. 1)

$$\tan \alpha = \frac{a_n}{a_t} = \frac{3.10 \text{ ft/s}^2}{2.75 \text{ ft/s}^2} \quad \alpha = 48.4^\circ$$

$$a = \frac{a_n}{\sin \alpha} = \frac{3.10 \text{ ft/s}^2}{\sin 48.4^\circ} \quad \mathbf{a} = 4.14 \text{ ft/s}^2$$

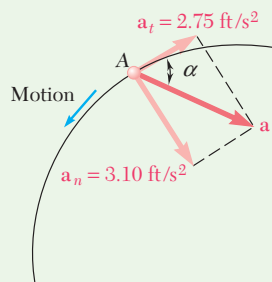


Fig. 1 Acceleration of the car.

REFLECT and THINK: The tangential component of acceleration is opposite the direction of motion, and the normal component of acceleration points to the center of curvature, which is what you would expect for slowing down on a curved path. Attempting to do this problem in Cartesian coordinates is quite difficult.

Sample Problem 11.17

Determine the minimum radius of curvature of the trajectory described by the projectile considered in Sample Prob. 11.10.

STRATEGY: You are asked to find the radius of curvature, so you should use normal and tangential coordinates.

MODELING and ANALYSIS: Since $a_n = v^2/\rho$, you have $\rho = v^2/a_n$. Therefore, the radius is small when v is small or when a_n is large. The speed v is minimum at the top of the trajectory, since $v_y = 0$ at that point; a_n is maximum at that same point, since the direction of the vertical coincides with the direction of the normal (Fig. 1). Therefore, the minimum radius of curvature occurs at the top of the trajectory. At this point, you have

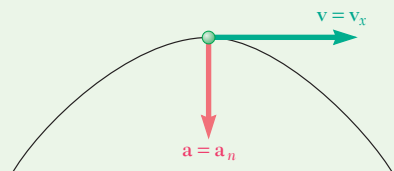


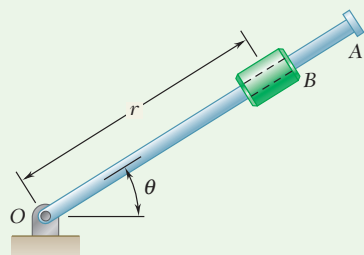
Fig. 1 Acceleration and velocity of the projectile.

$$v = v_x = 155.9 \text{ m/s} \quad a_n = a = 9.81 \text{ m/s}^2$$

$$\rho = \frac{v^2}{a_n} = \frac{(155.9 \text{ m/s})^2}{9.81 \text{ m/s}^2} \quad \rho = 2480 \text{ m} \quad \blacktriangleleft$$

REFLECT and THINK: The top of the trajectory is the easiest point to determine the radius of curvature. At any other point in the trajectory, you need to find the normal component of acceleration. You can do this easily at the top, because you know that the total acceleration is pointed vertically downward and the normal component is simply the component perpendicular to the tangent to the path. Once you have the normal acceleration, it is straightforward to find the radius of curvature if you know the speed.

Sample Problem 11.18



The rotation of the 0.9-m arm OA about O is defined by the relation $\theta = 0.15t^2$, where θ is expressed in radians and t in seconds. Collar B slides along the arm in such a way that its distance from O is $r = 0.9 - 0.12t^2$, where r is expressed in meters and t in seconds. After the arm OA has rotated through 30° , determine (a) the total velocity of the collar, (b) the total acceleration of the collar, (c) the relative acceleration of the collar with respect to the arm.

STRATEGY: You are given information in terms of r and θ , so you should use polar coordinates.

MODELING and ANALYSIS: Model the collar as a particle.

Time t at which $\theta = 30^\circ$. Substitute $\theta = 30^\circ = 0.524 \text{ rad}$ into the expression for θ . You obtain

$$\theta = 0.15t^2 \quad 0.524 = 0.15t^2 \quad t = 1.869 \text{ s}$$

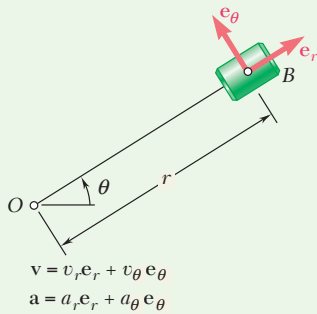


Fig. 1 Radial and transverse coordinates for collar B.

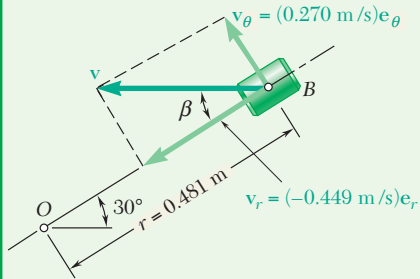


Fig. 2 Velocity of collar B.

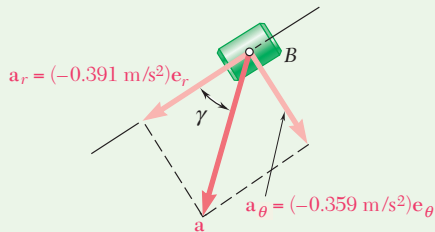


Fig. 3 Acceleration of collar B.

Equations of Motion. Substituting $t = 1.869 \text{ s}$ in the expressions for r , θ , and their first and second derivatives, you have

$$\begin{aligned}
 r &= 0.9 - 0.12t^2 = 0.481 \text{ m} & \theta &= 0.15t^2 = 0.524 \text{ rad} \\
 \dot{r} &= -0.24t = -0.449 \text{ m/s} & \dot{\theta} &= 0.30t = 0.561 \text{ rad/s} \\
 \ddot{r} &= -0.24 = -0.240 \text{ m/s}^2 & \ddot{\theta} &= 0.30 = 0.300 \text{ rad/s}^2
 \end{aligned}$$

a. Velocity of B. Using Eqs. (11.44), you can obtain the values of v_r and v_θ when $t = 1.869 \text{ s}$ (Fig. 1).

$$\begin{aligned}
 v_r &= \dot{r} = -0.449 \text{ m/s} \\
 v_\theta &= r\dot{\theta} = 0.481(0.561) = 0.270 \text{ m/s}
 \end{aligned}$$

Solve the right triangle shown in Fig. 2 to obtain the magnitude and direction of the velocity,

$$v = 0.524 \text{ m/s} \quad \beta = 31.0^\circ \quad \blacktriangleleft$$

b. Acceleration of B. Using Eqs. (11.45), you obtain (Fig. 3)

$$\begin{aligned}
 a_r &= \ddot{r} - r\dot{\theta}^2 \\
 &= -0.240 - 0.481(0.561)^2 = -0.391 \text{ m/s}^2 \\
 a_\theta &= r\ddot{\theta} + 2\dot{r}\dot{\theta} \\
 &= 0.481(0.300) + 2(-0.449)(0.561) = -0.359 \text{ m/s}^2
 \end{aligned}$$

$$a = 0.531 \text{ m/s}^2 \quad \gamma = 42.6^\circ \quad \blacktriangleleft$$

c. Acceleration of B with Respect to Arm OA. Note that the motion of the collar with respect to the arm is rectilinear and defined by the coordinate r (Fig. 4). You have

$$\begin{aligned}
 a_{B/OA} &= \ddot{r} = -0.240 \text{ m/s}^2 \\
 &= 0.240 \text{ m/s}^2 \text{ toward } O. \quad \blacktriangleleft
 \end{aligned}$$

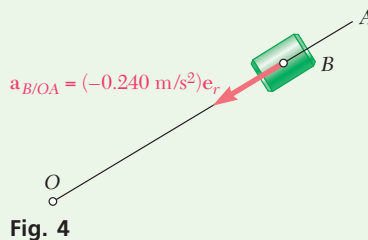


Fig. 4

REFLECT and THINK: You should consider polar coordinates for any kind of rotational motion. They turn this problem into a straightforward solution, whereas any other coordinate system would make this problem much more difficult. One way to make this problem harder would be to ask you to find the radius of curvature in addition to the velocity and acceleration. To do this, you would have to find the normal component of the acceleration; that is, the component of acceleration that is perpendicular to the tangential direction defined by the velocity vector.

Sample Problem 11.19

A boy is flying a kite that is 60 m high with 75 m of cord out. The kite moves horizontally from this position at a constant 6 km/h that is directly away from the boy. Ignoring the sag in the cord, determine how fast the cord is being let out at this instant and how fast this rate is increasing.

STRATEGY: The most natural way to describe the position of the kite is using a radial vector and angle, as shown in Fig. 1. The distance r is changing, so use polar coordinates.

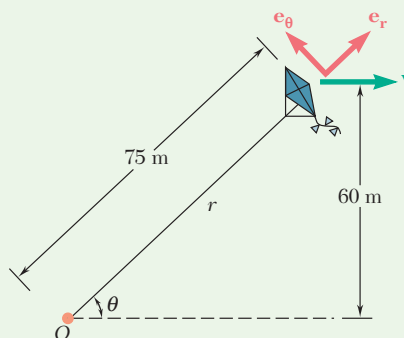
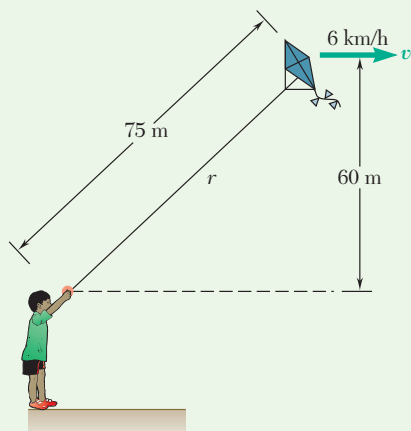


Fig. 1 Radial and transverse coordinates for the kite.

MODELING and ANALYSIS: The angle and the speed of the kite in m/s are found by

$$\theta = \sin^{-1}\left(\frac{60}{75}\right) = 53.13^\circ \quad \text{and} \quad v = 6\left(\frac{\text{km}}{\text{hr}}\right)\left(\frac{\text{hr}}{3600 \text{ s}}\right)\left(\frac{1000 \text{ m}}{\text{km}}\right) = \frac{5}{3} \text{ m/s}$$

Velocity in Polar Coordinates: You know that in polar coordinates the velocity is $\mathbf{v} = \dot{r}\mathbf{e}_r + r\dot{\theta}\mathbf{e}_\theta$. Using Fig. 1, you can resolve the velocity vector into polar coordinates, giving

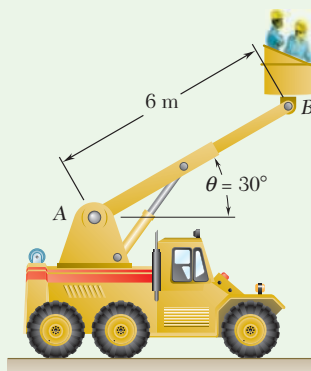
$$\dot{r} = v \cos \theta = \left(\frac{5}{3} \text{ m/s}\right) \cos 53.13^\circ \quad \dot{r} = 1.000 \text{ m/s} \quad \blacktriangleleft$$

$$r\dot{\theta} = -v \sin \theta \quad \dot{\theta} = -\frac{v \sin \theta}{r} = -\frac{(5/3 \text{ m/s}) \sin 53.13^\circ}{75 \text{ m}} = 0.01778 \text{ rad/s}$$

Acceleration in Polar Coordinates: You know that the acceleration is zero, because the kite is traveling at a constant speed. This means that both components of the acceleration need to be zero. You know the radial component is $a_r = \ddot{r} - r\dot{\theta}^2 = 0$. So

$$\ddot{r} = r\dot{\theta}^2 = (75 \text{ m})(-0.01778 \text{ rad/s})^2 \quad \ddot{r} = 0.0237 \text{ m/s}^2 \quad \blacktriangleleft$$

REFLECT and THINK: When the angle is 90° , then $\dot{r}$ will be zero. When the angle is very small—that is, when the kite is far away—you would expect the cord to increase at a rate of 6 m/s, which is the speed of the kite. Our answer is reasonable since it is between these two limits.



Sample Problem 11.20

At the instant shown, the length of the boom AB is being *decreased* at the constant rate of 0.2 m/s , and the boom is being lowered at the constant rate of 0.08 rad/s . Determine (a) the velocity of point B , (b) the acceleration of point B .

STRATEGY: Use polar coordinates, since that is the most natural way to describe the position of point B .

MODELING and ANALYSIS: From the problem statement, you know

$$\dot{r} = -0.2 \text{ m/s} \quad \ddot{r} = 0 \quad \dot{\theta} = -0.08 \text{ rad/s} \quad \ddot{\theta} = 0$$

a. Velocity of B. Using Eqs.(11.44), you can determine the values of v_r and v_θ at this instant to be

$$v_r = \dot{r} = -0.2 \text{ m/s}$$

$$v_\theta = r\dot{\theta} = (6 \text{ m})(-0.08 \text{ rad/s}) = -0.48 \text{ m/s}$$

Therefore, you can write the velocity vector as

$$\mathbf{v} = (-0.200 \text{ m/s})\mathbf{e}_r + (-0.480 \text{ m/s})\mathbf{e}_\theta \quad \blacktriangleleft$$

b. Acceleration of B. Using Eqs. (11.45), you find

$$a_r = \ddot{r} - r\dot{\theta}^2 = 0 - (6 \text{ m})(-0.08 \text{ rad/s})^2 = -0.0384 \text{ m/s}^2$$

$$a_\theta = r\ddot{\theta} + 2\dot{r}\dot{\theta} = 0 + 2(-0.2 \text{ m/s})(-0.08 \text{ rad/s}) = 0.00320 \text{ m/s}^2$$

or

$$\mathbf{a} = (-0.0384 \text{ m/s}^2)\mathbf{e}_r + (0.00320 \text{ m/s}^2)\mathbf{e}_\theta \quad \blacktriangleleft$$

REFLECT and THINK: Once you identify what you are given in the problem statement, this problem is quite straightforward. Sometimes you will be asked to express your answer in terms of a magnitude and direction. The easiest way is to first determine the x and y components and then to find the magnitude and direction. From Fig. 1,

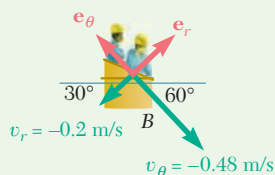


Fig. 1 Velocity of B .

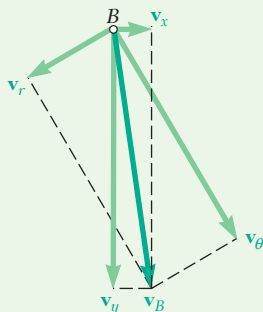


Fig. 2 Resultant velocity of collar B in Cartesian and in radial and transverse coordinates.

$$\rightarrow: (v_B)_x = 0.48 \cos 60^\circ - 0.2 \cos 30^\circ = 0.06680 \text{ m/s}$$

$$+\uparrow: (v_B)_y = -0.48 \sin 60^\circ - 0.2 \sin 30^\circ = -0.5157 \text{ m/s}$$

So the magnitude and direction are

$$v_B = \sqrt{0.06680^2 + 0.5157^2}$$

$$= 0.520 \text{ m/s} \quad \tan \beta = \frac{0.51569}{0.06680}, \quad \beta = 82.6^\circ$$

So, an alternative way of expressing the velocity of B is $\mathbf{v}_B = 0.520 \text{ m/s} \searrow 82.6^\circ$

You could also find the magnitude and direction of the acceleration if you needed it expressed in this way. It is important to note that no matter what coordinate system we choose, the resultant velocity vector is the same. You can choose to express this vector in whatever coordinate system is most useful. Figure 2 shows the velocity vector $\mathbf{v}_B$ resolved into x and y components and r and θ coordinates.

SOLVING PROBLEMS ON YOUR OWN

In the following problems, you will be asked to express the velocity and the acceleration of particles in terms of either their **tangential and normal components** or their **radial and transverse components**. Although these components may not be as familiar to you as rectangular components, you will find that they can simplify the solution of many problems and that certain types of motion are more easily described when they are used.

1. Using tangential and normal components. These components are most often used when the particle of interest travels along a known curvilinear path or when the radius of curvature of the path is to be determined [Sample Prob. 11.16]. Remember that the unit vector $\mathbf{e}_t$ is tangent to the path of the particle (and thus aligned with the velocity), whereas the unit vector $\mathbf{e}_n$ is directed along the normal to the path and always points toward its center of curvature. It follows that the directions of the two unit vectors are constantly changing as the particle moves.

2. Acceleration in terms of tangential and normal components. We derived in Sec. 11.5A the following equation, which is applicable to both the two-dimensional and the three-dimensional motion of a particle:

$$\mathbf{a} = \frac{dv}{dt}\mathbf{e}_t + \frac{v^2}{\rho}\mathbf{e}_n \quad (11.38)$$

The following observations may help you in solving the problems of this section.

a. The tangential component of the acceleration measures the rate of change of the speed as $a_t = dv/dt$. It follows that, when a_t is constant, you can use the equations for uniformly accelerated motion with the acceleration equal to a_t . Furthermore, when a particle moves at a constant speed, we have $a_t = 0$, and the acceleration of the particle reduces to its normal component.

b. The normal component of the acceleration is always directed toward the center of curvature of the path of the particle, and its magnitude is $a_n = v^2/\rho$. Thus, you can determine the normal component if you know the speed of the particle and the radius of curvature ρ of the path. Conversely, if you know the speed and normal acceleration of the particle, you can find the radius of curvature of the path by solving this equation for ρ [Sample Prob. 11.17].

3. Using radial and transverse components. These components are used to analyze the planar motion of a particle P when the position of P is defined by its polar coordinates r and θ . As shown in Fig. 11.23, the unit vector $\mathbf{e}_r$, which defines the **radial** direction, is attached to P and points away from the fixed point O , whereas the unit vector $\mathbf{e}_\theta$, which defines the **transverse** direction, is obtained by rotating $\mathbf{e}_r$ *counterclockwise* through 90° . The velocity and acceleration of a particle are expressed in terms of their radial and transverse components in Eqs. (11.42) and (11.43), respectively. Note that the expressions obtained contain the first and second derivatives with respect to t of both coordinates r and θ .

In the problems of this section, you will encounter the following types of problems involving radial and transverse components.

a. Both r and θ are known functions of t . In this case, you compute the first and second derivatives of r and θ and substitute the resulting expressions into Eqs. (11.42) and (11.43).

b. A certain relationship exists between r and θ . First, you should determine this relationship from the geometry of the given system and use it to express r as a function of θ . Once you know the function $r = f(\theta)$, you can apply the chain rule to determine $\dot{r}$ in terms of θ and $\dot{\theta}$, and $\ddot{r}$ in terms of θ , $\dot{\theta}$, and $\ddot{\theta}$:

$$\begin{aligned}\dot{r} &= f'(\theta)\dot{\theta} \\ \ddot{r} &= f''(\theta)\dot{\theta}^2 + f'(\theta)\ddot{\theta}\end{aligned}$$

You can then substitute these expressions into Eqs. (11.42) and (11.43).

c. The three-dimensional motion of a particle, as indicated at the end of Sec. 11.5B, often can be described effectively in terms of the **cylindrical coordinates** R , θ , and z (Fig. 11.24). The unit vectors then should consist of $\mathbf{e}_R$, $\mathbf{e}_\theta$, and $\mathbf{k}$. The corresponding components of the velocity and the acceleration are given in Eqs. (11.48) and (11.49). Note that the radial distance R is always measured in a plane parallel to the xy plane, and be careful not to confuse the position vector $\mathbf{r}$ with its radial component $R\mathbf{e}_R$.

Problems

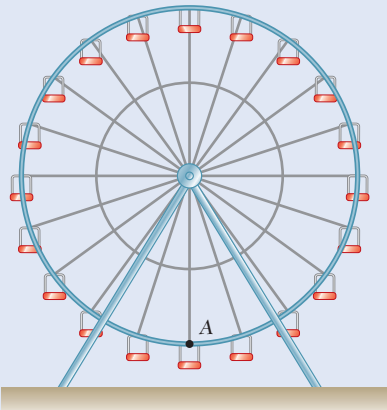


Fig. P11.CQ8

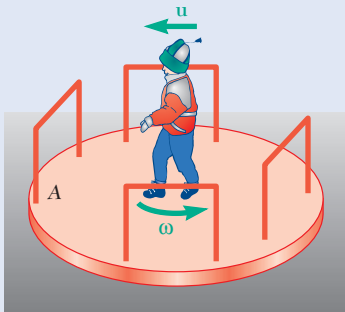


Fig. P11.CQ10

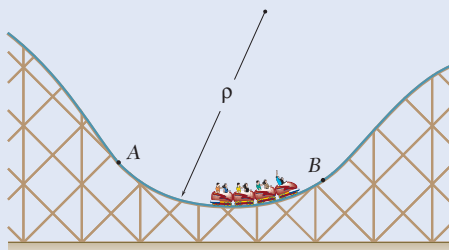


Fig. P11.134

CONCEPT QUESTIONS

- 11.CQ8** The Ferris wheel is rotating with a constant angular velocity ω . What is the direction of the acceleration of point A?
 a. $\rightarrow$ b. $\uparrow$ c. $\downarrow$ d. $\leftarrow$ e. The acceleration is zero.
- 11.CQ9** A race car travels around the track shown at a constant speed. At which point will the race car have the largest acceleration?
 a. A. b. B. c. C. d. D. e. The acceleration will be zero at all the points.

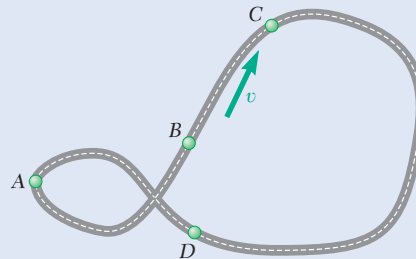


Fig. P11.CQ9

- 11.CQ10** A child walks across merry-go-round A with a constant speed u relative to A. The merry-go-round undergoes fixed-axis rotation about its center with a constant angular velocity ω counterclockwise. When the child is at the center of A, as shown, what is the direction of his acceleration when viewed from above?
 a. $\rightarrow$ b. $\leftarrow$ c. $\uparrow$ d. $\downarrow$ e. The acceleration is zero.

END-OF-SECTION PROBLEMS

- 11.133** Determine the smallest radius that should be used for a highway if the normal component of the acceleration of a car traveling at 72 km/h is not to exceed 0.8 m/s^2 .

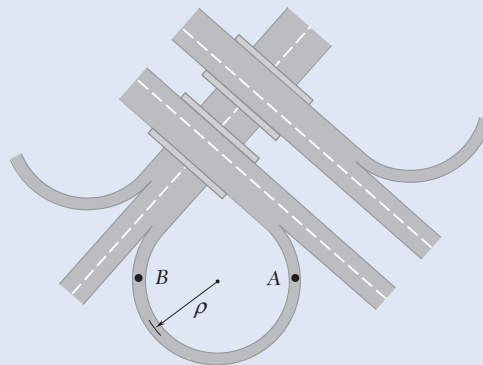


Fig. P11.133

- 11.134** Determine the maximum speed that the cars of the roller-coaster can reach along the circular portion AB of the track if $\rho = 25 \text{ m}$ and the normal component of their acceleration cannot exceed $3g$.

- 11.135** Human centrifuges are often used to simulate different acceleration levels for pilots and astronauts. Space shuttle pilots typically face inwards towards the center of the gondola in order to experience a simulated 3-g forward acceleration. Knowing that the astronaut sits 5 m from the axis of rotation and experiences 3 g's inward, determine her velocity.

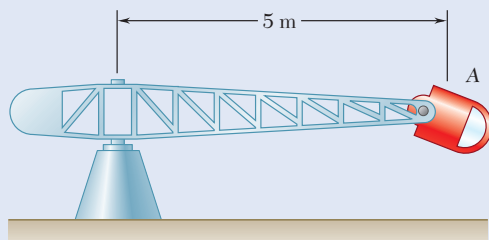


Fig. P11.135

- 11.136** The diameter of the eye of a stationary hurricane is 20 mi and the maximum wind speed is 100 mi/h at the eye wall with $r = 10$ mi. Assuming that the wind speed is constant for constant r and decreases uniformly with increasing r to 40 mi/h at $r = 110$ mi, determine the magnitude of the acceleration of the air at (a) $r = 10$ mi, (b) $r = 60$ mi, (c) $r = 110$ mi.

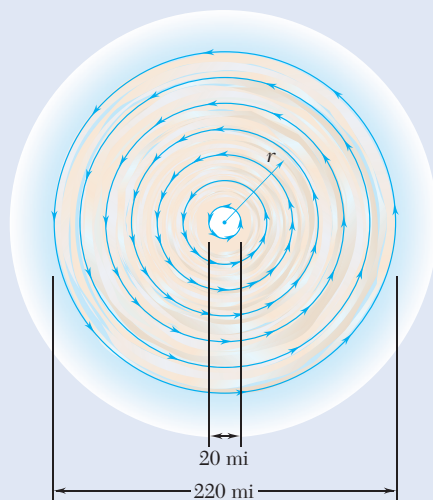


Fig. P11.136

- 11.137** The peripheral speed of the tooth of a 10-in.-diameter circular saw blade is 150 ft/s when the power to the saw is turned off. The speed of the tooth decreases at a constant rate, and the blade comes to rest in 9 s. Determine the time at which the total acceleration of the tooth is 130 ft/s².
- 11.138** A robot arm moves so that P travels in a circle about point B , which is not moving. Knowing that P starts from rest, and its speed increases at a constant rate of 10 mm/s², determine (a) the magnitude of the acceleration when $t = 4$ s, (b) the time for the magnitude of the acceleration to be 80 mm/s².

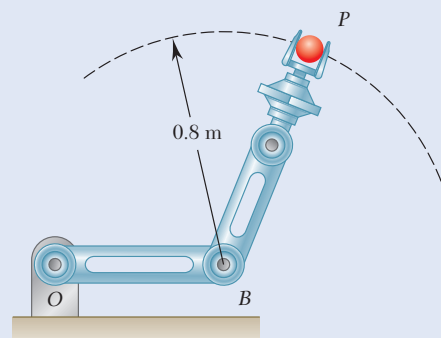


Fig. P11.138

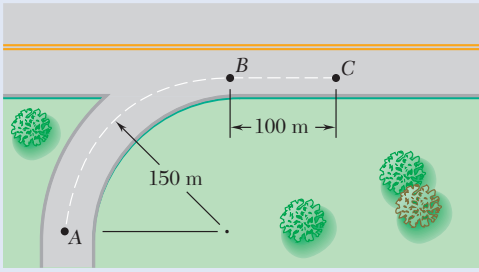


Fig. P11.140

11.139 A monorail train starts from rest on a curve of radius 400 m and accelerates at the constant rate a_t . If the maximum total acceleration of the train must not exceed 1.5 m/s^2 , determine (a) the shortest distance in which the train can reach a speed of 72 km/h, (b) the corresponding constant rate of acceleration a_t .

11.140 A motorist starts from rest at point A on a circular entrance ramp when $t = 0$, increases the speed of her automobile at a constant rate and enters the highway at point B. Knowing that her speed continues to increase at the same rate until it reaches 100 km/h at point C, determine (a) the speed at point B, (b) the magnitude of the total acceleration when $t = 20 \text{ s}$.

11.141 Race car A is traveling on a straight portion of the track while race car B is traveling on a circular portion of the track. At the instant shown, the speed of A is increasing at the rate of 10 m/s^2 , and the speed of B is decreasing at the rate of 6 m/s^2 . For the position shown, determine (a) the velocity of B relative to A, (b) the acceleration of B relative to A.

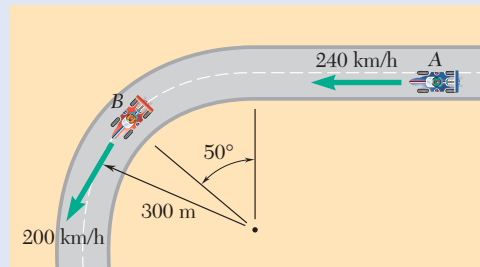


Fig. P11.141

11.142 At a given instant in an airplane race, airplane A is flying horizontally in a straight line, and its speed is being increased at the rate of 8 m/s^2 . Airplane B is flying at the same altitude as airplane A and, as it rounds a pylon, is following a circular path of 300-m radius. Knowing that at the given instant the speed of B is being decreased at the rate of 3 m/s^2 , determine, for the positions shown, (a) the velocity of B relative to A, (b) the acceleration of B relative to A.

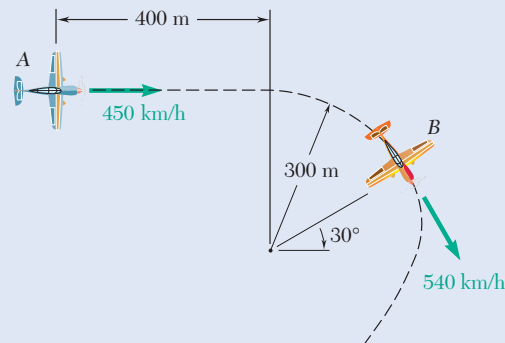


Fig. P11.142

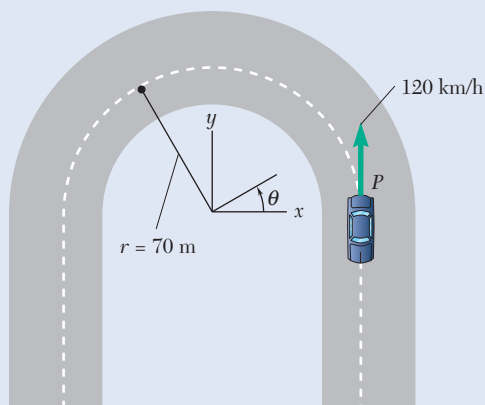


Fig. P11.143

11.143 A race car enters the circular portion of a track that has a radius of 70 m. When the car enters the curve at point P, it is travelling with a speed of 120 km/h that is increasing at 5 m/s^2 . Three seconds later, determine the x and y components of velocity and acceleration of the car.

11.144 An airplane flying at a constant speed of 240 m/s makes a banked horizontal turn. What is the minimum allowable radius of the turn if the structural specifications require that the acceleration of the airplane shall never exceed 4 g?

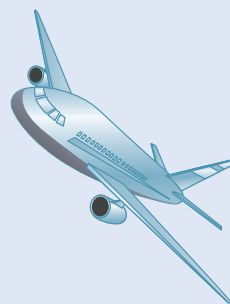


Fig. P11.144

11.145 A golfer hits a golf ball from point A with an initial velocity of 50 m/s at an angle of 25° with the horizontal. Determine the radius of curvature of the trajectory described by the ball (a) at point A, (b) at the highest point of the trajectory.

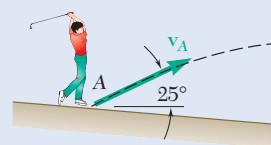


Fig. P11.145

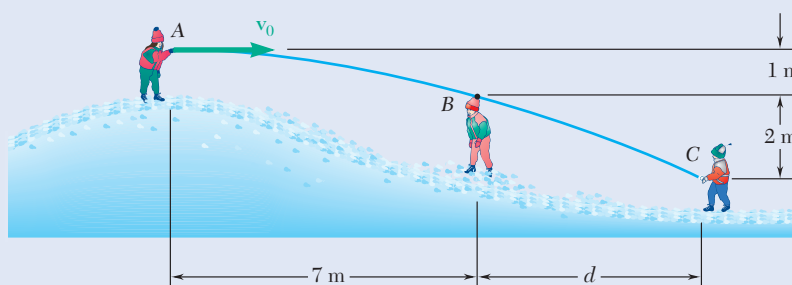


Fig. P11.146

11.147 Coal is discharged from the tailgate A of a dump truck with an initial velocity $v_A = 2 \text{ m/s}$ at 50° . Determine the radius of curvature of the trajectory described by the coal (a) at point A, (b) at the point of the trajectory 1 m below point A.

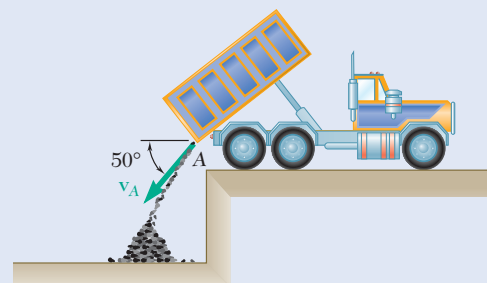


Fig. P11.147

11.148 From measurements of a photograph, it has been found that as the stream of water shown left the nozzle at A, it had a radius of curvature of 25 m. Determine (a) the initial velocity v_A of the stream, (b) the radius of curvature of the stream as it reaches its maximum height at B.

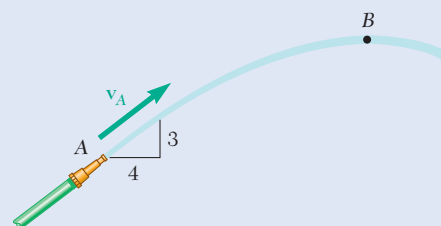


Fig. P11.148

11.149 A child throws a ball from point A with an initial velocity v_0 at an angle of 3° with the horizontal. Knowing that the ball hits a wall at point B, determine (a) the magnitude of the initial velocity, (b) the minimum radius of curvature of the trajectory.

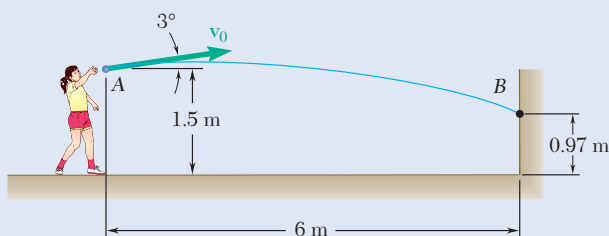


Fig. P11.149

- 11.150** A projectile is fired from point A with an initial velocity v_0 . (a) Show that the radius of curvature of the trajectory of the projectile reaches its minimum value at the highest point B of the trajectory. (b) Denoting by θ the angle formed by the trajectory and the horizontal at a given point C , show that the radius of curvature of the trajectory at C is $\rho = \rho_{\min}/\cos^3\theta$.

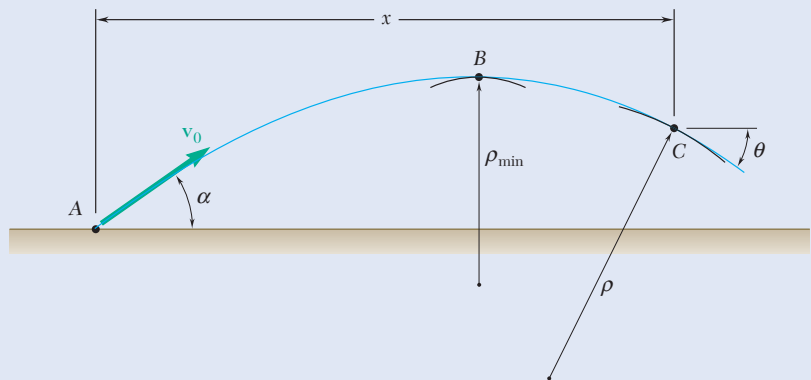


Fig. P11.150

- *11.151** Determine the radius of curvature of the path described by the particle of Prob. 11.95 when $t = 0$.
- *11.152** Determine the radius of curvature of the path described by the particle of Prob. 11.96 when $t = 0$, $A = 3$, and $B = 1$.
- 11.153 and 11.154** A satellite will travel indefinitely in a circular orbit around a planet if the normal component of the acceleration of the satellite is equal to $g(R/r)^2$, where g is the acceleration of gravity at the surface of the planet, R is the radius of the planet, and r is the distance from the center of the planet to the satellite. Knowing that the diameter of the sun is 1.39 Gm and that the acceleration of gravity at its surface is 274 m/s^2 , determine the radius of the orbit of the indicated planet around the sun assuming that the orbit is circular.
- 11.153** Earth: $(v_{\text{mean}})_{\text{orbit}} = 107 \text{ Mm/h}$.
- 11.154** Saturn: $(v_{\text{mean}})_{\text{orbit}} = 34.7 \text{ Mm/h}$.
- 11.155 through 11.157** Determine the speed of a satellite relative to the indicated planet if the satellite is to travel indefinitely in a circular orbit 100 mi above the surface of the planet. (See information given in Probs. 11.153–11.154.)
- 11.155** Venus: $g = 29.20 \text{ ft/s}^2$, $R = 3761 \text{ mi}$.
- 11.156** Mars: $g = 12.17 \text{ ft/s}^2$, $R = 2102 \text{ mi}$.
- 11.157** Jupiter: $g = 75.35 \text{ ft/s}^2$, $R = 44,432 \text{ mi}$.
- 11.158** A satellite will travel indefinitely in a circular orbit around the earth if the normal component of its acceleration is equal to $g(R/r)^2$, where $g = 9.81 \text{ m/s}^2$, $R = \text{radius of the earth} = 6370 \text{ km}$, and $r = \text{distance from the center of the earth to the satellite}$. Assuming that the orbit of the moon is a circle with a radius of $384 \times 10^3 \text{ km}$, determine the speed of the moon relative to the earth.

11.159 Knowing that the radius of the earth is 6370 km, determine the time of one orbit of the Hubble Space Telescope if the telescope travels in a circular orbit 590 km above the surface of the earth. (See information given in Probs. 11.153–11.154.)

11.160 Satellites *A* and *B* are traveling in the same plane in circular orbits around the earth at altitudes of 120 and 200 mi, respectively. If at $t = 0$ the satellites are aligned as shown and knowing that the radius of the earth is $R = 3960$ mi, determine when the satellites will next be radially aligned. (See information given in Probs. 11.153–11.154.)

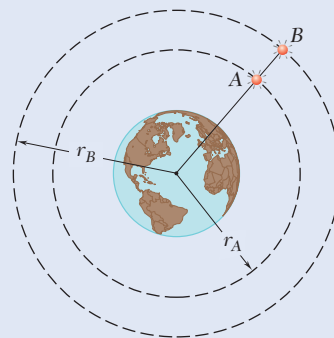


Fig. P11.160

11.161 The oscillation of rod *OA* about *O* is defined by the relation $\theta = (3/\pi)(\sin \pi t)$, where θ and t are expressed in radians and seconds, respectively. Collar *B* slides along the rod so that its distance from *O* is $r = 6(1 - e^{-2t})$ where r and t are expressed in inches and seconds, respectively. When $t = 1$ s, determine (a) the velocity of the collar, (b) the acceleration of the collar, (c) the acceleration of the collar relative to the rod.

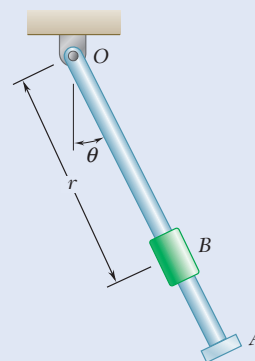


Fig. P11.161

11.162 The path of a particle *P* is a limaçon. The motion of the particle is defined by the relations $r = b(2 + \cos \pi t)$ and $\theta = \pi t$ where t and θ are expressed in seconds and radians, respectively. Determine (a) the velocity and the acceleration of the particle when $t = 2$ s, (b) the value of θ for which the magnitude of the velocity is maximum.

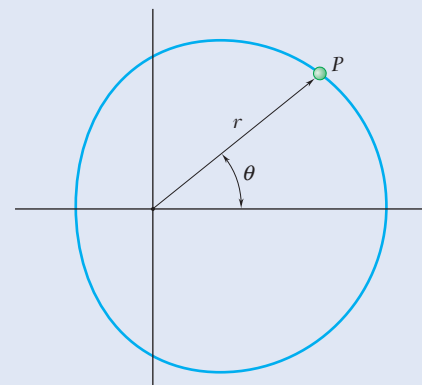


Fig. P11.162

11.163 During a parasailing ride, the boat is traveling at a constant 30 km/hr with a 200-m long tow line. At the instant shown, the angle between the line and the water is 30° and is increasing at a constant rate of $2^\circ/\text{s}$. Determine the velocity and acceleration of the parasailer at this instant.



Fig. P11.163

11.164 Some parasailing systems use a winch to pull the rider back to the boat. During the interval when θ is between 20° and 40° (where $t = 0$ at $\theta = 20^\circ$), the angle increases at the constant rate of $2^\circ/\text{s}$. During this time, the length of the rope is defined by the relationship $r = 600 - \frac{1}{8}t^{5/2}$, where r and t are expressed in feet and seconds, respectively. Knowing that the boat is travelling at a constant rate of 15 knots (where 1 knot = 1.15 mi/h), (a) plot the magnitude of the velocity of the parasailer as a function of time, (b) determine the magnitude of the acceleration of the parasailer when $t = 5$ s.

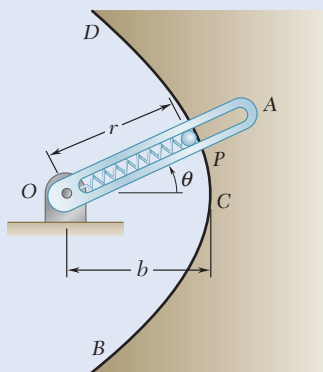


Fig. P11.165

11.165 As rod OA rotates, pin P moves along the parabola BCD . Knowing that the equation of this parabola is $r = 2b/(1 + \cos \theta)$ and that $\theta = kt$, determine the velocity and acceleration of P when (a) $\theta = 0$, (b) $\theta = 90^\circ$.

11.166 The pin at B is free to slide along the circular slot DE and along the rotating rod OC . Assuming that the rod OC rotates at a constant rate $\dot{\theta}$, (a) show that the acceleration of pin B is of constant magnitude, (b) determine the direction of the acceleration of pin B .

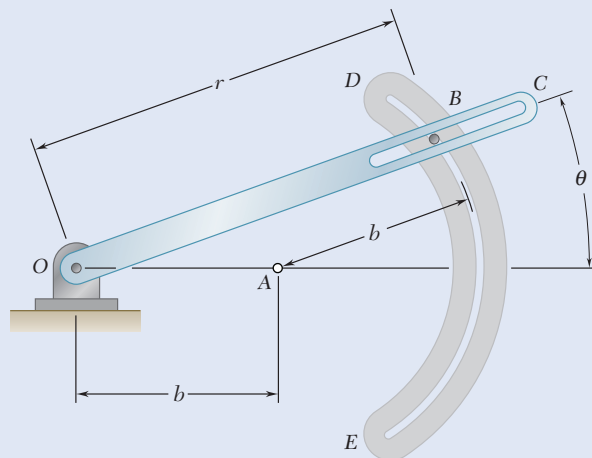


Fig. P11.166

11.167 To study the performance of a race car, a high-speed camera is positioned at point A . The camera is mounted on a mechanism which permits it to record the motion of the car as the car travels on straightaway BC . Determine (a) the speed of the car in terms of b , θ , and $\dot{\theta}$, (b) the magnitude of the acceleration in terms of b , θ , $\dot{\theta}$, and $\ddot{\theta}$.

11.168 After taking off, a helicopter climbs in a straight line at a constant angle β . Its flight is tracked by radar from point A . Determine the speed of the helicopter in terms of d , β , θ , and $\dot{\theta}$.

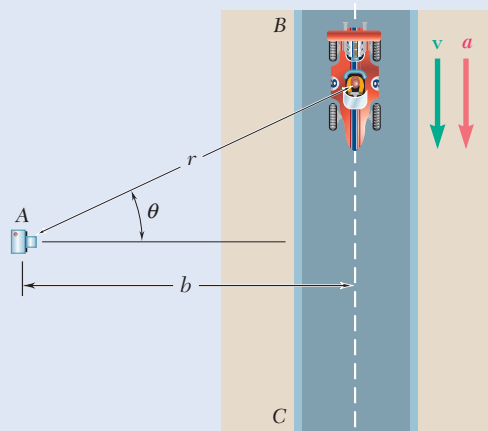


Fig. P11.167

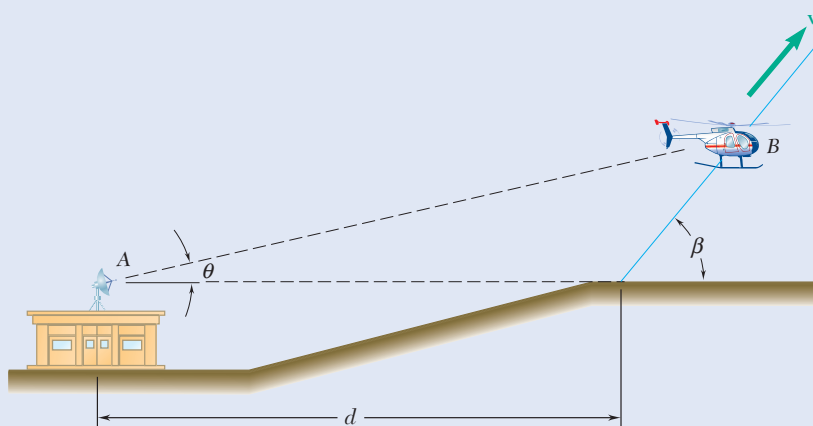


Fig. P11.168

- 11.169** At the bottom of a loop in the vertical plane an airplane has a horizontal velocity of 315 mi/h and is speeding up at a rate of 10 ft/s^2 . The radius of curvature of the loop is 1 mi. The plane is being tracked by radar at O . What are the recorded values of $\dot{r}$, $\ddot{r}$, $\dot{\theta}$, and $\ddot{\theta}$ for this instant?

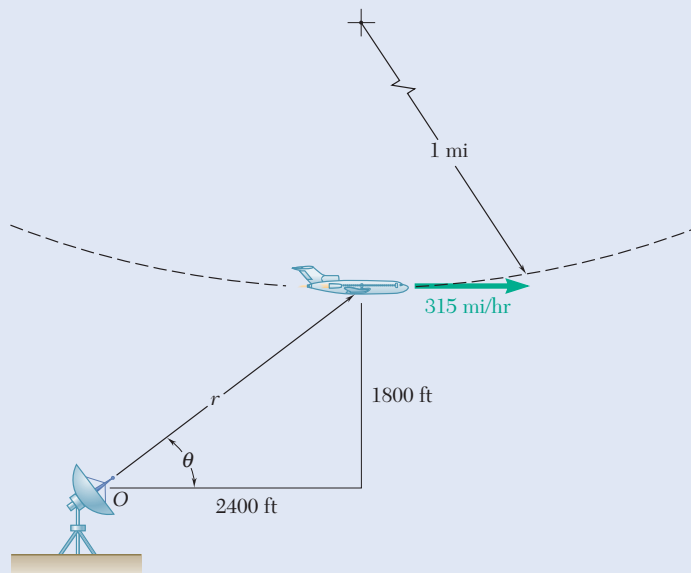


Fig. P11.169

- 11.170** Pin C is attached to rod BC and slides freely in the slot of rod OA which rotates at the constant rate ω . At the instant when $\beta = 60^\circ$, determine (a) $\dot{r}$ and $\dot{\theta}$, (b) $\ddot{r}$ and $\ddot{\theta}$. Express your answers in terms of d and ω .

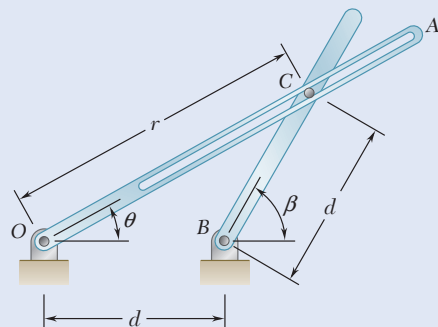


Fig. P11.170

- 11.171** For the race car of Prob. 11.167, it was found that it took 0.5 s for the car to travel from the position $\theta = 60^\circ$ to the position $\theta = 35^\circ$. Knowing that $b = 25 \text{ m}$, determine the average speed of the car during the 0.5-s interval.
- 11.172** For the helicopter of Prob. 11.168, it was found that when the helicopter was at B , the distance and the angle of elevation of the helicopter were $r = 3000 \text{ ft}$ and $\theta = 20^\circ$, respectively. Four seconds later, the radar station sighted the helicopter at $r = 3320 \text{ ft}$ and $\theta = 23.1^\circ$. Determine the average speed and the angle of climb β of the helicopter during the 4-s interval.

11.173 and 11.174 A particle moves along the spiral shown. Determine the magnitude of the velocity of the particle in terms of b , θ , and $\dot{\theta}$.

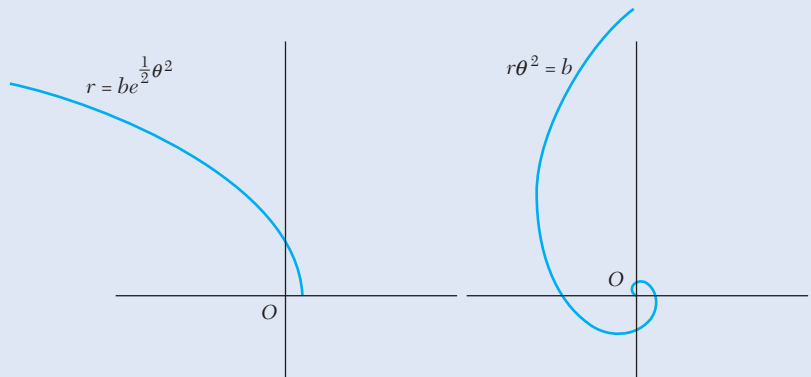


Fig. P11.173 and P11.175

Fig. P11.174 and P11.176

11.175 and 11.176 A particle moves along the spiral shown. Knowing that $\dot{\theta}$ is constant and denoting this constant by ω , determine the magnitude of the acceleration of the particle in terms of b , θ , and $\dot{\theta}$.

11.177 The motion of a particle on the surface of a right circular cylinder is defined by the relations $R = A$, $\theta = 2\pi t$, and $z = B \sin 2\pi n t$, where A and B are constants and n is an integer. Determine the magnitudes of the velocity and acceleration of the particle at any time t .

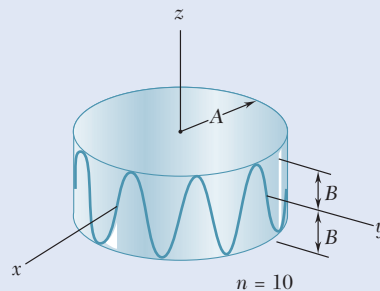


Fig. P11.177

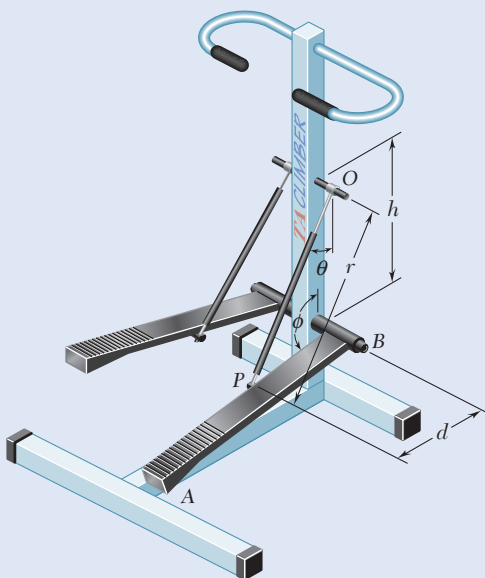


Fig. P11.178

11.178 Show that $\dot{r} = h\dot{\phi} \sin \theta$ knowing that at the instant shown, step AB of the step exerciser is rotating counterclockwise at a constant rate $\dot{\phi}$.

11.179 The three-dimensional motion of a particle is defined by the relations $R = A(1 - e^{-t})$, $\theta = 2\pi t$, and $z = B(1 - e^{-t})$. Determine the magnitudes of the velocity and acceleration when (a) $t = 0$, (b) $t = \infty$.

***11.180** For the conic helix of Prob. 11.95, determine the angle that the osculating plane forms with the y axis.

***11.181** Determine the direction of the binormal of the path described by the particle of Prob. 11.96 when (a) $t = 0$, (b) $t = \pi/2$ s.

Review and Summary

Position Coordinate of a Particle in Rectilinear Motion

In the first half of this chapter, we analyzed the **rectilinear motion of a particle**, i.e., the motion of a particle along a straight line. To define the position P of the particle on that line, we chose a fixed origin O and a positive direction (Fig. 11.25). The distance x from O to P , with the appropriate sign, completely defines the position of the particle on the line and is called the **position coordinate** of the particle [Sec. 11.1A].

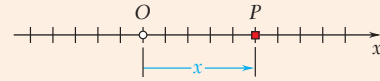


Fig. 11.25

Velocity and Acceleration in Rectilinear Motion

The **velocity** v of the particle was shown to be equal to the time derivative of the position coordinate x , so

$$v = \frac{dx}{dt} \quad (11.1)$$

And we obtained the **acceleration** a by differentiating v with respect to t , as

$$a = \frac{dv}{dt} \quad (11.2)$$

or

$$a = \frac{d^2x}{dt^2} \quad (11.3)$$

We also noted that a could be expressed as

$$a = v \frac{dv}{dx} \quad (11.4)$$

We observed that the velocity v and the acceleration a are represented by algebraic numbers that can be positive or negative. A positive value for v indicates that the particle moves in the positive direction, and a negative value shows that it moves in the negative direction. A positive value for a , however, may mean that the particle is truly accelerated (i.e., moves faster) in the positive direction or that it is decelerated (i.e., moves more slowly) in the negative direction. A negative value for a is subject to a similar interpretation [Sample Prob. 11.1].

Determination of the Velocity and Acceleration by Integration

In most problems, the conditions of motion of a particle are defined by the type of acceleration that the particle possesses and by the initial conditions [Sec. 11.1B]. Then we can obtain the velocity and position of the particle by integrating two of the equations (11.1) to (11.4). The selection of these equations depends upon the type of acceleration involved [Sample Probs. 11.2 through 11.4].

Uniform Rectilinear Motion

Two types of motion are frequently encountered. **Uniform rectilinear motion** [Sec. 11.2A], in which the velocity v of the particle is constant, is described by

$$x = x_0 + vt \quad (11.5)$$

Uniformly Accelerated Rectilinear Motion

Uniformly accelerated rectilinear motion [Sec. 11.2B], in which the acceleration a of the particle is constant, is described by

$$v = v_0 + at \quad (11.6)$$

$$x = x_0 + v_0t + \frac{1}{2}at^2 \quad (11.7)$$

$$v^2 = v_0^2 + 2a(x - x_0) \quad (11.8)$$

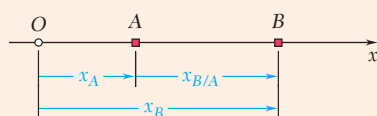


Fig. 11.26

Relative Motion of Two Particles

When two particles A and B (such as two aircraft) move, we may wish to consider the **relative motion** of B with respect to A [Sec. 11.2C]. Denoting the **relative position coordinate** of B with respect to A by $x_{B/A}$ (Fig. 11.26), we have

$$x_B = x_A + x_{B/A} \quad (11.9)$$

Differentiating Eq. (11.9) twice with respect to t , we obtained successively

$$v_B = v_A + v_{B/A} \quad (11.10)$$

$$a_B = a_A + a_{B/A} \quad (11.11)$$

where $v_{B/A}$ and $a_{B/A}$ represent, respectively, the **relative velocity** and the **relative acceleration** of B with respect to A .

Dependent Motion

When several blocks are **connected by inextensible cords**, it is possible to write a linear relation between their position coordinates. We can then write similar relations between their velocities and between their accelerations, which we can use to analyze their motion [Sample Probs. 11.7 and 11.8].

Graphical Solutions

It is sometimes convenient to use a **graphical solution** for problems involving the rectilinear motion of a particle [Sec. 11.3]. The graphical solution most commonly used involves the $x-t$, $v-t$, and $a-t$ curves [Sample Prob. 11.10]. It was shown at any given time t that

$$v = \text{slope of } x-t \text{ curve}$$

$$a = \text{slope of } v-t \text{ curve}$$

Also, over any given time interval from t_1 to t_2 , we have

$$v_2 - v_1 = \text{area under } a-t \text{ curve}$$

$$x_2 - x_1 = \text{area under } v-t \text{ curve}$$

Position Vector and Velocity in Curvilinear Motion

In the second half of this chapter, we analyzed the **curvilinear motion of a particle**, i.e., the motion of a particle along a curved path. We defined the position P of the particle at a given time [Sec. 11.4A] by the **position vector** $\mathbf{r}$

joining the O of the coordinates and point P (Fig. 11.27). We defined the **velocity** $\mathbf{v}$ of the particle by the relation

$$\mathbf{v} = \frac{d\mathbf{r}}{dt} \quad (11.14)$$

The velocity is a **vector tangent to the path of the particle** with a magnitude v (called the **speed** of the particle) equal to the time derivative of the length s of the arc described by the particle. Thus,

$$v = \frac{ds}{dt} \quad (11.15)$$

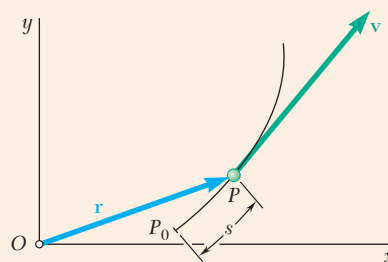


Fig. 11.27

Acceleration in Curvilinear Motion

We defined the **acceleration** $\mathbf{a}$ of the particle by the relation

$$\mathbf{a} = \frac{d\mathbf{v}}{dt} \quad (11.17)$$

and we noted that, in general, *the acceleration is not tangent to the path of the particle*.

Derivative of a Vector Function

Before proceeding to the consideration of the components of velocity and acceleration, we reviewed the formal definition of the derivative of a vector function and established a few rules governing the differentiation of sums and products of vector functions. We then showed that the rate of change of a vector is the same with respect both to a fixed frame and to a frame in translation [Sec. 11.4B].

Rectangular Components of Velocity and Acceleration

Denoting the rectangular coordinates of a particle P by x , y , and z , we found that the rectangular components of the velocity and acceleration of P equal, respectively, the first and second derivatives with respect to t of the corresponding coordinates. Thus,

$$v_x = \dot{x} \quad v_y = \dot{y} \quad v_z = \dot{z} \quad (11.28)$$

$$a_x = \ddot{x} \quad a_y = \ddot{y} \quad a_z = \ddot{z} \quad (11.29)$$

Component Motions

When the component a_x of the acceleration depends only upon t , x , and/or v_x ; when, similarly, a_y depends only upon t , y , and/or v_y ; and a_z upon t , z , and/or v_z , Eq. (11.29) can be integrated independently. The analysis of the given curvilinear motion then reduces to the analysis of three independent rectilinear component motions [Sec. 11.4C]. This approach is particularly effective in the study of the motion of projectiles [Sample Probs. 11.10 and 11.11].

Relative Motion of Two Particles

For two particles A and B moving in space (Fig. 11.28), we considered the relative motion of B with respect to A , or more precisely, with respect to a moving frame attached to A and in translation with A [Sec. 11.4D]. Denoting the **relative position vector** of B with respect to A by $\mathbf{r}_{B/A}$ (Fig. 11.28), we have

$$\mathbf{r}_B = \mathbf{r}_A + \mathbf{r}_{B/A} \quad (11.30)$$

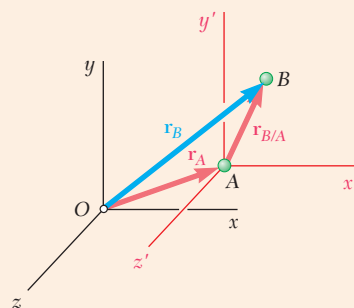


Fig. 11.28

Denoting the **relative velocity** and the **relative acceleration** of B with respect to A by $\mathbf{v}_{B/A}$ and $\mathbf{a}_{B/A}$, respectively, we also showed that

$$\mathbf{v}_B = \mathbf{v}_A + \mathbf{v}_{B/A} \quad (11.32)$$

and

$$\mathbf{a}_B = \mathbf{a}_A + \mathbf{a}_{B/A} \quad (11.33)$$

Tangential and Normal Components

It is sometimes convenient to resolve the velocity and acceleration of a particle P into components other than the rectangular x , y , and z components. For a particle P moving along a path contained in a plane, we attached to P unit vectors $\mathbf{e}_t$ tangent to the path and $\mathbf{e}_n$ normal to the path and directed toward the center of curvature of the path [Sec. 11.5A]. We then express the velocity and acceleration of the particle in terms of tangential and normal components. We have

$$\mathbf{v} = v\mathbf{e}_t \quad (11.35)$$

and

$$\mathbf{a} = \frac{dv}{dt}\mathbf{e}_t + \frac{v^2}{\rho}\mathbf{e}_n \quad (11.38)$$

where v is the speed of the particle and ρ is the radius of curvature of its path [Sample Probs. 11.16, ,and 11.17]. We observed that, while the velocity $\mathbf{v}$ is directed along the tangent to the path, the acceleration $\mathbf{a}$ consists of a component $\mathbf{a}_t$ directed along the tangent to the path and a component $\mathbf{a}_n$ directed toward the center of curvature of the path (Fig. 11.29).

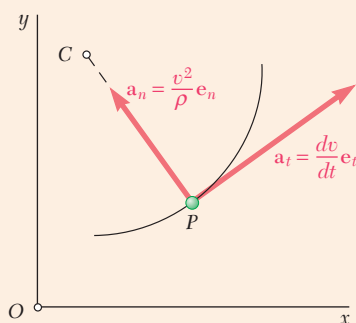


Fig. 11.29

Motion Along a Space Curve

For a particle P moving along a space curve, we defined the plane that most closely fits the curve in the neighborhood of P as the **osculating plane**. This plane contains the unit vectors $\mathbf{e}_t$ and $\mathbf{e}_n$ that define the tangent and principal normal to the curve, respectively. The unit vector $\mathbf{e}_b$, which is perpendicular to the osculating plane, defines the **binormal**.

Radial and Transverse Components

When the position of a particle P moving in a plane is defined by its polar coordinates r and θ , it is convenient to use radial and transverse components directed, respectively, along the position vector $\mathbf{r}$ of the particle and in the direction obtained by rotating $\mathbf{r}$ through 90° counterclockwise [Sec. 11.5B]. We attached to P unit vectors $\mathbf{e}_r$ and $\mathbf{e}_\theta$ directed in the radial and transverse directions, respectively (Fig. 11.30). We then expressed the velocity and acceleration of the particle in terms of radial and transverse components as

$$\mathbf{v} = \dot{r}\mathbf{e}_r + r\dot{\theta}\mathbf{e}_\theta \quad (11.42)$$

$$\mathbf{a} = (\ddot{r} - r\dot{\theta}^2)\mathbf{e}_r + (r\ddot{\theta} + 2\dot{r}\dot{\theta})\mathbf{e}_\theta \quad (11.43)$$

where dots are used to indicate differentiation with respect to time. The scalar components of the velocity and acceleration in the radial and transverse directions are therefore

$$v_r = \dot{r} \quad v_\theta = r\dot{\theta} \quad (11.44)$$

$$a_r = \ddot{r} - r\dot{\theta}^2 \quad a_\theta = r\ddot{\theta} + 2\dot{r}\dot{\theta} \quad (11.45)$$

It is important to note that a_r is *not* equal to the time derivative of v_r and that a_θ is *not* equal to the time derivative of v_θ [Sample Probs. 11.18, 11.19, and 11.20].

This chapter ended with a discussion of the use of cylindrical coordinates to define the position and motion of a particle in space.

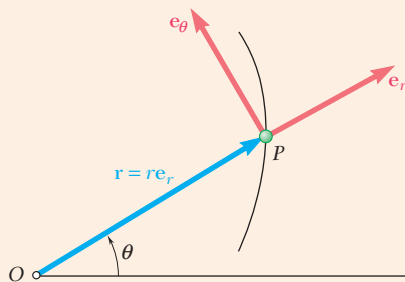


Fig. 11.30

Review Problems

- 11.182** The motion of a particle is defined by the relation $x = 2t^3 - 15t^2 + 24t + 4$, where x and t are expressed in meters and seconds, respectively. Determine (a) when the velocity is zero, (b) the position and the total distance traveled when the acceleration is zero.
- 11.183** A drag car starts from rest and moves down the racetrack with an acceleration defined by $a = 50 - 10t$, where a and t are in m/s^2 and seconds, respectively. After reaching a speed of 125 m/s, a parachute is deployed to help slow down the dragster. Knowing that this deceleration is defined by the relationship $a = -0.02v^2$, where v is the velocity in m/s, determine (a) the total time from the beginning of the race until the car slows back down to 10 m/s, (b) the total distance the car travels during this time.
- 11.184** A particle moves in a straight line with the acceleration shown in the figure. Knowing that the particle starts from the origin with $v_0 = -2$ m/s, (a) construct the $v-t$ and $x-t$ curves for $0 < t < 18$ s, (b) determine the position and the velocity of the particle and the total distance traveled when $t = 18$ s.

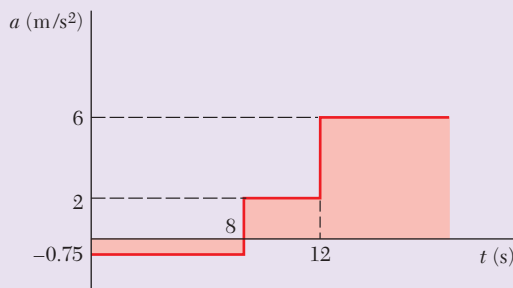


Fig. P11.184

- 11.185** The velocities of commuter trains A and B are as shown. Knowing that the speed of each train is constant and that B reaches the crossing 10 min after A passed through the same crossing, determine (a) the relative velocity of B with respect to A, (b) the distance between the fronts of the engines 3 min after A passed through the crossing.

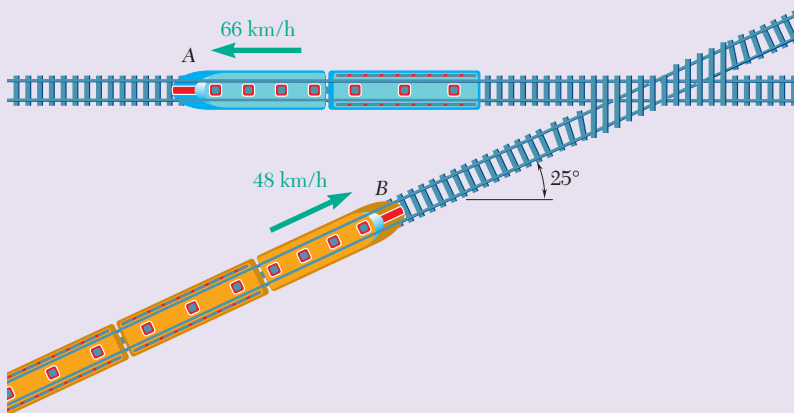


Fig. P11.185

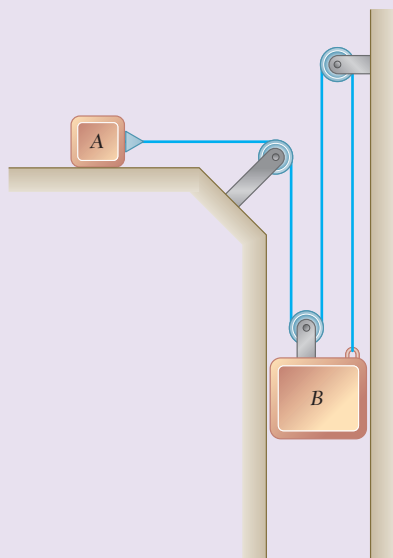


Fig. P11.186

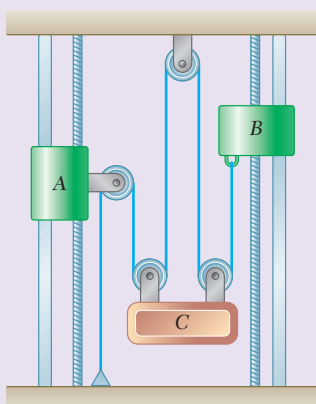


Fig. P11.187

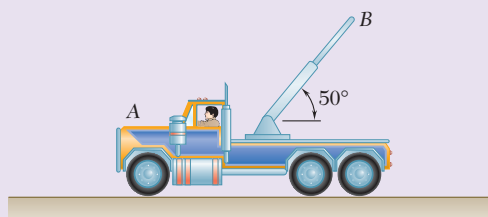


Fig. P11.189

11.186 Knowing that slider block A starts from rest and moves to the left with a constant acceleration of 1 ft/s^2 , determine (a) the relative acceleration of block A with respect to block B , (b) the velocity of block B after 2 s.

11.187 Collar A starts from rest at $t = 0$ and moves downward with a constant acceleration of 7 in./s^2 . Collar B moves upward with a constant acceleration, and its initial velocity is 8 in./s . Knowing that collar B moves through 20 in. between $t = 0$ and $t = 2 \text{ s}$, determine (a) the accelerations of collar B and block C , (b) the time at which the velocity of block C is zero, (c) the distance through which block C will have moved at that time.

11.188 A golfer hits a ball with an initial velocity of magnitude v_0 at an angle α with the horizontal. Knowing that the ball must clear the tops of two trees and land as close as possible to the flag, determine v_0 and the distance d when the golfer uses (a) a six-iron with $\alpha = 31^\circ$, (b) a five-iron with $\alpha = 27^\circ$.

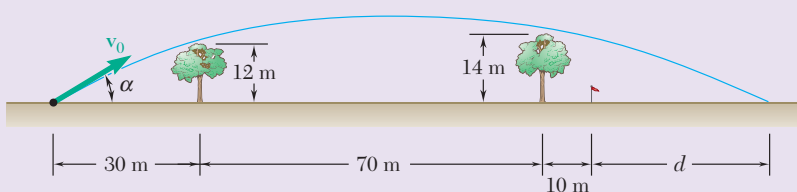


Fig. P11.188

11.189 As the truck shown begins to back up with a constant acceleration of 4 ft/s^2 , the outer section B of its boom starts to retract with a constant acceleration of 1.6 ft/s^2 relative to the truck. Determine (a) the acceleration of section B , (b) the velocity of section B when $t = 2 \text{ s}$.

11.190 A velodrome is a specially designed track used in bicycle racing that has constant radius curves at each end. Knowing that a rider starts from rest $a_t = (11.46 - 0.01878v^2) \text{ m/s}^2$, determine her acceleration at point B .

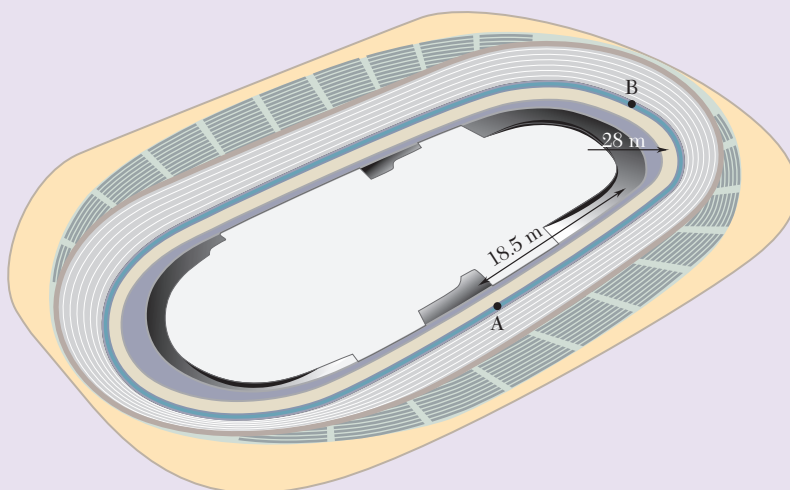


Fig. P11.190

11.191 Sand is discharged at A from a conveyor belt and falls onto the top of a stockpile at B . Knowing that the conveyor belt forms an angle $\alpha = 25^\circ$ with the horizontal, determine (a) the speed v_0 of the belt, (b) the radius of curvature of the trajectory described by the sand at point B .

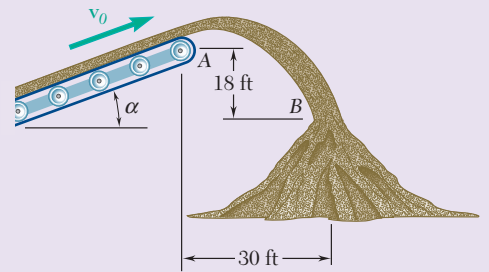


Fig. P11.191

11.192 The end point B of a boom is originally 5 m from fixed point A when the driver starts to retract the boom with a constant radial acceleration of $\ddot{r} = -1.0 \text{ m/s}^2$ and lower it with a constant angular acceleration $\ddot{\theta} = -0.5 \text{ rad/s}^2$. At $t = 2 \text{ s}$, determine (a) the velocity of point B , (b) the acceleration of point B , (c) the radius of curvature of the path.

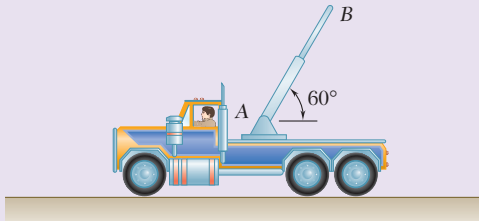


Fig. P11.192

11.193 A telemetry system is used to quantify kinematic values of a ski jumper immediately before she leaves the ramp. According to the system $r = .500 \text{ ft}$, $\dot{r} = -105 \text{ ft/s}$, $\ddot{r} = -10 \text{ ft/s}^2$, $\theta = 25^\circ$, $\dot{\theta} = 0.07 \text{ rad/s}$, $\ddot{\theta} = 0.06 \text{ rad/s}^2$. Determine (a) the velocity of the skier immediately before she leaves the jump, (b) the acceleration of the skier at this instant, (c) the distance of the jump d neglecting lift and air resistance.

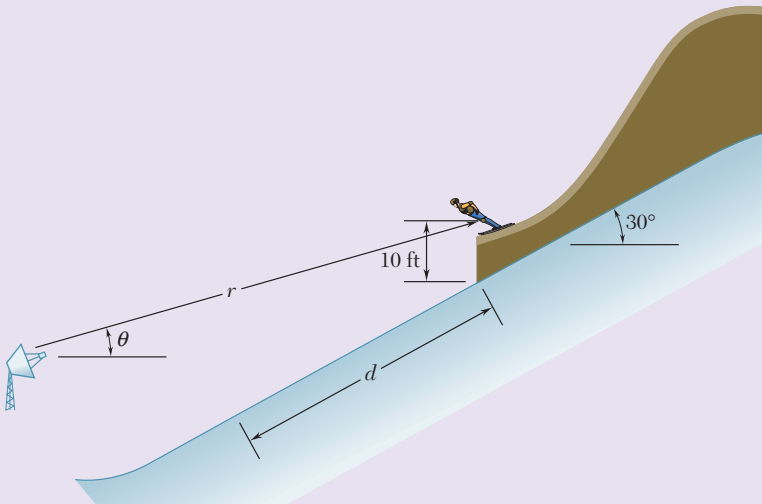


Fig. P11.193



12

Kinetics of Particles: Newton's Second Law

The forces experienced by the passengers on a roller coaster will depend on whether the roller-coaster car is traveling up a hill or down a hill, in a straight line, or along a horizontal or vertical curved path. The relation existing among force, mass, and acceleration will be studied in this chapter.

Introduction

12.1 NEWTON'S SECOND LAW AND LINEAR MOMENTUM

12.1A Newton's Second Law of Motion

12.1B Linear Momentum of a Particle and Its Rate of Change

12.1C Systems of Units

12.1D Equations of Motion

12.2 ANGULAR MOMENTUM AND ORBITAL MOTION

12.2A A. Angular Momentum of a Particle and Its Rate of Change

12.2B Motion Under a Central Force and Conservation of Angular Momentum

12.2C Newton's Law of Gravitation

12.3 APPLICATIONS OF CENTRAL-FORCE MOTION

12.3A Trajectory of a Particle Under a Central Force

12.3B Application to Space Mechanics

12.3C Kepler's Laws of Planetary Motion

Objectives

- **Explain** the relationships between mass, force, and acceleration.
- **Model** physical systems by drawing complete free-body diagrams and kinetic diagrams.
- **Apply** Newton's second law of motion to solve particle kinetics problems using different coordinate systems.
- **Analyze** central force motion problems using principles of angular momentum and Newton's law of gravitation.

Introduction

In statics, we used Newton's first and third laws of motion extensively to study bodies at rest and the forces acting upon them. We also use these two laws in dynamics; in fact, they are sufficient for analyzing the motion of bodies that have no acceleration. However, when a body is accelerated—that is, when the magnitude or the direction of its velocity changes—it is necessary to use Newton's second law of motion to relate the motion of the body to the forces acting on it.

In this chapter, we discuss Newton's second law and apply it to analyzing the motion of particles. According to the second law, if the resultant of the forces acting on a particle is not zero, the particle has an acceleration proportional to the magnitude of the resultant and in the direction of this resultant force. Moreover, we use the ratio of the magnitudes of the resultant force and of the acceleration to define the *mass* of the particle. In Sec. 12.1B, we define the *linear momentum* of a particle as the product $\mathbf{L} = m\mathbf{v}$ of the mass m and velocity $\mathbf{v}$ of the particle. Then we can express Newton's second law in an alternative form, relating the rate of change of the linear momentum to the resultant of the forces acting on that particle.

In the Sample Problems, we apply Newton's second law to the solution of engineering problems using either rectangular components, tangential and normal components, or radial and transverse coordinates of the forces and accelerations involved. Recall that we can consider an actual body—including bodies as large as a car, rocket, or airplane—as a particle for the purpose of analyzing its motion, as long as the effect of a rotation of the body about its center of mass can be ignored. We stress the need for consistent units in solving these problems, briefly reviewing the International System of Units (SI units) and the system of U.S. customary units.

The second part of this chapter is devoted to the study of the motion of particles under central forces. We define the *angular momentum* $\mathbf{H}_O$ of a particle about a point O as the moment about O of the linear momentum of the particle: $\mathbf{H}_O = \mathbf{r} \times m\mathbf{v}$. It then follows from Newton's second law that the rate of change of the angular momentum $\mathbf{H}_O$ of a particle is equal to the sum of the moments about O of the forces acting on that particle.

We can use this form of the second law to deal with the motion of a particle under a *central force*, i.e., under a force directed toward or away from a fixed point O . Since such a force has zero moment about O , it follows that the angular momentum of the particle about O is conserved. This property greatly simplifies the analysis of the motion, as we show by solving problems involving the orbital motion of bodies under gravitational attraction.

In Sec.12.3, which is optional, we present a more extensive discussion of orbital motion, including several problems related to space mechanics.

12.1 NEWTON'S SECOND LAW AND LINEAR MOMENTUM

In statics, we dealt with forces acting on particles that led to a state of equilibrium. Now we study forces acting on particles that lead to a state of motion. The key relationship connecting force and motion is Newton's second law.

12.1A Newton's Second Law of Motion

We can state Newton's second law as follows:

If the resultant force acting on a particle is not zero, the particle has an acceleration proportional to the magnitude of the resultant and in the direction of this resultant force.

Newton's second law of motion is best understood by imagining the following experiment: A particle is subjected to a force $\mathbf{F}_1$ of constant direction and constant magnitude F_1 . Under the action of that force, the particle moves in a straight line and *in the direction of the force* (Fig. 12.1a). By determining the position of the particle at various instants, we find that its acceleration has a constant magnitude a_1 . If we repeat the experiment with forces $\mathbf{F}_2, \mathbf{F}_3, \dots$ of a different magnitude or direction (Fig. 12.1b and c), we find each time that the particle moves in the direction of the force acting on it and that the magnitudes $a_1, a_2, a_3, \dots$ of the accelerations are proportional to the magnitudes $F_1, F_2, F_3, \dots$ of the corresponding forces. Thus,

$$\frac{F_1}{a_1} = \frac{F_2}{a_2} = \frac{F_3}{a_3} = \dots = \text{constant}$$

The constant value obtained for the ratio of the magnitudes of the forces and accelerations is a characteristic of the particle under consideration; it is called the **mass** of the particle and is denoted by m . When a particle of mass m is acted upon by a force $\mathbf{F}$, the force $\mathbf{F}$ and the acceleration $\mathbf{a}$ of the particle must therefore satisfy the relation

Newton's second law

$$\mathbf{F} = m\mathbf{a}$$

(12.1)

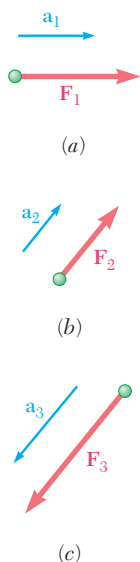


Fig. 12.1 Experiments show that a force applied to a particle gives the particle an acceleration proportional to the magnitude of the force and in the same direction as the force.

This relation provides a complete formulation of Newton's second law; it states not only that the magnitudes of $\mathbf{F}$ and $\mathbf{a}$ are proportional, but also (since m is a positive scalar) that the vectors $\mathbf{F}$ and $\mathbf{a}$ have the same direction (Fig. 12.2). Note that Eq. (12.1) still holds when $\mathbf{F}$ is not constant, but varies with time in magnitude or direction. The magnitudes of $\mathbf{F}$ and $\mathbf{a}$ remain proportional, and the two vectors have the same direction at any given instant. However, they are not, in general, tangent to the path of the particle.

When a particle is subjected simultaneously to several forces, Eq. (12.1) should be replaced by

Newton's second law, multiple forces

$$\Sigma \mathbf{F} = m\mathbf{a} \quad (12.2)$$

where $\Sigma \mathbf{F}$ represents the sum or resultant of all the forces acting on the particle.

Note that the system of axes with respect to which we determine the acceleration $\mathbf{a}$ is not arbitrary. These axes must have a constant orientation with respect to the stars, and their origin either must be attached to the sun (more accurately, to the center of mass of the solar system) or move with a constant velocity with respect to the sun. Such a system of axes is called a **newtonian frame of reference**[†]. A system of axes attached to the earth does *not* constitute a newtonian frame of reference, since the earth rotates with respect to the stars and is accelerated with respect to the sun. However, in most engineering applications, we can determine the acceleration $\mathbf{a}$ with respect to axes attached to the earth and use Eqs. (12.1) and (12.2) without any appreciable error. However, these equations do not hold if $\mathbf{a}$ represents a relative acceleration measured with respect to moving axes, such as axes attached to an accelerated car or to a rotating piece of machinery.

If the resultant $\Sigma \mathbf{F}$ of the forces acting on the particle is zero, it follows from Eq. (12.2) that the acceleration $\mathbf{a}$ of the particle is also zero. If the particle is initially at rest ($\mathbf{v}_0 = 0$) with respect to the newtonian frame of reference used, it will thus remain at rest ($\mathbf{v} = 0$). If originally moving with a velocity $\mathbf{v}_0$, the particle will maintain a constant velocity $\mathbf{v} = \mathbf{v}_0$; that is, it will move with the constant speed v_0 in a straight line. This, we recall, is the statement of Newton's first law (Sec. 2.3B); thus, Newton's first law is a particular case of Newton's second law.

12.1B Linear Momentum of a Particle and its Rate of Change

Suppose we replace the acceleration $\mathbf{a}$ in Eq. (12.2) by the derivative $d\mathbf{v}/dt$. We have

$$\Sigma \mathbf{F} = m \frac{d\mathbf{v}}{dt}$$

Since the mass m of the particle is constant, we can write this as

$$\Sigma \mathbf{F} = \frac{d}{dt}(m\mathbf{v}) \quad (12.3)$$

[†]Stars are not actually fixed, so a more rigorous definition of a newtonian frame of reference (also called an *inertial system*) is one with respect to which Eq. (12.2) holds.

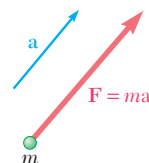


Fig. 12.2 By Newton's second law, the proportionality constant between an applied force and the resulting acceleration is the particle's mass m .



Photo 12.1 When the racecar accelerates forward, the rear tires have a friction force acting on them in the direction the car is moving.

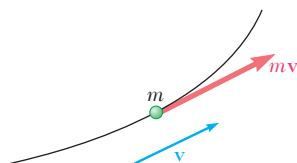


Fig. 12.3 Linear momentum is the product of the mass m and the velocity $\mathbf{v}$ of a particle. It is a vector in the same direction as the velocity.

The product $m\mathbf{v}$ is called the **linear momentum**, or simply the **momentum**, of the particle. It has the same direction as the velocity of the particle, and its magnitude is equal to the product of the mass m and the speed v of the particle (Fig. 12.3). Equation (12.3) says:

The resultant of the forces acting on the particle is equal to the rate of change of the linear momentum of the particle.

The second law of motion was originally stated by Newton in this form. Denoting the linear momentum of the particle by $\mathbf{L}$, we have

$$\text{Linear momentum} \quad \mathbf{L} = m\mathbf{v} \quad (12.4)$$

If we denote its derivative with respect to t as $\dot{\mathbf{L}}$, we can write Eq. (12.3) in the alternative form as

**Newton's second law,
momentum form**

$$\Sigma \mathbf{F} = \dot{\mathbf{L}} \quad (12.5)$$

We assumed that the mass m of the particle is constant in Eqs. (12.3) through (12.5). Therefore, you should not use Equation (12.3) or (12.5) to solve problems involving the motion of bodies, such as rockets, that gain or lose mass. We will consider problems of that type in Sec. 14.3B.[†]

It follows from Eq. (12.3) that the rate of change of the linear momentum $m\mathbf{v}$ is zero when $\Sigma \mathbf{F} = 0$. Thus, we have the statement:

If the resultant force acting on a particle is zero, the linear momentum of the particle remains constant in both magnitude and direction.

This is the principle of **conservation of linear momentum** for a particle.

12.1C Systems of Units

In using the fundamental equation $\mathbf{F} = m\mathbf{a}$, the units of force, mass, length, and time cannot be chosen arbitrarily. If they are, the magnitude of the force $\mathbf{F}$ required to give an acceleration $\mathbf{a}$ to the mass m will *not* be numerically equal to the product ma ; it will only be proportional to this product. Thus, we can choose three of the four units arbitrarily, but we must choose the fourth unit so that the equation $\mathbf{F} = m\mathbf{a}$ is satisfied. The units are then said to form a system of consistent kinetic units.

Two systems of consistent kinetic units are currently used by American engineers: the International System of Units (SI units[‡]) and the system of U.S. customary units. Both systems were discussed in detail in Sec. 1.3, so we describe them only briefly in this section.

International System of Units (SI Units). In this system, the base units are the units of length, mass, and time and are called, respectively, the *meter* (m), the *kilogram* (kg), and the *second* (s). All three are

[†]Note that Eqs. (12.3) and (12.5) do hold in relativistic mechanics, where the mass m of the particle is assumed to vary with the speed of the particle.

[‡]SI stands for *Système International d'Unités* (French).

arbitrarily defined (Sec. 1.3). The unit of force is a derived unit. It is called the *newton* (N) and is defined as the force that gives an acceleration of 1 m/s^2 to a mass of 1 kg (Fig. 12.4). From Eq. (12.1), we have

$$1 \text{ N} = (1 \text{ kg})(1 \text{ m/s}^2) = 1 \text{ kg}\cdot\text{m/s}^2$$

The SI units are said to form an *absolute* system of units. This means that the three base units chosen are independent of the location where measurements are made. The meter, the kilogram, and the second may be used anywhere on the earth; they may even be used on another planet. They always have the same meaning.

The *weight* W of a body, or the *force of gravity* exerted on that body, should, like any other force, be expressed in newtons. A body subjected only to its own weight acquires an acceleration equal to the acceleration due to gravity g . (Be careful using the term *acceleration due to gravity*, since the only time an object accelerates with a magnitude g is during free-fall in the absence of drag.) It follows from Newton's second law that the magnitude W of the weight of a body of mass m is

$$W = mg \quad (12.6)$$

Recall that $g = 9.81 \text{ m/s}^2$, so the weight of a body of mass 1 kg (Fig. 12.5) is

$$W = (1 \text{ kg})(9.81 \text{ m/s}^2) = 9.81 \text{ N}$$

This value would be much less on the moon, where the acceleration due to gravity is 1.6249 m/s^2 .

Multiples and submultiples of the units of length, mass, and force are frequently used in engineering practice. They are, respectively, the *kilometer* (km) and the *millimeter* (mm); the *megagram* (Mg, which is also called the metric ton) and the *gram* (g); and the *kilonewton* (kN). By definition,

$$\begin{aligned} 1 \text{ km} &= 1000 \text{ m} & 1 \text{ mm} &= 0.001 \text{ m} \\ 1 \text{ Mg} &= 1000 \text{ kg} & 1 \text{ g} &= 0.001 \text{ kg} \\ 1 \text{ kN} &= 1000 \text{ N} \end{aligned}$$

You can convert these units to meters, kilograms, and newtons, respectively, simply by moving the decimal point three places to the right or to the left.

Units other than those of mass, length, and time all can be expressed in terms of these three base units. For example, we can obtain the unit of linear momentum by recalling the definition and writing

$$mv = (\text{kg})(\text{m/s}) = \text{kg}\cdot\text{m/s}$$

U.S. Customary Units. Most practicing American engineers still commonly use a system in which the base units are those of length, force, and time. These units are, respectively, the *foot* (ft), the *pound* (lb), and the *second* (s). The second is the same as the corresponding SI unit. The foot is equal to 0.3048 m . The pound is defined as the *weight* of a platinum standard, called the *standard pound*, which is kept at the National Institute of Standards and Technology outside Washington, D.C. The mass of this standard is $0.453\,592\,43 \text{ kg}$. Since the weight of a body depends upon the gravitational attraction of the earth, which varies with location, the standard

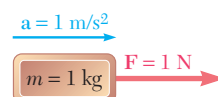


Fig. 12.4 A force of 1 newton gives a 1-kilogram mass an acceleration of 1 m/s^2 .

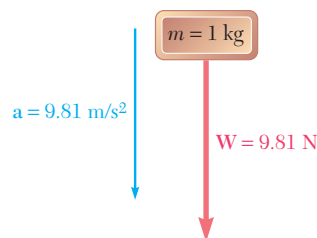


Fig. 12.5 In the SI system, a block with mass 1 kg has a weight of 9.81 N .

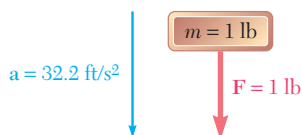


Fig. 12.6 In the U.S. customary system, a block with a weight of 1 lb in free fall has an acceleration of 32.2 ft/s^2 .

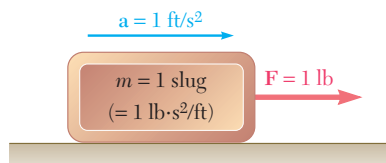


Fig. 12.7 In the U.S. customary system, a force of 1 lb applied to a block with a mass of 1 slug produces an acceleration of 1 ft/s^2 .

pound should be placed at sea level and at a latitude of 45° to properly define a force of 1 lb. Clearly, the U.S. customary units do not form an absolute system of units. Because of their dependence upon the earth's gravitational attraction, they are said to form a *gravitational* system of units.

Although the standard pound also serves as the unit of mass in commercial transactions in the United States, it cannot be used that way in engineering computations because such a unit would not be consistent with the base units defined in this system. Indeed, when acted upon by a force of 1 lb—that is, when subjected to its own weight—the standard pound receives the acceleration of gravity, $g = 32.2 \text{ ft/s}^2$ (Fig. 12.6), and not the unit acceleration required by Eq. (12.1). The unit of mass consistent with the foot, the pound, and the second is the mass that receives an acceleration of 1 ft/s^2 when a force of 1 lb is applied to it (Fig. 12.7). This unit, sometimes called a *slug*, can be derived from the equation $F = ma$ after substituting 1 lb and 1 ft/s^2 for F and a , respectively. We have

$$F = ma \qquad 1 \text{ lb} = (1 \text{ slug})(1 \text{ ft/s}^2)$$

From this, we obtain

$$1 \text{ slug} = \frac{1 \text{ lb}}{1 \text{ ft/s}^2} = 1 \text{ lb} \cdot \text{s}^2/\text{ft}$$

Comparing Figs. 12.6 and 12.7, we conclude that the slug is a mass 32.2 times larger than the mass of the standard pound. (On a horizontal surface, when acted on by a force of 1 pound, the motion of the larger mass is relatively “sluggish.”)

The fact that bodies are characterized in the U.S. customary system of units by their weight in pounds rather than by their mass in slugs was convenient in the study of statics, where we were dealing (for the most part) with weights and other forces and only seldom with masses. However, in the study of kinetics, which involves forces, masses, and accelerations, we will often have to express the mass m of a body in slugs, the weight W of which is given in pounds. Recalling Eq. (12.6), we have

$$m = \frac{W}{g} \qquad (12.7)$$

where g is the acceleration due to gravity ($g = 32.2 \text{ ft/s}^2$).

Units other than the units of force, length, and time all can be expressed in terms of these three base units. For example, we can obtain the unit of linear momentum from its definition as

$$mv = (\text{slug})(\text{ft/s}) = (\text{lb} \cdot \text{s}^2/\text{ft})(\text{ft/s}) = \text{lb} \cdot \text{s}$$

Conversion from One System of Units to Another. The conversion from U.S. customary units to SI units, and vice versa, was discussed in Sec. 1.4. Recall that the conversion factors obtained for the units of length, force, and mass are, respectively,

$$\begin{array}{ll} \text{Length:} & 1 \text{ ft} = 0.3048 \text{ m} \\ \text{Force:} & 1 \text{ lb} = 4.448 \text{ N} \\ \text{Mass:} & 1 \text{ slug} = 1 \text{ lb} \cdot \text{s}^2/\text{ft} = 14.59 \text{ kg} \end{array}$$

Thermodynamicists often use a unit called the pound-mass (lbm); this is not a unit that is consistent with Newton's second law, and whenever we use pounds in dynamics, it will refer to pounds-force (lbf).

Although it cannot be used as a consistent unit of mass, the mass of the standard pound is, by definition,

$$1 \text{ pound-mass} = 0.4536 \text{ kg}$$

This constant can be used to determine the *mass* in SI units (kilograms) of a body that has been characterized by its *weight* in U.S. customary units (pounds).

12.1D Equations of Motion

Consider a particle of mass m acted upon by several forces. Recall that we can express Newton's second law by the equation

$$\Sigma \mathbf{F} = m\mathbf{a} \quad (12.2)$$

which relates the forces acting on the particle to the vector $m\mathbf{a}$ (Fig. 12.8).[†] Two of the most important tools you will use in solving dynamics problems, particularly those involving Newton's second law, are the free-body diagram and the kinetic diagram. These diagrams will help you to model dynamic systems and apply appropriate equations of motion. The free-body diagram shown on the left side of Fig. 12.9 is no different from what you did in statics in Chapter 4 and consists of the following steps:

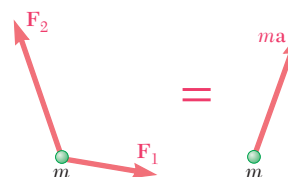


Fig. 12.8 The sum of forces applied to a particle of mass m produces a vector $m\mathbf{a}$ in the direction of the resultant force.

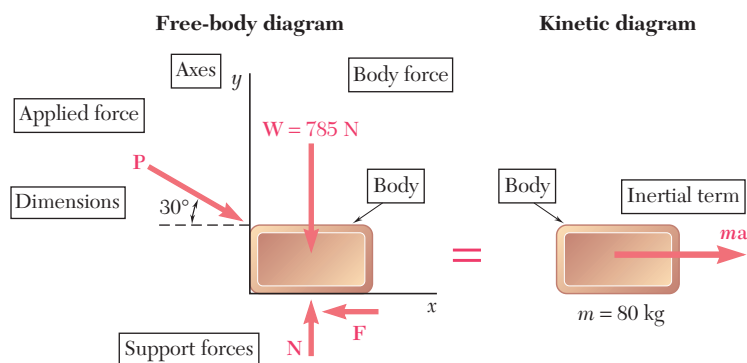


Fig. 12.9 Steps in drawing a free-body diagram and a kinetic diagram for solving dynamics problems.

Body: Define your system by isolating the body (or bodies) of interest. If a problem has multiple bodies (such as in Sample Problems 12.3 through 12.5), you may have to draw multiple free-body diagrams and kinetic diagrams.

Axes: Draw an appropriate coordinate system (e.g., Cartesian, normal and tangential, or radial and transverse).

Support Forces: Replace supports or constraints with appropriate forces (e.g., two perpendicular forces for a pin, normal forces, friction forces).

[†]In the 1700s, Jean-Baptiste le Rond d'Alembert expressed Newton's second law as $\Sigma \mathbf{F} - m\mathbf{a} = 0$ so he could solve dynamics problems using the principles of statics. The $-m\mathbf{a}$ term has been called a fictitious *inertial force*, but it is important for you to realize that there is no such thing as inertial forces (or centrifugal forces that “push” you outward when going around a curve). D'Alembert's principle (also called dynamic equilibrium) is seldom used in modern engineering.

Applied Forces and Body Forces: Draw any applied forces and body forces (also sometimes called field forces) on your diagram (e.g., weight, magnetic forces, a known pulling force).

Dimensions: Add any angles or distances that are important for solving the problem.

In statics problems, we deal with bodies in equilibrium, and the inertial term in Newton's second law is zero. For dynamics problems, this is not the case. We utilize the kinetic diagram to visualize this term.

Body: This is the same body as in the free-body diagram; place this beside the free-body diagram.

Inertial terms: Draw the $m\mathbf{a}$ term to be consistent with the coordinate system. Generally, draw this term in different components (e.g., ma_x and ma_y or ma_n and ma_t). If they are unknown quantities, it is best to draw them in the positive directions as defined by your coordinates.

Drawing these two diagrams clarifies how to develop your equations of motion. The free-body diagram is a visual representation of the $\Sigma\mathbf{F}$ term, and the kinetic diagram is a visual representation of the $m\mathbf{a}$ term. Since Newton's second law is a vector equation, you can use the free-body diagram and kinetic diagram to write $\Sigma\mathbf{F} = m\mathbf{a}$ directly in component form. Examples of using these diagrams to help you write your equations of motion are shown in the Sample Problems, and you can get extra practice by solving the Free-Body Problems 12.F1 through 12.F12.

As mentioned, it is usually more convenient to replace Eq. (12.2) with equivalent equations involving scalar quantities. As we saw in Chapter 11, we can resolve these vectors into components using several different coordinate systems (e.g., Cartesian, tangential and normal, or radial and transverse), depending on the type of problem we are solving.

Rectangular Components. Resolving each force $\mathbf{F}$ and the acceleration $\mathbf{a}$ into rectangular components, we have

$$\Sigma(F_x\mathbf{i} + F_y\mathbf{j} + F_z\mathbf{k}) = m(a_x\mathbf{i} + a_y\mathbf{j} + a_z\mathbf{k})$$

It follows from this equation that

$$\Sigma F_x = ma_x \quad \Sigma F_y = ma_y \quad \Sigma F_z = ma_z \quad (12.8)$$

Recall from Sec. 11.4C that the components of the acceleration are equal to the second derivatives of the coordinates of the particle. This gives us

$$\Sigma F_x = m\ddot{x} \quad \Sigma F_y = m\ddot{y} \quad \Sigma F_z = m\ddot{z} \quad (12.8')$$

Consider, as an example, the motion of a projectile. If we neglect air resistance, the only force acting on the projectile after it has been fired is its weight $\mathbf{W} = -W\mathbf{j}$. The equations defining the motion of the projectile are therefore

$$m\ddot{x} = 0 \quad m\ddot{y} = -W \quad m\ddot{z} = 0$$

and the components of the acceleration of the projectile are

$$\ddot{x} = 0 \quad \ddot{y} = -\frac{W}{m} = -g \quad \ddot{z} = 0$$

where g is 9.81 m/s^2 or 32.2 ft/s^2 . You can integrate these equations independently, as shown in Sec. 11.4C, to obtain the velocity and displacement of the projectile at any instant.



Photo 12.2 Biomechanics researchers use video analysis and force plate measurements in Cartesian coordinates to analyze human motion.

When a problem involves two or more bodies, you should write equations of motion for each of the bodies (see Sample Probs. 12.3 through 12.5). Recall from Sec. 12.1A that all accelerations should be measured with respect to a newtonian frame of reference. In most engineering applications, you can determine accelerations with respect to axes attached to the earth, but relative accelerations measured with respect to moving axes, such as axes attached to an accelerated body, cannot be substituted for $\mathbf{a}$ in the equations of motion.

Tangential and Normal Components. We can also resolve the forces and the acceleration of the particle into components along the tangent to the path (in the direction of motion) and the normal (toward the inside of the path) (Fig. 12.10). Substituting into Eq. (12.2), we obtain the two scalar equations of

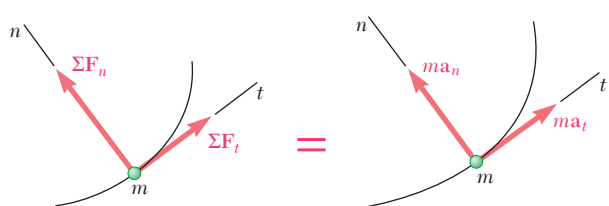
$$\Sigma F_t = ma_t \quad \Sigma F_n = ma_n \quad (12.9)$$


Fig. 12.10 The net force acting on a particle moving in a curvilinear path can be resolved into components tangent to the path and normal to the path, producing tangential and normal components of acceleration.

Now substituting for a_t and a_n from Eqs. (11.39), we have

$$\Sigma F_t = m \frac{dv}{dt} \quad \Sigma F_n = m \frac{v^2}{\rho} \quad (12.9')$$

We can solve these equations for two unknowns.

Radial and Transverse Components. Consider a particle P , with polar coordinates r and θ , that moves in a plane under the action of several forces. Resolving the forces and the acceleration of the particle into radial and transverse components (Fig. 12.11) and substituting into Eq. (12.2), we obtain the two scalar equations of

$$\Sigma F_r = ma_r \quad \Sigma F_\theta = ma_\theta \quad (12.10)$$

Substituting for a_r and a_θ from Eqs. (11.45), we have

$$\Sigma F_r = m(\ddot{r} - r\dot{\theta}^2) \quad (12.11)$$

$$\Sigma F_\theta = m(r\ddot{\theta} + 2\dot{r}\dot{\theta}) \quad (12.12)$$

We can solve these equations for two unknowns.

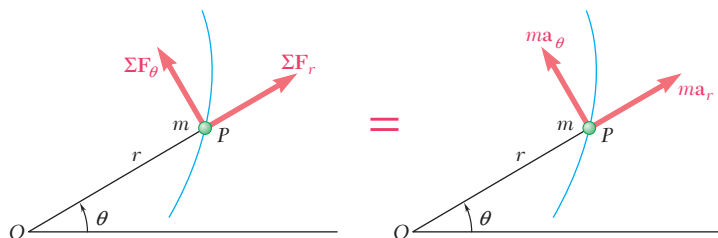


Fig. 12.11 Pictorial representation of Newton's second law in radial and transverse components.

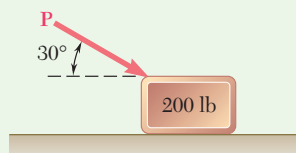


Photo 12.3 A fighter jet making a sharp turn has a large normal component of acceleration, often equal to several g . As a result, the pilot experiences a large normal force, which in extreme cases, can cause blackouts.



Photo 12.4 The forces on the specimens used in a high speed centrifuge can be described in terms of radial and transverse components.

Sample Problem 12.1



A 200-lb block rests on a horizontal plane. Find the magnitude of the force $\mathbf{P}$ required to give the block an acceleration of 10 ft/s^2 to the right. The coefficient of kinetic friction between the block and the plane is $\mu_k = 0.25$.

STRATEGY: You are given an acceleration and want to find the applied force. Therefore, you need to use Newton's second law.

MODELING: Pick the block as your system and model it as a particle. Drawing its free-body and kinetic diagrams, you obtain Fig. 1.

ANALYSIS: Before using Fig. 1, it is convenient to determine the mass of the object.

$$m = \frac{W}{g} = \frac{200 \text{ lb}}{32.2 \text{ ft/s}^2} = 6.21 \text{ lb}\cdot\text{s}^2/\text{ft}$$

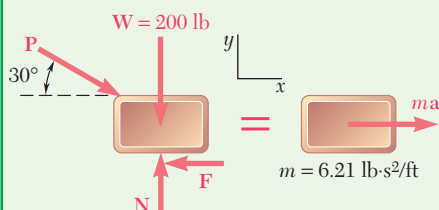


Fig. 1 Free-body diagram and kinetic diagram for the block.

From Fig. 1, it is clear that the forces acting on the block shown in the free-body diagram need to be equal to the vector $m\mathbf{a}$, as shown in the kinetic diagram. Using these diagrams, you can write

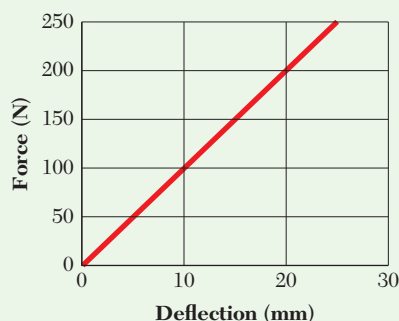
$$\begin{aligned} \rightarrow \Sigma F_x = ma: \quad & P \cos 30^\circ - 0.25N = (6.21 \text{ lb}\cdot\text{s}^2/\text{ft})(10 \text{ ft/s}^2) \\ & P \cos 30^\circ - 0.25N = 62.1 \text{ lb} \end{aligned} \quad (1)$$

$$+\uparrow \Sigma F_y = 0: \quad N - P \sin 30^\circ - 200 \text{ lb} = 0 \quad (2)$$

Solving Eq. (2) for N and substituting the result into Eq. (1), you obtain

$$\begin{aligned} N &= P \sin 30^\circ + 200 \text{ lb} \\ P \cos 30^\circ - 0.25(P \sin 30^\circ + 200 \text{ lb}) &= 62.1 \text{ lb} \quad P = 151 \text{ lb} \quad \blacktriangleleft \end{aligned}$$

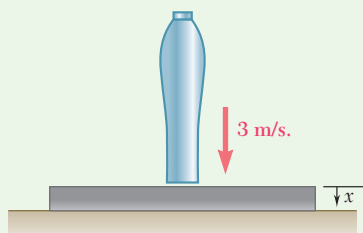
REFLECT and THINK: When you begin pushing on an object, you first have to overcome the static friction force ($F = \mu_s N$) before the object will move. Also note that the downward component of force $\mathbf{P}$ increases the normal force N , which in turn increases the friction force $\mathbf{F}$ that you must overcome.



Sample Problem 12.2

A 0.5-kg fragile glass vase is dropped onto a thick pad that has a force-deflection relationship as shown. Knowing that the vase has a speed of 3 m/s when it first contacts the pad, determine the maximum downward displacement of the vase.

STRATEGY: Use Newton's second law to find the acceleration of the vase and then integrate it to find the displacement.



MODELING: Choose the vase to be your system and model it as a particle. Because the force is a linear function of displacement, you can write the force acting on the vase as

$$F_p = \frac{200 \text{ N}}{0.02 \text{ m}} y = (10\,000 \text{ N/m})x$$

Draw its free-body diagram and kinetic diagram (Fig. 1).

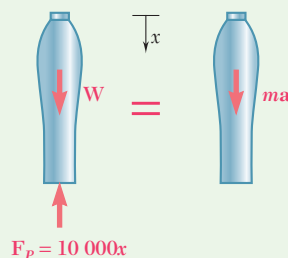


Fig. 1 Free-body diagram and kinetic diagram for the vase.

ANALYSIS: You can obtain a scalar equation by applying Newton's second law in the vertical direction. Thus,

$$+\downarrow \Sigma F_x = ma \quad W - (10\,000)x = ma$$

Substituting in values and solving for a gives

$$a = 9.81 - 20\,000x$$

Maximum Displacement. Now that you have the acceleration as a function of displacement, you need to use the basic kinematic relationships to find the maximum compression of the pad. Substituting $a = 9.81 - 20\,000x$ into $a = v dv/dx$ gives

$$a = 9.81 - 20\,000x = \frac{v dv}{dx}$$

Separating variables and integrating, you find

$$v dv = (9.81 - 20\,000x)dx \longrightarrow \int_{v_0}^0 v dv = \int_0^{x_{\max}} (9.81 - 20\,000x)dx$$

$$0 - \frac{1}{2}v_0^2 = 9.81x_{\max} - 10\,000x_{\max}^2 \quad (1)$$

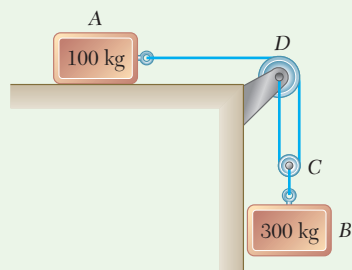
Substituting $v_0 = 3 \text{ m/s}$ into Eq. (1) and solving for $x_{\max}$ using the quadratic formula gives $x_{\max} = 0.0217 \text{ m}$.

$$x_{\max} = 21.7 \text{ mm} \quad \blacktriangleleft$$

REFLECT and THINK: A distance of 21.7 mm indicates that the pad must be relatively thick. For a real pad, the assumption that it acts as a linear spring may not be an accurate model. For the numbers given in this problem, the maximum acceleration the vase experiences is

$$a = 9.81 - (20\,000)(0.0217) = -207.3 \text{ m/s}^2 \text{ or about } 21 \text{ g's}$$

Sample Problem 12.3



The two blocks shown start from rest. The horizontal plane and the pulley are frictionless, and the pulley is assumed to be of negligible mass. Determine the acceleration of each block and the tension in each cord.

STRATEGY: You are interested in finding the tension in the rope and the acceleration of the two blocks, so use Newton's second law. The two blocks are connected by a cable, indicating that you need to relate their accelerations using the techniques discussed in Chapter 11 for objects with dependent motion.

MODELING: Treat both blocks as particles and assume that the pulley is massless and frictionless. Since there are two masses, you need two systems: block A by itself and block B by itself. The free-body and kinetic diagrams for these objects are shown in Figs. 1 and 2. To help determine the forces acting on block B, you can also isolate the massless pulley C as a system (Fig. 3).

ANALYSIS: You can start with either kinetics or kinematics. The key is to make sure you keep track of your equations and unknowns.

Kinetics. Apply Newton's second law successively to block A, block B, and pulley C.

Block A. Denote the tension in cord ACD by T_1 (Fig. 1). Then you have

$$+\rightarrow \Sigma F_x = m_A a_A: \quad T_1 = 100a_A \quad (1)$$

Block B. Observe that the weight of block B is

$$W_B = m_B g = (300 \text{ kg})(9.81 \text{ m/s}^2) = 2940 \text{ N}$$

Denote the tension in cord BC by T_2 (Fig. 2). Then

$$+\downarrow \Sigma F_y = m_B a_B: \quad 2940 - T_2 = 300a_B \quad (2)$$

Pulley C. Assuming m_C is zero, you have (Fig. 3)

$$+\downarrow \Sigma F_y = m_C a_C = 0: \quad T_2 - 2T_1 = 0 \quad (3)$$

At this point, you have three equations, (1), (2), and (3), and four unknowns, T_1 , T_2 , a_B , and a_A . Therefore, you need one more equation, which you can get from kinematics.

Kinematics. It is important to make sure that the directions you assumed for the kinetic diagrams are consistent with the kinematic analysis. Note that if block A moves through a distance x_A to the right, block B moves down through a distance

$$x_B = \frac{1}{2}x_A$$

Differentiating twice with respect to t , you have

$$a_B = \frac{1}{2}a_A \quad (4)$$

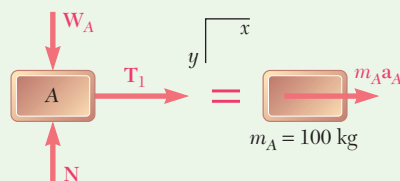


Fig. 1 Free-body diagram and kinetic diagram for A.

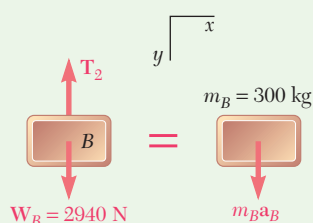


Fig. 2 Free-body diagram and kinetic diagram for B.

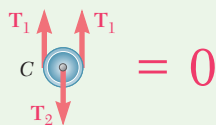


Fig. 3 Free-body diagram and kinetic diagram for the pulley.

You now have four equations and four unknowns, so you can solve this problem. You can do this using a computer, a calculator, or by hand. To solve these equations by hand, you can substitute for a_B from Eq. (4) into Eq. (2) for

$$\begin{aligned} 2940 - T_2 &= 300\left(\frac{1}{2}a_A\right) \\ T_2 &= 2940 - 150a_A \end{aligned} \quad (5)$$

Now substitute for T_1 and T_2 from Eqs. (1) and (5), respectively, into Eq. (3).

$$\begin{aligned} 2940 - 150a_A - 2(100a_A) &= 0 \\ 2940 - 350a_A &= 0 \quad a_A = 8.40 \text{ m/s}^2 \quad \blacktriangleleft \end{aligned}$$

Then substitute the value obtained for a_A into Eqs. (4) and (1).

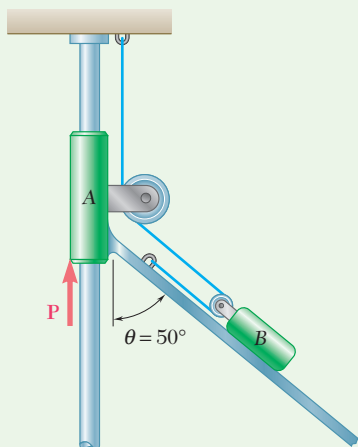
$$\begin{aligned} a_B &= \frac{1}{2}a_A = \frac{1}{2}(8.40 \text{ m/s}^2) & a_B &= 4.20 \text{ m/s}^2 \quad \blacktriangleleft \\ T_1 &= 100a_A = (100 \text{ kg})(8.40 \text{ m/s}^2) & T_1 &= 840 \text{ N} \quad \blacktriangleleft \end{aligned}$$

Recalling Eq. (3), you have

$$T_2 = 2T_1 \quad T_2 = 2(840 \text{ N}) \quad T_2 = 1680 \text{ N} \quad \blacktriangleleft$$

REFLECT and THINK: Note that the value obtained for T_2 is *not* equal to the weight of block B . Rather than choosing B and the pulley as separate systems, you could have chosen the system to be B and the pulley. In this case, T_2 would have been an internal force.

Sample Problem 12.4



Collar A has a ramp that is welded to it and a force $P = 5 \text{ lb}$ applied as shown. Collar A and the ramp weigh 3 lb , and block B weighs 0.8 lb . Neglecting friction, determine the tension in the cable.

STRATEGY: The principle you need to use is Newton's second law. Since a block is sliding down an incline and a cable is connecting A and B , you also need to use relative motion and dependent motion.

MODELING: Model A and B as particles and assume all surfaces are smooth. As usual, start by choosing a system and then drawing a free-body diagram and a kinetic diagram. This problem has two systems, and you need to be careful with how you define them. The easiest systems to use are (a) collar A with its pulley and the ramp welded to it (system 1) and (b) block B and the pulley attached to it (system 2), as shown in Fig. 1. The free-body and kinetic diagrams for system 1 are shown in Fig. 2. The free-body and kinetic diagrams for B are a little trickier, because you don't know the direction of the acceleration of B .

(continued)

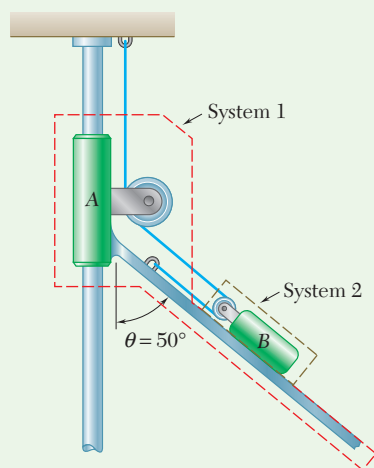


Fig. 1 System boundaries.

Kinematics for Block B. Express the acceleration $\mathbf{a}_B$ of block B as the sum of the acceleration of A and the acceleration of B relative to A. Hence,

$$\mathbf{a}_B = \mathbf{a}_A + \mathbf{a}_{B/A}$$

Here $\mathbf{a}_{B/A}$ is directed along the inclined surface of the wedge. Now you can draw the appropriate diagrams (Fig. 3). Note that you do not need to use the same x - y coordinate system for each mass, since these directions are simply used for obtaining the scalar equations.

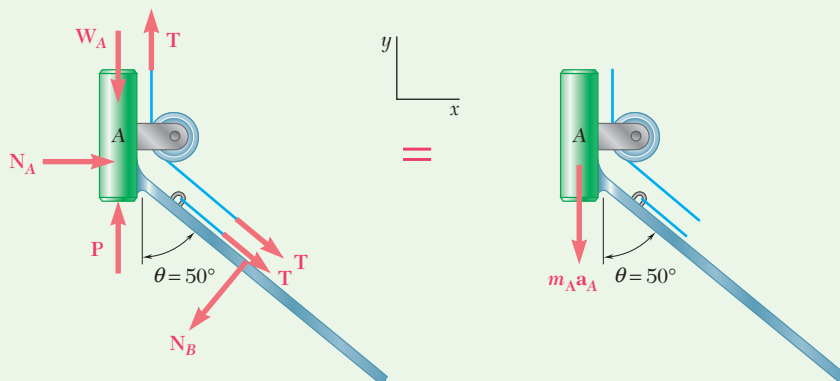


Fig. 2 Free-body diagram and kinetic diagram for system 1.

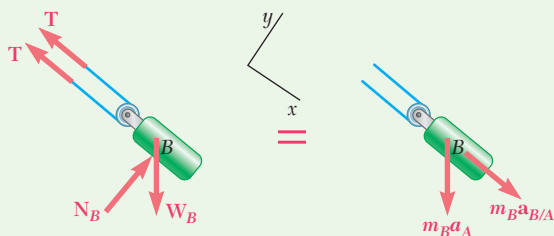


Fig. 3 Free-body diagram and kinetic diagram for B.

ANALYSIS: You can obtain a scalar equation by applying Newton's second law to each of these systems.

a. System 1:

$$+\rightarrow \Sigma F_x = m_A a_{A_x} \quad N_A - N_B \cos 50^\circ + 2T \cos 40^\circ = 0 \quad (1)$$

$$+\uparrow \Sigma F_y = m_A a_{A_y} \quad -W_A + P + T - 2T \sin 40^\circ - N_B \sin 50^\circ = -m_A a_A \quad (2)$$

b. Block B:

$$+\searrow \Sigma F_x = m_B a_{B_x} \quad -2T + W_B \sin 40^\circ = m_B a_{B/A} + m_B a_A \sin 40^\circ \quad (3)$$

$$+\nearrow \Sigma F_y = m_B a_{B_y} \quad N_B - W_B \cos 40^\circ = -m_B a_A \cos 40^\circ \quad (4)$$

You now have four equations and five unknowns (T , N_A , N_B , a_A , and $a_{B/A}$), so you need one more equation. The motion of A and B are related because they are connected by a cable.

Constraint Equations. Define position vectors as shown in Fig. 4. Note that the positive directions for the position vectors for A and B are

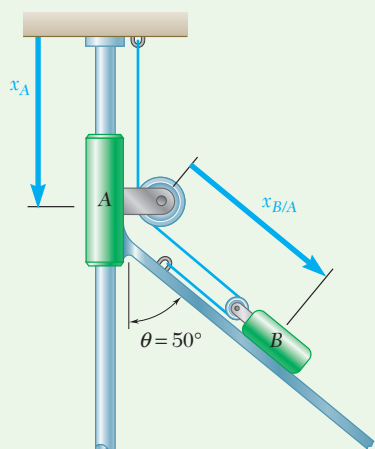


Fig. 4 Position vectors for dependent motion.

defined from the kinetic diagrams in Figs. 2 and 3. Assuming the cable is inextensible, you can write the lengths in terms of the coordinates and then differentiate.

Constraint equation for the cable: $x_A + 2x_{B/A} = \text{constant}$

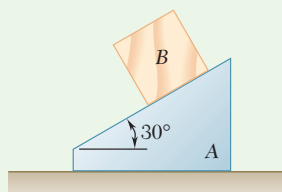
Differentiating this twice gives

$$a_A = -2a_{B/A} \quad (5)$$

You now have five equations and five unknowns, so all that remains is to substitute the known values and solve for the unknowns. The results are $N_A = -0.1281 \text{ lb}$, $N_B = 0.869 \text{ lb}$, $T = 0.281 \text{ lb}$, $a_A = -13.46 \text{ ft/s}^2$, and $a_{B/A} = 6.73 \text{ ft/s}^2$.

$$T = 0.281 \text{ lb} \quad \blacktriangleleft$$

REFLECT and THINK: In this problem, we focused on the problem formulation and assumed that you can solve the resulting equations either by hand or by using a calculator/computer. It is important to note that you are given the weights of A and B , so you need to calculate the masses in slugs using $m = W/g$. The solution required multiple systems and multiple concepts, including Newton's second law, relative motion, and dependent motion. If friction occurred between B and the ramp, you would first need to determine whether or not the system would move under the applied force by assuming that it does not move and calculating the friction force. Then you would compare this force to the maximum allowable force $\mu_s N$.



Sample Problem 12.5

The 12-lb block B starts from rest and slides on the 30-lb wedge A , which is supported by a horizontal surface. Neglecting friction, determine (a) the acceleration of the wedge, (b) the acceleration of the block relative to the wedge.

STRATEGY: You are given the forces (weights) of the two objects and want to find their accelerations. You can use Newton's second law, but you have to take into account relative motion as well.

MODELING: Treat both objects as particles. Since you have two objects, you will need two systems: wedge A and block B . In order to draw the kinetic diagrams for each of these systems, you need to know the direction of the accelerations. Therefore, before drawing the free-body and kinetic diagrams, look at the kinematics.

Kinematics. First examine the acceleration of the wedge and the acceleration of the block.

Wedge A. Since the wedge is constrained to move on the horizontal surface, its acceleration $\mathbf{a}_A$ is horizontal (Fig. 1). Assume that it is directed to the right.

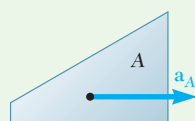


Fig. 1 Acceleration of A .

(continued)

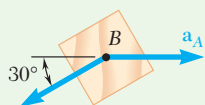


Fig. 2 Acceleration of B .

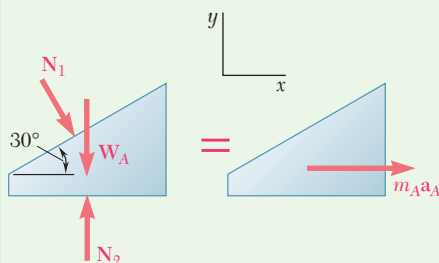


Fig. 3 Free-body diagram and kinetic diagram for A .

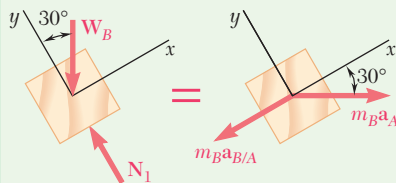


Fig. 4 Free-body diagram and kinetic diagram for B .

Block B. You can express the acceleration $\mathbf{a}_B$ of block B as the sum of the acceleration of A and the acceleration of B relative to A (Fig. 2), so

$$\mathbf{a}_B = \mathbf{a}_A + \mathbf{a}_{B/A}$$

Here $\mathbf{a}_{B/A}$ is directed along the inclined surface of the wedge. Now you can draw the appropriate diagrams. The free-body diagrams and kinetic diagrams for A and B are shown in Figs. 3 and 4, respectively. The forces exerted by the block and the horizontal surface on wedge A are represented by $\mathbf{N}_1$ and $\mathbf{N}_2$, respectively.

ANALYSIS:

Kinetics. Recall that Figs. 3 and 4 are visual representations of Newton's second law. Therefore, you can use them to obtain scalar equations.

Wedge A. For Wedge A , the positive x -direction is defined to be to the right. Applying Newton's second law in the x -direction gives

$$\begin{aligned} \rightarrow \Sigma F_x = m_A a_A: \quad N_1 \sin 30^\circ &= m_A a_A \\ 0.5N_1 &= (W_A/g)a_A \end{aligned} \quad (1)$$

Block B. Using the coordinate axes shown in Fig. 4 and resolving $\mathbf{a}_B$ into its components $\mathbf{a}_A$ and $\mathbf{a}_{B/A}$, you have

$$\begin{aligned} + \nearrow \Sigma F_x = m_B a_x: \quad -W_B \sin 30^\circ &= m_B a_A \cos 30^\circ - m_B a_{B/A} \\ -W_B \sin 30^\circ &= (W_B/g)(a_A \cos 30^\circ - a_{B/A}) \\ a_{B/A} &= a_A \cos 30^\circ + g \sin 30^\circ \end{aligned} \quad (2)$$

$$\begin{aligned} + \nwarrow \Sigma F_y = m_B a_y: \quad N_1 - W_B \cos 30^\circ &= -m_B a_A \sin 30^\circ \\ N_1 - W_B \cos 30^\circ &= -(W_B/g)a_A \sin 30^\circ \end{aligned} \quad (3)$$

You now have three equations, (1), (2), and (3), and three unknowns, N_1 , a_A , and $a_{B/A}$, so you can solve these with your calculator or by hand as shown here.

a. Acceleration of Wedge A. Substitute for N_1 from Eq. (1) into Eq. (3).

$$2(W_A/g)a_A - W_B \cos 30^\circ = -(W_B/g)a_A \sin 30^\circ$$

Then solve for a_A and substitute the numerical data.

$$\begin{aligned} a_A &= \frac{W_B \cos 30^\circ}{2W_A + W_B \sin 30^\circ} g = \frac{(12 \text{ lb}) \cos 30^\circ}{2(30 \text{ lb}) + (12 \text{ lb}) \sin 30^\circ} (32.2 \text{ ft/s}^2) \\ a_A &= +5.07 \text{ ft/s}^2 \quad \mathbf{a_A = 5.07 \text{ ft/s}^2 \rightarrow} \end{aligned}$$

b. Acceleration of Block B Relative to A. Now substitute the value obtained for a_A into Eq. (2).

$$\begin{aligned} a_{B/A} &= (5.07 \text{ ft/s}^2) \cos 30^\circ + (32.2 \text{ ft/s}^2) \sin 30^\circ \\ a_{B/A} &= +20.5 \text{ ft/s}^2 \quad \mathbf{a_{B/A} = 20.5 \text{ ft/s}^2 \nearrow 30^\circ} \end{aligned}$$

REFLECT and THINK: Many students are tempted to draw the acceleration of block B down the incline in the kinetic diagram. It is important to recognize that this is the direction of the *relative* acceleration. Rather than the kinetic diagram you used for block B , you could have simply put unknown accelerations in the x and y directions and then used your relative motion equation to obtain more scalar equations.

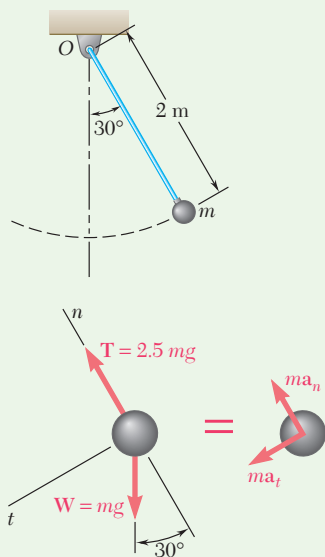


Fig. 1 Free-body diagram and kinetic diagram for the bob.

Sample Problem 12.6

The bob of a 2-m pendulum describes an arc of a circle in a vertical plane. If the tension in the cord is 2.5 times the weight of the bob for the position shown, find the velocity and the acceleration of the bob in that position.

STRATEGY: The most direct approach is to use Newton's law with tangential and normal components.

MODELING: Choose the bob as your system; if its radius is small, you can model it as a particle. Draw the free-body and kinetic diagrams for the bob knowing that the weight of the bob is $W = mg$; the tension in the cord is $2.5mg$. The normal acceleration $\mathbf{a}_n$ is directed toward O , and you can assume that $\mathbf{a}_t$ is in the direction shown in Fig. 1.

ANALYSIS: You can obtain scalar equations by applying Newton's second law in the normal and tangential directions. Hence,

$$+\swarrow \Sigma F_t = ma_t: \quad mg \sin 30^\circ = ma_t$$

$$a_t = g \sin 30^\circ = +4.90 \text{ m/s}^2 \quad \mathbf{a}_t = 4.90 \text{ m/s}^2 \swarrow$$

$$+\nwarrow \Sigma F_n = ma_n: \quad 2.5mg - mg \cos 30^\circ = ma_n$$

$$a_n = 1.634g = +16.03 \text{ m/s}^2 \quad \mathbf{a}_n = 16.03 \text{ m/s}^2 \nwarrow$$

Since $a_n = v^2/\rho$, you have $v^2 = \rho a_n = (2 \text{ m})(16.03 \text{ m/s}^2)$. Thus,

$$v = \pm 5.66 \text{ m/s} \quad \mathbf{v} = 5.66 \text{ m/s} \nearrow (\text{up or down})$$

REFLECT and THINK: If you look at these equations for an angle of zero instead of 30° , you will see that when the bob is straight below point O , the tangential acceleration is zero, and the velocity is a maximum. The normal acceleration is not zero because the bob has a velocity at this point.

Sample Problem 12.7

Determine the rated speed of a highway curve with a radius of $\rho = 400 \text{ ft}$ banked through an angle $\theta = 18^\circ$. The *rated speed* of a banked highway curve is the speed at which a car should travel to have no lateral friction force exerted on its wheels.

STRATEGY: You are given information about the lateral friction force—that is, it is equal to zero—so use Newton's second law. Use normal and tangential components, since the car is traveling in a curved path and the problem involves speed and a radius of curvature.

MODELING: Choose the car to be the system. Assuming you can neglect the rotation of the car about its center of mass, treat it as a particle.

(continued)

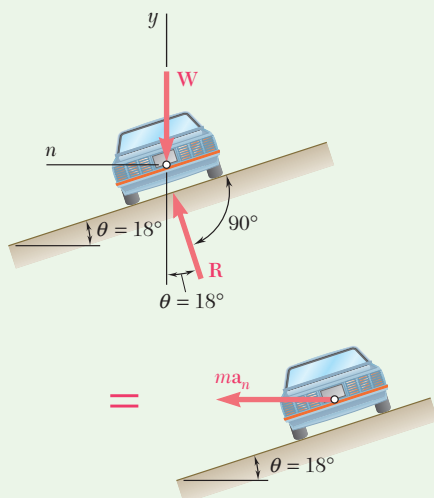


Fig. 1 Free-body diagram and kinetic diagram of the car.

The car travels in a *horizontal* circular path with a radius of ρ . The normal component $\mathbf{a}_n$ of the acceleration is directed toward the center of the path, as shown in the kinetic diagram (Fig. 1); its magnitude is $a_n = v^2/\rho$, where v is the speed of the car in ft/s. The mass m of the car is W/g , where W is the weight of the car. Since no lateral friction force is exerted on the car, the reaction $\mathbf{R}$ of the road is perpendicular to the roadway, as shown in the free-body diagram (Fig. 1).

ANALYSIS: You can obtain scalar equations by applying Newton's second law in the vertical and normal directions. Thus,

$$+\uparrow \Sigma F_y = 0: \quad R \cos \theta - W = 0 \quad R = \frac{W}{\cos \theta} \quad (1)$$

$$\leftarrow \Sigma F_n = ma_n: \quad R \sin \theta = \frac{W}{g} a_n \quad (2)$$

Substituting R from Eq. (1) into Eq. (2), and recalling that $a_n = v^2/\rho$, you obtain

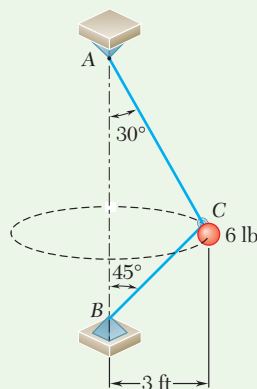
$$\frac{W}{\cos \theta} \sin \theta = \frac{W}{g} \frac{v^2}{\rho} \quad v^2 = g \rho \tan \theta$$

Finally, substituting $\rho = 400$ ft and $\theta = 18^\circ$ into this equation, you get $v^2 = (32.2 \text{ ft/s}^2)(400 \text{ ft}) \tan 18^\circ$. Hence,

$$v = 64.7 \text{ ft/s} \quad v = 44.1 \text{ mi/h} \quad \blacktriangleleft$$

REFLECT and THINK: For a highway curve, this seems like a reasonable speed for avoiding a spin-out. For this problem, the tangential direction is into the page; since you were not asked about forces or accelerations in this direction, you did not need to analyze motion in the tangential direction. If the roadway were banked at a larger angle, would the rated speed be larger or smaller than this calculated value?

Sample Problem 12.8



Two wires AC and BC are tied at C to a sphere that revolves at the constant speed v in the horizontal circle shown. Knowing that the wires will break if their tension exceeds 15 lb, determine the range of values of v for which both wires remain taut and the wires do not break.

STRATEGY: You are given information about the forces in the wires, so use Newton's second law. The sphere is moving along a curved path, so use normal and tangential coordinates.

MODELING: Choose the sphere for the system and assume you can treat it as a particle. Draw the free-body and kinetic diagrams as shown in Fig. 1. The tensions act in the direction of the wires, and the normal direction is toward the center of the circular path.

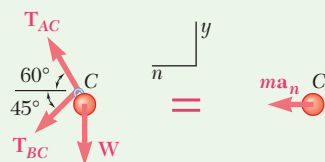


Fig. 1 Free-body diagram and kinetic diagram for the sphere.

ANALYSIS: You can obtain a scalar equation by applying Newton's second law in the normal and vertical directions. Thus,

$$+\leftarrow \Sigma F_n = ma_n \quad T_{AC} \cos 60^\circ + T_{BC} \cos 45^\circ = ma_n = m \frac{v^2}{\rho} \quad (1)$$

$$+\uparrow \Sigma F_y = ma_y \quad -W + T_{AC} \sin 60^\circ - T_{BC} \sin 45^\circ = 0 \quad (2)$$

where $m = W/g = 6 \text{ lb}/(32.2 \text{ ft/s}^2) = 0.1863 \text{ lb}\cdot\text{s}^2/\text{ft}$ and $\rho = 3 \text{ ft}$. In these two equations, you have three unknowns, T_{AC} , T_{BC} , and v , so you need a third equation. The problem statement indicates that you want the range of speeds when the both wires remain taut (that is, the tension is positive) and that this tension must be less than 15 lb. To find this range, first set each tension equal to zero and solve the resulting set of equations.

For $T_{AC} = 0$, you find $v = 9.83 \text{ ft/s}$ and $T_{BC} = -8.485 \text{ lb}$, which is impossible for a wire.

For $T_{BC} = 0$, you find $v = 7.468 \text{ ft/s}$ and $T_{AC} = 6.928 \text{ lb}$.

Thus, the minimum speed is 7.47 ft/s. Now set the tensions equal to 15 lb to find the maximum speed.

For $T_{AC} = 15 \text{ lb}$, you find $v = 15.29 \text{ ft/s}$ and $T_{BC} = 9.886 \text{ lb}$.

For $T_{BC} = 15 \text{ lb}$, you find $v = 18.03 \text{ ft/s}$ and $T_{AC} = 19.18 \text{ lb}$.

Therefore, the maximum speed is 15.29 ft/s. Combining these results gives you

$$7.47 \text{ ft/s} \leq v \leq 15.29 \text{ ft/s} \quad \blacktriangleleft$$

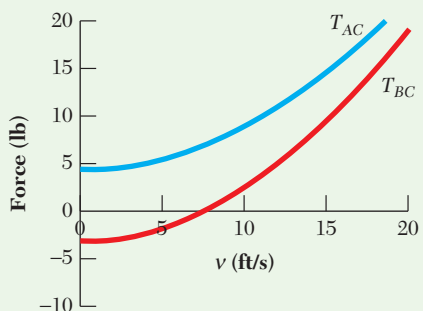
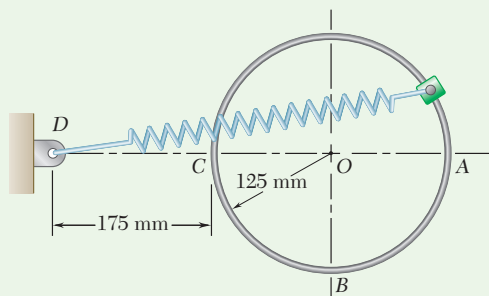


Fig. 2 Tension in cables as a function of speed.

REFLECT and THINK: In this problem, you needed to use the information in the problem statement to obtain additional equations so that you could determine the range of speeds. Another way to look at the solution is to solve Eqs. (1) and (2) for T_{AC} and T_{BC} in terms of v and to plot these as shown in Fig. 2. It is easy to see from this graph that T_{AC} determines the maximum speed and T_{BC} determines the minimum speed if both wires are to remain taut and also have tensions less than 15 lb.

Sample Problem 12.9



A 0.5-kg collar is attached to a spring and slides without friction along a circular rod in a vertical plane. The spring has an undeformed length of 150 mm and a constant $k = 200 \text{ N/m}$. Knowing that the collar has a speed of 3 m/s as it passes through point B, determine the tangential acceleration of the collar and the force of the rod on the collar at this instant.

STRATEGY: This problem deals with forces and accelerations, so you need to use Newton's second law. The collar moves along a curved path, so you should use normal and tangential coordinates.

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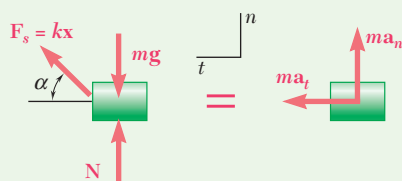


Fig. 1 Free-body diagram and kinetic diagram for the collar.

MODELING: Choose the collar as your system and assume you can treat it as a particle. Draw the free-body and kinetic diagrams as shown in Fig. 1. The spring force acts in the direction of the spring, and the force is drawn assuming that the spring is stretched and not compressed. Check this using geometry.

$$\sin \alpha = \frac{125 \text{ mm}}{300 \text{ mm}} = 0.4167 \rightarrow \alpha = 24.62^\circ$$

$$L_{BD} = \sqrt{(300 \text{ mm})^2 + (125 \text{ mm})^2} = 325 \text{ mm}$$

Thus, when the collar is at B , the spring is extended as $x = L_{BD} - L_0 = 325 \text{ mm} - 150 \text{ mm} = 175 \text{ mm}$.

ANALYSIS: You can obtain scalar equations by applying Newton's second law in the normal and tangential directions. Hence,

$$+\uparrow \Sigma F_n = ma_n \quad kx \sin \alpha + N - mg = ma_n = m \frac{v^2}{\rho} \quad (1)$$

$$\leftarrow \Sigma F_t = ma_t \quad Fx \cos \alpha = ma_t \quad (2)$$

You now have two equations (1) and (2) and two unknown a_t and N . You can solve for these by hand or using your calculator/computer. You can solve for the normal force in Eq. (1) as

$$N = mg + m \frac{v^2}{\rho} - kx \cos \alpha$$

Substituting values gives

$$N = (0.5 \text{ kg})(9.81 \text{ m/s}^2) + (0.5 \text{ kg}) \frac{(3 \text{ m/s})^2}{0.125 \text{ m}} - (200 \text{ N/m})(0.175 \text{ m}) \sin(24.62^\circ)$$

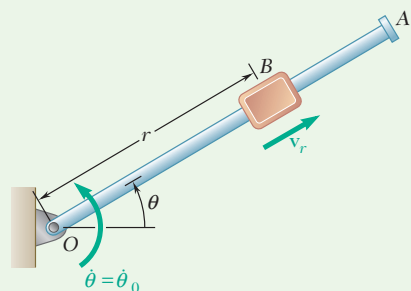
$$N = 26.3 \text{ N} \quad \blacktriangleleft$$

$$a_t = \frac{Fx \cos \alpha}{m} = \frac{(200 \text{ N/m})(0.175) \cos(24.62^\circ)}{0.5 \text{ kg}}$$

$$a_t = 63.6 \text{ m/s}^2 \quad \blacktriangleleft$$

REFLECT and THINK: How would this problem have changed if you had been told friction was acting between the rod and the collar? You would have had one additional term in your free-body diagram, $\mu_k N$, in the direction opposite to the velocity. Thus, you would need to be told the direction the collar was moving as well as the coefficient of kinetic friction.

Sample Problem 12.10



A block B with a mass m can slide freely on a frictionless arm OA that rotates in a horizontal plane at a constant rate $\dot{\theta}_0$. Knowing that B is released at a distance r_0 from O , express as a function of r : (a) the component v_r of the velocity of B along OA , (b) the magnitude of the horizontal force $\mathbf{F}$ exerted on B by the arm OA .

STRATEGY: You want to find a force, so use Newton's second law. The radial distance r of the mass is changing, as is the angular displacement θ , so use radial and transverse coordinates.

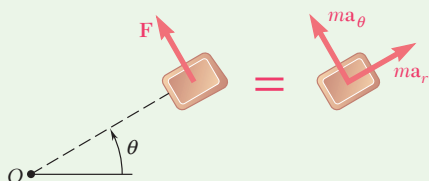


Fig. 1 Free-body diagram and kinetic diagram for the block.

MODELING: Choose block B as your system and assume you can model it as a particle. Since all other forces are perpendicular to the plane of the figure, the only force shown acting on B is the force $\mathbf{F}$ perpendicular to OA . Draw free-body and kinetic diagrams for block B as shown in Fig. 1.

ANALYSIS:

Equations of Motion. You can obtain scalar equations by applying Newton's second law in the radial and transverse directions. Hence,

$$+\nearrow \Sigma F_r = ma_r: \quad 0 = m(\ddot{r} - r\dot{\theta}^2) \quad (1)$$

$$+\searrow \Sigma F_\theta = ma_\theta: \quad F = m(r\ddot{\theta} + 2\dot{r}\dot{\theta}) \quad (2)$$

a. Component v_r of Velocity. Since $v_r = \dot{r}$, you have

$$\ddot{r} = \dot{v}_r = \frac{dv_r}{dt} = \frac{dv_r}{dr} \frac{dr}{dt} = v_r \frac{dv_r}{dr}$$

After using Eq. (1) to obtain $\ddot{r} = r\dot{\theta}^2$ and recalling that $\dot{\theta} = \dot{\theta}_0$, you can separate the variables to obtain

$$v_r dv_r = \dot{\theta}_0^2 r dr$$

Multiply by 2 and integrate from 0 to v_r and from r_0 to r . The result is

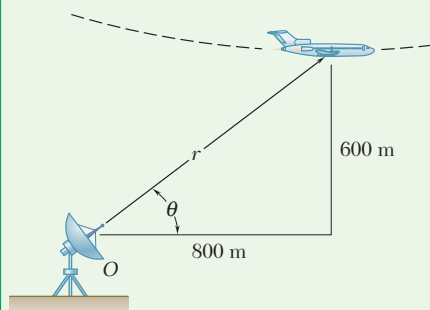
$$v_r^2 = \dot{\theta}_0^2(r^2 - r_0^2) \quad v_r = \dot{\theta}_0(r^2 - r_0^2)^{1/2} \quad \blacktriangleleft$$

b. Horizontal Force F . Set $\dot{\theta} = \dot{\theta}_0$, $\ddot{\theta} = 0$, and $\dot{r} = v_r$ in Eq. (2). Then substitute for v_r the expression obtained in part a. The result is

$$F = 2m\dot{\theta}_0(r^2 - r_0^2)^{1/2}\dot{\theta}_0 \quad F = 2m\dot{\theta}_0^2(r^2 - r_0^2)^{1/2} \quad \blacktriangleleft$$

REFLECT and THINK: Introducing radial and transverse components of force and acceleration involves using components of velocity as well in the computations. But this is still much simpler and more direct than trying to use other coordinate systems. Even though the radial acceleration is zero, the block accelerates relative to the rod with acceleration $\ddot{r}$.

Sample Problem 12.11



NASA flies a reduced-gravity aircraft (affectionately known as the Vomit Comet) in an elliptic flight to train astronauts in a microgravity environment. The plane is being tracked by radar located at O . When the plane is near the bottom of its trajectory, as shown, values from the radar tracking station are $\dot{r} = 120$ m/s, $\dot{\theta} = -0.900$ rad/s, $\ddot{r} = 34.8$ m/s², and $\ddot{\theta} = 0.0156$ rad/s². At the instant shown, determine the force exerted on the 80-kg pilot by his seat.

STRATEGY: You want to find the force the pilot experiences at this instant and you can calculate the accelerations, so you should use Newton's second law. Since you know that the radial distance and the angle are changing with time, use radial and transverse components.

(continued)

MODELING: Choosing the pilot as the system, draw the free-body and kinetic diagrams as shown in Fig. 1. You could choose to put the forces and the pilot in the r and θ direction or the x and y direction (we chose F_x and F_y to represent the forces from the seat back and bottom, respectively).

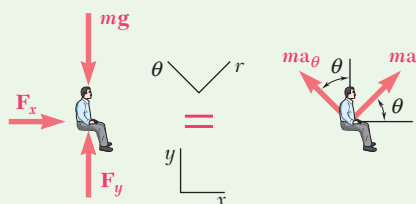


Fig. 1

ANALYSIS: Before you apply Newton's second law, determine r and θ from the geometry.

$$r = \sqrt{800^2 + 600^2} = 1000 \text{ m} \quad \theta = \tan^{-1}(600/800) = 36.87^\circ$$

Kinematics. Determine the components of the accelerations as

$$a_r = \ddot{r} - r\dot{\theta}^2 = 34.8 \text{ m/s}^2 - (100 \text{ m})(-0.090 \text{ rad/s})^2 = 26.7 \text{ m/s}^2$$

$$a_\theta = r\ddot{\theta} + 2\dot{r}\dot{\theta} = (1000 \text{ m})(0.0156 \text{ rad/s}^2) + 2(120 \text{ m/s})(-0.090 \text{ rad/s}) = -6.00 \text{ m/s}^2$$

Kinetics. Obtain scalar equations by applying Newton's second law in the horizontal and vertical directions. Thus,

$$\rightarrow \Sigma F_x = ma_x \quad F_x = ma_r \cos \theta - ma_\theta \sin \theta \quad (1)$$

$$+\uparrow \Sigma F_y = ma_y \quad F_y - mg = ma_r \sin \theta + ma_\theta \cos \theta \quad (2)$$

You have two equations, (1) and (2), and two unknowns, F_x and F_y . Substituting the known values into Eqs. (1) and (2) gives

$$F_x = (80 \text{ kg})(26.7 \text{ m/s}^2) \cos 36.87^\circ - (80 \text{ kg})(-6.00 \text{ m/s}^2) \sin 36.87^\circ$$

$$F_x = 1997 \text{ N} \rightarrow \blacktriangleleft$$

$$F_y = (80 \text{ kg})(9.81 \text{ m/s}^2) + (80 \text{ kg})(26.7 \text{ m/s}^2) \sin 36.87^\circ + (80 \text{ kg})(-6.00 \text{ m/s}^2) \cos 36.87^\circ$$

$$F_y = 1682 \text{ N} \uparrow \blacktriangleleft$$

REFLECT and THINK: These forces correspond to a forward acceleration of 2.54 g and a vertical acceleration of 2.14 g . Although this is a bit high for a passenger aircraft, it is within the flight characteristics for the Vomit Comet. If you had been asked to determine whether the plane was speeding up or slowing down, you would need to find the component of the acceleration in the tangential direction, which is defined by the direction of the velocity vector.

SOLVING PROBLEMS ON YOUR OWN

In the problems for this section, you will apply **Newton's second law of motion**, $\Sigma \mathbf{F} = m\mathbf{a}$, to relate the forces acting on a particle to its motion.

1. Writing the equations of motion. When applying Newton's second law to the types of motion discussed in this section, you will find it most convenient to express the vectors $\mathbf{F}$ and $\mathbf{a}$ in terms of either their rectangular components, their tangential and normal components, or their radial and transverse components.

a. When using rectangular components [Sample Probs. 12.1 through 12.5], recall from Sec. 11.4C the expressions found for a_x , a_y , and a_z . Then you can write

$$\Sigma F_x = m\ddot{x} \quad \Sigma F_y = m\ddot{y} \quad \Sigma F_z = m\ddot{z}$$

b. When using tangential and normal components [Sample Probs. 12.6 and 12.9], recall from Sec. 11.5A the expressions found for a_t and a_n . Then you can write

$$\Sigma F_t = m \frac{dv}{dt} \quad \Sigma F_n = m \frac{v^2}{\rho}$$

c. When using radial and transverse components [Sample Probs. 12.10 and 12.11], recall from Sec. 11.5B the expressions found for a_r and a_θ . Then you can write

$$\Sigma F_r = m(\ddot{r} - r\dot{\theta}^2) \quad \Sigma F_\theta = m(r\ddot{\theta} + 2\dot{r}\dot{\theta})$$

2. Drawing a free-body diagram and a kinetic diagram. Drawing a free-body diagram showing the applied forces and a kinetic diagram showing the vector $m\mathbf{a}$ or its components will provide you with a pictorial representation of Newton's second law [Sample Probs. 12.1 through 12.11]. These diagrams will be of great help to you when writing the equations of motion. Note that when a problem involves two or more bodies, it is usually best to consider each body separately.

3. Applying Newton's second law. As we observed in Sec. 12.1A, the acceleration used in the equation $\Sigma \mathbf{F} = m\mathbf{a}$ always should be the absolute acceleration of the particle (that is, it should be measured with respect to a newtonian frame of reference). Also, if the sense of the acceleration $\mathbf{a}$ is unknown or is not easily deduced, assume an arbitrary sense for $\mathbf{a}$ (usually the positive direction of a coordinate axis), and then let the solution provide the correct sense. Finally, note how the solutions of Sample Probs. 12.3 through 12.5 were divided into a *kinematics* portion and a *kinetics* portion, and how in Sample Probs. 12.4 and 12.5 we used two systems of coordinate axes to simplify the equations of motion.

4. When a problem involves dry friction, be sure to review the relevant section of *Statics* [Sec. 8.1] before attempting to solve that problem. In particular, you should know when to use each of the equations $F = \mu_s N$ and $F = \mu_k N$. You should also recognize that if the motion of a system is not specified, it is necessary first to assume a possible motion and then to check the validity of that assumption. For example, you can assume that the motion is impending, then check to see if the friction force is greater than $\mu_s N$ (if it is, then your assumption was wrong and the particle is moving).

5. Solving problems involving relative motion. When a body B moves with respect to a body A , as in Sample Probs. 12.4 and 12.5, it is often convenient to express the acceleration of B as

$$\mathbf{a}_B = \mathbf{a}_A + \mathbf{a}_{B/A}$$

where $\mathbf{a}_{B/A}$ is the acceleration of B relative to A , that is, the acceleration of B as observed from a frame of reference attached to A and in translation. If B is observed to move in a straight line, $\mathbf{a}_{B/A}$ is directed along that line. On the other hand, if B is observed to move along a circular path, you should resolve the relative acceleration $\mathbf{a}_{B/A}$ into components tangential and normal to that path.

6. Finally, always consider the implications of any assumption you make. Thus, in a problem involving two cords, if you assume that the tension in one of the cords is equal to its maximum allowable value, check whether any requirements set for the other cord will be satisfied. For instance, will the tension T in that cord satisfy the relation $0 \leq T \leq T_{\max}$? That is, will the cord remain taut and will its tension be less than its maximum allowable value?

Problems

CONCEPT QUESTIONS

- 12.CQ1** A 1000-lb boulder B is resting on a 200-lb platform A when truck C accelerates to the left with a constant acceleration. Which of the following statements are true (more than one may be true)?
- The tension in the cord connected to the truck is 200 lb.
 - The tension in the cord connected to the truck is 1200 lb.
 - The tension in the cord connected to the truck is greater than 1200 lb.
 - The normal force between A and B is 1000 lb.
 - The normal force between A and B is 1200 lb.
 - None of the above are true.

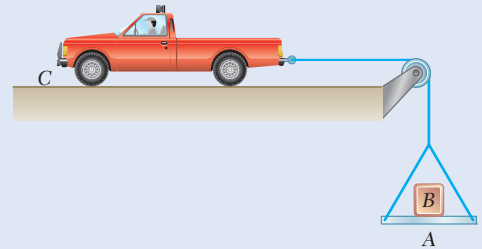


Fig. P12.CQ1

- 12.CQ2** Marble A is placed in a hollow tube, and the tube is swung in a horizontal plane causing the marble to be thrown out. As viewed from the top, which of the following choices best describes the path of the marble after leaving the tube?
- 1
 - 2
 - 3
 - 4
 - 5

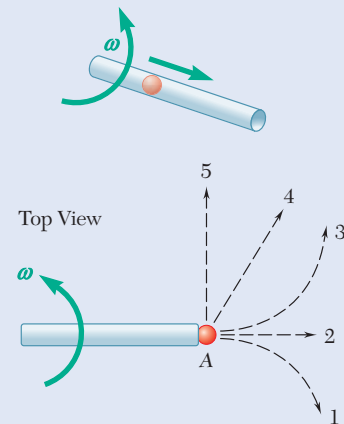


Fig. P12.CQ2

- 12.CQ3** The two systems shown start from rest. On the left, two 40-lb weights are connected by an inextensible cord, and on the right, a constant 40-lb force pulls on the cord. Neglecting all frictional forces, which of the following statements is true?
- Blocks A and C will have the same acceleration.
 - Block C will have a larger acceleration than block A .
 - Block A will have a larger acceleration than block C .
 - Block A will not move.
 - None of the above are true.

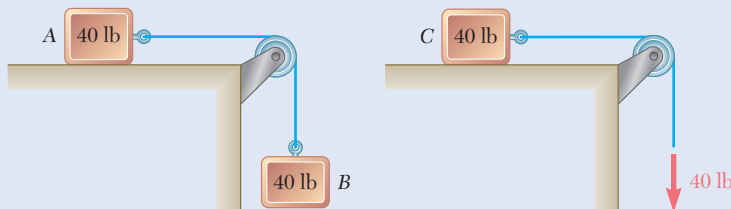


Fig. P12.CQ3

- 12.CQ4** Blocks A and B are released from rest in the position shown. Neglecting friction, the normal force between block A and the ground is:
- Less than the weight of A plus the weight of B .
 - Equal to the weight of A plus the weight of B .
 - Greater than the weight of A plus the weight of B .

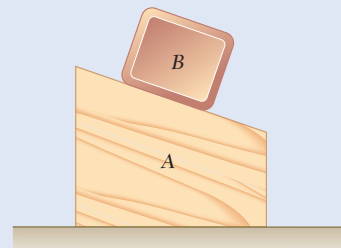


Fig. P12.CQ4

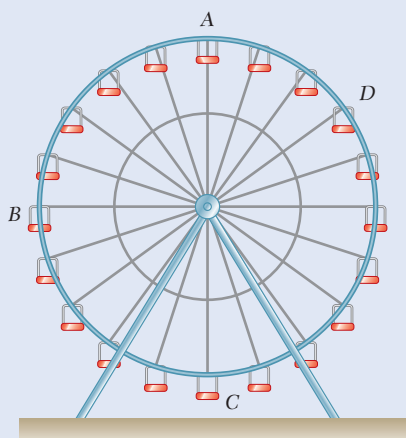


Fig. P12.CQ5

12.CQ5 People sit on a Ferris wheel at points A , B , C , and D . The Ferris wheel travels at a constant angular velocity. At the instant shown, which person experiences the largest force from his or her chair (back and seat)? Assume you can neglect the size of the chairs—that is, the people are located the same distance from the axis of rotation.

- A
- B
- C
- D
- The force is the same for all the passengers.

FREE-BODY PRACTICE PROBLEMS

12.F1 Crate A is gently placed with zero initial velocity onto a moving conveyor belt. The coefficient of kinetic friction between the crate and the belt is μ_k . Draw the FBD and KD for A immediately after it contacts the belt.

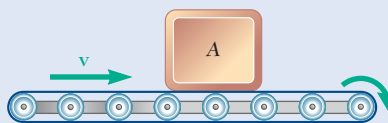


Fig. P12.F1

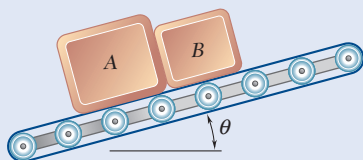


Fig. P12.F2

12.F2 Two blocks weighing W_A and W_B are at rest on a conveyor that is initially at rest. The belt is suddenly started in an upward direction so that slipping occurs between the belt and the boxes. Assuming the coefficient of friction between the boxes and the belt is μ_k , draw the FBDs and KDs for blocks A and B . How would you determine if A and B remain in contact?

12.F3 Objects A , B , and C have masses m_A , m_B , and m_C , respectively. The coefficient of kinetic friction between A and B is μ_k , and the friction between A and the ground is negligible and the pulleys are massless and frictionless. Assuming B slides on A , draw the FBD and KD for each of the three masses A , B , and C .

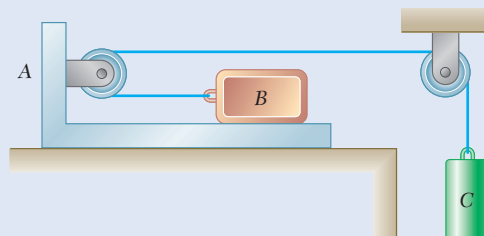


Fig. P12.F3

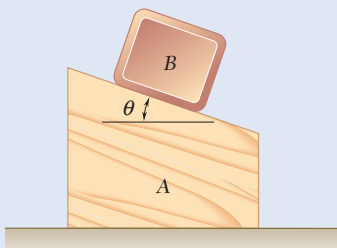


Fig. P12.F4

12.F4 Blocks A and B have masses m_A and m_B , respectively. Neglecting friction between all surfaces, draw the FBD and KD for each mass.

- 12.F5** Blocks A and B have masses m_A and m_B , respectively. Neglecting friction between all surfaces, draw the FBD and KD for the two systems shown.

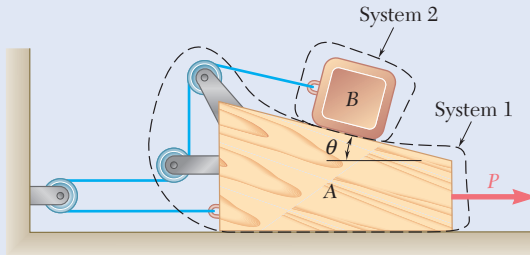


Fig. P12.F5

- 12.F6** A pilot of mass m flies a jet in a half-vertical loop of radius R so that the speed of the jet, v , remains constant. Draw a FBD and KD of the pilot at points A , B , and C .
- 12.F7** Wires AC and BC are attached to a sphere which revolves at a constant speed v in the horizontal circle of radius r as shown. Draw a FBD and KD of C .

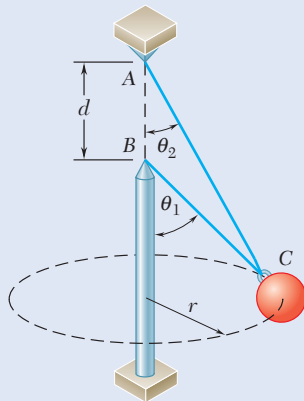


Fig. P12.F7

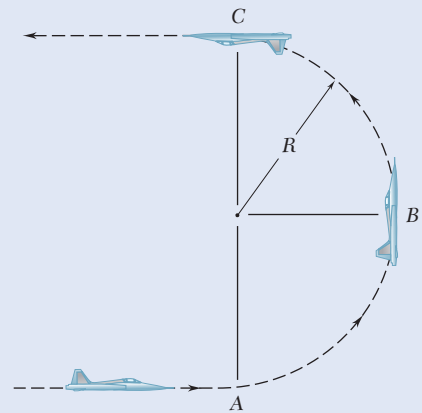


Fig. P12.F6

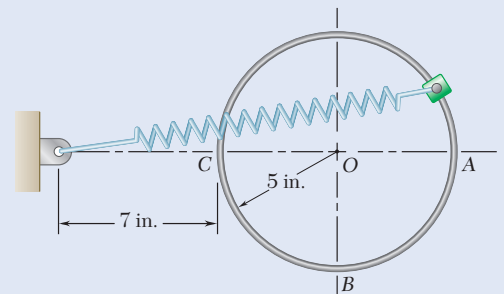


Fig. P12.F8

- 12.F8** A collar of mass m is attached to a spring and slides without friction along a circular rod in a vertical plane. The spring has an undeformed length of 5 in. and a constant k . Knowing that the collar has a speed v at point C , draw the FBD and KD of the collar at this point.
- 12.F9** Four pins slide in four separate slots cut in a horizontal circular plate as shown. When the plate is at rest, each pin has a velocity directed as shown and of the same constant magnitude u . Each pin has a mass m and maintains the same velocity relative to the plate when the plate rotates about O with a constant counterclockwise angular velocity ω . Draw the FBDs and KDs to determine the forces on pins P_1 and P_2 .

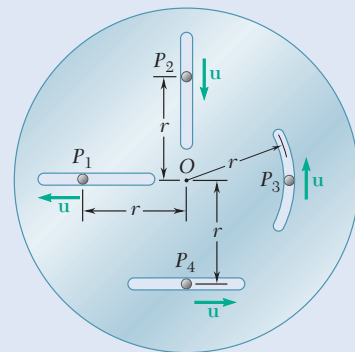


Fig. P12.F9

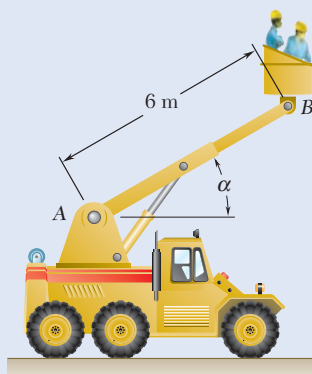


Fig. P12.F10

12.F10 At the instant shown, the length of the boom AB is being *decreased* at the constant rate of 0.2 m/s , and the boom is being lowered at the constant rate of 0.08 rad/s . If the mass of the men and lift connected to the boom at point B is m , draw the FBD and KD that could be used to determine the horizontal and vertical forces at B .

12.F11 Disk A rotates in a horizontal plane about a vertical axis at the constant rate $\dot{\theta}_0$. Slider B has a mass m and moves in a frictionless slot cut in the disk. The slider is attached to a spring of constant k , which is undeformed when $r = 0$. Knowing that the slider is released with no radial velocity in the position $r = r_0$, draw a FBD and KD at an arbitrary distance r from O .

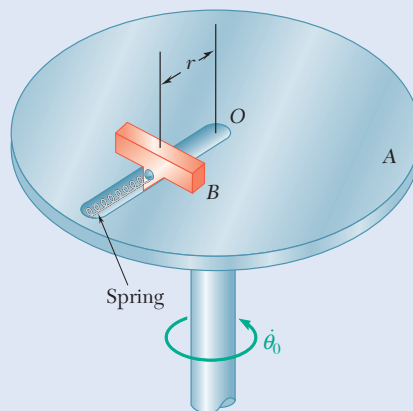


Fig. P12.F11

12.F12 Pin B has a mass m and slides along the slot in the rotating arm OC and along the slot DE which is cut in a fixed horizontal plate. Neglecting friction and knowing that rod OC rotates at the constant rate $\dot{\theta}_0$, draw a FBD and KD that can be used to determine the forces $\mathbf{P}$ and $\mathbf{Q}$ exerted on pin B by rod OC and the wall of slot DE , respectively.

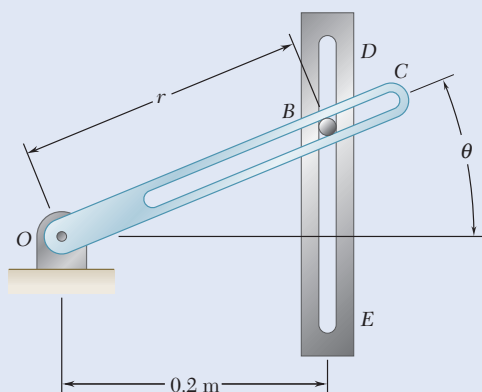


Fig. P12.F12

END-OF-SECTION PROBLEMS

12.1 Astronauts who landed on the moon during the Apollo 15, 16, and 17 missions brought back a large collection of rocks to the earth. Knowing the rocks weighed 139 lb when they were on the moon, determine (a) the weight of the rocks on the earth, (b) the mass of the rocks in slugs. The acceleration due to gravity on the moon is 5.30 ft/s^2 .

12.2 The value of g at any latitude ϕ may be obtained from the formula

$$g = 32.09(1 + 0.0053 \sin^2 \phi) \text{ ft/s}^2$$

which takes into account the effect of the rotation of the earth, as well as the fact that the earth is not truly spherical. Knowing that the weight of a silver bar has been officially designated as 5 lb , determine to four significant figures, (a) the mass in slugs, (b) the weight in pounds at the latitudes of 0° , 45° , and 60° .

12.3 A 400-kg satellite has been placed in a circular orbit 1500 km above the surface of the earth. The acceleration of gravity at this elevation is 6.43 m/s^2 . Determine the linear momentum of the satellite, knowing that its orbital speed is $25.6 \times 10^3 \text{ km/h}$.

12.4 A spring scale *A* and a lever scale *B* having equal lever arms are fastened to the roof of an elevator, and identical packages are attached to the scales as shown. Knowing that when the elevator moves downward with an acceleration of 1 m/s^2 the spring scale indicates a load of 60 N , determine (a) the weight of the packages, (b) the load indicated by the spring scale and the mass needed to balance the lever scale when the elevator moves upward with an acceleration of 1 m/s^2 .

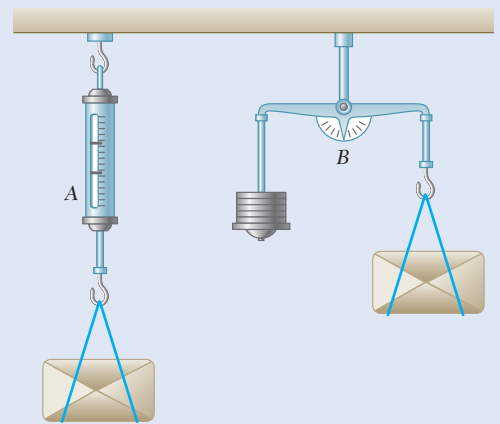


Fig. P12.4

12.5 In anticipation of a long 7° upgrade, a bus driver accelerates at a constant rate of 3 ft/s^2 while still on a level section of the highway. Knowing that the speed of the bus is 60 mi/h as it begins to climb the grade and that the driver does not change the setting of his throttle or shift gears, determine the distance traveled by the bus up the grade when its speed has decreased to 50 mi/h .

12.6 A 0.2-lb model rocket is launched vertically from rest at time $t = 0$ with a constant thrust of 2 lb for one second and no thrust for $t > 1 \text{ s}$. Neglecting air resistance and the decrease in mass of the rocket, determine (a) the maximum height h reached by the rocket, (b) the time required to reach this maximum height.

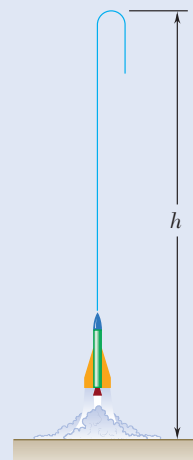


Fig. P12.6

12.7 A tugboat pulls a small barge through a harbor. The propeller thrust minus the drag produces a net thrust that varies linearly with speed. Knowing that the combined weight of the tug and barge is 3600 kN , determine (a) the time required to increase the speed from an initial value $v_1 = 1.0 \text{ m/s}$ to a final value $v_2 = 2.5 \text{ m/s}$, (b) the distance traveled during this time interval.

12.8 Determine the maximum theoretical speed that may be achieved over a distance of 60 m by a car starting from rest, knowing that the coefficient of static friction is 0.80 between the tires and the pavement and that 60 percent of the weight of the car is distributed over its front wheels and 40 percent over its rear wheels. Assume (a) four-wheel drive, (b) front-wheel drive, (c) rear-wheel drive.

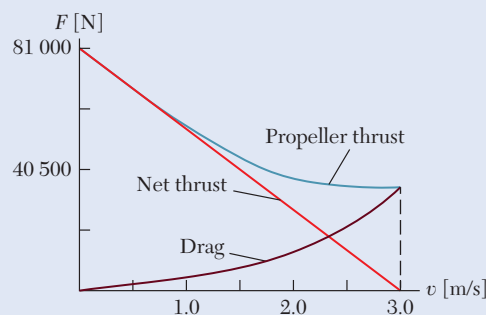
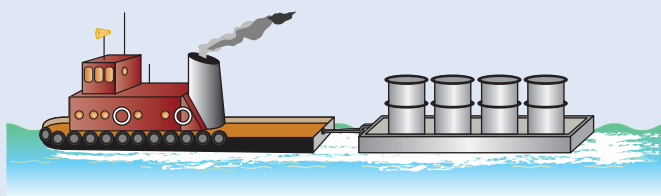


Fig. P12.7

12.9 If an automobile's braking distance from 90 km/h is 45 m on level pavement, determine the automobile's braking distance from 90 km/h when it is (a) going up a 5° incline, (b) going down a 3-percent incline. Assume the braking force is independent of grade.

12.10 A mother and her child are skiing together, and the mother is holding the end of a rope tied to the child's waist. They are moving at a speed of 7.2 km/h on a gently sloping portion of the ski slope when the mother observes that they are approaching a steep descent. She pulls on the rope with an average force of 7 N. Knowing the coefficient of friction between the child and the ground is 0.1 and the angle of the rope does not change, determine (a) the time required for the child's speed to be cut in half, (b) the distance traveled in this time.

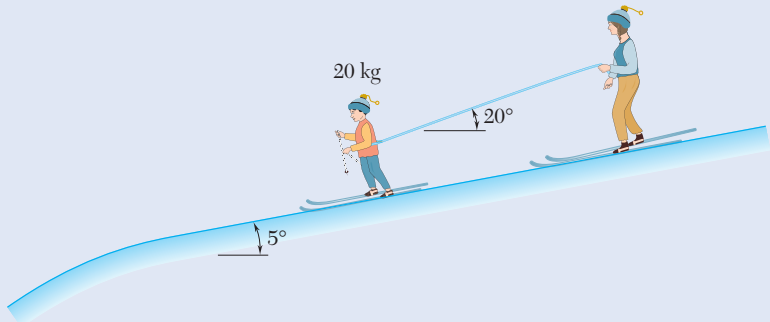


Fig. P12.10

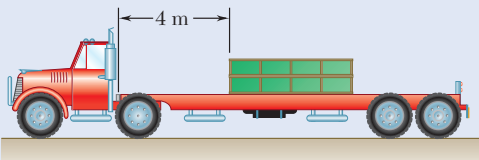


Fig. P12.11

12.11 The coefficients of friction between the load and the flatbed trailer shown are $\mu_s = 0.40$ and $\mu_k = 0.30$. Knowing that the speed of the rig is 72 km/h, determine the shortest distance in which the rig can be brought to a stop if the load is not to shift.

12.12 A light train made up of two cars is traveling at 90 km/h when the brakes are applied to both cars. Knowing that car A has a mass of 25 Mg and car B a mass of 20 Mg, and that the braking force is 30 kN on each car, determine (a) the distance traveled by the train before it comes to a stop, (b) the force in the coupling between the cars while the train is slowing down.

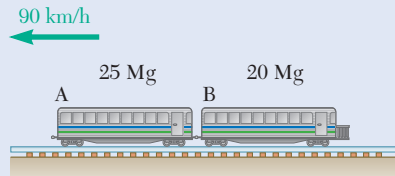


Fig. P12.12

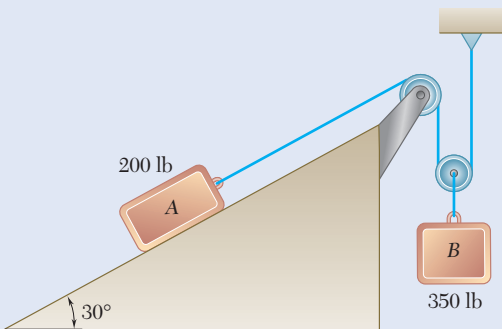


Fig. P12.13

12.13 The two blocks shown are originally at rest. Neglecting the masses of the pulleys and the effect of friction in the pulleys and between block A and the incline, determine (a) the acceleration of each block, (b) the tension in the cable.

12.14 Solve Prob. 12.13, assuming that the coefficients of friction between block A and the incline are $\mu_s = 0.25$ and $\mu_k = 0.20$.

- 12.15** Each of the systems shown is initially at rest. Neglecting axle friction and the masses of the pulleys, determine for each system (a) the acceleration of block A, (b) the velocity of block A after it has moved through 10 ft, (c) the time required for block A to reach a velocity of 20 ft/s.

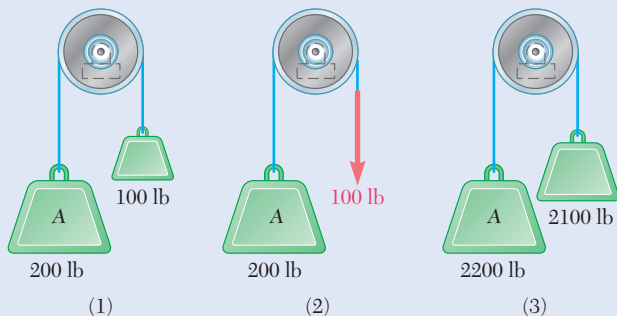


Fig. P12.15

- 12.16** Boxes A and B are at rest on a conveyor belt that is initially at rest. The belt is suddenly started in an upward direction so that slipping occurs between the belt and the boxes. Knowing that the coefficients of kinetic friction between the belt and the boxes are $(\mu_k)_A = 0.30$ and $(\mu_k)_B = 0.32$, determine the initial acceleration of each box.

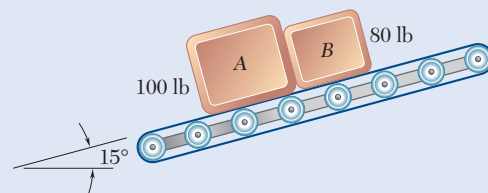


Fig. P12.16

- 12.17** A 5000-lb truck is being used to lift a 1000-lb boulder B that is on a 200-lb pallet A. Knowing the acceleration of the truck is 1 ft/s^2 , determine (a) the horizontal force between the tires and the ground, (b) the force between the boulder and the pallet.

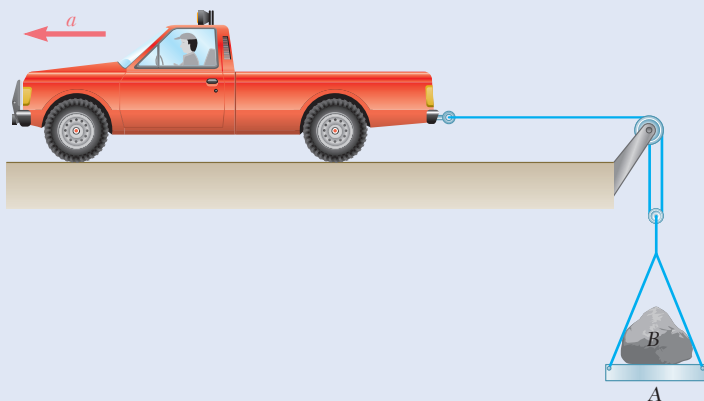


Fig. P12.17

- 12.18** Block A has a mass of 40 kg, and block B has a mass of 8 kg. The coefficients of friction between all surfaces of contact are $\mu_s = 0.20$ and $\mu_k = 0.15$. If $P = 0$, determine (a) the acceleration of block B, (b) the tension in the cord.
- 12.19** Block A has a mass of 40 kg, and block B has a mass of 8 kg. The coefficients of friction between all surfaces of contact are $\mu_s = 0.20$ and $\mu_k = 0.15$. If $P = 40 \text{ N}$, determine (a) the acceleration of block B, (b) the tension in the cord.

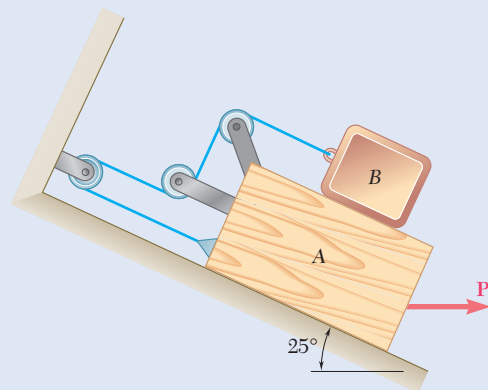


Fig. P12.18 and P12.19

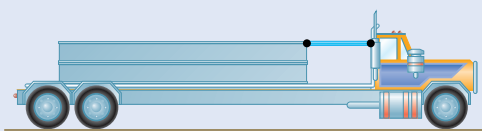


Fig. P12.20

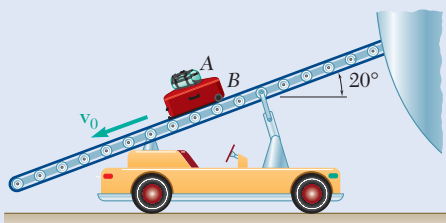


Fig. P12.21

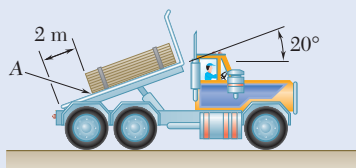


Fig. P12.22

12.20 The flat-bed trailer carries two 1500-kg beams with the upper beam secured by a cable. The coefficients of static friction between the two beams and between the lower beam and the bed of the trailer are 0.25 and 0.30, respectively. Knowing that the load does not shift, determine (a) the maximum acceleration of the trailer and the corresponding tension in the cable, (b) the maximum deceleration of the trailer.

12.21 A baggage conveyor is used to unload luggage from an airplane. The 10-kg duffel bag *A* is sitting on top of the 20-kg suitcase *B*. The conveyor is moving the bags down at a constant speed of 0.5 m/s when the belt suddenly stops. Knowing that the coefficient of friction between the belt and *B* is 0.3 and that bag *A* does not slip on suitcase *B*, determine the smallest allowable coefficient of static friction between the bags.

12.22 To unload a bound stack of plywood from a truck, the driver first tilts the bed of the truck and then accelerates from rest. Knowing that the coefficients of friction between the bottom sheet of plywood and the bed are $\mu_s = 0.40$ and $\mu_k = 0.30$, determine (a) the smallest acceleration of the truck which will cause the stack of plywood to slide, (b) the acceleration of the truck which causes corner *A* of the stack to reach the end of the bed in 0.9 s.

12.23 To transport a series of bundles of shingles *A* to a roof, a contractor uses a motor-driven lift consisting of a horizontal platform *BC* which rides on rails attached to the sides of a ladder. The lift starts from rest and initially moves with a constant acceleration $\mathbf{a}_1$ as shown. The lift then decelerates at a constant rate $\mathbf{a}_2$ and comes to rest at *D*, near the top of the ladder. Knowing that the coefficient of static friction between a bundle of shingles and the horizontal platform is 0.30, determine the largest allowable acceleration $\mathbf{a}_1$ and the largest allowable deceleration $\mathbf{a}_2$ if the bundle is not to slide on the platform.

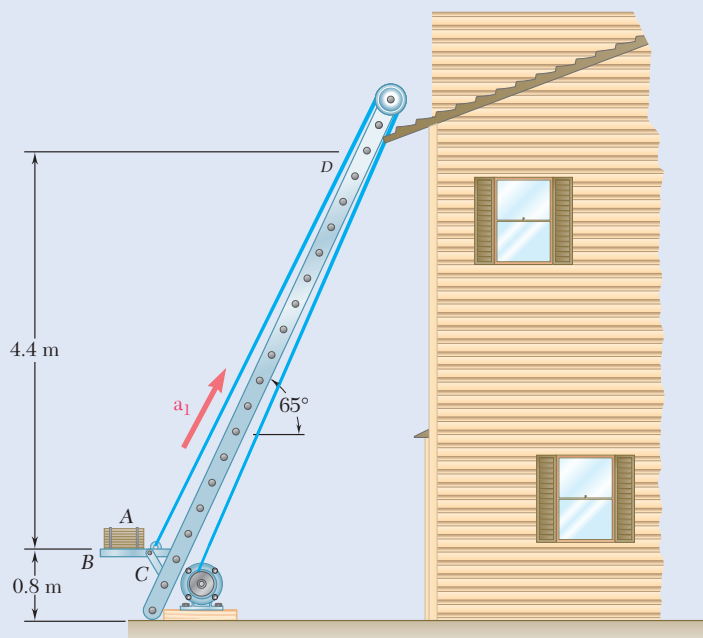


Fig. P12.23

- 12.24** An airplane has a mass of 25 Mg and its engines develop a total thrust of 40 kN during take-off. If the drag **D** exerted on the plane has a magnitude $D = 2.25 v^2$, where v is expressed in meters per second and D in newtons, and if the plane becomes airborne at a speed of 240 km/h, determine the length of runway required for the plane to take off.
- 12.25** A 4-kg projectile is fired vertically with an initial velocity of 90 m/s, reaches a maximum height, and falls to the ground. The aerodynamic drag **D** has a magnitude $D = 0.0024 v^2$ where D and v are expressed in newtons and m/s, respectively. Knowing that the direction of the drag is always opposite to the direction of the velocity, determine (a) the maximum height of the trajectory, (b) the speed of the projectile when it reaches the ground.
- 12.26** A constant force **P** is applied to a piston and rod of total mass m to make them move in a cylinder filled with oil. As the piston moves, the oil is forced through orifices in the piston and exerts on the piston a force of magnitude kv in a direction opposite to the motion of the piston. Knowing that the piston starts from rest at $t = 0$ and $x = 0$, show that the equation relating x , v , and t , where x is the distance traveled by the piston and v is the speed of the piston, is linear in each of these variables.



Fig. P12.26

- 12.27** A spring AB of constant k is attached to a support at A and to a collar of mass m . The unstretched length of the spring is l . Knowing that the collar is released from rest at $x = x_0$ and neglecting friction between the collar and the horizontal rod, determine the magnitude of the velocity of the collar as it passes through point C .
- 12.28** Block A has a mass of 10 kg, and blocks B and C have masses of 5 kg each. Knowing that the blocks are initially at rest and that B moves through 3 m in 2 s, determine (a) the magnitude of the force **P**, (b) the tension in the cord AD . Neglect the masses of the pulleys and axle friction.

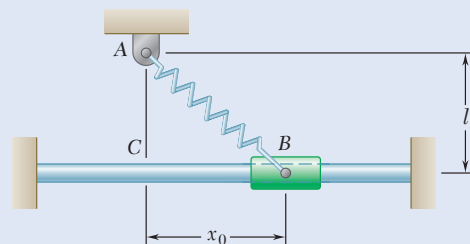


Fig. P12.27

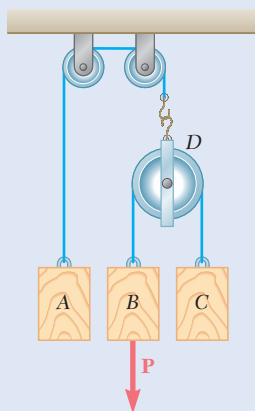


Fig. P12.28

- 12.29** A 40-lb sliding panel is supported by rollers at B and C . A 25-lb counterweight A is attached to a cable as shown and, in cases a and c , is initially in contact with a vertical edge of the panel. Neglecting friction, determine in each case shown the acceleration of the panel and the tension in the cord immediately after the system is released from rest.

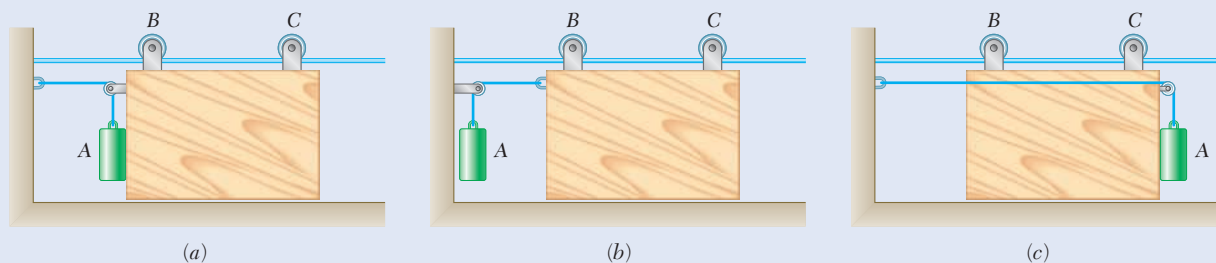


Fig. P12.29

- 12.30** An athlete pulls handle A to the left with a constant force of $P = 100$ N. Knowing that after the handle A has been pulled 30 cm and its velocity is 3 m/s, determine the mass of the weight stack B .

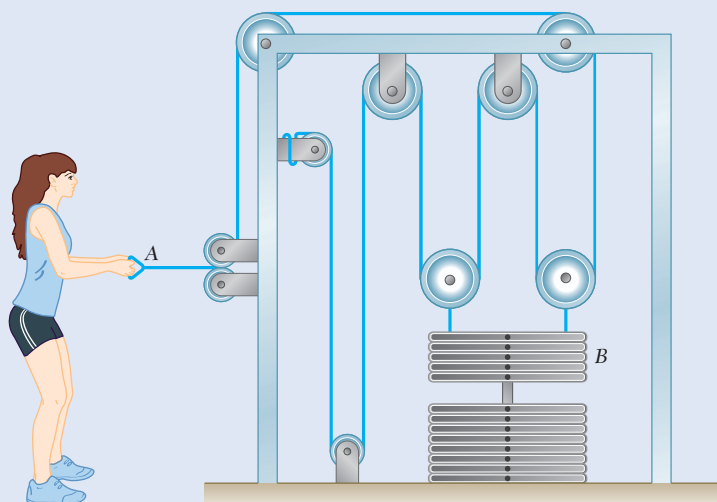


Fig. P12.30

- 12.31** A 10-lb block B rests as shown on a 20-lb bracket A . The coefficients of friction are $\mu_s = 0.30$ and $\mu_k = 0.25$ between block B and bracket A , and there is no friction in the pulley or between the bracket and the horizontal surface. (a) Determine the maximum weight of block C if block B is not to slide on bracket A . (b) If the weight of block C is 10 percent larger than the answer found in a , determine the accelerations of A , B , and C .

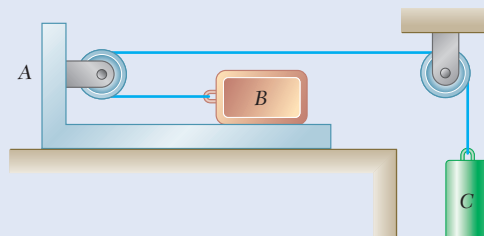


Fig. P12.31

- 12.32** Knowing that $\mu_k = 0.30$, determine the acceleration of each block when $m_A = m_B = m_C$.
- 12.33** Knowing that $\mu_k = 0.30$, determine the acceleration of each block when $m_A = 5$ kg, $m_B = 30$ kg, and $m_C = 15$ kg.
- 12.34** A 25-kg block A rests on an inclined surface, and a 15-kg counterweight B is attached to a cable as shown. Neglecting friction, determine the acceleration of A and the tension in the cable immediately after the system is released from rest.

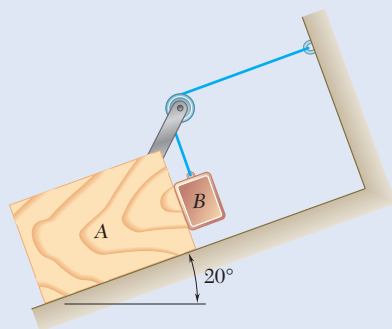


Fig. P12.34

- 12.35** Block B of mass 10 kg rests as shown on the upper surface of a 22-kg wedge A . Knowing that the system is released from rest and neglecting friction, determine (a) the acceleration of B , (b) the velocity of B relative to A at $t = 0.5$ s.
- 12.36** A 450-g tetherball A is moving along a horizontal circular path at a constant speed of 4 m/s. Determine (a) the angle θ that the cord forms with pole BC , (b) the tension in the cord.
- 12.37** During a hammer thrower's practice swings, the 7.1-kg head A of the hammer revolves at a constant speed v in a horizontal circle as shown. If $\rho = 0.93$ m and $\theta = 60^\circ$, determine (a) the tension in wire BC , (b) the speed of the hammer's head.

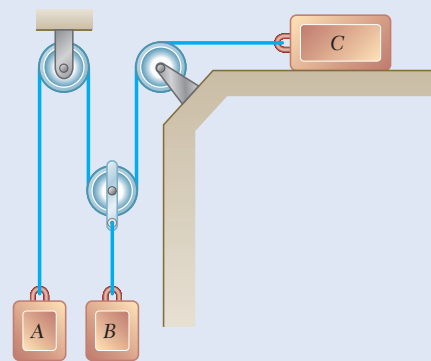


Fig. P12.32 and P12.33

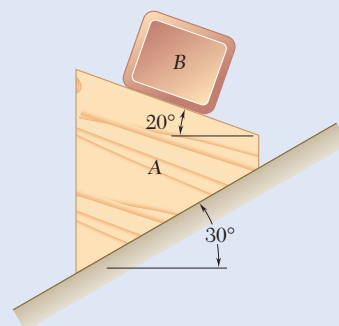


Fig. P12.35

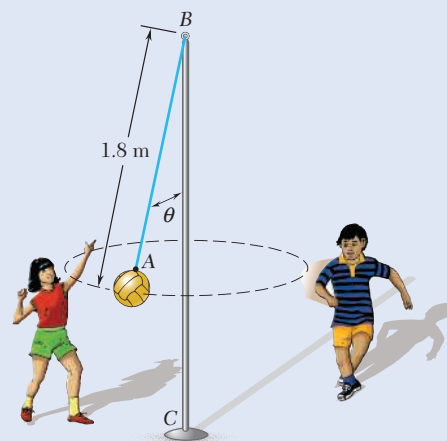


Fig. P12.36

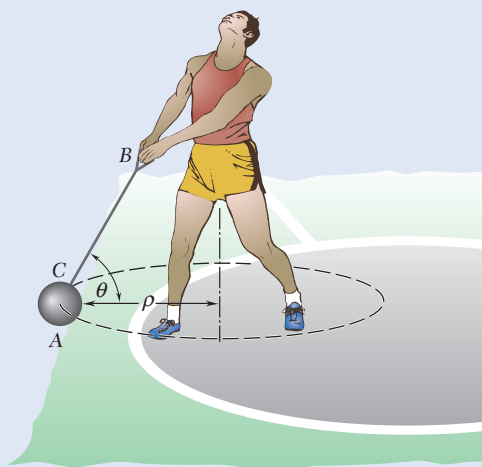


Fig. P12.37

- 12.38** Human centrifuges are often used to simulate different acceleration levels for pilots. When aerospace physiologists say that a pilot is pulling $9\ g$'s, they mean that the resultant normal force on the pilot from the bottom of the seat is nine times their weight. Knowing that the centrifuge starts from rest and has a constant angular acceleration of $1.5\ \text{RPM per second}$ until the pilot is pulling $9\ g$'s and then continues with a constant angular velocity, determine (a) how long it will take for the pilot to reach $9\ g$'s (b) the angle θ of the normal force once the pilot reaches $9\ g$'s. Assume that the force parallel to the seat is zero.

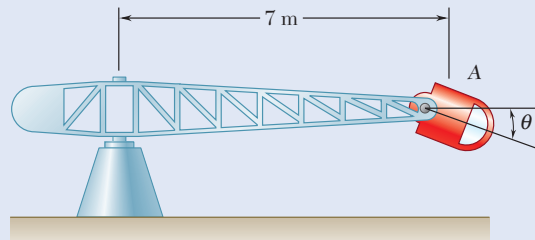


Fig. P12.38

- 12.39** A single wire ACB passes through a ring at C attached to a sphere that revolves at a constant speed v in the horizontal circle shown. Knowing that the tension is the same in both portions of the wire, determine the speed v .

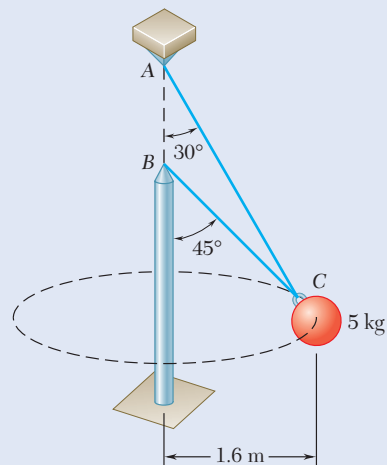


Fig. P12.39 and P12.40

- *12.40** Two wires AC and BC are tied at C to a sphere that revolves at a constant speed v in the horizontal circle shown. Determine the range of the allowable values of v if both wires are to remain taut and if the tension in either of the wires is not to exceed $60\ \text{N}$.

- 12.41** A 1-kg sphere is at rest relative to a parabolic dish that rotates at a constant rate about a vertical axis. Neglecting friction and knowing that $r = 1\ \text{m}$, determine (a) the speed v of the sphere, (b) the magnitude of the normal force exerted by the sphere on the inclined surface of the dish.

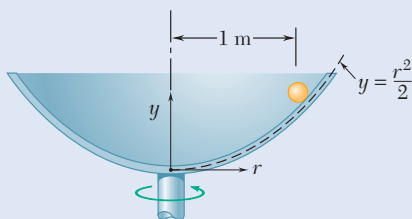


Fig. P12.41

***12.42** As part of an outdoor display, a 12-lb model C of the earth is attached to wires AC and BC and revolves at a constant speed v in the horizontal circle shown. Determine the range of the allowable values of v if both wires are to remain taut and if the tension in either of the wires is not to exceed 26 lb.

***12.43** The 1.2-lb flyballs of a centrifugal governor revolve at a constant speed v in the horizontal circle of 6-in. radius shown. Neglecting the weights of links AB , BC , AD , and DE and requiring that the links support only tensile forces, determine the range of the allowable values of v so that the magnitudes of the forces in the links do not exceed 17 lb.

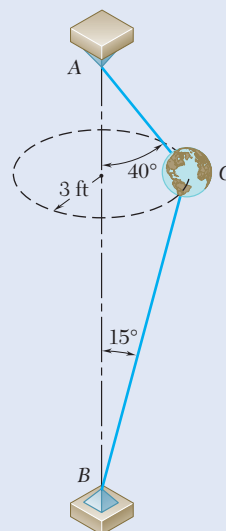


Fig. P12.42

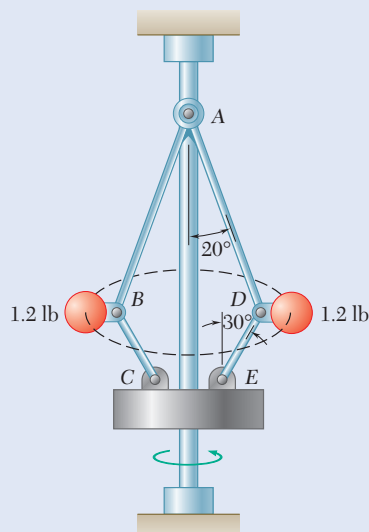


Fig. P12.43

12.44 A 130-lb wrecking ball B is attached to a 45-ft-long steel cable AB and swings in the vertical arc shown. Determine the tension in the cable (a) at the top C of the swing, (b) at the bottom D of the swing, where the speed of B is 13.2 ft/s.

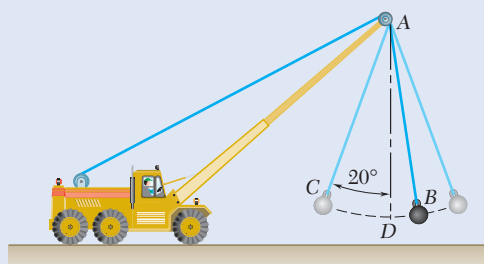


Fig. P12.44

12.45 During a high-speed chase, a 2400-lb sports car traveling at a speed of 100 mi/h just loses contact with the road as it reaches the crest A of a hill. (a) Determine the radius of curvature ρ of the vertical profile of the road at A . (b) Using the value of ρ found in part a , determine the force exerted on a 160-lb driver by the seat of his 3100-lb car as the car, traveling at a constant speed of 50 mi/h, passes through A .

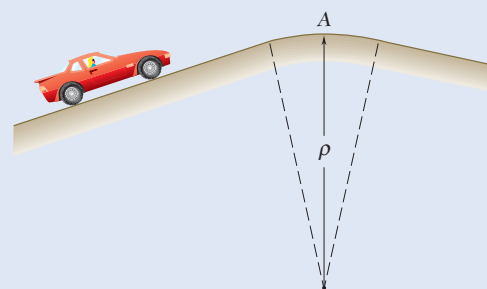


Fig. P12.45

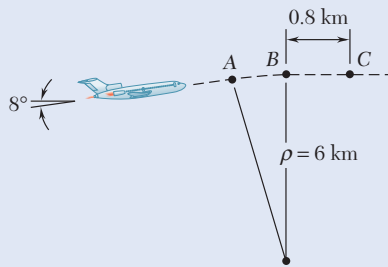


Fig. P12.46

12.46 An airline pilot climbs to a new flight level along the path shown. Knowing that the speed of the airplane decreases at a constant rate from 180 m/s at point A to 160 m/s at point C, determine the magnitude of the abrupt change in the force exerted on a 90-kg passenger as the airplane passes point B.

12.47 The roller-coaster track shown is contained in a vertical plane. The portion of track between A and B is straight and horizontal, while the portions to the left of A and to the right of B have radii of curvature as indicated. A car is traveling at a speed of 72 km/h when the brakes are suddenly applied, causing the wheels of the car to slide on the track ($\mu_k = 0.20$). Determine the initial deceleration of the car if the brakes are applied as the car (a) has almost reached A, (b) is traveling between A and B, (c) has just passed B.

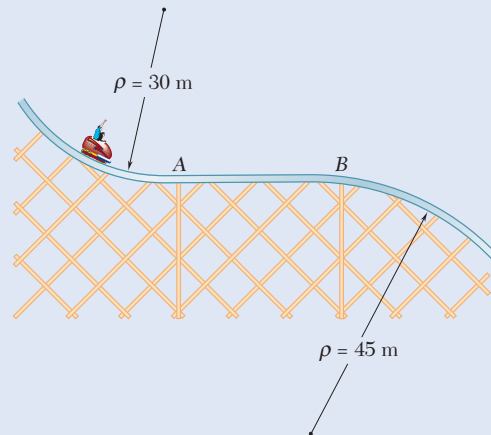


Fig. P12.47

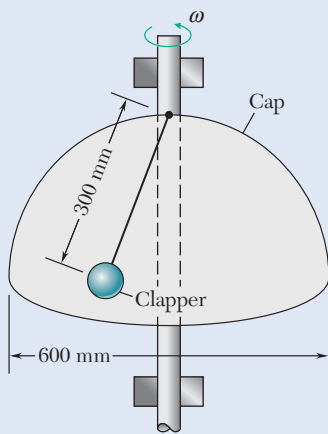


Fig. P12.48

12.48 A spherical-cap governor is fixed to a vertical shaft that rotates with angular velocity ω . When the string-supported clapper of mass m touches the cap, a cutoff switch is operated electrically to reduce the speed of the shaft. Knowing that the radius of the clapper is small relative to the cap, determine the minimum angular speed at which the cutoff switch operates.

12.49 A series of small packages, each with a mass of 0.5 kg, are discharged from a conveyor belt as shown. Knowing that the coefficient of static friction between each package and the conveyor belt is 0.4, determine (a) the force exerted by the belt on the package just after it has passed point A, (b) the angle θ defining the point B where the packages first slip relative to the belt.

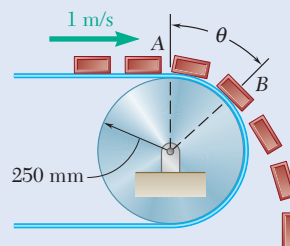


Fig. P12.49

- 12.50** A 54-kg pilot flies a jet trainer in a half-vertical loop of 1200-m radius so that the speed of the trainer decreases at a constant rate. Knowing that the pilot's apparent weights at points *A* and *C* are 1680 N and 350 N, respectively, determine the force exerted on her by the seat of the trainer when the trainer is at point *B*.

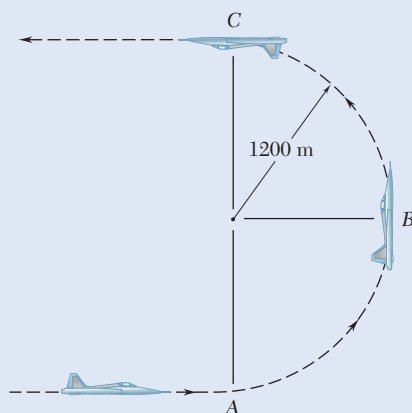


Fig. P12.50

- 12.51** A carnival ride is designed to allow the general public to experience high-acceleration motion. The ride rotates about point *O* in a horizontal circle such that the rider has a speed v_0 . The rider reclines on a platform *A* which rides on rollers such that friction is negligible. A mechanical stop prevents the platform from rolling down the incline. Determine (a) the speed v_0 at which the platform *A* begins to roll upward, (b) the normal force experienced by an 80-kg rider at this speed.

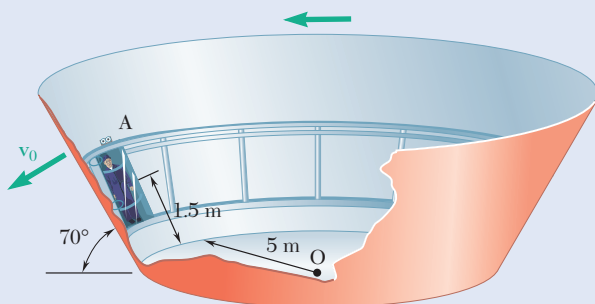


Fig. P12.51

- 12.52** A curve in a speed track has a radius of 1000 ft and a rated speed of 120 mi/h. (See Sample Prob. 12.7 for the definition of rated speed.) Knowing that a racing car starts skidding on the curve when traveling at a speed of 180 mi/h, determine (a) the banking angle θ , (b) the coefficient of static friction between the tires and the track under the prevailing conditions, (c) the minimum speed at which the same car could negotiate the curve.

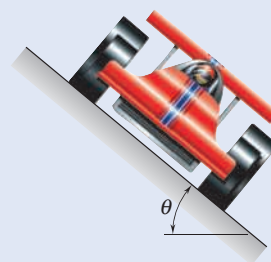


Fig. P12.52

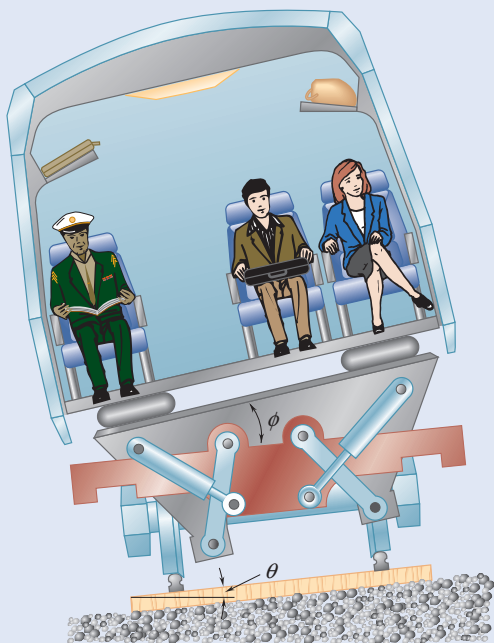


Fig. P12.53 and P12.54

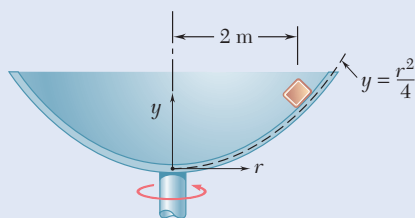


Fig. P12.55

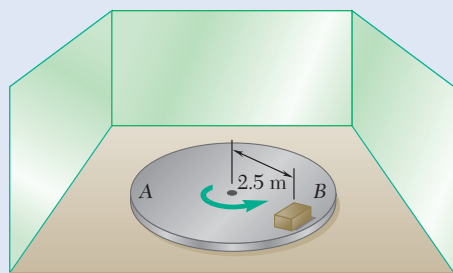


Fig. P12.57

12.53 Tilting trains, such as the *American Flyer* which will run from Washington to New York and Boston, are designed to travel safely at high speeds on curved sections of track which were built for slower, conventional trains. As it enters a curve, each car is tilted by hydraulic actuators mounted on its trucks. The tilting feature of the cars also increases passenger comfort by eliminating or greatly reducing the side force $\mathbf{F}_s$ (parallel to the floor of the car) to which passengers feel subjected. For a train traveling at 100 mi/h on a curved section of track banked through an angle $\theta = 6^\circ$ and with a rated speed of 60 mi/h, determine (a) the magnitude of the side force felt by a passenger of weight W in a standard car with no tilt ($\phi = 0$), (b) the required angle of tilt ϕ if the passenger is to feel no side force. (See Sample Prob. 12.7 for the definition of rated speed.)

12.54 Tests carried out with the tilting trains described in Prob. 12.53 revealed that passengers feel queasy when they see through the car windows that the train is rounding a curve at high speed, yet do not feel any side force. Designers, therefore, prefer to reduce, but not eliminate that force. For the train of Prob. 12.53, determine the required angle of tilt ϕ if passengers are to feel side forces equal to 10 percent of their weights.

12.55 A 3-kg block is at rest relative to a parabolic dish which rotates at a constant rate about a vertical axis. Knowing that the coefficient of static friction is 0.5 and that $r = 2$ m, determine the maximum allowable velocity v of the block.

12.56 A polisher is started so that the fleece along the circumference undergoes a constant tangential acceleration of 4 m/s^2 . Three seconds after it is started, small tufts of fleece from along the circumference of the 225-mm-diameter polishing pad are observed to fly free of the pad. At this instant, determine (a) the speed v of a tuft as it leaves the pad, (b) the magnitude of the force required to free a tuft if the average mass of a tuft is 1.6 mg.

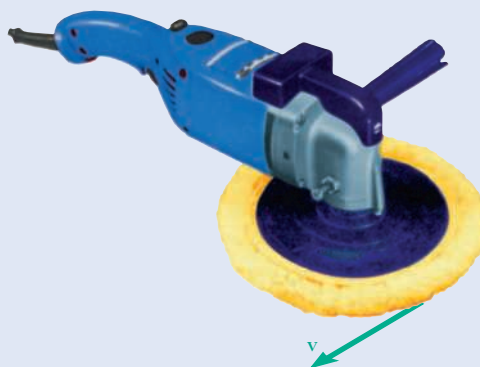


Fig. P12.56

12.57 A turntable A is built into a stage for use in a theatrical production. It is observed during a rehearsal that a trunk B starts to slide on the turntable 10 s after the turntable begins to rotate. Knowing that the trunk undergoes a constant tangential acceleration of 0.24 m/s^2 , determine the coefficient of static friction between the trunk and the turntable.

- 12.58** The carnival ride from Prob. 12.51 is modified so that the 80-kg riders can move up and down the inclined wall as the speed of the ride increases. Assuming that the friction between the wall and the carriage is negligible, determine the position h of the rider if the speed $v_0 = 13$ m/s.

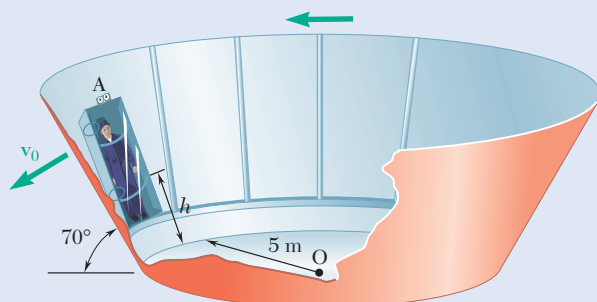


Fig. P12.58 and P12.59

- 12.59** The carnival ride from Prob 12.51 is modified so that the 80-kg riders can move up and down the inclined wall as the speed of the ride increases. Knowing that the coefficient of static friction between the wall and the platform is 0.2, determine the range of values of the constant speed v_0 for which the platform will remain at $h = 1.5$ m.

- 12.60** A semicircular slot of 10-in. radius is cut in a flat plate that rotates about the vertical AD at a constant rate of 14 rad/s. A small, 0.8-lb block E is designed to slide in the slot as the plate rotates. Knowing that the coefficients of friction are $\mu_s = 0.35$ and $\mu_k = 0.25$, determine whether the block will slide in the slot if it is released in the position corresponding to (a) $\theta = 80^\circ$, (b) $\theta = 40^\circ$. Also determine the magnitude and the direction of the friction force exerted on the block immediately after it is released.

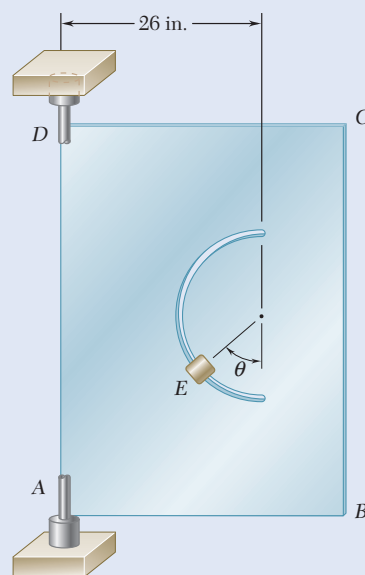


Fig. P12.60

- 12.61** A small block B fits inside a slot cut in arm OA that rotates in a vertical plane at a constant rate. The block remains in contact with the end of the slot closest to A and its speed is 1.4 m/s for $0 \leq \theta \leq 150^\circ$. Knowing that the block begins to slide when $\theta = 150^\circ$, determine the coefficient of static friction between the block and the slot.

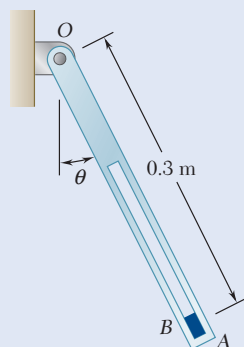


Fig. P12.61

- 12.62** The parallel-link mechanism $ABCD$ is used to transport a component I between manufacturing processes at stations E , F , and G by picking it up at a station when $\theta = 0$ and depositing it at the next station when $\theta = 180^\circ$. Knowing that member BC remains horizontal throughout its motion and that links AB and CD rotate at a constant rate in a vertical plane in such a way that $v_B = 2.2$ ft/s, determine (a) the minimum value of the coefficient of static friction between the component and BC if the component is not to slide on BC while being transferred, (b) the values of θ for which sliding is impending.

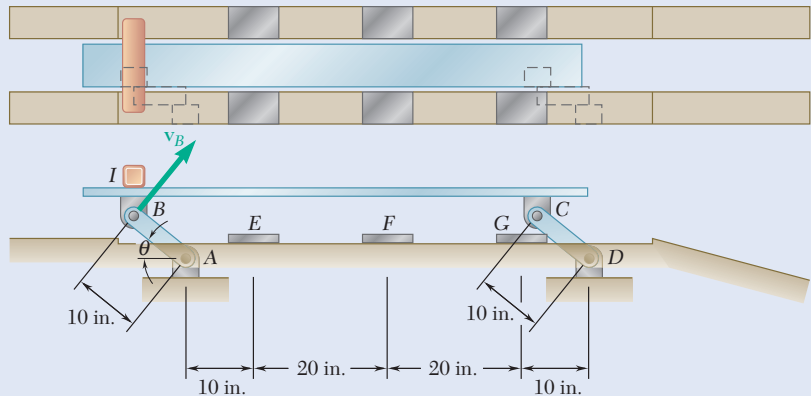


Fig. P12.62

- 12.63** Knowing that the coefficients of friction between the component I and member BC of the mechanism of Prob. 12.62 are $\mu_s = 0.35$ and $\mu_k = 0.25$, determine (a) the maximum allowable constant speed v_B if the component is not to slide on BC while being transferred, (b) the values of θ for which sliding is impending.

- 12.64** A small 250-g collar C can slide on a semicircular rod which is made to rotate about the vertical AB at a constant rate of 7.5 rad/s. Determine the three values of θ for which the collar will not slide on the rod, assuming no friction between the collar and the rod.

- 12.65** A small 250-g collar C can slide on a semicircular rod which is made to rotate about the vertical AB at a constant rate of 7.5 rad/s. Knowing that the coefficients of friction are $\mu_s = 0.25$ and $\mu_k = 0.20$, indicate whether the collar will slide on the rod if it is released in the position corresponding to (a) $\theta = 75^\circ$, (b) $\theta = 40^\circ$. Also, determine the magnitude and direction of the friction force exerted on the collar immediately after release.

- 12.66** An advanced spatial disorientation trainer allows the cab to rotate around multiple axes as well as to extend inwards and outwards. It can be used to simulate driving, fixed-wing aircraft flying, and helicopter maneuvering. In one training scenario, the trainer rotates and translates in the horizontal plane where the location of the pilot is defined by the relationships $r = 10 + 2 \cos\left(\frac{\pi}{3}t\right)$ and $\theta = 0.1(2t^2 - t)$, where r , θ , and t are expressed in feet, radians, and seconds, respectively. Knowing that the pilot has a weight of 175 lbs, (a) determine the magnitude of the resulting force acting on the pilot at $t = 5$ s, (b) plot the magnitudes of the radial and transverse components of the force exerted on the pilot from 0 to 10 seconds.

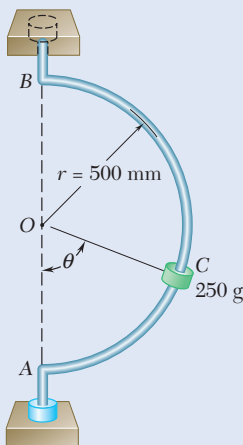


Fig. P12.64 and P12.65

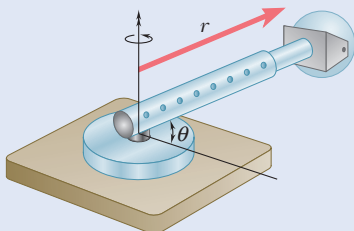


Fig. P12.66 and P12.67

- 12.67** An advanced spatial disorientation trainer is programmed to only rotate and translate in the horizontal plane. The pilot's location is defined by the relationships $r = 8(1 - e^{-t})$ and $\theta = 2/\pi(\sin \frac{\pi}{2}t)$, where r , θ , and t are expressed in feet, radians, and seconds, respectively. Determine the radial and transverse components of the force exerted on the 175-lb pilot at $t = 3$ s.

- 12.68** The 3-kg collar B slides on the frictionless arm AA' . The arm is attached to drum D and rotates about O in a horizontal plane at the rate $\dot{\theta} = 0.75t$, where $\dot{\theta}$ and t are expressed in rad/s and seconds, respectively. As the arm-drum assembly rotates, a mechanism within the drum releases cord so that the collar moves outward from O with a constant speed of 0.5 m/s. Knowing that at $t = 0$, $r = 0$, determine the time at which the tension in the cord is equal to the magnitude of the horizontal force exerted on B by arm AA' .

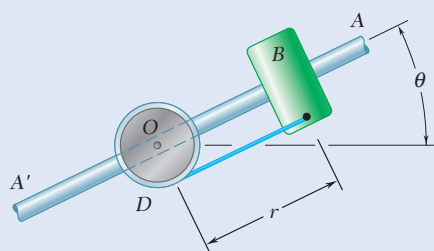


Fig. P12.68

- 12.69** A 0.5-kg block B slides without friction inside a slot cut in arm OA that rotates in a vertical plane. The rod has a constant angular acceleration $\ddot{\theta} = 10 \text{ rad/s}^2$. Knowing that when $\theta = 45^\circ$ and $r = 0.8$ m the velocity of the block is zero, determine at this instant, (a) the force exerted on the block by the arm, (b) the relative acceleration of the block with respect to the arm.

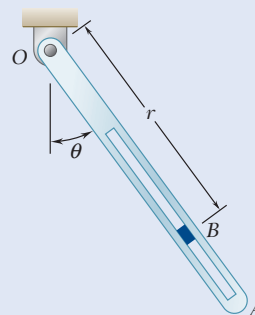


Fig. P12.69

- 12.70** Pin B weighs 4 oz and is free to slide in a horizontal plane along the rotating arm OC and along the fixed circular slot DE of radius $b = 20$ in. Neglecting friction and assuming that $\dot{\theta} = 15 \text{ rad/s}$ and $\ddot{\theta} = 250 \text{ rad/s}^2$ for the position $\theta = 20^\circ$, determine for that position (a) the radial and transverse components of the resultant force exerted on pin B , (b) the forces $\mathbf{P}$ and $\mathbf{Q}$ exerted on pin B , respectively, by rod OC and the wall of slot DE .

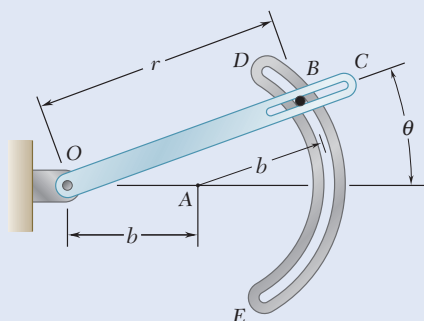


Fig. P12.70

- 12.71** The two blocks are released from rest when $r = 0.8$ m and $\theta = 30^\circ$. Neglecting the mass of the pulley and the effect of friction in the pulley and between block A and the horizontal surface, determine (a) the initial tension in the cable, (b) the initial acceleration of block A, (c) the initial acceleration of block B.

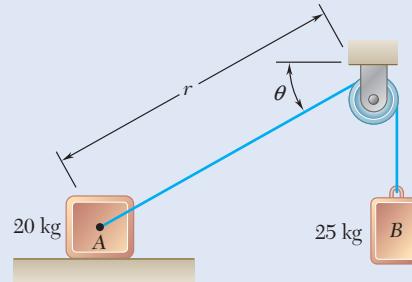


Fig. P12.71 and P12.72

- 12.72** The velocity of block A is 2 m/s to the right at the instant when $r = 0.8$ m and $\theta = 30^\circ$. Neglecting the mass of the pulley and the effect of friction in the pulley and between block A and the horizontal surface, determine, at this instant, (a) the tension in the cable, (b) the acceleration of block A, (c) the acceleration of block B.

- *12.73** Slider C has a weight of 0.5 lb and may move in a slot cut in arm AB, which rotates at the constant rate $\dot{\theta}_0 = 10$ rad/s in a horizontal plane. The slider is attached to a spring of constant $k = 2.5$ lb/ft, which is unstretched when $r = 0$. Knowing that the slider is released from rest with no radial velocity in the position $r = 18$ in. and neglecting friction, determine for the position $r = 12$ in. (a) the radial and transverse components of the velocity of the slider, (b) the radial and transverse components of its acceleration, (c) the horizontal force exerted on the slider by arm AB.

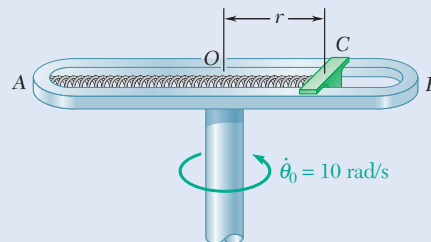


Fig. P12.73

12.2 ANGULAR MOMENTUM AND ORBITAL MOTION

In Sec. 12.1, we introduced the idea of linear momentum and showed how Newton's second law could be expressed as the rate of change of linear momentum. Angular momentum, or the moment of linear momentum, is another useful quantity. In this section, we define angular momentum for a particle and discuss the motion of a particle under a central force, which is applicable to many types of orbital motion.

12.2A Angular Momentum of a Particle and Its Rate of Change

Consider a particle P with a mass m moving with respect to a newtonian frame of reference $Oxyz$. As we saw in Sec. 12.1B, the linear momentum of the particle at a given instant is defined as the vector $m\mathbf{v}$ that is obtained by multiplying the velocity $\mathbf{v}$ of the particle by its mass m . The moment about O of the vector $m\mathbf{v}$ is called the *moment of momentum*, or the **angular momentum**, of the particle about O at that instant and is denoted by $\mathbf{H}_O$. Recall the definition of the moment of a vector (Sec. 3.1E) and denote the position vector of P by $\mathbf{r}$. Then we have

**Angular momentum
of a particle**

$$\mathbf{H}_O = \mathbf{r} \times m\mathbf{v} \quad (12.13)$$

Note that $\mathbf{H}_O$ is a vector perpendicular to the plane containing $\mathbf{r}$ and $m\mathbf{v}$ and has a magnitude

$$H_O = rmv \sin \phi \quad (12.14)$$

where ϕ is the angle between $\mathbf{r}$ and $m\mathbf{v}$ (Fig. 12.12). We can determine the sense of $\mathbf{H}_O$ from the sense of $m\mathbf{v}$ by applying the right-hand rule. The unit of angular momentum is obtained by multiplying the units of length and of linear momentum (Sec. 12.1C). In SI units, we have

$$(\text{m})(\text{kg}\cdot\text{m/s}) = \text{kg}\cdot\text{m}^2/\text{s}$$

In U.S. customary units, we have

$$(\text{ft})(\text{slug})(\text{ft/s}) = (\text{ft})(\text{lb}\cdot\text{s}) = \text{ft}\cdot\text{lb}\cdot\text{s}$$

Resolving the vectors $\mathbf{r}$ and $m\mathbf{v}$ into components and applying formula (3.10), we obtain

$$\mathbf{H}_O = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ x & y & z \\ mv_x & mv_y & mv_z \end{vmatrix} \quad (12.15)$$

The components of $\mathbf{H}_O$, which also represent the moments of the linear momentum $m\mathbf{v}$ about the coordinate axes, can be obtained by expanding the determinant in Eq. (12.15). The results are

$$\begin{aligned} H_x &= m(yv_z - zv_y) \\ H_y &= m(zv_x - xv_z) \\ H_z &= m(xv_y - yv_x) \end{aligned} \quad (12.16)$$

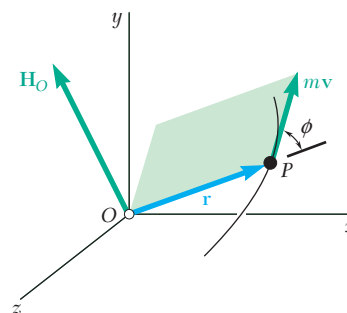


Fig. 12.12 The angular momentum vector of a particle is the vector product of the position vector $\mathbf{r}$ and the linear momentum vector $m\mathbf{v}$.

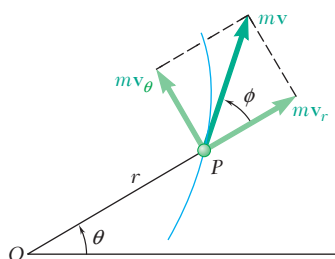


Fig. 12.13 In polar coordinates, angular momentum of a particle is the product of the position r and the transverse component of linear momentum.

In the case of a particle moving in the xy plane, we have $z = v_z = 0$ and the components H_x and H_y reduce to zero. The angular momentum is thus perpendicular to the xy plane; it is then completely defined by the scalar

$$H_O = H_z = m(xv_y - yv_x) \quad (12.17)$$

This value can be positive or negative, according to the sense in which the particle is observed to move from O . If we use polar coordinates, we resolve the linear momentum of the particle into radial and transverse components (Fig. 12.13), which gives us

$$H_O = rmv \sin \phi = rmv_\theta \quad (12.18)$$

Alternatively, recalling from Eq. (11.44) that $v_\theta = r\dot{\theta}$ we have

**Angular momentum
in polar coordinates**

$$H_O = mr^2\dot{\theta} \quad (12.19)$$

Let us now compute the derivative with respect to t of the angular momentum $\mathbf{H}_O$ of a particle P moving in space. Differentiating both sides of Eq. (12.13) and recalling the rule for the differentiation of a vector product (Sec. 11.4B), we have

$$\dot{\mathbf{H}}_O = \dot{\mathbf{r}} \times m\mathbf{v} + \mathbf{r} \times m\dot{\mathbf{v}} = \mathbf{v} \times m\mathbf{v} + \mathbf{r} \times m\mathbf{a}$$

Since the vectors $\mathbf{v}$ and $m\mathbf{v}$ are collinear, the first term of this expression is zero; by Newton's second law, $m\mathbf{a}$ is equal to the sum $\Sigma \mathbf{F}$ of the forces acting on P . Noting that $\mathbf{r} \times \Sigma \mathbf{F}$ represents the sum $\Sigma \mathbf{M}_O$ of the moments about O of these forces, we obtain

$$\Sigma \mathbf{M}_O = \dot{\mathbf{H}}_O \quad (12.20)$$

Equation (12.20), which results directly from Newton's second law, states:

The sum of the moments about O of the forces acting on the particle is equal to the rate of change of angular momentum (or moment of momentum) of the particle about O .

12.2B Motion Under a Central Force and Conservation of Angular Momentum

When the only force acting on a particle P is a force $\mathbf{F}$ directed toward or away from a fixed point O , the particle is said to be moving under a **central force**, and the point O is referred to as the **center of force** (Fig. 12.14). Since the line of action of $\mathbf{F}$ passes through O , we must have $\Sigma \mathbf{M}_O = 0$ at any given instant. Substituting into Eq. (12.20), we obtain

$$\dot{\mathbf{H}}_O = 0$$

for all values of t and, integrating in t ,

$$\mathbf{H}_O = \text{constant} \quad (12.21)$$

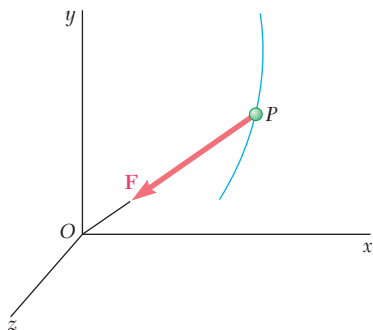


Fig. 12.14 The central force $\mathbf{F}$ acts towards the center of force O .

We thus conclude that

The angular momentum of a particle moving under a central force is constant in both magnitude and direction.

Recall the definition of the angular momentum of a particle (Sec. 12.2A). From that, we have

Conservation of angular momentum

$$\mathbf{r} \times m\mathbf{v} = \mathbf{H}_O = \text{constant} \quad (12.22)$$

It follows that the position vector $\mathbf{r}$ of the particle P must be perpendicular to the constant vector $\mathbf{H}_O$. Thus, a particle under a central force moves in a fixed plane perpendicular to $\mathbf{H}_O$. The vector $\mathbf{H}_O$ and the fixed plane are defined by the initial position vector $\mathbf{r}_0$ and the initial velocity $\mathbf{v}_0$ of the particle. For convenience, let us assume that the plane of the figure coincides with the fixed plane of motion (Fig. 12.15).

Since the magnitude H_O of the angular momentum of the particle P is constant, the right-hand side in Eq. (12.14) must be constant. Therefore, we have

$$rmv \sin \phi = r_0 m v_0 \sin \phi_0 \quad (12.23)$$

This is another way to express the conservation of angular momentum; this relation applies to the motion of any particle under a central force. Since the gravitational force exerted by the sun on a planet is a central force directed toward the center of the sun, Eq. (12.23) is fundamental to the study of planetary motion. For a similar reason, it is also fundamental to studying the motion of space vehicles in orbit about the earth.

Alternatively, from Eq. (12.19), we can express the fact that the magnitude H_O of the angular momentum of the particle P is constant by writing

$$mr^2\dot{\theta} = H_O = \text{constant} \quad (12.24)$$

Dividing by m and using h to denote the angular momentum per unit mass H_O/m , we have

$$r^2\dot{\theta} = h \quad (12.25)$$

Equation (12.25) has an interesting geometric interpretation. Note from Fig. 12.16 that the radius vector OP sweeps across an infinitesimal area $dA = \frac{1}{2}r^2d\theta$ as it rotates through an angle $d\theta$. Then, defining the **areal velocity** of the particle as the quotient dA/dt , we see that the left-hand side of Eq. (12.25) represents twice the areal velocity of the particle. We thus conclude that

When a particle moves under a central force, its areal velocity is constant.

12.2C Newton's Law of Gravitation

As you saw in the preceding section, the gravitational force exerted by the sun on a planet or by the earth on an orbiting satellite is an important example of a central force. In this section, you will learn how to determine the magnitude of a gravitational force.

Newton's **law of universal gravitation** states that two particles of masses M and m at a distance r from each other have a mutual attraction

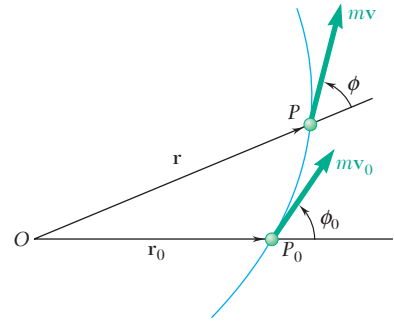


Fig. 12.15 Angular momentum of a particle moving in a fixed plane under the action of a central force.

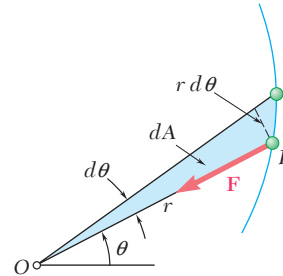


Fig. 12.16 When a particle moves under a central force, its areal velocity is constant.

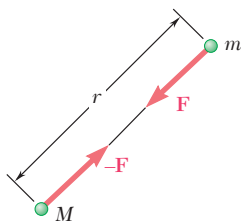


Fig. 12.17 By Newton's law of gravitation, two masses attract each other with equal force.

of equal and opposite forces $\mathbf{F}$ and $-\mathbf{F}$ directed along the line joining the particles (Fig. 12.17). The common magnitude F of the two forces is

Newton's law of universal gravitation

$$F = G \frac{Mm}{r^2} \quad (12.26)$$

where G is a universal constant, called the **constant of gravitation**. Experiments show that the value of G is $(66.73 \pm 0.03) \times 10^{-12} \text{ m}^3/\text{kg}\cdot\text{s}^2$ in SI units or approximately $34.4 \times 10^{-9} \text{ ft}^4/\text{lb}\cdot\text{s}^4$ in U.S. customary units. Gravitational forces exist between any pair of bodies, but their effect is appreciable only when one of the bodies has a very large mass. The effect of gravitational forces is apparent in the cases of the motion of a planet about the sun, of satellites orbiting about the earth, or of bodies falling on the surface of the earth.

Since the force exerted by the earth on a body of mass m located on or near its surface is defined as the weight $\mathbf{W}$ of the body, we can substitute the magnitude $W = mg$ of the weight for F , and the earth's radius R for r in Eq. (12.26). We obtain

$$W = mg = \frac{GM}{R^2} m \quad \text{or} \quad g = \frac{GM}{R^2} \quad (12.27)$$

where M is the mass of the earth. Since the earth is not truly spherical, the distance R from the center of the earth depends upon the point selected on its surface. Thus, the values of W and g vary with the altitude and latitude of the point considered. Another reason for the variation of W and g with latitude is that a system of axes attached to the earth does not constitute a newtonian frame of reference (see Sec. 12.1A). A more accurate definition of the weight of a body should therefore include a component representing the effects of this centripetal acceleration due to the earth's rotation. Values of g at sea level vary from 9.781 m/s^2 (or 32.09 ft/s^2) at the equator to 9.833 m/s^2 (or 32.26 ft/s^2) at the poles.[†]

We can use Eq. (12.26) to find the force exerted by the earth on a body of mass m located in space at a distance r from its center. The computations are somewhat simplified by noting that, according to Eq. (12.27), we can express the product of the constant of gravitation G and the mass M of the earth as

$$GM = gR^2 \quad (12.28)$$

Here we give g and the earth's radius R as their average values $g = 9.81 \text{ m/s}^2$ and $R = 6.37 \times 10^6 \text{ m}$ in SI units[‡] and $g = 32.2 \text{ ft/s}^2$ and $R = (3960 \text{ mi})(5280 \text{ ft/mi})$ in U.S. customary units.

The discovery of the law of universal gravitation often has been attributed to the belief that, after observing an apple falling from a tree, Newton realized that the earth must attract an apple in much the same way as the moon. It is doubtful that this incident actually took place, but we can say that Newton would not have formulated his law if he had not first perceived that the acceleration of a falling body must have the same cause as the acceleration that keeps the moon in its orbit.

[†]A formula expressing g in terms of the latitude ϕ was given in Prob. 12.2.

[‡]You can find the value of R simply by relating the earth's circumference to its radius as $2\pi R = 40 \times 10^6 \text{ m}$.

Sample Problem 12.12

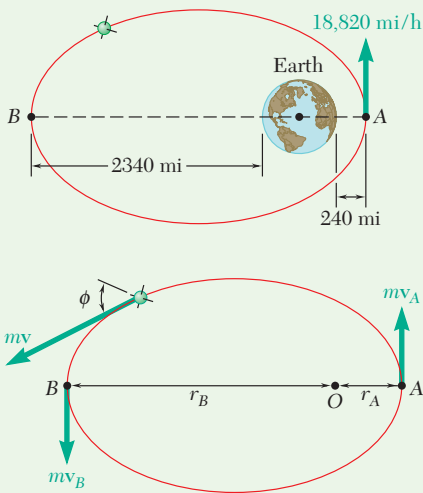


Fig. 1 The satellite at various positions. A and B, we have

A satellite is launched in a direction parallel to the surface of the earth with a velocity of 18,820 mi/h from an altitude of 240 mi. Determine the velocity of the satellite as it reaches its maximum altitude of 2340 mi. Recall that the earth's radius is 3960 mi.

STRATEGY: The satellite is acted on by a central force, so angular momentum is conserved. You can use the principle of conservation of angular momentum to determine the velocity of the satellite.

MODELING and ANALYSIS: Since the satellite is moving under a central force directed toward the center O of the earth, its angular momentum $\mathbf{H}_O$ is constant. From Eq. (12.14), you have

$$rmv \sin \phi = H_O = \text{constant}$$

This equation shows that v is at a minimum at B , where both r and $\sin \phi$ are maximum. Expressing the conservation of angular momentum between

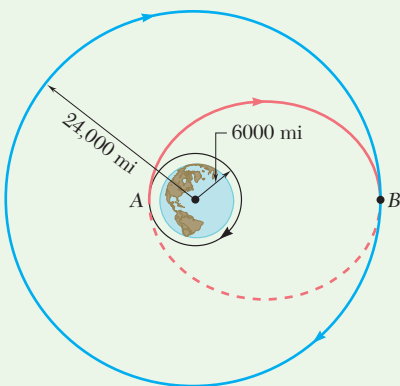
$$r_A m v_A = r_B m v_B$$

Hence,

$$v_B = v_A \frac{r_A}{r_B} = (18,820 \text{ mi/h}) \frac{3960 \text{ mi} + 240 \text{ mi}}{3960 \text{ mi} + 2340 \text{ mi}} \\ v_B = 12,550 \text{ mi/h} \quad \blacktriangleleft$$

REFLECT and THINK: Note that in order to increase velocity, you could choose to apply thrusters pushing the spacecraft closer to the earth. Since this is a central force, the spacecraft's angular momentum remains constant. Therefore, its speed v increases as the radial distance r decreases.

Sample Problem 12.13



A space tug travels a circular orbit with a 6000-mi radius around the earth. In order to transfer it to a larger orbit with a 24,000-mi radius, the tug is first placed on an elliptical path AB by firing its engines as it passes through A , thus increasing its velocity by 3810 mi/h. Determine how much the tug's velocity should be increased as it reaches B to insert it into the larger circular orbit.

STRATEGY: Use Newton's second law and conservation of angular momentum.

MODELING: Choose the space tug as the system, and assume you can treat it as a particle. Draw free-body and kinetic diagrams of the system at A as shown in Fig. 1.

(continued)

ANALYSIS:

Circular Orbit through A. Applying Newton's second law in the normal direction when the tug is at A gives

$$\overset{+}{\rightarrow} \Sigma F_n = ma_n \qquad \frac{GMm}{r_A^2} = \frac{mv_A^2}{r_A} \quad (1)$$

Solve Eq. (1) for v_A and substitute in numbers to find

$$v_A = \sqrt{\frac{GM}{r_A}} = \sqrt{\frac{gR^2}{r_A}} = \sqrt{\frac{(32.2 \text{ ft/s}^2)((3960 \text{ mi})(5280 \text{ ft/mi}))^2}{(600 \text{ mi})(5280 \text{ ft/mi})}} = 21,080 \text{ ft/s}$$

Converting this to mi/h gives $v_A = 14,370 \text{ mi/h}$. Thus, the velocity to put the space tug into an elliptic orbit is $(v_A)_{\text{ell}} = 14,370 \text{ mi/h} + 3810 \text{ mi/h} = 18,180 \text{ mi/h}$.

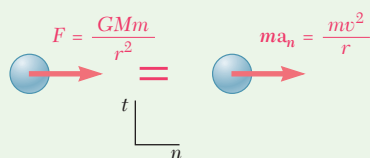


Fig. 1 Free-body diagram and kinetic diagram of satellite at point A.

Elliptic Path AB. To find the velocity at B, use the conservation of angular momentum between A and B. The velocity is perpendicular to r at both A and B, so you have

$$H_O = r_A m v_A = r_B m v_B \quad (2)$$

Solving Eq. (2) for v_B and substituting in numbers give

$$(v_B)_{\text{ell}} = \frac{r_A}{r_B} (v_A)_{\text{ell}} = \frac{6000 \text{ mi}}{24,000 \text{ mi}} (18,180 \text{ mi/h}) = 4545 \text{ mi/h}$$

Circular Orbit through B. Applying Newton's second law in the normal direction when the tug is at B gives

$$\overset{+}{\leftarrow} \Sigma F_n = ma_n \qquad \frac{GMm}{r_B^2} = \frac{mv_B^2}{r_B} \quad (3)$$

By solving Eq. (3) for v_B and substituting in numbers, you find

$$v_B = \sqrt{\frac{GM}{r_B}} = \sqrt{\frac{gR^2}{r_B}} = \sqrt{\frac{(32.2 \text{ ft/s}^2)((3960 \text{ mi})(5280 \text{ ft/mi}))^2}{(24,000 \text{ mi})(5280 \text{ ft/mi})}} = 10,540 \text{ ft/s}$$

This is the speed of the space tug at B for it to have a circular orbit. Converting this to mi/h gives $v_B = 7186 \text{ mi/h}$. Therefore, the required increase in velocity is

$$\Delta v_B = 7186 \text{ mi/h} - 4545 \text{ mi/h}$$

$$\Delta v_B = 2640 \text{ mi/h} \quad \blacktriangleleft$$

REFLECT and THINK: The speeds of satellites and orbiting vehicles are quite large, as seen in this problem. The next type of question we could ask is what force is required to impart this change in speed.

SOLVING PROBLEMS ON YOUR OWN

In this section, we introduced the *angular momentum* or the *moment of the momentum*, $\mathbf{H}_O$, of a particle about O as

$$\mathbf{H}_O = \mathbf{r} \times m\mathbf{v} \quad (12.13)$$

and found that $\mathbf{H}_O$ is constant when the particle moves under a **central force** with its center located at O .

1. Solving problems involving the motion of a particle under a central force. In problems of this type, the angular momentum $\mathbf{H}_O$ of the particle about the center of force O is conserved. Therefore, we can express the conservation of angular momentum of particle P about O by $rmv \sin \phi = r_0mv_0 \sin \phi_0$.

2. In space mechanics problems involving the orbital motion of a planet about the sun or of a satellite about the earth, the moon, or some other planet, the central force $\mathbf{F}$ is the force of gravitational attraction. This force is directed *toward* the center of force O and has the magnitude

$$F = G \frac{Mm}{r^2} \quad (12.26)$$

Note that in the particular case of the gravitational force exerted by the earth, the product GM can be replaced by gR^2 , where R is the earth's radius [Eq. 12.28].

The following two cases of orbital motion are frequently encountered:

a. For a satellite in a circular orbit, the force $\mathbf{F}$ is normal to the orbit and you can write $F = ma_n$ [Sample Prob. 12.13]. Substituting for F from Eq. (12.26) and observing that $a_n = v^2/\rho = v^2/r$, you obtain

$$G \frac{Mm}{r^2} = m \frac{v^2}{r} \quad \text{or} \quad v^2 = \frac{GM}{r}$$

b. For a satellite in an elliptical orbit, the radius vector $\mathbf{r}$ and the velocity $\mathbf{v}$ of the satellite are perpendicular to each other at points A and B , which are closest and farthest to the center of force O , respectively [Sample Prob. 12.12]. Thus, the conservation of angular momentum of the satellite between these two points can be expressed as

$$r_A m v_A = r_B m v_B$$

Problems

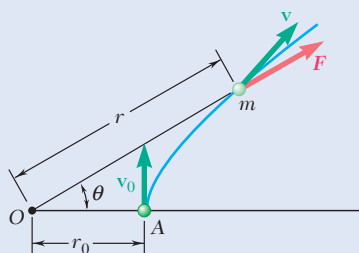


Fig. P12.74

12.74 A particle of mass m is projected from point A with an initial velocity $\mathbf{v}_0$ perpendicular to line OA and moves under a central force $\mathbf{F}$ directed away from the center of force O . Knowing that the particle follows a path defined by the equation $r = r_0 / \sqrt{\cos 2\theta}$ and using Eq. (12.25), express the radial and transverse components of the velocity $\mathbf{v}$ of the particle as functions of θ .

12.75 For the particle of Prob. 12.74, show (a) that the velocity of the particle and the central force $\mathbf{F}$ are proportional to the distance r from the particle to the center of force O , (b) that the radius of curvature of the path is proportional to r^3 .

12.76 A particle of mass m is projected from point A with an initial velocity $\mathbf{v}_0$ perpendicular to line OA and moves under a central force $\mathbf{F}$ along a semicircular path of diameter OA . Observing that $r = r_0 \cos \theta$ and using Eq. (12.25), show that the speed of the particle is $v = v_0 / \cos^2 \theta$.

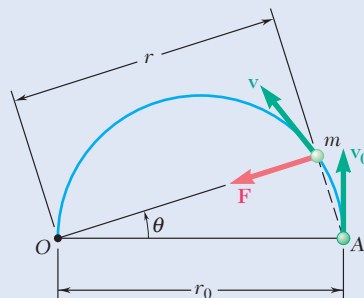


Fig. P12.76

12.77 For the particle of Prob. 12.76, determine the tangential component F_t of the central force $\mathbf{F}$ along the tangent to the path of the particle for (a) $\theta = 0$, (b) $\theta = 45^\circ$.

12.78 Determine the mass of the earth knowing that the mean radius of the moon's orbit about the earth is 238,910 mi and that the moon requires 27.32 days to complete one full revolution about the earth.

12.79 Show that the radius r of the moon's orbit can be determined from the radius R of the earth, the acceleration of gravity g at the surface of the earth, and the time τ required for the moon to complete one full revolution about the earth. Compute r knowing that $\tau = 27.3$ days, giving the answer in both SI and U.S. customary units.

12.80 Communication satellites are placed in a geosynchronous orbit, i.e., in a circular orbit such that they complete one full revolution about the earth in one sidereal day (23.934 h), and thus appear stationary with respect to the ground. Determine (a) the altitude of these satellites above the surface of the earth, (b) the velocity with which they describe their orbit. Give the answers in both SI and U.S. customary units.

- 12.81** Show that the radius r of the orbit of a moon of a given planet can be determined from the radius R of the planet, the acceleration of gravity at the surface of the planet, and the time τ required by the moon to complete one full revolution about the planet. Determine the acceleration of gravity at the surface of the planet Jupiter knowing that $R = 71\,492\text{ km}$ and that $\tau = 3.551\text{ days}$ and $r = 670.9 \times 10^3\text{ km}$ for its moon Europa.
- 12.82** The orbit of the planet Venus is nearly circular with an orbital velocity of $126.5 \times 10^3\text{ km/h}$. Knowing that the mean distance from the center of the sun to the center of Venus is $108 \times 10^6\text{ km}$ and that the radius of the sun is $695.5 \times 10^3\text{ km}$, determine (a) the mass of the sun, (b) the acceleration of gravity at the surface of the sun.
- 12.83** A satellite is placed into a circular orbit about the planet Saturn at an altitude of 2100 mi . The satellite describes its orbit with a velocity of $54.7 \times 10^3\text{ mi/h}$. Knowing that the radius of the orbit about Saturn and the periodic time of Atlas, one of Saturn's moons, are $85.54 \times 10^3\text{ mi}$ and 0.6017 days , respectively, determine (a) the radius of Saturn, (b) the mass of Saturn. (The *periodic time* of a satellite is the time it requires to complete one full revolution about the planet.)
- 12.84** The periodic time (see Prob. 12.83) of an earth satellite in a circular polar orbit is 120 minutes . Determine (a) the altitude h of the satellite, (b) the time during which the satellite is above the horizon for an observer located at the north pole.
- 12.85** A 500-kg spacecraft first is placed into a circular orbit about the earth at an altitude of 4500 km and then is transferred to a circular orbit about the moon. Knowing that the mass of the moon is 0.01230 times the mass of the earth and that the radius of the moon is 1737 km , determine (a) the gravitational force exerted on the spacecraft as it was orbiting the earth, (b) the required radius of the orbit of the spacecraft about the moon if the periodic times (see Prob. 12.83) of the two orbits are to be equal, (c) the acceleration of gravity at the surface of the moon.
- 12.86** A space vehicle is in a circular orbit of 2200-km radius around the moon. To transfer it to a smaller circular orbit of 2080-km radius, the vehicle is first placed on an elliptic path AB by reducing its speed by 26.3 m/s as it passes through A . Knowing that the mass of the moon is $73.49 \times 10^{21}\text{ kg}$, determine (a) the speed of the vehicle as it approaches B on the elliptic path, (b) the amount by which its speed should be reduced as it approaches B to insert it into the smaller circular orbit.
- 12.87** As a first approximation to the analysis of a space flight from the earth to the planet Mars, assume the orbits of the earth and Mars are circular and coplanar. The mean distances from the sun to the earth and to Mars are $149.6 \times 10^6\text{ km}$ and $227.8 \times 10^6\text{ km}$, respectively. To place the spacecraft into an elliptical transfer orbit at point A , its speed is increased over a short interval of time to v_A , which is 2.94 km/s faster than the earth's orbital speed. When the spacecraft reaches point B on the elliptical transfer orbit, its speed v_B is increased to the orbital speed of Mars. Knowing that the mass of the sun is 332.8×10^3 times the mass of the earth, determine the increase in speed required at B .

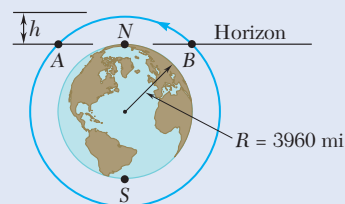


Fig. P12.84

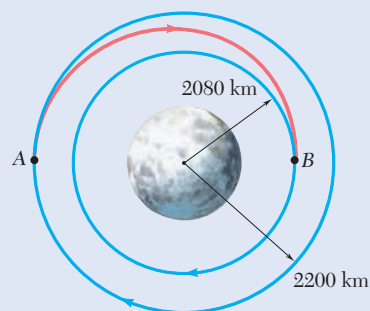


Fig. P12.86

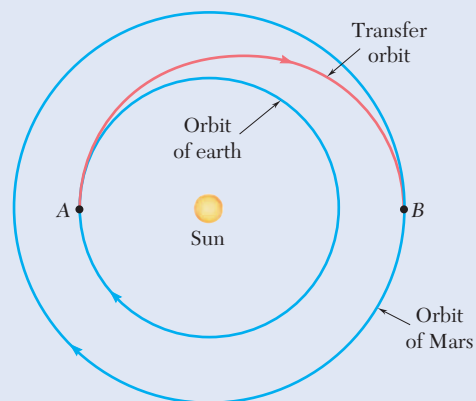


Fig. P12.87

- 12.88** To place a communications satellite into a geosynchronous orbit (see Prob. 12.80) at an altitude of 22,240 mi above the surface of the earth, the satellite first is released from a space shuttle, which is in a circular orbit at an altitude of 185 mi, and then is propelled by an upper-stage booster to its final altitude. As the satellite passes through *A*, the booster's motor is fired to insert the satellite into an elliptic transfer orbit. The booster is again fired at *B* to insert the satellite into a geosynchronous orbit. Knowing that the second firing increases the speed of the satellite by 4810 ft/s, determine (a) the speed of the satellite as it approaches *B* on the elliptic transfer orbit, (b) the increase in speed resulting from the first firing at *A*.

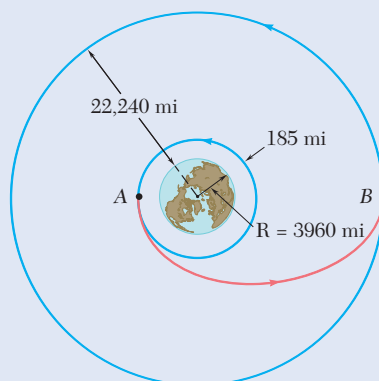


Fig. P12.88

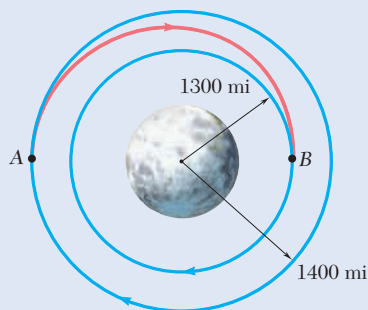


Fig. P12.89

- 12.89** A space vehicle is in a circular orbit with a 1400-mi radius around the moon. To transfer to a smaller orbit with a 1300-mi radius, the vehicle is first placed in an elliptic path *AB* by reducing its speed by 86 ft/s as it passes through *A*. Knowing that the mass of the moon is 5.03×10^{21} lb·s²/ft, determine (a) the speed of the vehicle as it approaches *B* on the elliptic path, (b) the amount by which its speed should be reduced as it approaches *B* to insert it into the smaller circular orbit.
- 12.90** A 1-kg collar can slide on a horizontal rod that is free to rotate about a vertical shaft. The collar is initially held at *A* by a cord attached to the shaft. A spring of constant 30 N/m is attached to the collar and to the shaft and is undeformed when the collar is at *A*. As the rod rotates at the rate $\dot{\theta} = 16$ rad/s, the cord is cut and the collar moves out along the rod. Neglecting friction and the mass of the rod, determine (a) the radial and transverse components of the acceleration of the collar at *A*, (b) the acceleration of the collar relative to the rod at *A*, (c) the transverse component of the velocity of the collar at *B*.

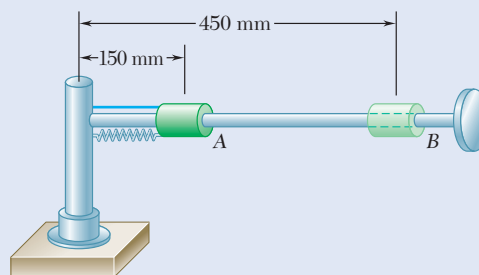


Fig. P12.90

- 12.91** A 1-lb ball A and a 2-lb ball B are mounted on a horizontal rod that rotates freely about a vertical shaft. The balls are held in the positions shown by pins. The pin holding B is suddenly removed and the ball moves to position C as the rod rotates. Neglecting friction and the mass of the rod and knowing that the initial speed of A is $v_A = 8$ ft/s, determine (a) the radial and transverse components of the acceleration of ball B immediately after the pin is removed, (b) the acceleration of ball B relative to the rod at that instant, (c) the speed of ball A after ball B has reached the stop at C .

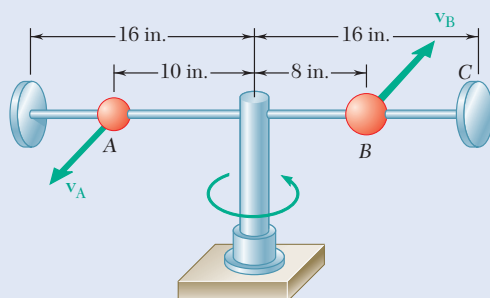


Fig. P12.91

- 12.92** Two 2.6-lb collars A and B can slide without friction on a frame, consisting of the horizontal rod OE and the vertical rod CD , which is free to rotate about CD . The two collars are connected by a cord running over a pulley that is attached to the frame at O and a stop prevents collar B from moving. The frame is rotating at the rate $\dot{\theta} = 12$ rad/s and $r = 0.6$ ft when the stop is removed allowing collar A to move out along rod OE . Neglecting friction and the mass of the frame, determine, for the position $r = 1.2$ ft, (a) the transverse component of the velocity of collar A , (b) the tension in the cord and the acceleration of collar A relative to the rod OE .

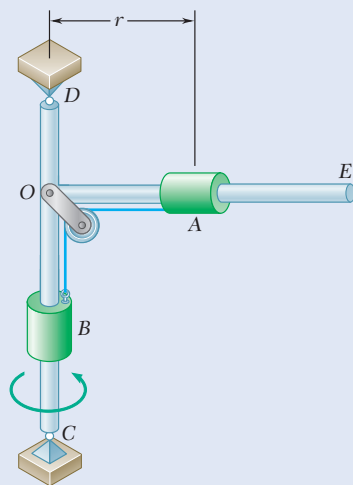


Fig. P12.92

- 12.93** A small ball swings in a horizontal circle at the end of a cord of length l_1 , which forms an angle θ_1 with the vertical. The cord is then slowly drawn through the support at O until the length of the free end is l_2 . (a) Derive a relation among l_1 , l_2 , θ_1 , and θ_2 . (b) If the ball is set in motion so that initially $l_1 = 0.8$ m and $\theta_1 = 35^\circ$, determine the angle θ_2 when $l_2 = 0.6$ m.

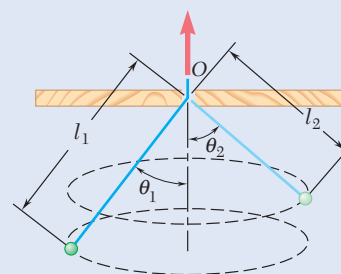


Fig. P12.93

*12.3 APPLICATIONS OF CENTRAL-FORCE MOTION

The most important examples of a particle moving under the action of a central force occur in space mechanics, where gravity is the central force. In this section, we examine some of the basic ideas of this motion, concentrating on the motions of satellites around the earth and planets around a star.

12.3A Trajectory of a Particle Under a Central Force

Consider a particle P moving under a central force $\mathbf{F}$. In order to fully characterize the motion of particle P (which could represent a satellite, a moon, etc.), we must develop a differential equation that defines its trajectory.

Assuming that the force $\mathbf{F}$ is directed toward the center of force O , we note that ΣF_r and ΣF_θ reduce, respectively, to $-F$ and zero in Eqs. (12.11) and (12.12). Therefore, we have

$$m(\ddot{r} - r\dot{\theta}^2) = -F \quad (12.29)$$

$$m(r\ddot{\theta} + 2\dot{r}\dot{\theta}) = 0 \quad (12.30)$$

These equations define the motion of P . We can also use Eq. (12.25) to analyze the motion of P , obtaining

$$r^2\dot{\theta} = h \quad \text{or} \quad r^2 \frac{d\theta}{dt} = h \quad (12.31)$$

We can use Eq. (12.31) to eliminate the independent variable t from Eq. (12.29). Solving Eq. (12.31) for $\dot{\theta}$, or $d\theta/dt$, we have

$$\dot{\theta} = \frac{d\theta}{dt} = \frac{h}{r^2} \quad (12.32)$$

It follows that

$$\dot{r} = \frac{dr}{dt} = \frac{dr}{d\theta} \frac{d\theta}{dt} = \frac{h}{r^2} \frac{dr}{d\theta} = -h \frac{d}{d\theta} \left(\frac{1}{r} \right) \quad (12.33)$$

$$\ddot{r} = \frac{d\dot{r}}{dt} = \frac{d\dot{r}}{d\theta} \frac{d\theta}{dt} = \frac{h}{r^2} \frac{d\dot{r}}{d\theta}$$

If we substitute for $\dot{r}$ from Eq. (12.33) into the expression for $\ddot{r}$, we obtain

$$\begin{aligned} \ddot{r} &= \frac{h}{r^2} \frac{d}{d\theta} \left[-h \frac{d}{d\theta} \left(\frac{1}{r} \right) \right] \\ \ddot{r} &= -\frac{h^2}{r^2} \frac{d^2}{d\theta^2} \left(\frac{1}{r} \right) \end{aligned} \quad (12.34)$$

Now, substituting for $\dot{\theta}$ and $\ddot{r}$ from Eqs. (12.32) and (12.34), respectively, in Eq. (12.29) and introducing the function $u = 1/r$, we obtain, after reductions,

$$\frac{d^2u}{d\theta^2} + u = \frac{F}{mh^2u^2} \quad (12.35)$$

In deriving Eq. (12.35), we assumed force $\mathbf{F}$ to be directed toward O . The magnitude F therefore should be positive if $\mathbf{F}$ is actually directed toward O (attractive force) and negative if $\mathbf{F}$ is directed away from O (repulsive force). If F is a known function of r and thus of u , Eq. (12.35) is a differential equation in u and θ . This differential equation defines the trajectory followed by the particle under the central force $\mathbf{F}$. We can obtain the equation of the trajectory by solving the differential equation (12.35) for u as a function of θ and determining the constants of integration from the initial conditions.

*12.3B Application to Space Mechanics

After the last stages of their launching rockets have burned out, earth satellites and other space vehicles are subject to only the gravitational pull of the earth. We can therefore determine their motion from Eqs. (12.31) and (12.35), which govern the motion of a particle under a central force, after replacing F by the expression for the force of gravitational attraction.[†] We set F in Eq. (12.35) as

$$F = \frac{GMm}{r^2} = GMmu^2$$

where M = mass of the earth

m = mass of space vehicle

r = distance from center of the earth to vehicle

$u = 1/r$

Then we obtain the differential equation

$$\frac{d^2u}{d\theta^2} + u = \frac{GM}{h^2} \quad (12.36)$$

Note that the right-hand side is a constant.

To solve the differential equation (12.36), we add the particular solution $u = GM/h^2$ to the general solution $u = C \cos(\theta - \theta_0)$ of the corresponding homogeneous equation (i.e., the equation obtained by setting the right-hand side equal to zero). Choosing the polar axis so that $\theta_0 = 0$, we have

$$\frac{1}{r} = u = \frac{GM}{h^2} + C \cos \theta \quad (12.37)$$

Equation (12.37) is the equation of a *conic section* (ellipse, parabola, or hyperbola) in the polar coordinates r and θ . The origin O of the coordinates, which is located at the center of the earth, is a *focus* of this conic section, and the polar axis is one of its axes of symmetry (Fig. 12.18).

The ratio of the constants C and GM/h^2 defines the **eccentricity** ε of the conic section. If we set

$$\varepsilon = \frac{C}{GM/h^2} = \frac{Ch^2}{GM} \quad (12.38)$$

[†]We assume that the space vehicles considered here are attracted by the earth only and that their masses are negligible compared to the mass of the earth. If a vehicle travels very far from the earth, its path may be affected by the gravitational attraction of the sun, the moon, or another planet.



Photo 12.5 The Hubble telescope was carried into orbit by the space shuttle in 1990.

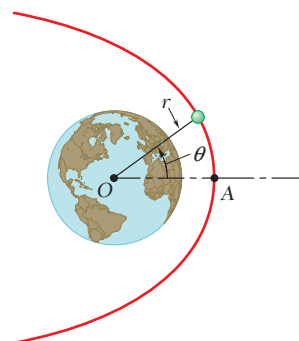


Fig. 12.18 The trajectory of an earth satellite is a conic section with the center of the earth as one of its foci.

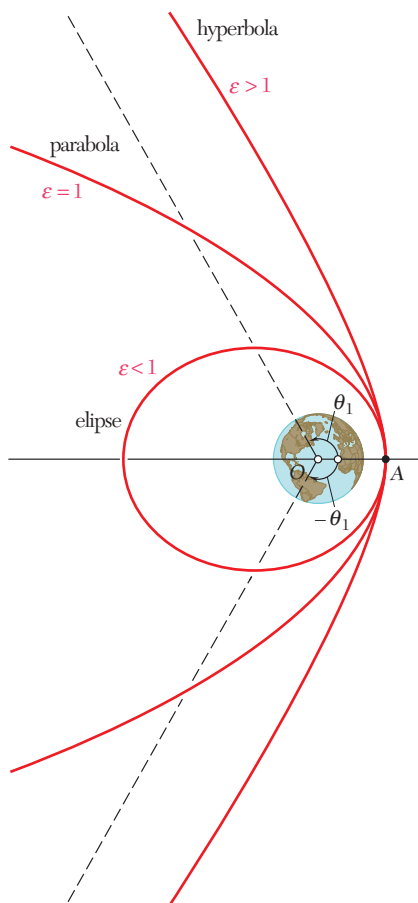


Fig. 12.19 Depending on the eccentricity, the orbit of an earth satellite can be a hyperbola, a parabola, or an ellipse.

we can write Eq. (12.37) in the form

$$\frac{1}{r} = \frac{GM}{h^2} (1 + \varepsilon \cos \theta) \quad (12.37')$$

This equation represents three possible trajectories.

1. $\varepsilon > 1$, or $C > GM/h^2$: There are two values θ_1 and $-\theta_1$ of the polar angle, defined by $\cos \theta_1 = -GM/Ch^2$, for which the right-hand side of Eq. (12.37) becomes zero. For both these values, the radius vector r becomes infinite; the conic section is a *hyperbola* (Fig. 12.19).
2. $\varepsilon = 1$, or $C = GM/h^2$: The radius vector becomes infinite for $\theta = 180^\circ$; the conic section is a *parabola*.
3. $\varepsilon < 1$, or $C < GM/h^2$: The radius vector remains finite for every value of θ ; the conic section is an *ellipse*. In the particular case when $\varepsilon = C = 0$, the length of the radius vector is constant; the conic section is a circle.

Let's now see how we can determine the constants C and GM/h^2 , which characterize the trajectory of a space vehicle, from the vehicle's position and velocity at the beginning of its free flight. We assume that, as is generally the case, the powered phase of its flight has been programmed in such a way that as the last stage of the launching rocket burns out, the vehicle has a velocity parallel to the surface of the earth (Fig. 12.20). In other words, we assume that the space vehicle begins its free flight at the vertex A of its trajectory. (In Sec. 13.2D, we consider problems involving oblique launchings.)

Denoting the radius and speed of the vehicle at the beginning of its free flight by r_0 and v_0 , respectively, we observe that the velocity reduces to its transverse component. Thus, $v_0 = r_0 \dot{\theta}_0$. Recalling Eq. (12.25), we express the angular momentum per unit mass h as

$$h = r_0^2 \dot{\theta}_0 = r_0 v_0 \quad (12.39)$$

The value obtained for h can be used to determine the constant GM/h^2 . We also note that the computation of this constant is simplified if we use the relation obtained in Sec. 12.2C.

$$GM = gR^2 \quad (12.28)$$

where R is the radius of the earth ($R = 6.37 \times 10^6$ m or 3960 mi) and g is the acceleration due to gravity at the earth's surface.

We obtain the constant C by setting $\theta = 0$, $r = r_0$ in Eq. (12.37). Hence,

$$C = \frac{1}{r_0} - \frac{GM}{h^2} \quad (12.40)$$

Substituting for h from Eq. (12.39), we can easily express C in terms of r_0 and v_0 .

Initial Conditions. Now we can determine the initial conditions corresponding to each of the three fundamental trajectories indicated. Considering first the parabolic trajectory, we set C equal to GM/h^2 in Eq. (12.40) and eliminate h between Eqs. (12.39) and (12.40). Solving for v_0 , we obtain

$$v_0 = \sqrt{\frac{2GM}{r_0}}$$

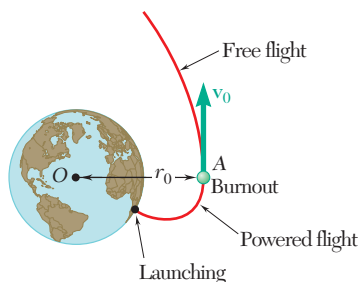


Fig. 12.20 Typically, a space vehicle has a velocity parallel to the surface of the earth after the powered portion of its flight.

We can check that a larger value of the initial velocity corresponds to a hyperbolic trajectory and a smaller value corresponds to an elliptic orbit. Since the value of v_0 obtained for the parabolic trajectory is the smallest value for which the space vehicle does not return to its starting point, it is called the **escape velocity**. Therefore, making use of Eq. (12.28), we have

$$v_{\text{esc}} = \sqrt{\frac{2GM}{r_0}} \quad \text{or} \quad v_{\text{esc}} = \sqrt{\frac{2gR^2}{r_0}} \quad (12.41)$$

Note that the trajectory is (1) hyperbolic if $v_0 > v_{\text{esc}}$, (2) parabolic if $v_0 = v_{\text{esc}}$, and (3) elliptic if $v_0 < v_{\text{esc}}$.

Among the various possible elliptic orbits, the one obtained when $C = 0$, the *circular orbit*, is of special interest. Taking into account Eq. (12.28), the value of the initial velocity corresponding to a circular orbit is

$$v_{\text{circ}} = \sqrt{\frac{GM}{r_0}} \quad \text{or} \quad v_{\text{circ}} = \sqrt{\frac{gR^2}{r_0}} \quad (12.42)$$

Note from Fig. 12.21 that, for values of v_0 larger than v_{circ} but smaller than v_{esc} , point A is the point of the orbit closest to the earth where free flight begins. This point is called the *perigee*, whereas point A', which is farthest away from the earth, is known as the *apogee*. For values of v_0 smaller than v_{circ} , point A is the apogee and point A'', which is on the other side of the orbit, is the perigee. For values of v_0 much smaller than v_{circ} , the trajectory of the space vehicle intersects the surface of the earth; in such a case, the vehicle does not go into orbit.

Ballistic missiles, which were designed to hit the surface of the earth, also travel along elliptic trajectories. In fact, you should now realize that any object projected in vacuum with an initial velocity v_0 smaller than v_{esc} moves along an elliptic path. Only when the distances involved are small enough that we can assume the gravitational field of the earth is uniform can we approximate the elliptic path by a parabolic path, as we did earlier (Sec. 11.4C) in the case of conventional projectiles.

Periodic Time. An important characteristic of the motion of an earth satellite is the time required by the satellite to travel through one complete orbit. This time, which is known as the satellite's **periodic time**, is denoted by τ . We first observe, in view of the definition of areal velocity (Sec. 12.2B), that we can obtain τ by dividing the area inside the orbit by the areal velocity. The area of an ellipse is equal to πab , where a and b denote the semimajor and semiminor axes, respectively. Since the areal velocity is equal to $h/2$, we have

$$\tau = \frac{2\pi ab}{h} \quad (12.43)$$

Although we can readily determine h from r_0 and v_0 in the case of a satellite launched in a direction parallel to the earth's surface, the semiaxes a and b are not directly related to the initial conditions. However, the values r_0 and r_1 of r corresponding to the perigee and apogee of the orbit can be determined from Eq. (12.37), so we can express the semiaxes a and b in terms of r_0 and r_1 .

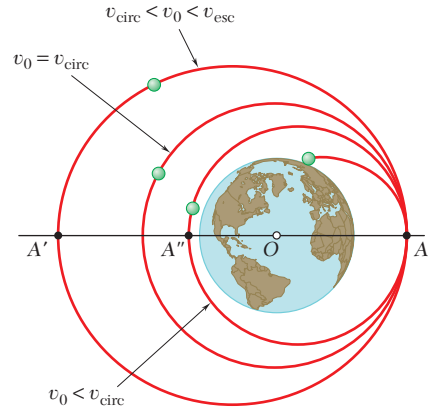


Fig. 12.21 Various elliptic orbits are possible for earth satellites, depending on the initial velocity.

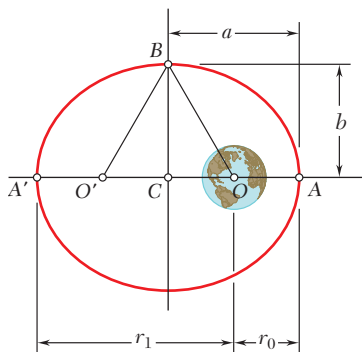


Fig. 12.22 For an elliptic orbit, the distances to apogee (A') and perigee (A) are related to the semimajor and semiminor axes.

Consider the elliptic orbit shown in Fig. 12.22. The earth's center is located at O and coincides with one of the two foci of the ellipse, and the points A and A' represent, respectively, the perigee and apogee of the orbit. We easily check that

$$r_0 + r_1 = 2a$$

and thus

$$a = \frac{1}{2}(r_0 + r_1) \quad (12.44)$$

Recall that the sum of the distances from each of the foci to any point of the ellipse is constant, so we have

$$O'B + BO = O'A + OA = 2a \quad \text{or} \quad BO = a$$

On the other hand, we have $CO = a - r_0$. We can therefore write

$$\begin{aligned} b^2 &= (BC)^2 = (BO)^2 - (CO)^2 = a^2 - (a - r_0)^2 \\ b^2 &= r_0(2a - r_0) = r_0 r_1 \end{aligned}$$

and thus

$$b = \sqrt{r_0 r_1} \quad (12.45)$$

Formulas (12.44) and (12.45) indicate that the semimajor and semiminor axes of the orbit are equal, respectively, to the arithmetic and geometric means of the maximum and minimum values of the radius vector. Once you have determined r_0 and r_1 , you can compute the lengths of the semiaxes and substitute for a and b in Eq. (12.43).

*12.3C Kepler's Laws of Planetary Motion

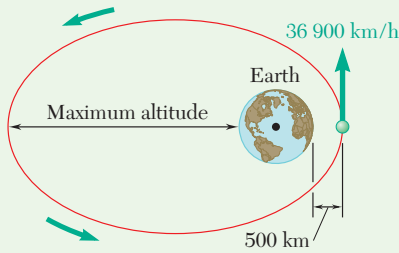
We can use the equations governing the motion of an earth satellite to describe the motion of the moon around the earth. In that case, however, the mass of the moon is not negligible compared with the earth's mass, and the results are not entirely accurate.

We can also apply the theory developed in the preceding sections to the study of the motion of the planets around the sun. Although another error is introduced by neglecting the forces exerted by the planets on one another, the approximation obtained is excellent. Indeed, even before Newton had formulated his fundamental theory, the properties expressed by Eq. (12.37), where M now represents the mass of the sun, and by Eq. (12.31) had been discovered by the German astronomer Johannes Kepler (1571–1630) from astronomical observations of the motion of the planets.

Kepler's three **laws of planetary motion** can be stated as follows.

1. The path of each planet describes an ellipse, with the sun located at one of its foci.
2. The radius vector drawn from the sun to a planet sweeps equal areas in equal times.
3. The squares of the periodic times of the planets are proportional to the cubes of the semimajor axes of their orbits.

The first law states a particular case of the result established in Sec. 12.3B, and the second law expresses that the areal velocity of each planet is constant (see Sec. 12.2B). Kepler's third law also can be derived from the results obtained in Sec. 12.3B. (See also Prob. 12.120.)



Sample Problem 12.14

A satellite is launched in a direction parallel to the earth's surface with a velocity of 36 900 km/h from an altitude of 500 km. Determine (a) the maximum altitude reached by the satellite, (b) the periodic time of the satellite.

STRATEGY: After the satellite is launched, it is subjected to the earth's gravitational attraction only and undergoes central-force motion. Knowing this, you can determine the satellite's trajectory, maximum altitude, and periodic time.

MODELING and ANALYSIS: The satellite can be modeled as a particle.

a. Maximum Altitude. After the satellite is launched, it is subject only to the earth's gravitational attraction. Thus, its motion is governed by Eq. (12.37), so

$$\frac{1}{r} = \frac{GM}{h^2} + C \cos \theta \quad (1)$$

Since the radial component of the velocity is zero at the point of launching A, you have $h = r_0 v_0$. Recalling that for the earth, $R = 6370$ km, you can compute

$$r_0 = 6370 \text{ km} + 500 \text{ km} = 6870 \text{ km} = 6.87 \times 10^6 \text{ m}$$

$$v_0 = 36\,900 \text{ km/h} = \frac{36.9 \times 10^6 \text{ m}}{3.6 \times 10^3 \text{ s}} = 10.25 \times 10^3 \text{ m/s}$$

$$h = r_0 v_0 = (6.87 \times 10^6 \text{ m})(10.25 \times 10^3 \text{ m/s}) = 70.4 \times 10^9 \text{ m}^2/\text{s}$$

$$h^2 = 4.96 \times 10^{21} \text{ m}^4/\text{s}^2$$

Since $GM = gR^2$, where R is the radius of the earth, you also have

$$GM = gR^2 = (9.81 \text{ m/s}^2)(6.37 \times 10^6 \text{ m})^2 = 398 \times 10^{12} \text{ m}^3/\text{s}^2$$

$$\frac{GM}{h^2} = \frac{398 \times 10^{12} \text{ m}^3/\text{s}^2}{4.96 \times 10^{21} \text{ m}^4/\text{s}^2} = 80.3 \times 10^{-9} \text{ m}^{-1}$$

Substituting this value into Eq. (1) gives

$$\frac{1}{r} = 80.3 \times 10^{-9} \text{ m}^{-1} + C \cos \theta \quad (2)$$

Note that at point A, $\theta = 0$ and $r = r_0 = 6.87 \times 10^6$ m (Fig. 1). From this, you can compute the constant C as

$$\frac{1}{6.87 \times 10^6 \text{ m}} = 80.3 \times 10^{-9} \text{ m}^{-1} + C \cos 0^\circ \quad C = 65.3 \times 10^{-9} \text{ m}^{-1}$$

At A', which is the point on the orbit farthest from the earth, you have $\theta = 180^\circ$ (Fig. 1). Using Eq. (2), you can compute the corresponding distance r_1 to be

$$\frac{1}{r_1} = 80.3 \times 10^{-9} \text{ m}^{-1} + (65.3 \times 10^{-9} \text{ m}^{-1}) \cos 180^\circ$$

$$r_1 = 66.7 \times 10^6 \text{ m} = 66\,700 \text{ km}$$

$$\text{Maximum altitude} = 66\,700 \text{ km} - 6370 \text{ km} = 60\,300 \text{ km} \quad \blacktriangleleft$$

(continued)

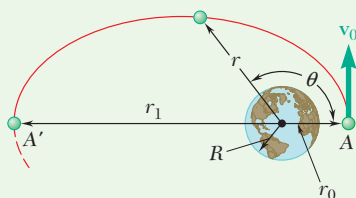


Fig. 1 Satellite orbit after launch velocity v_0 .

b. Periodic Time. Since A and A' are the perigee and apogee, respectively, of the elliptic orbit, use Eqs. (12.44) and (12.45) to compute the semimajor and semiminor axes of the orbit (Fig. 2):

$$a = \frac{1}{2}(r_0 + r_1) = \frac{1}{2}(6.87 + 66.7)(10^6) \text{ m} = 36.8 \times 10^6 \text{ m}$$

$$b = \sqrt{r_0 r_1} = \sqrt{(6.87)(66.7)} \times 10^6 \text{ m} = 21.4 \times 10^6 \text{ m}$$

$$\tau = \frac{2\pi ab}{h} = \frac{2\pi(36.8 \times 10^6 \text{ m})(21.4 \times 10^6 \text{ m})}{70.4 \times 10^9 \text{ m}^2/\text{s}}$$

$$\tau = 70.3 \times 10^3 \text{ s} = 1171 \text{ min} = 19 \text{ h } 31 \text{ min} \quad \blacktriangleleft$$

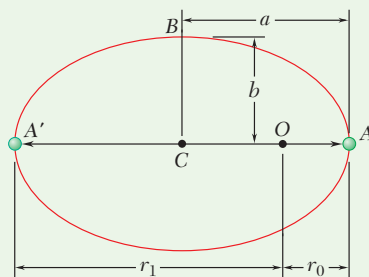


Fig. 2 Semimajor and semiminor axes of the orbit.

REFLECT and THINK: The satellite takes less than one day to travel over 60 000 km around the earth. In this problem, you started with Eq. 12.37, but it is important to remember that this formula was the solution to a differential equation that was derived using Newton's second law.

SOLVING PROBLEMS ON YOUR OWN

In this section, we continued our study of the motion of a particle under a central force and applied the results to problems in space mechanics. We found that the trajectory of a particle under a central force is defined by the differential equation

$$\frac{d^2u}{d\theta^2} + u = \frac{F}{mh^2u^2} \quad (12.35)$$

where u is the reciprocal of the distance r of the particle from the center of force ($u = 1/r$), F is the magnitude of the central force $\mathbf{F}$, and h is a constant equal to the angular momentum per unit mass of the particle. In space mechanics problems, $\mathbf{F}$ is the force of gravitational attraction exerted on the satellite or spacecraft by the sun, the earth, or other planet about which it travels. Substituting $F = GMm/r^2 = GMmu^2$ into Eq. (12.35), we obtain for that case

$$\frac{d^2u}{d\theta^2} + u = \frac{GM}{h^2} \quad (12.36)$$

where the right-hand side is a constant.

1. Analyzing the motion of satellites and spacecraft. The solution of the differential equation (12.36) defines the trajectory of a satellite or spacecraft. We obtained it in Sec. 12.3B in the alternative forms

$$\frac{1}{r} = \frac{GM}{h^2} + C \cos \theta \quad \text{or} \quad \frac{1}{r} = \frac{GM}{h^2} (1 + \varepsilon \cos \theta) \quad (12.37, 12.37')$$

Remember when applying these equations that $\theta = 0$ always corresponds to the perigee (the point of closest approach) of the trajectory (Fig. 12.18) and that h is a constant for a given trajectory. Depending on the value of the eccentricity ε , the trajectory is either a hyperbola, a parabola, or an ellipse.

a. $\varepsilon > 1$: The trajectory is a hyperbola. For this case, the spacecraft never returns to its starting point.

b. $\varepsilon = 1$: The trajectory is a parabola. This is the limiting case between open (hyperbolic) and closed (elliptic) trajectories. We had observed for this case that the velocity v_0 at the perigee is equal to the escape velocity v_{esc} . Hence,

$$v_0 = v_{\text{esc}} = \sqrt{\frac{2GM}{r_0}} \quad (12.41)$$

Note that the escape velocity is the smallest velocity for which the spacecraft does not return to its starting point.

c. $\varepsilon < 1$: The trajectory is an elliptic orbit. For problems involving elliptic orbits, you may find that the relation derived in Prob. 12.102

$$\frac{1}{r_0} + \frac{1}{r_1} = \frac{2GM}{h^2}$$

is useful in the solution of subsequent problems. When you apply this equation, remember that r_0 and r_1 are the distances from the center of force to the perigee ($\theta = 0$) and apogee ($\theta = 180^\circ$), respectively; that $h = r_0 v_0 = r_1 v_1$; and that, for a satellite orbiting the earth, $GM_{\text{earth}} = gR^2$, where R is the radius of the earth. Also recall that the trajectory is a circle when $\varepsilon = 0$.

2. Determining the point of impact of a descending spacecraft. For problems of this type, you may assume that the trajectory is elliptic and that the initial point of the descent trajectory is the apogee of the path (Fig. 12.21). Note that at the point of impact, the distance r in Eqs. (12.37) and (12.37') is equal to the radius R of the body on which the spacecraft lands or crashes. In addition, we have $h = Rv_1 \sin \phi_1$, where v_1 is the speed of the spacecraft at impact and ϕ_1 is the angle that its path forms with the vertical at the point of impact.

3. Calculating the time to travel between two points on a trajectory. For central-force motion, you can determine the time t required for a particle to travel along a portion of its trajectory by recalling from Sec. 12.2B that the rate at which area is swept per unit time by the position vector $\mathbf{r}$ is equal to one-half of the angular momentum per unit mass h of the particle: $dA/dt = h/2$. Since h is a constant for a given trajectory, it follows that

$$t = \frac{2A}{h}$$

where A is the total area swept in the time t .

a. In the case of an elliptic trajectory, the time required to complete one orbit is called the **periodic time** and is expressed as

$$\tau = \frac{2(\pi ab)}{h} \quad (12.43)$$

where a and b are the semimajor and semiminor axes, respectively, of the ellipse and are related to the distances r_0 and r_1 by

$$a = \frac{1}{2}(r_0 + r_1) \quad \text{and} \quad b = \sqrt{r_0 r_1} \quad (12.44, 12.45)$$

b. Kepler's third law provides a convenient relation between the periodic times of two satellites describing elliptic orbits about the same body [Sec. 12.3C]. Denoting the semimajor axes of the two orbits by a_1 and a_2 , respectively, and the corresponding periodic times by τ_1 and τ_2 , we have

$$\frac{\tau_1^2}{\tau_2^2} = \frac{a_1^3}{a_2^3}$$

c. In the case of a parabolic trajectory, you may be able to use the expression given on the inside of the front cover of this book for a parabolic or a semiparabolic area to calculate the time required to travel between two points of the trajectory.

Problems

CONCEPT QUESTIONS

12.CQ6 A uniform crate C with mass m is being transported to the left by a forklift with a constant speed v_1 . What is the magnitude of the angular momentum of the crate about point D , that is, the upper left corner of the crate?

- a. 0
- b. mv_1a
- c. mv_1b
- d. $mv_1\sqrt{a^2 + b^2}$

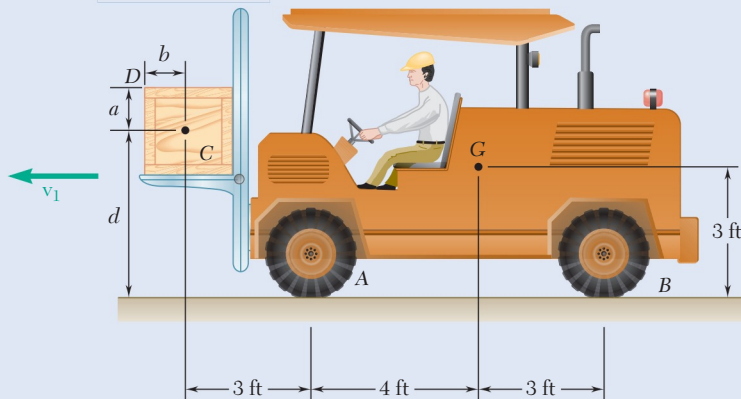


Fig. P12.CQ6 and P12.CQ7

12.CQ7 A uniform crate C with mass m is being transported to the left by a forklift with a constant speed v_1 . What is the magnitude of the angular momentum of the crate about point A , that is, the point of contact between the front tire of the forklift and the ground?

- a. 0
- b. mv_1d
- c. $3mv_1$
- d. $mv_1\sqrt{3^2 + d^2}$

END-OF-SECTION PROBLEMS

12.94 A particle of mass m is projected from point A with an initial velocity v_0 perpendicular to OA and moves under a central force $\mathbf{F}$ along an elliptic path defined by the equation $r = r_0/(2 - \cos \theta)$. Using Eq. (12.35), show that $\mathbf{F}$ is inversely proportional to the square of the distance r from the particle to the center of force O .

12.95 A particle of mass m describes the logarithmic spiral $r = r_0 e^{b\theta}$ under a central force $\mathbf{F}$ directed toward the center of force O . Using Eq. (12.35), show that $\mathbf{F}$ is inversely proportional to the cube of the distance r from the particle to O .

12.96 A particle with a mass m describes the path defined by the equation $r = r_0/(6 \cos \theta - 5)$ under a central force $\mathbf{F}$ directed away from the center of force O . Using Eq. (12.35), show that $\mathbf{F}$ is inversely proportional to the square of the distance r from the particle to O .

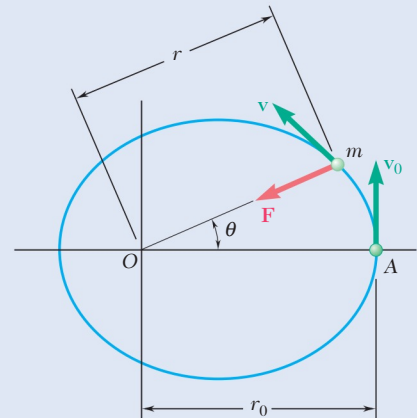


Fig. P12.94

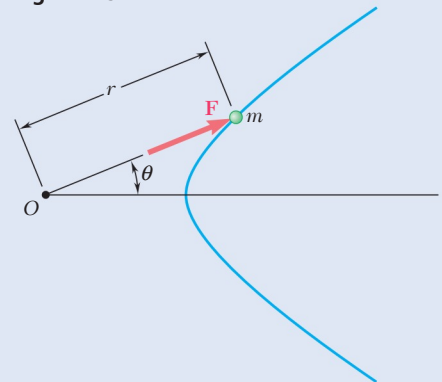


Fig. P12.96

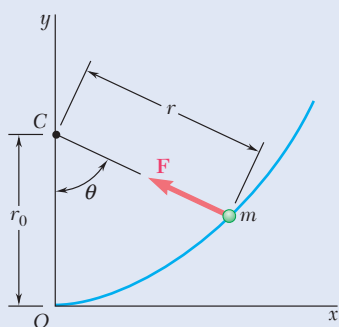


Fig. P12.97

12.97 A particle of mass m describes the parabola $y = x^2/4r_0$ under a central force $\mathbf{F}$ directed toward the center of force C . Using Eq. (12.35) and Eq. (12.37') with $\varepsilon = 1$, show that $\mathbf{F}$ is inversely proportional to the square of the distance r from the particle to the center of force and that the angular momentum per unit mass $h = \sqrt{2GMr_0}$.

12.98 It was observed that during its second flyby of the earth, the Galileo spacecraft had a velocity of 14.1 km/s as it reached its minimum altitude of 303 km above the surface of the earth. Determine the eccentricity of the trajectory of the spacecraft during this portion of its flight.

12.99 It was observed that during the Galileo spacecraft's first flyby of the earth, its minimum altitude was 600 mi above the surface of the earth. Assuming that the trajectory of the spacecraft was parabolic, determine the maximum velocity of Galileo during its first flyby of the earth.

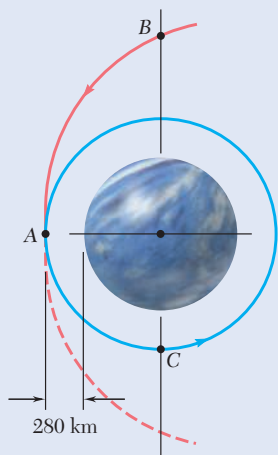


Fig. P12.100

12.100 As a space probe approaching the planet Venus on a parabolic trajectory reaches point A closest to the planet, its velocity is decreased to insert it into a circular orbit. Knowing that the mass and the radius of Venus are 4.87×10^{24} kg and 6052 km, respectively, determine (a) the velocity of the probe as it approaches A, (b) the decrease in velocity required to insert it into the circular orbit.

12.101 It was observed that as the Voyager I spacecraft reached the point of its trajectory closest to the planet Saturn, it was at a distance of 185×10^3 km from the center of the planet and had a velocity of 21.0 km/s. Knowing that Tethys, one of Saturn's moons, describes a circular orbit of radius 295×10^3 km at a speed of 11.35 km/s, determine the eccentricity of the trajectory of Voyager I on its approach to Saturn.

12.102 A satellite describes an elliptic orbit about a planet of mass M . Denoting by r_0 and r_1 , respectively, the minimum and maximum values of the distance r from the satellite to the center of the planet, derive the relation

$$\frac{1}{r_0} + \frac{1}{r_1} = \frac{2GM}{h^2}$$

where h is the angular momentum per unit mass of the satellite.

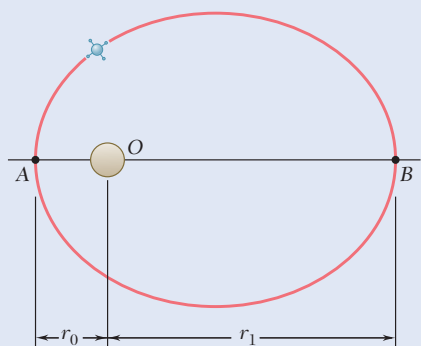


Fig. P12.102

12.103 A space probe is describing a circular orbit about a planet of radius R . The altitude of the probe above the surface of the planet is αR and its speed is v_0 . To place the probe in an elliptic orbit which will bring it closer to the planet, its speed is reduced from v_0 to βv_0 , where $\beta < 1$, by firing its engine for a short interval of time. Determine the smallest permissible value of β if the probe is not to crash on the surface of the planet.

- 12.104** A satellite describes a circular orbit at an altitude of 19 110 km above the surface of the earth. Determine (a) the increase in speed required at point A for the satellite to achieve the escape velocity and enter a parabolic orbit, (b) the decrease in speed required at point A for the satellite to enter an elliptic orbit with a minimum altitude of 6370 km, (c) the eccentricity ϵ of the elliptic orbit.

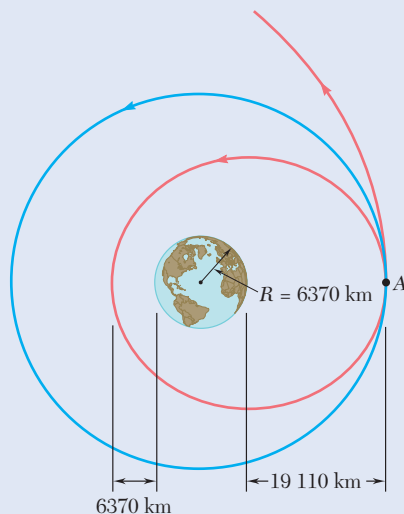


Fig. P12.104

- 12.105** A space probe is to be placed in a circular orbit of 5600-mi radius about the planet Venus in a specified plane. As the probe reaches A, the point of its original trajectory closest to Venus, it is inserted in a first elliptic transfer orbit by reducing its speed by Δv_A . This orbit brings it to point B with a much reduced velocity. There the probe is inserted in a second transfer orbit located in the specified plane by changing the direction of its velocity and further reducing its speed by Δv_B . Finally, as the probe reaches point C, it is inserted in the desired circular orbit by reducing its speed by Δv_C . Knowing that the mass of Venus is 0.82 times the mass of the earth, that $r_A = 9.3 \times 10^3$ mi and $r_B = 190 \times 10^3$ mi, and that the probe approaches A on a parabolic trajectory, determine by how much the velocity of the probe should be reduced (a) at A, (b) at B, (c) at C.

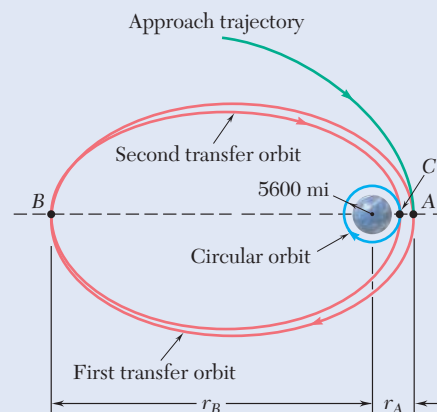


Fig. P12.105

- 12.106** For the space probe of Prob. 12.105, it is known that $r_A = 9.3 \times 10^3$ mi and that the velocity of the probe is reduced to 20,000 ft/s as it passes through A. Determine (a) the distance from the center of Venus to point B, (b) the amounts by which the velocity of the probe should be reduced at B and C, respectively.

- 12.107** As it describes an elliptic orbit about the sun, a spacecraft reaches a maximum distance of 202×10^6 mi from the center of the sun at point A (called the aphelion) and a minimum distance of 92×10^6 mi at point B (called the perihelion). To place the spacecraft in a smaller elliptic orbit with aphelion at A' and perihelion at B', where A' and B' are located 164.5×10^6 mi and 85.5×10^6 mi, respectively, from the center of the sun, the speed of the spacecraft is first reduced as it passes through A and then is further reduced as it passes through B'. Knowing that the mass of the sun is 332.8×10^3 times the mass of the earth, determine (a) the speed of the spacecraft at A, (b) the amounts by which the speed of the spacecraft should be reduced at A and B' to insert it into the desired elliptic orbit.

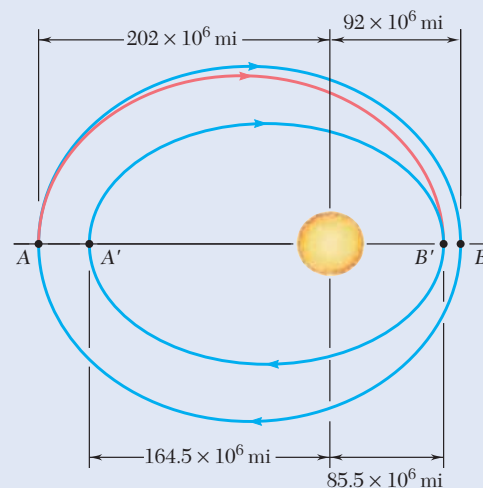


Fig. P12.107

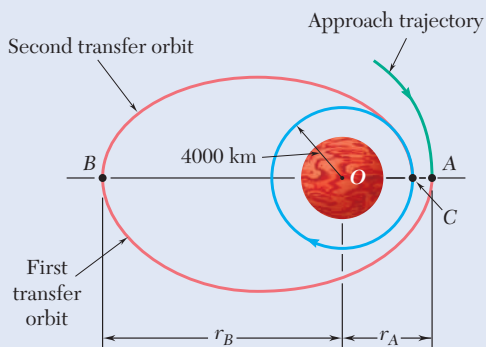


Fig. P12.110

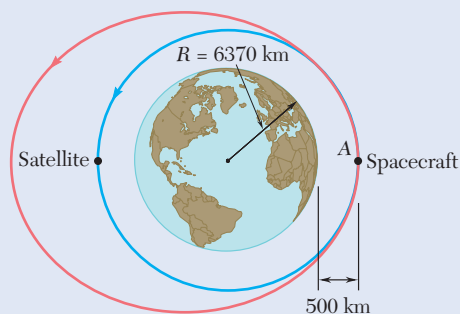


Fig. P12.111

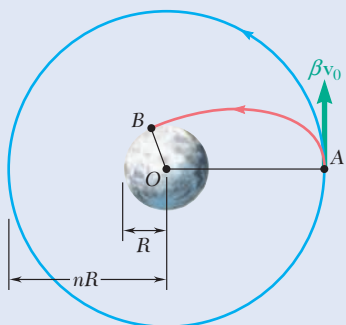


Fig. P12.114

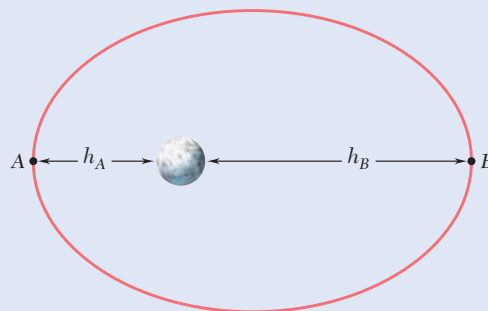


Fig. P12.112

12.108 Halley's comet travels in an elongated elliptic orbit for which the minimum distance from the sun is approximately $\frac{1}{2} r_E$, where $r_E = 150 \times 10^6$ km is the mean distance from the sun to the earth. Knowing that the periodic time of Halley's comet is about 76 years, determine the maximum distance from the sun reached by the comet.

12.109 Based on observations made during the 1996 sighting of comet Hyakutake, it was concluded that the trajectory of the comet is a highly elongated ellipse for which the eccentricity is approximately $\epsilon = 0.999887$. Knowing that for the 1996 sighting the minimum distance between the comet and the sun was $0.230R_E$, where R_E is the mean distance from the sun to the earth, determine the periodic time of the comet.

12.110 A space probe is to be placed in a circular orbit of radius 4000 km about the planet Mars. As the probe reaches A, the point of its original trajectory closest to Mars, it is inserted into a first elliptic transfer orbit by reducing its speed. This orbit brings it to point B with a much-reduced velocity. There the probe is inserted into a second transfer orbit by further reducing its speed. Knowing that the mass of Mars is 0.1074 times the mass of the earth, that $r_A = 9000$ km and $r_B = 180\,000$ km, and that the probe approaches A on a parabolic trajectory, determine the time needed for the space probe to travel from A to B on its first transfer orbit.

12.111 A spacecraft and a satellite are at diametrically opposite positions in the same circular orbit of altitude 500 km above the earth. As it passes through point A, the spacecraft fires its engine for a short interval of time to increase its speed and enter an elliptic orbit. Knowing that the spacecraft returns to A at the same time the satellite reaches A after completing one and a half orbits, determine (a) the increase in speed required, (b) the periodic time for the elliptic orbit.

12.112 The Clementine spacecraft described an elliptic orbit of minimum altitude $h_A = 400$ km and maximum altitude $h_B = 2940$ km above the surface of the moon. Knowing that the radius of the moon is 1737 km and that the mass of the moon is 0.01230 times the mass of the earth, determine the periodic time of the spacecraft.

12.113 Determine the time needed for the space probe of Prob. 12.100 to travel from B to C.

12.114 A space probe is describing a circular orbit of radius nR with a velocity v_0 about a planet of radius R and center O. As the probe passes through point A, its velocity is reduced from v_0 to βv_0 , where $\beta < 1$, to place the probe on a crash trajectory. Express in terms of n and β the angle AOB, where B denotes the point of impact of the probe on the planet.

12.115 A long-range ballistic trajectory between points A and B on the earth's surface consists of a portion of an ellipse with the apogee at point C . Knowing that point C is 1500 km above the surface of the earth and the range $R\phi$ of the trajectory is 6000 km, determine (a) the velocity of the projectile at C , (b) the eccentricity ε of the trajectory.

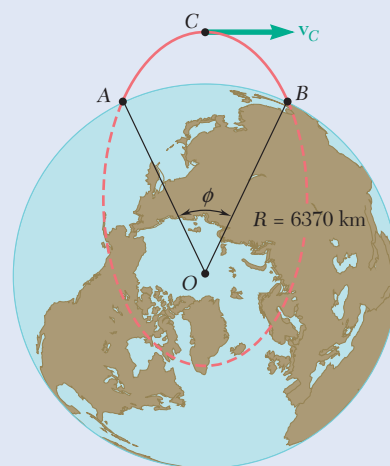


Fig. P12.115

12.116 A space shuttle is describing a circular orbit at an altitude of 563 km above the surface of the earth. As it passes through point A , it fires its engine for a short interval of time to reduce its speed by 152 m/s and begin its descent toward the earth. Determine the angle AOB so that the altitude of the shuttle at point B is 121 km. (Hint: Point A is the apogee of the elliptic descent trajectory.)

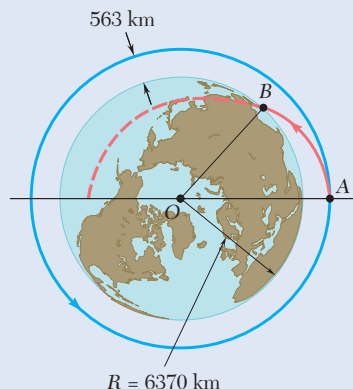


Fig. P12.116

12.117 As a spacecraft approaches the planet Jupiter, it releases a probe which is to enter the planet's atmosphere at point B at an altitude of 280 mi above the surface of the planet. The trajectory of the probe is a hyperbola of eccentricity $\varepsilon = 1.031$. Knowing that the radius and the mass of Jupiter are 44,423 mi and 1.30×10^{26} slug, respectively, and that the velocity $\mathbf{v}_B$ of the probe at B forms an angle of 82.9° with the direction of OA , determine (a) the angle AOB , (b) the speed v_B of the probe at B .

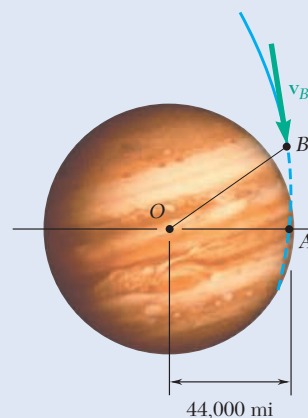


Fig. P12.117

12.118 A satellite describes an elliptic orbit about a planet. Denoting by r_0 and r_1 the distances corresponding, respectively, to the perigee and apogee of the orbit, show that the curvature of the orbit at each of these two points can be expressed as

$$\frac{1}{\rho} = \frac{1}{2} \left(\frac{1}{r_0} + \frac{1}{r_1} \right)$$

12.119 (a) Express the eccentricity ε of the elliptic orbit described by a satellite about a planet in terms of the distances r_0 and r_1 corresponding, respectively, to the perigee and apogee of the orbit. (b) Use the result obtained in part a and the data given in Prob. 12.109, where $R_E = 149.6 \times 10^6$ km, to determine the approximate maximum distance from the sun reached by comet Hyakutake.

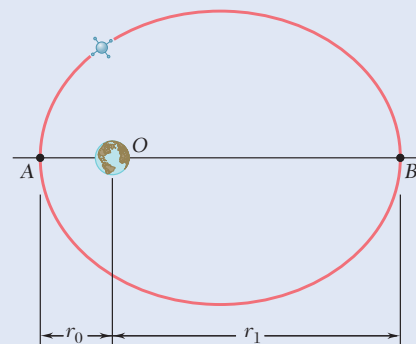


Fig. P12.118 and P12.119

12.120 Derive Kepler's third law of planetary motion from Eqs 12.37 and 12.43.

12.121 Show that the angular momentum per unit mass h of a satellite describing an elliptic orbit of semimajor axis a and eccentricity ε about a planet of mass M can be expressed as

$$h = \sqrt{GMa(1 - \varepsilon^2)}$$

Review and Summary

This chapter was devoted to Newton's second law and its application to analyzing the motion of particles.

Newton's Second Law

Denote the mass of a particle by m , the sum (or resultant) of the forces acting on the particle by $\Sigma \mathbf{F}$, and the acceleration of the particle relative to a newtonian frame of reference by $\mathbf{a}$ [Sec. 12.1A]. Then we have

$$\Sigma \mathbf{F} = m\mathbf{a} \quad (12.2)$$

Linear Momentum

Introducing the **linear momentum** of a particle, $\mathbf{L} = m\mathbf{v}$ [Sec. 12.1B], we saw that Newton's second law also can be written in the form

$$\Sigma \mathbf{F} = \dot{\mathbf{L}} \quad (12.5)$$

This equation states that **the resultant of the forces acting on a particle is equal to the rate of change of the linear momentum of the particle.**

Consistent Systems of Units

Equation (12.2) holds only if we use a consistent system of units. With SI units, the forces should be expressed in newtons, the masses in kilograms, and the accelerations in m/s^2 ; with U.S. customary units, the forces should be expressed in pounds, the masses in $\text{lb}\cdot\text{s}^2/\text{ft}$ (also referred to as *slugs*), and the accelerations in ft/s^2 [Sec. 12.1C].

Free-Body Diagram and Kinetic Diagram

A **free-body diagram** for a system shows the applied forces and a **kinetic diagram** shows the vector $m\mathbf{a}$ or its components. These diagrams provide a pictorial representation of Newton's second law. Drawing them will be of great help to you when writing the equations of motion. Note that when a problem involves two or more bodies, it is usually best to consider each body separately.

Equations of Motion for a Particle

To solve a problem involving the motion of a particle, we should first draw the free-body diagram and kinetic diagram for each particle in the system. Then we can use these diagrams to help us write equations containing scalar quantities (Sec. 12.1D). Using **rectangular components** of $\mathbf{F}$ and $\mathbf{a}$, we have

$$\Sigma F_x = ma_x \quad \Sigma F_y = ma_y \quad \Sigma F_z = ma_z \quad (12.8)$$

Using **tangential and normal components**, we have

$$\Sigma F_t = m \frac{dv}{dt} \quad \Sigma F_n = m \frac{v^2}{\rho} \quad (12.9')$$

Using **radial and transverse components**, we have

$$\Sigma F_r = m(\ddot{r} - r\dot{\theta}^2) \quad (12.11)$$

$$\Sigma F_\theta = m(r\ddot{\theta} + 2\dot{r}\dot{\theta}) \quad (12.12)$$

Sample Probs. 12.1 through 12.5 used rectangular components, Sample Probs. 12.6 through 12.9 used tangential and normal coordinates, and Sample Probs. 12.10 and 12.11 used radial and transverse coordinates.

Angular Momentum

In the second part of this chapter, we defined the **angular momentum** $\mathbf{H}_O$ of a particle about a point O as the moment about O of the linear momentum $m\mathbf{v}$ of that particle [Sec. 12.2A]. Thus,

$$\mathbf{H}_O = \mathbf{r} \times m\mathbf{v} \quad (12.13)$$

We noted that $\mathbf{H}_O$ is a vector perpendicular to the plane containing $\mathbf{r}$ and $m\mathbf{v}$ (Fig. 12.23) and has a magnitude of

$$H_O = rmv \sin \phi \quad (12.14)$$

Resolving the vectors $\mathbf{r}$ and $m\mathbf{v}$ into rectangular components, we expressed the angular momentum $\mathbf{H}_O$ in the determinant form

$$\mathbf{H}_O = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ x & y & z \\ mv_x & mv_y & mv_z \end{vmatrix} \quad (12.15)$$

In the case of a particle moving in the xy plane we have $z = v_z = 0$. The angular momentum is perpendicular to the xy plane and is completely defined by its magnitude. We have

$$H_O = H_z = m(xv_y - yv_x) \quad (12.17)$$

Rate of Change of Angular Momentum

Computing the rate of change $\dot{\mathbf{H}}_O$ of the angular momentum $\mathbf{H}_O$ and applying Newton's second law, we obtain the equation

$$\Sigma \mathbf{M}_O = \dot{\mathbf{H}}_O \quad (12.20)$$

This equation states that **the sum of the moments about O of the forces acting on a particle is equal to the rate of change of the angular momentum of the particle about O .**

Motion Under a Central Force

When the only force acting on a particle P is a force $\mathbf{F}$ directed toward or away from a fixed point O , the particle is said to be moving **under a central force** [Sec. 12.2B]. Since $\Sigma \mathbf{M}_O = 0$ at any given instant, it follows from Eq. (12.20) that $\dot{\mathbf{H}}_O = 0$ for all values of t and thus

$$\mathbf{H}_O = \text{constant} \quad (12.21)$$

We concluded that **the angular momentum of a particle moving under a central force is constant, both in magnitude and direction**, and that the particle moves in a plane perpendicular to the vector $\mathbf{H}_O$.

Recalling Eq. (12.14), we wrote the relation

$$rmv \sin \phi = r_0 m v_0 \sin \phi_0 \quad (12.23)$$

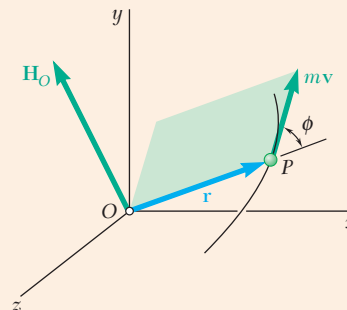


Fig. 12.23

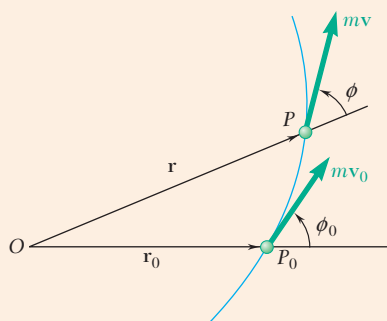


Fig. 12.24

for the motion of any particle under a central force (Fig. 12.24). Using polar coordinates and recalling Eq. (12.19), we also had

$$r^2 \dot{\theta} = h \quad (12.25)$$

where h is a constant representing the angular momentum per unit mass, H_O/m , of the particle. We observed (Fig. 12.25) that the infinitesimal area dA swept by the radius vector OP as it rotates through $d\theta$ is equal to $dA = \frac{1}{2}r^2 d\theta$ and, thus, that the left-hand side of Eq. (12.25) represents twice the **areal velocity** dA/dt of the particle. Therefore, **the areal velocity of a particle moving under a central force is constant.**

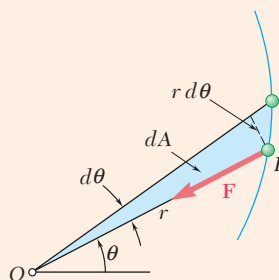


Fig. 12.25

Newton's Law of Universal Gravitation

An important application of the motion under a central force is provided by the orbital motion of bodies under gravitational attraction [Sec. 12.2C]. According to **Newton's law of universal gravitation**, two particles at a distance r from each other and of masses M and m , respectively, attract each other with equal and opposite forces $\mathbf{F}$ and $-\mathbf{F}$ directed along the line joining the particles (Fig. 12.26). The common magnitude F of the two forces is

$$F = G \frac{Mm}{r^2} \quad (12.26)$$

where G is the **constant of gravitation**. In the case of a body of mass m subjected to the gravitational attraction of the earth, we can express the product GM , where M is the mass of the earth, as

$$GM = gR^2 \quad (12.28)$$

where $g = 9.81 \text{ m/s}^2 = 32.2 \text{ ft/s}^2$ and R is the radius of the earth.

Orbital Motion

We showed in Sec. 12.3A that a particle moving under a central force describes a trajectory defined by the differential equation

$$\frac{d^2 u}{d\theta^2} + u = \frac{F}{mh^2 u^2} \quad (12.35)$$

where $F > 0$ corresponds to an attractive force and $u = 1/r$. In the case of a particle moving under a force of gravitational attraction [Sec. 12.12C], we substituted for F the expression given in Eq. (12.26). Measuring θ from the axis OA joining the focus O to the point A of the trajectory closest to O (Fig. 12.27), we found that the solution to Eq. (12.35) is

$$\frac{1}{r} = u = \frac{GM}{h^2} + C \cos \theta \quad (12.37)$$

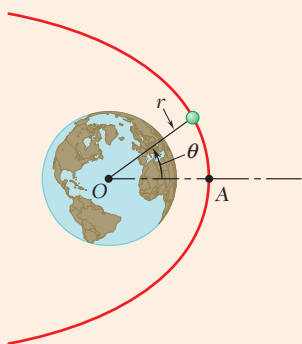


Fig. 12.27

This is the equation of a conic of eccentricity $\varepsilon = Ch^2/GM$. The conic is an **ellipse** if $\varepsilon < 1$, a **parabola** if $\varepsilon = 1$, and a **hyperbola** if $\varepsilon > 1$. We can determine the constants C and h from the initial conditions; if the particle is projected from point A ($\theta = 0$, $r = r_0$) with an initial velocity $\mathbf{v}_0$ that is perpendicular to OA , we have $h = r_0 v_0$ [Sample Prob. 12.14].

Escape Velocity

We also showed that the values of the initial velocity corresponding, respectively, to a parabolic and a circular trajectory are

$$v_{\text{esc}} = \sqrt{\frac{2GM}{r_0}} \quad (12.41)$$

$$v_{\text{circ}} = \sqrt{\frac{GM}{r_0}} \quad (12.42)$$

The first of these values, called the **escape velocity**, is the smallest value of v_0 for which the particle will not return to its starting point.

Periodic Time

The **periodic time** τ of a planet or satellite is defined as the time required by that body to describe its orbit. We showed that

$$\tau = \frac{2\pi ab}{h} \quad (12.43)$$

where $h = r_0 v_0$ and where a and b represent the semimajor and semiminor axes of the orbit. We further showed that these semiaxes are respectively equal to the arithmetic and geometric means of the maximum and minimum values of the radius r .

Kepler's Laws

The last part of the chapter [Sec. 12.3C] presented **Kepler's laws of planetary motion** and showed that these empirical laws, obtained from early astronomical observations, confirm Newton's laws of motion as well as his law of gravitation.

Review Problems

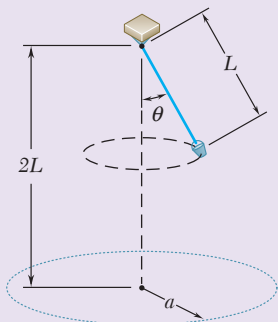


Fig. P12.123

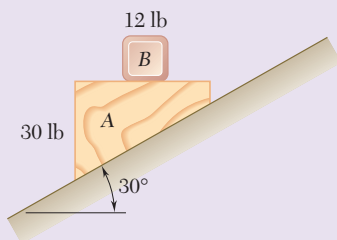


Fig. P12.124

12.122 In the braking test of a sports car its velocity is reduced from 70 mi/h to zero in a distance of 170 ft with slipping impending. Knowing that the coefficient of kinetic friction is 80 percent of the coefficient of static friction, determine (a) the coefficient of static friction, (b) the stopping distance for the same initial velocity if the car skids. Ignore air resistance and rolling resistance.

12.123 A bucket is attached to a rope of length $L = 1.2$ m and is made to revolve in a horizontal circle. Drops of water leaking from the bucket fall and strike the floor along the perimeter of a circle of radius a . Determine the radius a when $\theta = 30^\circ$.

12.124 A 12-lb block B rests as shown on the upper surface of a 30-lb wedge A . Neglecting friction, determine immediately after the system is released from rest (a) the acceleration of A , (b) the acceleration of B relative to A .

12.125 A 500-lb crate B is suspended from a cable attached to a 40-lb trolley A which rides on an inclined I-beam as shown. Knowing that at the instant shown the trolley has an acceleration of 1.2 ft/s^2 up and to the right, determine (a) the acceleration of B relative to A , (b) the tension in cable CD .

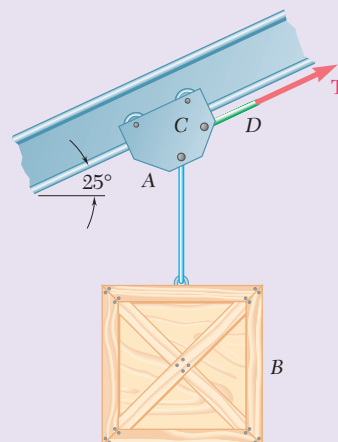


Fig. P12.125

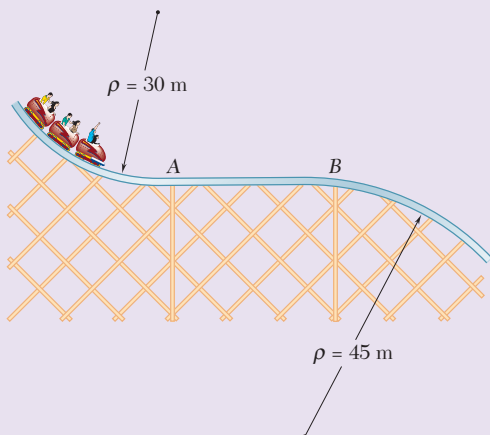


Fig. P12.126

12.126 The roller-coaster track shown is contained in a vertical plane. The portion of track between A and B is straight and horizontal, while the portions to the left of A and to the right of B have radii of curvature as indicated. A car is traveling at a speed of 72 km/h when the brakes are suddenly applied, causing the wheels of the car to slide on the track ($\mu_k = 0.25$). Determine the initial deceleration of the car if the brakes are applied as the car (a) has almost reached A , (b) is traveling between A and B , (c) has just passed B .

- 12.127** The parasailing system shown uses a winch to pull the rider in towards the boat, which is travelling with a constant velocity. During the interval when θ is between 20° and 40° (where $t = 0$ at $\theta = 20^\circ$), the angle increases at the constant rate of $2^\circ/\text{s}$. During this time, the length of the rope is defined by the relationship $r = 125 - \frac{1}{3}t^{3/2}$, where r and t are expressed in meters and seconds, respectively. At the instant when the rope makes a 30° angle with the water, the tension in the rope is 18 kN. At this instant, what is the magnitude and direction of the force of the parasail on the 75 kg parasailor?



Fig. P12.127

- 12.128** A small 200-g collar C can slide on a semicircular rod that is made to rotate about the vertical AB at the constant rate of 6 rad/s. Determine the minimum required value of the coefficient of static friction between the collar and the rod if the collar is not to slide when (a) $\theta = 90^\circ$, (b) $\theta = 75^\circ$, (c) $\theta = 45^\circ$. Indicate in each case the direction of the impending motion.
- 12.129** Telemetry technology is used to quantify kinematic values of a 200-kg roller-coaster cart as it passes overhead. According to the system, $r_c = 25$ m, $\dot{r} = -10$ m/s, $\ddot{r} = -2$ m/s², $\theta = 90^\circ$, $\dot{\theta} = -0.4$ rad/s, $\ddot{\theta} = -0.32$ rad/s². At this instant, determine (a) the normal force between the cart and the track, (b) the radius of curvature of the track.

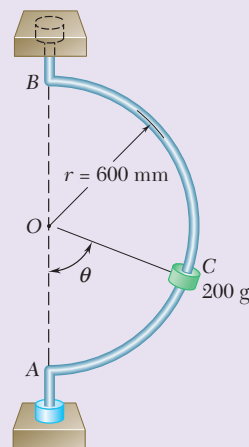


Fig. P12.128

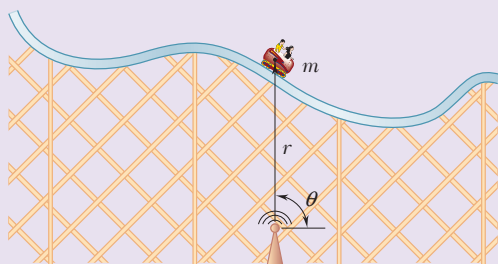


Fig. P12.129

12.130 The radius of the orbit of a moon of a given planet is equal to twice the radius of that planet. Denoting by ρ the mean density of the planet, show that the time required by the moon to complete one full revolution about the planet is $(24\pi/G\rho)^{1/2}$, where G is the constant of gravitation.

12.131 At engine burnout on a mission, a shuttle had reached point A at an altitude of 40 mi above the surface of the earth and had a horizontal velocity v_0 . Knowing that its first orbit was elliptic and that the shuttle was transferred to a circular orbit as it passed through point B at an altitude of 170 mi, determine (a) the time needed for the shuttle to travel from A to B on its original elliptic orbit, (b) the periodic time of the shuttle on its final circular orbit.

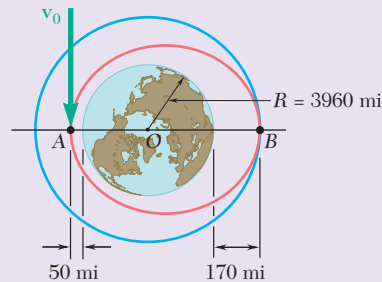


Fig. P12.131

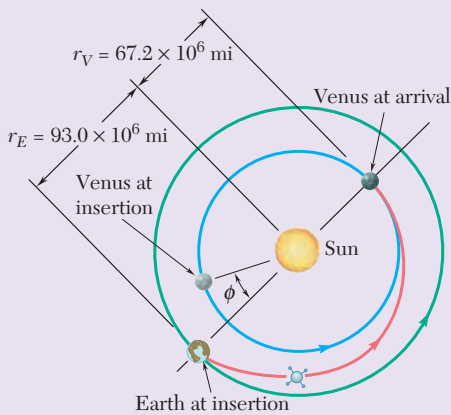


Fig. P12.132

12.132 A space probe in a low earth orbit is inserted into an elliptic transfer orbit to the planet Venus. Knowing that the mass of the sun is 332.8×10^3 times the mass of the earth and assuming that the probe is subjected only to the gravitational attraction of the sun, determine the value of ϕ , which defines the relative position of Venus with respect to the earth at the time the probe is inserted into the transfer orbit.

***12.133** Disk A rotates in a horizontal plane about a vertical axis at the constant rate $\dot{\theta}_0 = 10$ rad/s. Slider B has mass 1 kg and moves in a frictionless slot cut in the disk. The slider is attached to a spring of constant k , which is undeformed when $r = 0$. Knowing that the slider is released with no radial velocity in the position $r = 500$ mm, determine the position of the slider and the horizontal force exerted on it by the disk at $t = 0.1$ s for (a) $k = 100$ N/m, (b) $k = 200$ N/m.

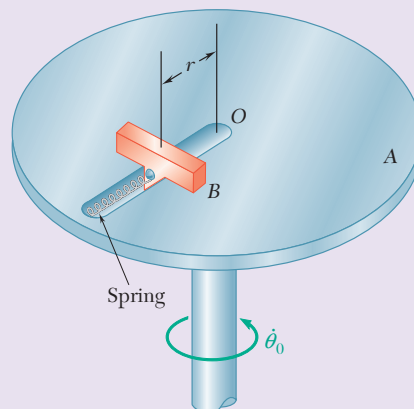


Fig. P12.133



13

Kinetics of Particles: Energy and Momentum Methods

A golf ball will deform upon impact as shown by this high-speed photo. The maximum deformation will occur when the club head velocity and the ball velocity are the same. In this chapter impacts will be analyzed using the coefficient of restitution and conservation of linear momentum. The kinetics of particles using energy and momentum methods is the subject of this chapter.

Introduction

13.1 WORK AND ENERGY

13.1A Work of a Force

13.1B Principle of Work and Energy

13.1C Applications of the Principle of Work and Energy

13.1D Power and Efficiency

13.2 CONSERVATION OF ENERGY

13.2A Potential Energy

*13.2B Conservative Forces

13.2C The Principle of Conservation of Energy

13.2D Application to Space Mechanics: Motion Under a Conservative Central Force

13.3 IMPULSE AND MOMENTUM

13.3A Principle of Impulse and Momentum

13.2B Impulsive Motion

13.4 IMPACTS

13.4A Direct Central Impact

13.4B Oblique Central Impact

13.4C Problems Involving Multiple Principles

Objectives

- **Calculate** the work done by a force.
- **Calculate** the kinetic energy of a particle.
- **Calculate** the gravitational and elastic potential energy of a system.
- **Solve** particle kinetics problems using the principle of work and energy.
- **Calculate** the power and efficiency of a mechanical system.
- **Solve** particle kinetics problems using conservation of energy.
- **Solve** particle kinetic problems involving conservative central forces.
- **Draw** complete and accurate impulse-momentum diagrams.
- **Solve** particle kinetics problems using the principle of impulse and momentum.
- **Solve** particle kinetics problems using conservation of linear momentum.
- **Solve** impact problems using the principle of impact and momentum and the coefficient of restitution.
- **Determine** the appropriate principle(s) to apply when solving a particle dynamics problem.
- **Solve** multi-step dynamics problems using multiple kinetics principles.

Introduction

In the preceding chapter, we solved most problems dealing with the motion of particles through the use of the fundamental equation of motion $\mathbf{F} = m\mathbf{a}$. Given a particle acted upon by a force $\mathbf{F}$, we could solve this equation for the acceleration $\mathbf{a}$; then by applying the principles of kinematics, we could determine from $\mathbf{a}$ the velocity and position of the particle at any time.

However, using the general equation $\mathbf{F} = m\mathbf{a}$ together with kinematics allows us to obtain two additional concepts: the **principle of work and energy** and the **principle of impulse and momentum**. The advantage of these ideas lies in the fact that they make the determination of the acceleration unnecessary. Indeed, the principle of work and energy directly relates force, mass, velocity, and displacement, whereas the principle of impulse and momentum relates force, mass, velocity, and time.

We present work and energy first. In Sec. 13.1, we define the *work of a force* and the *kinetic energy of a particle*. Then we apply the principle of work and energy to the solution of engineering problems. We also introduce the concepts of *power* and *efficiency* of a machine, which are important in engineering applications such as motors and hydraulic actuators.

In Sec. 13.2, we examine the concept of *potential energy* of a conservative force, and we apply the principle of conservation of energy to various problems of practical interest. In Sec. 13.2D, we use the principles of conservation of energy and of conservation of angular momentum jointly to solve problems of space mechanics.

The second part of this chapter deals with the principle of impulse and momentum and its application to the study of the motion of a particle. You will see in Sec. 13.3B that this principle is particularly effective in the study of the *impulsive motion* of a particle, where very large forces act for a very short time interval—like hitting a nail with a hammer.

We also consider the *central impact* of two bodies. We will show that a relation exists between the relative velocities of the two colliding bodies before and after impact. We can use this relation, together with the fact that the total momentum of the two bodies is conserved, to solve several types of practical problems.

Finally, we will discuss how to choose the best principle for solving a given problem from among Newton's second law, work and energy, or impulse and momentum. You may even need to apply multiple principles in order to solve some dynamics problems.

13.1 WORK AND ENERGY

Work and energy have very specific meanings in science and engineering. In everyday speech, you might say that holding up a concrete block is a lot of work, but in science, if the block doesn't move, you don't do any work at all while holding it. Similarly, people talk about energy all the time, from how you feel on a particular day ("I don't seem to have much energy today") to national and international policy ("The high cost of energy is affecting our trade balance with other countries."). In science and engineering, work and energy have very specific definitions that involve forces, displacements, masses, and velocities. These two concepts are of great value in analyzing a wide range of engineering problems.

13.1A Work of a Force

We first define the terms *displacement* and *work* as they are used in mechanics.[†] Consider a particle that moves from a point A to a neighboring point A' (Fig. 13.1). If $\mathbf{r}$ denotes the position vector corresponding to point A , we can denote the small vector joining A and A' by the differential $d\mathbf{r}$; the vector $d\mathbf{r}$ is called the **displacement** of the particle. Now, let us assume that a force $\mathbf{F}$ is acting on the particle. We define the **work of the force $\mathbf{F}$ corresponding to the displacement $d\mathbf{r}$** as the quantity

$$dU = \mathbf{F} \cdot d\mathbf{r} \quad (13.1)$$

We obtain dU by taking the scalar product of the force $\mathbf{F}$ and the displacement $d\mathbf{r}$. We denote the magnitudes of the force and of the displacement by F and ds , respectively, and the angle formed by $\mathbf{F}$ and

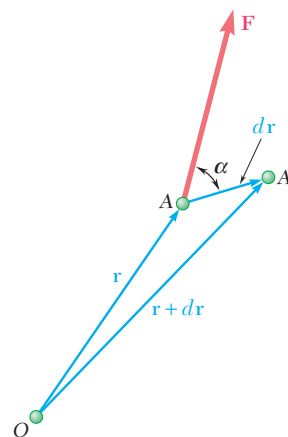


Fig. 13.1 The work of a force acting on a particle is the scalar product of the force $\mathbf{F}$ and the displacement $d\mathbf{r}$ of the particle.

[†]We defined work in Sec. 10.1A and outlined its basic properties in Secs. 10.1A and 10.2A. For convenience, we repeat here the portions of this material that relate to the kinetics of particles.

$d\mathbf{r}$ by α . Then from the definition of the scalar product of two vectors (Sec. 3.2A), we have

$$dU = F ds \cos \alpha \quad (13.1')$$

Using Eq. (3.30), we can also express the work dU in terms of the rectangular components of the force and of the displacement:

$$dU = F_x dx + F_y dy + F_z dz \quad (13.1'')$$

Work is a scalar quantity, so it has a magnitude and a sign but no direction.

Note that work is expressed in units obtained by multiplying units of length by units of force. Thus, if we use U.S. customary units, work is expressed in ft·lb or in·lb. If we use SI units, work is expressed in N·m. The unit of work N·m is called a **joule** (J).[†] Recalling the conversion factors indicated in Sec. 12.1C, we have

$$1 \text{ ft} \cdot \text{lb} = (1 \text{ ft})(1 \text{ lb}) = (0.3048 \text{ m})(4.448 \text{ N}) = 1.356 \text{ J}$$

It follows from Eq. (13.1') that the work dU is positive if angle α is acute and negative if α is obtuse. Three particular cases are of special interest. If the force $\mathbf{F}$ has the same direction as $d\mathbf{r}$, the work dU reduces to $F ds$. If $\mathbf{F}$ has a direction opposite to that of $d\mathbf{r}$, the work is $dU = -F ds$. Finally, if $\mathbf{F}$ is perpendicular to $d\mathbf{r}$, the work dU is zero.

We can obtain the work of $\mathbf{F}$ during a *finite* displacement of the particle from A_1 to A_2 (Fig. 13.2a) by integrating Eq. (13.1) along the path described by the particle. This work, denoted by $U_{1 \rightarrow 2}$, is

Work of a force

$$U_{1 \rightarrow 2} = \int_{A_1}^{A_2} \mathbf{F} \cdot d\mathbf{r} \quad (13.2)$$

Using the alternative expression of Eq. (13.1') for the elementary work dU , and observing that $F \cos \alpha$ represents the tangential component F_t of the force, we can also express the work $U_{1 \rightarrow 2}$ as

$$U_{1 \rightarrow 2} = \int_{s_1}^{s_2} (F \cos \alpha) ds = \int_{s_1}^{s_2} F_t ds \quad (13.2')$$

where the variable of integration s measures the distance traveled by the particle along the path. The work $U_{1 \rightarrow 2}$ is represented by the area under the curve obtained by plotting $F_t = F \cos \alpha$ against s (Fig. 13.2b).

When the force $\mathbf{F}$ is defined by its rectangular components, we can use the expression of Eq. (13.1'') for the elementary work. We have

$$U_{1 \rightarrow 2} = \int_{A_1}^{A_2} (F_x dx + F_y dy + F_z dz) \quad (13.2'')$$

where the integration is performed along the path described by the particle.

[†]The joule (J) is the SI unit of energy, whether in mechanical form (work, potential energy, or kinetic energy) or in chemical, electrical, or thermal form. Note that even though $\text{N} \cdot \text{m} = \text{J}$, the moment of a force must be expressed in $\text{N} \cdot \text{m}$ and not in joules, since the moment of a force is not a form of energy.

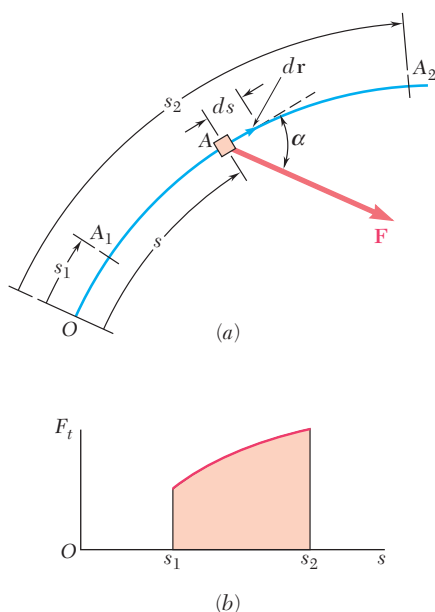


Fig. 13.2 (a) The work of force $\mathbf{F}$ over a finite displacement is the integral of Eq. (13.1) from point A_1 to point A_2 . (b) The work is represented by the area under the graph of F_t versus s from s_1 to s_2 .

We can use these equations to derive formulas for the work done by a force in several common and important situations, as we now show. These formulas can simplify the calculations needed to solve many common problems. For other situations, you can return to the basic Equations (13.1) and (13.2) and their variants.

Work of a Constant Force in Rectilinear Motion. When a particle moving in a straight line is acted upon by a force $\mathbf{F}$ of constant magnitude and of constant direction (Fig. 13.3), formula (13.2') yields

$$U_{1 \rightarrow 2} = (F \cos \alpha) \Delta x \quad (13.3)$$

where α = angle the force forms with direction of motion

Δx = displacement from A_1 to A_2

Work of the Force of Gravity. We can obtain the work of the weight $\mathbf{W}$ of a body—i.e., of the force of gravity exerted on that body—by substituting the components of $\mathbf{W}$ into Eqs. (13.1'') and (13.2''). Choosing the y axis upward (Fig. 13.4), we have $F_x = 0$, $F_y = -W$, and $F_z = 0$. This gives us

$$\begin{aligned} dU &= -W dy \\ U_{1 \rightarrow 2} &= - \int_{y_1}^{y_2} W dy = Wy_1 - Wy_2 \end{aligned} \quad (13.4)$$

or

$$U_{1 \rightarrow 2} = -W(y_2 - y_1) = -W \Delta y \quad (13.4')$$

where Δy is the vertical displacement from A_1 to A_2 . The work of the weight $\mathbf{W}$ is thus equal to **the product of W and the vertical displacement of the center of gravity of the body**. The work is positive when $\Delta y < 0$, that is, when the body moves down. When the body moves up (and $\Delta y > 0$), the force and displacement are in opposite directions, and the work is negative.

Work of the Force Exerted by a Spring. Consider a body A attached to a fixed point B by a spring; we assume that the spring is undeformed when the body is at A_0 (Fig. 13.5a). For a linear spring, the magnitude of the force $\mathbf{F}$ exerted by the spring on body A is proportional to the deflection x of the spring measured from the unstretched position A_0 (i.e., $x = L_{\text{stretched}} - L_{\text{unstretched}}$). We have

$$F = kx \quad (13.5)$$

where k is the **spring constant** expressed in N/m or kN/m if SI units are used and in lb/ft or lb/in. if U.S. customary units are used.[†]

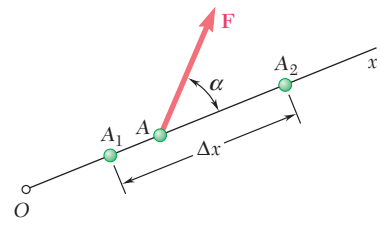


Fig. 13.3 For a constant force in rectilinear motion, the work equals the displacement times the component of force in the direction of the displacement.

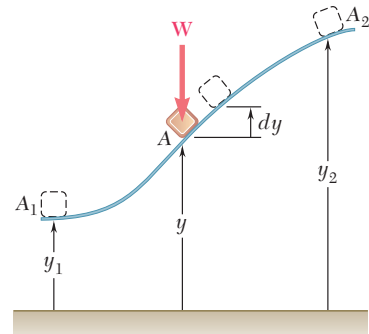


Fig. 13.4 The work done by the force of gravity is the product of the weight and the vertical displacement of the object's center of gravity. If the object moves up, the work done by gravity is negative.

[†]The relation $F = kx$ is correct under static conditions only. Under dynamic conditions, Eq. (13.5) should be modified to take into account the inertia of the spring. However, the error introduced by using $F = kx$ in the solution of kinetics problems is small if the mass of the spring is small compared with the other masses in motion.

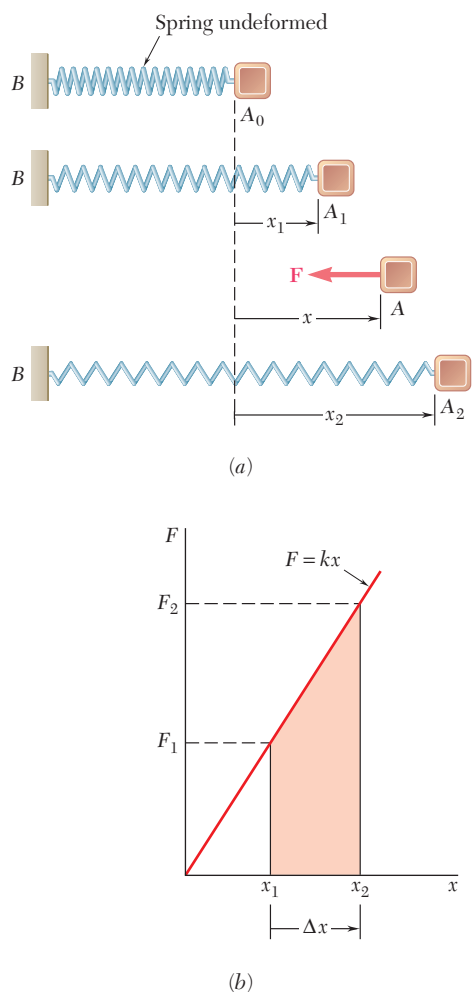


Fig. 13.5 (a) The work of a force exerted by a spring depends on the spring constant and the initial and final positions of the spring. (b) The work is represented by the area under the graph of force versus position.

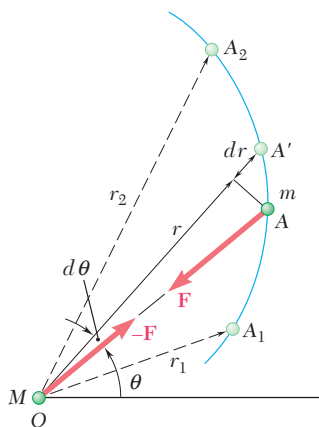


Fig. 13.6 The work of a gravitational force depends on the gravitational constant, the masses of the interacting bodies, and the radial distance between them.

We can obtain the work of force $\mathbf{F}$ exerted by the spring during a finite displacement of the body from $A_1(x = x_1)$ to $A_2(x = x_2)$ by writing

$$dU = -F dx = -kx dx$$

$$U_{1 \rightarrow 2} = - \int_{x_1}^{x_2} kx dx = \frac{1}{2} kx_1^2 - \frac{1}{2} kx_2^2 \quad (13.6)$$

You need to be careful in expressing k and x in consistent units. For example, if you use U.S. customary units, k should be expressed in lb/ft and x in feet, or k should be given in lb/in. and x in inches. In the first case, the work will have units of ft·lb; in the second case, it will have units of in·lb. Note that the work of force $\mathbf{F}$ exerted by the spring on the body is positive when $x_2 < x_1$; that is, when the spring is returning to its undeformed position. When the body is moved from x_1 to x_2 , the work of the force is negative, since the displacement and force are in opposite directions.

Since Eq. (13.5) is the equation of a straight line of slope k passing through the origin, we can also obtain the work $U_{1 \rightarrow 2}$ of $\mathbf{F}$ during the displacement from A_1 to A_2 by evaluating the area of the trapezoid shown in Fig. 13.5b. We can do this by computing F_1 and F_2 and multiplying the base Δx of the trapezoid by its mean height $\frac{1}{2}(F_1 + F_2)$. Since the work of the force $\mathbf{F}$ exerted by the spring is positive for a negative value of Δx , we have

$$U_{1 \rightarrow 2} = -\frac{1}{2}(F_1 + F_2) \Delta x \quad (13.6')$$

Work of a Gravitational Force. We saw in Sec. 12.2C that two particles of mass M and m separated by a distance r attract each other with equal and opposite forces $\mathbf{F}$ and $-\mathbf{F}$, directed along the line joining the particles and of magnitude as

$$F = G \frac{Mm}{r^2}$$

Let us assume that particle M occupies a fixed position O while particle m moves along the path shown in Fig. 13.6. We can obtain the work of force $\mathbf{F}$ exerted on particle m during an infinitesimal displacement of the particle from A to A' by multiplying the magnitude F of the force by the radial component dr of the displacement. Since $\mathbf{F}$ is directed toward O and dr is directed away from O , the work is negative, and we have

$$dU = -F dr = -G \frac{Mm}{r^2} dr$$

The work of the gravitational force $\mathbf{F}$ during a finite displacement from $A_1(r = r_1)$ to $A_2(r = r_2)$ is therefore

$$U_{1 \rightarrow 2} = - \int_{r_1}^{r_2} \frac{GMm}{r^2} dr = \frac{GMm}{r_2} - \frac{GMm}{r_1} \quad (13.7)$$

where M is the mass of the earth. We can use this formula to determine the work of the force exerted by the earth on a body of mass m at a distance r from the earth's center when r is larger than the radius R of the earth. Recalling the first of the relations in Eq. (12.27), we can replace the product GMm in Eq. (13.7) by WR^2 , where R is the earth's radius

($R = 6.37 \times 10^6$ m or 3960 mi) and W is the weight of the body at the earth's surface.

Some forces frequently encountered in kinetics problems *do no work*. They are forces applied to fixed points ($ds = 0$) or acting in a direction perpendicular to the displacement ($\cos \alpha = 0$). Forces that do no work include the reaction at a frictionless pin when the body supported rotates about the pin; the normal force at a frictionless fixed surface when the body in contact moves along the surface; the reaction at a roller moving along its track; and the weight of a body when its center of gravity moves horizontally.

13.1B Principle of Work and Energy

Consider a particle of mass m acted upon by a force $\mathbf{F}$ and moving along a path that is either rectilinear or curved (Fig. 13.7). Expressing Newton's second law in terms of the tangential components of the force and of the acceleration (see Sec. 12.1D), we have

$$F_t = ma_t \quad \text{or} \quad F_t = m \frac{dv}{dt}$$

where v is the speed of the particle. Recalling from Sec. 11.4A that $v = ds/dt$, we obtain

$$F_t = m \frac{dv}{ds} \frac{ds}{dt} = mv \frac{dv}{ds}$$

$$F_t ds = mv dv$$

Integrating from A_1 , where $s = s_1$ and $v = v_1$, to A_2 , where $s = s_2$ and $v = v_2$, we have

$$\int_{s_1}^{s_2} F_t ds = m \int_{v_1}^{v_2} v dv = \frac{1}{2}mv_2^2 - \frac{1}{2}mv_1^2 \quad (13.8)$$

The left-hand side of Eq. (13.8) represents the work $U_{1 \rightarrow 2}$ of the force $\mathbf{F}$ exerted on the particle during the displacement from A_1 to A_2 ; as indicated earlier, the work $U_{1 \rightarrow 2}$ is a scalar quantity. Thus, the expression $\frac{1}{2}mv^2$ is also a scalar quantity. We define it as the kinetic energy of the particle, denoted by T . That is,

**Kinetic energy
of a particle**

$$T = \frac{1}{2}mv^2 \quad (13.9)$$

Substituting into Eq. (13.8), we have

**Principle of work
and energy**

$$U_{1 \rightarrow 2} = T_2 - T_1 \quad (13.10)$$

This equation states that when a particle moves from A_1 to A_2 under the action of a force $\mathbf{F}$, **the work of the force $\mathbf{F}$ is equal to the change in kinetic energy of the particle**. This is known as the **principle of work and energy**. Rearranging the terms in Eq. (13.10) gives

$$T_1 + U_{1 \rightarrow 2} = T_2 \quad (13.11)$$

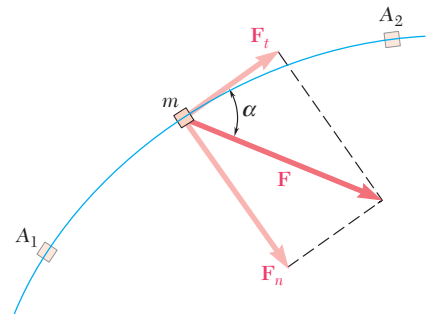


Fig. 13.7 A particle m acted upon by a force $\mathbf{F}$.

Like Newton's second law from which it is derived, the principle of work and energy applies only with respect to a newtonian frame of reference (Sec. 12.1A). The speed v used to determine the kinetic energy T therefore should be measured with respect to a newtonian frame of reference.

Since both work and kinetic energy are scalar quantities, we can compute their sum as an ordinary algebraic sum with the work $U_{1 \rightarrow 2}$ being positive or negative according to the direction of $\mathbf{F}$. When several forces act on the particle, the expression $U_{1 \rightarrow 2}$ represents the total work of the forces acting on the particle; it is obtained by adding algebraically the work of the various forces.

As just noted, the kinetic energy of a particle is a scalar quantity. It further appears from the definition $T = \frac{1}{2}mv^2$ that, regardless of the particle's direction of motion, the kinetic energy is always positive. Considering the particular case when $v_1 = 0$ and $v_2 = v$, and substituting $T_1 = 0$ and $T_2 = T$ into Eq. (13.10), we observe that the work done by the forces acting on the particle is equal to T . Thus, the kinetic energy of a particle moving with a speed v represents the work that must be done to bring the particle from rest to the speed v . Substituting $T_1 = T$ and $T_2 = 0$ into Eq. (13.10), we also note that when a particle moving with a speed v is brought to rest, the work done by the forces acting on the particle is $-T$. Assuming that no energy is dissipated into heat, we conclude that the work done by the forces exerted *by the particle* on the bodies that cause it to come to rest is equal to T . Thus, the kinetic energy of a particle also represents **the capacity to do work associated with the speed of the particle**.

The kinetic energy is measured in the same units as work, i.e., in joules if we use SI units and in ft·lb if we use U.S. customary units. We check that, in SI units,

$$T = \frac{1}{2}mv^2 = \text{kg}(\text{m/s})^2 = (\text{kg} \cdot \text{m/s}^2)\text{m} = \text{N} \cdot \text{m} = \text{J}$$

whereas in customary units,

$$T = \frac{1}{2}mv^2 = (\text{slug})(\text{ft/s})^2 = (\text{lb} \cdot \text{s}^2/\text{ft})(\text{ft/s})^2 = \text{ft} \cdot \text{lb}$$

13.1C Applications of the Principle of Work and Energy

Using the principle of work and energy greatly simplifies the solution of many problems involving forces, displacements, and velocities. Consider, for example, the pendulum OA consisting of a bob A of weight W attached to a cord of length l (Fig. 13.8a). The pendulum is released with no initial velocity from a horizontal position OA_1 and allowed to swing in a vertical plane. We wish to determine the speed of the bob as it passes through A_2 , directly under O .

We first determine the work done during the displacement from A_1 to A_2 by the forces acting on the bob. We draw a free-body diagram of the bob, showing all the *actual* forces acting on it; i.e., the weight $\mathbf{W}$ and the force $\mathbf{P}$ exerted by the cord (Fig. 13.8b). (Recall that an inertia vector is not an actual force and *should not* be included in the free-body diagram.) Note that force $\mathbf{P}$ does no work, since it is normal to the path; the only force that does work is thus the weight $\mathbf{W}$. We obtain the work of $\mathbf{W}$ by

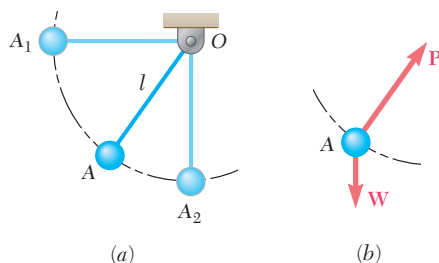


Fig. 13.8 (a) A bob of weight W swings from an initial position A_1 to a final position A_2 ; (b) free-body diagram of the bob at position A .

multiplying its magnitude W by the vertical displacement l (Sec. 13.1A); since the displacement is downward, the work is positive. We therefore have $U_{1 \rightarrow 2} = Wl$.

Now consider the kinetic energy of the bob. We have $T_1 = 0$ at A_1 and $T_2 = \frac{1}{2}(W/g)v_2^2$ at A_2 . We can now apply the principle of work and energy. From Eq. (13.11), we have

$$T_1 + U_{1 \rightarrow 2} = T_2 \quad 0 + Wl = \frac{1}{2} \frac{W}{g} v_2^2$$

Solving for v_2 , we find $v_2 = \sqrt{2gl}$. Note that this speed is also that of a body falling freely from a height l .

This example illustrates the following advantages of the method of work and energy:

1. In order to find the speed at A_2 , there is no need to determine the acceleration in an intermediate position A and to integrate the acceleration expression from A_1 to A_2 .
2. All quantities involved are scalars and can be added directly, without using x and y components.
3. Forces that do no work are eliminated from the solution of the problem.

What is an advantage in one problem, however, may be a disadvantage in another. It is evident, for instance, that the method of work and energy cannot be used to directly determine an acceleration. It is also evident that to determine a force that is normal to the path of the particle (i.e., a force that does no work) we must supplement the method of work and energy by the direct application of Newton's second law. Suppose, for example, that we wish to determine the tension in the cord of the pendulum of Fig. 13.8a as the bob passes through A_2 . We draw a free-body diagram and kinetic diagram of the bob in that position (Fig. 13.9) and express Newton's second law in terms of tangential and normal components. The equations $\Sigma F_t = ma_t$ and $\Sigma F_n = ma_n$ yield, respectively, $a_t = 0$ and

$$P - W = ma_n = \frac{W}{g} \frac{v_2^2}{l}$$

But earlier, we determined the speed at A_2 by the method of work and energy. Substituting $v_2^2 = 2gl$ and solving for P , we have

$$P = W + \frac{W}{g} \frac{2gl}{l} = 3W$$

If we used only statics principles and designed the cord to hold the weight of the bob (or even twice the weight of the bob), the cord would have failed.

When a problem involves two particles or more, we can apply the principle of work and energy to each particle separately. Adding the kinetic energies of the various particles and considering the work of all the forces acting on them, we can also write a single equation of work and energy for all the particles involved. We have

$$T_1 + U_{1 \rightarrow 2} = T_2 \quad (13.11)$$

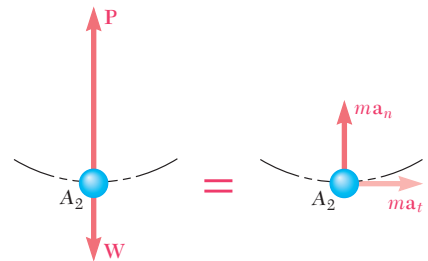


Fig. 13.9 Free-body diagram and kinetic diagram for determining the force on a pendulum bob.

where T_1 represents the arithmetic sum of the kinetic energies of the particles involved at position 1, T_2 represents the arithmetic sum of the kinetic energies of the particles involved at position 2, and $U_{1 \rightarrow 2}$ is the work of all the forces acting on the particles, including the forces of action and reaction exerted by the particles on each other. In problems involving bodies connected by inextensible cords or links, however, the work of the forces exerted by a given cord or link on the two bodies it connects cancels out, since the points of application of these forces move through equal distances (see Sample Prob. 13.2). (In Chapter 14 we discuss how to apply the method of work and energy to a system of particles.)

Friction forces have a direction opposite of that of the displacement of the body on which they act, so **the work of friction forces is always negative**. This work represents energy dissipated into heat and always results in a decrease in the kinetic energy of the body involved (see Sample Prob. 13.3).

13.1D Power and Efficiency

We define **power** as the time rate at which work is done. In the selection of a motor or engine, power is a much more important criterion than is the actual amount of work to be performed. Either a small motor or a large power plant can be used to do a given amount of work, but the small motor may require a month to do the work done by the power plant in a matter of minutes. If ΔU is the work done during the time interval Δt , the average power during that time interval is

$$\text{Average power} = \frac{\Delta U}{\Delta t}$$

Letting Δt approach zero, we obtain in the limit

$$\text{Power} = \frac{dU}{dt} \quad (13.12)$$

Substituting the scalar product $\mathbf{F} \cdot d\mathbf{r}$ for dU , we can also write

$$\text{Power} = \frac{dU}{dt} = \frac{\mathbf{F} \cdot d\mathbf{r}}{dt}$$

Then, recalling that $d\mathbf{r}/dt$ represents the velocity $\mathbf{v}$ of the point of application of $\mathbf{F}$, we have

$$\text{Power} = \mathbf{F} \cdot \mathbf{v} \quad (13.13)$$



Photo 13.1 The power used to operate a chair lift at a ski resort is the product of the force applied and the speed of the lift.

Since we defined power as the time rate at which work is done, we obtain its units by dividing units of work by the unit of time. Thus, if

we use SI units, power is expressed in J/s; this unit is called a *watt* (W). We have

$$1 \text{ W} = 1 \text{ J/s} = 1 \text{ N}\cdot\text{m/s}$$

If we use U.S. customary units, power is expressed in ft·lb/s or in *horsepower* (hp), where one horsepower is defined as

$$1 \text{ hp} = 550 \text{ ft}\cdot\text{lb/s}$$

Recall from Sec. 13.1A that $1 \text{ ft}\cdot\text{lb} = 1.356 \text{ J}$, so we can verify that

$$1 \text{ ft}\cdot\text{lb/s} = 1.356 \text{ J/s} = 1.356 \text{ W}$$

$$1 \text{ hp} = 550(1.356 \text{ W}) = 746 \text{ W} = 0.746 \text{ kW}$$

We defined the **mechanical efficiency** of a machine in Sec. 10.1D as the ratio of the output work to the input work:

$$\eta = \frac{\text{output work}}{\text{input work}} \quad (13.14)$$

This definition is based on the assumption that work is done at a constant rate. The ratio of the output to the input work is therefore equal to the ratio of the rates at which output and input work are done, and we have

Mechanical efficiency

$$\eta = \frac{\text{power output}}{\text{power input}} \quad (13.15)$$

Because of energy losses due to friction, the output work is always smaller than the input work, and consequently, the power output is always smaller than the power input. The mechanical efficiency of a machine is therefore always less than 1.

When we use a machine to transform mechanical energy into electrical energy or thermal energy into mechanical energy, we can obtain its *overall efficiency* from Eq. (13.15). The overall efficiency of a machine is always less than 1; it provides a measure of all the various energy losses involved (losses of electric or thermal energy as well as frictional losses). Note that you have to express the power output and the power input in the same units before using Eq. (13.15).

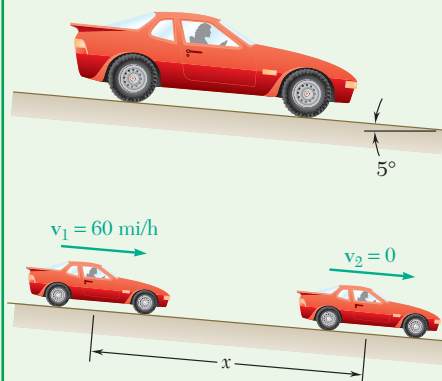


Fig. 1 Car at the two positions of interest.

Sample Problem 13.1

An automobile weighing 4000 lb is driven down a 5° incline at a speed of 60 mi/h when the brakes are applied, causing a constant total braking force (applied by the road on the tires) of 1500 lb. Determine the distance traveled by the automobile as it comes to a stop.

STRATEGY: You are given the velocity of the car at two positions along the road and need to determine the distance x between them (Fig. 1), so use the principle of work and energy.

MODELING: Choose the car as the system and assume it can be modeled as a particle.

ANALYSIS: To apply the principle of work and energy, you find the kinetic energy at each position of the car. The difference between the kinetic energies will be equal to the work done by the braking force.

Principle of Work and Energy.

$$T_1 + U_{1 \rightarrow 2} = T_2 \quad (1)$$

Therefore, you need to calculate each term in this equation.

Kinetic Energy.

Position 1.
$$v_1 = \left(60 \frac{\text{mi}}{\text{h}}\right) \left(\frac{5280 \text{ ft}}{1 \text{ mi}}\right) \left(\frac{1 \text{ h}}{3600 \text{ s}}\right) = 88 \text{ ft/s}$$

$$T_1 = \frac{1}{2}mv_1^2 = \frac{1}{2}(4000/32.2)(88)^2 = 481,000 \text{ ft}\cdot\text{lb}$$

Position 2.
$$v_2 = 0 \quad T_2 = 0$$

Work. The best way to identify which forces do work is to draw a free-body diagram, as shown in Fig. 2. It is clear that the only external forces that do work are the total braking force and the weight. The normal force does no work because it is perpendicular to the motion. Using the definition of work gives

$$U_{1 \rightarrow 2} = -1500x + (4000 \sin 5^\circ)x = -1151x$$

Note that the work of the gravitational force is positive since the automobile is moving down. Substituting into Eq. (1) gives

$$481,000 - 1151x = 0 \quad x = 418 \text{ ft} \quad \blacktriangleleft$$

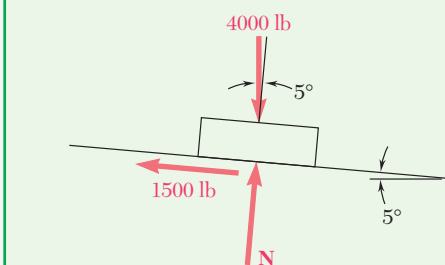
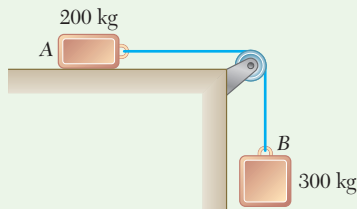


Fig. 2 Free-body diagram for the car.

REFLECT and THINK: Solving this problem using Newton's second law would require determining the car's deceleration from the free-body diagram (Fig. 2) and then integrating this using the given velocity information. Using the principle of work and energy allows you to avoid that calculation.



Sample Problem 13.2

Two blocks are joined by an inextensible cable as shown. If the system is released from rest, determine the velocity of block A after it has moved 2 m. Assume that the coefficient of kinetic friction between block A and the plane is $\mu_k = 0.25$ and that the pulley is weightless and frictionless.

STRATEGY: You are interested in determining the velocity and are given two locations in space, so use the principle of work and energy. You can apply this principle to each block and combine the resulting equations, or you can choose your system to be both blocks and the cable, thereby avoiding the need to determine the work of internal forces.

MODELING: Define two separate systems, one for each block, and model them as particles. As stated in the problem, assume the pulley is weightless and frictionless.

ANALYSIS:

Work and Energy for Block A. Denote the friction force by F_A and the force exerted by the cable by F_C . Then you have (Fig. 1)

$$m_A = 200 \text{ kg} \quad W_A = (200 \text{ kg})(9.81 \text{ m/s}^2) = 1962 \text{ N}$$

$$F_A = \mu_k N_A = \mu_k W_A = 0.25(1962 \text{ N}) = 490 \text{ N}$$

$$T_1 + U_{1 \rightarrow 2} = T_2: \quad 0 + F_C(2 \text{ m}) - F_A(2 \text{ m}) = \frac{1}{2} m_A v^2$$

$$F_C(2 \text{ m}) - (490 \text{ N})(2 \text{ m}) = \frac{1}{2}(200 \text{ kg})v^2 \quad (1)$$

Work and Energy for Block B. From the free-body diagram for Block B (Fig. 2), you have

$$m_B = 300 \text{ kg} \quad W_B = (300 \text{ kg})(9.81 \text{ m/s}^2) = 2940 \text{ N}$$

$$T_1 + U_{1 \rightarrow 2} = T_2: \quad 0 + W_B(2 \text{ m}) - F_C(2 \text{ m}) = \frac{1}{2} m_B v^2$$

$$(2940 \text{ N})(2 \text{ m}) - F_C(2 \text{ m}) = \frac{1}{2}(300 \text{ kg})v^2 \quad (2)$$

Now add the left-hand and right-hand sides of Eqs. (1) and (2). The work of the forces exerted by the cable on A and B cancels out. This is why when solving problems using work and energy, it is usually best to choose your system to include all the objects of interest, so you don't need to worry about the work of internal forces. Therefore, after combining Eqs. (1) and (2) or by choosing your system to be block A, block B, and the cable, you get

$$(2940 \text{ N})(2 \text{ m}) - (490 \text{ N})(2 \text{ m}) = \frac{1}{2}(200 \text{ kg} + 300 \text{ kg})v^2$$

$$4900 \text{ J} = \frac{1}{2}(500 \text{ kg})v^2 \quad v = 4.43 \text{ m/s} \quad \blacktriangleleft$$

REFLECT and THINK: When using the principle of work and energy, it usually saves time to choose your system to be everything that moves. Now that you know the velocity of the block, you could use Eq. (1) to determine the force in the cable. Only when you need to determine an internal force would you need to isolate part of a system.

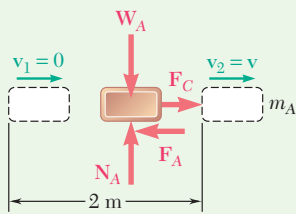


Fig. 1 Free-body diagram and two positions for block A.

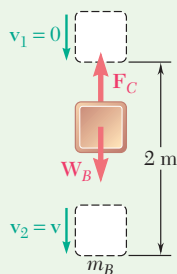


Fig. 2 Free-body diagram and two positions for block B.

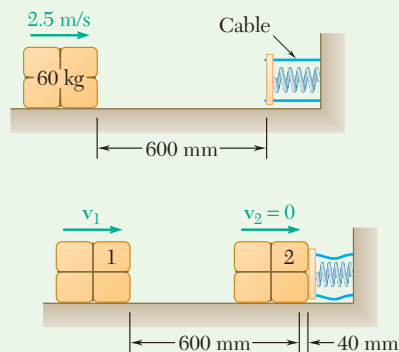


Fig. 1 The package at position 1 and position 2.

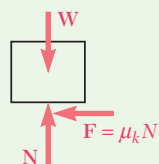


Fig. 2 Free-body diagram before spring is engaged.

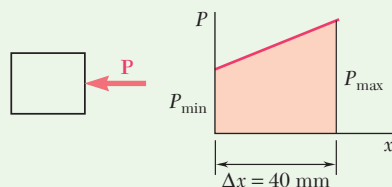


Fig. 3 Force P on the block after it hits the spring.

Sample Problem 13.3

A spring is used to stop a 60-kg package that is sliding on a horizontal surface. The spring has a constant $k = 20 \text{ kN/m}$ and is held by cables so that it is initially compressed 120 mm. The package has a velocity of 2.5 m/s in the position shown, and the maximum additional deflection of the spring is 40 mm. Determine (a) the coefficient of kinetic friction between the package and the surface, (b) the velocity of the package as it passes again through the position shown.

STRATEGY: You have velocity information and specific locations in space, so use the principle of work and energy. Break the motion into two segments: segment 1 is the initial position to the point where the spring has a maximum deflection (Fig. 1), and segment 2 is from the point the spring has a maximum deflection back to the original position.

MODELING: The system is the crate, which you can model as a particle. A free-body diagram for the crate when it is not in contact with the spring is shown in Fig. 2. After it hits the spring, it has an additional force P acting on it due to the compression of the spring (Fig. 3).

ANALYSIS: The principle of work and energy is

$$T_1 + U_{1 \rightarrow 2} = T_2 \quad (1)$$

Call the initial position of the package position 1 and the position where maximum spring deflection occurs position 2 (Fig. 1)

a. Motion from Position 1 to Position 2

Kinetic Energy. Position 1. $v_1 = 2.5 \text{ m/s}$

$$T_1 = \frac{1}{2}mv_1^2 = \frac{1}{2}(60 \text{ kg})(2.5 \text{ m/s})^2 = 187.5 \text{ N}\cdot\text{m} = 187.5 \text{ J}$$

Position 2. (maximum spring deflection): $v_2 = 0$ $T_2 = 0$

Work. Friction Force F . You have (Fig. 2)

$$F = \mu_k N = \mu_k W = \mu_k mg = \mu_k(60 \text{ kg})(9.81 \text{ m/s}^2) = (588.6 \text{ N})\mu_k$$

The work of F is negative and equal to

$$(U_{1 \rightarrow 2})_f = -Fx = -(588.6 \text{ N})\mu_k(0.600 \text{ m} + 0.040 \text{ m}) = -(377 \text{ J})\mu_k$$

Spring Force P . The variable force P exerted by the spring does an amount of negative work equal to the area under the force-deflection curve of the spring force. You have

$$\begin{aligned} P_{\min} &= kx_0 = (20 \text{ kN/m})(120 \text{ mm}) = (20\,000 \text{ N/m})(0.120 \text{ m}) = 2400 \text{ N} \\ P_{\max} &= P_{\min} + k\Delta x = 2400 \text{ N} + (20 \text{ kN/m})(40 \text{ mm}) = 3200 \text{ N} \\ (U_{1 \rightarrow 2})_e &= -\frac{1}{2}(P_{\min} + P_{\max})\Delta x = -\frac{1}{2}(2400 \text{ N} + 3200 \text{ N})(0.040 \text{ m}) = -112.0 \text{ J} \end{aligned}$$

The total work between positions 1 and 2 is thus

$$U_{1 \rightarrow 2} = (U_{1 \rightarrow 2})_f + (U_{1 \rightarrow 2})_e = -(377 \text{ J})\mu_k - 112.0 \text{ J}$$

Principle of Work and Energy. You can determine the coefficient of kinetic friction from the expression for the principle of work and energy in this segment of the motion.

$$T_1 + U_{1 \rightarrow 2} = T_2: \quad 187.5 \text{ J} - (377 \text{ J})\mu_k - 112.0 \text{ J} = 0 \quad \mu_k = 0.20 \quad \blacktriangleleft$$

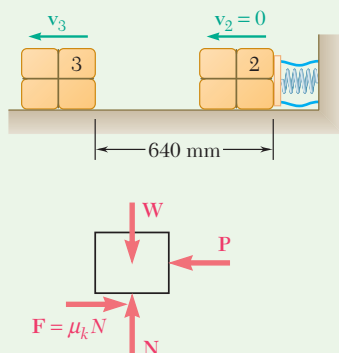


Fig. 4 Free-body diagram when the package is moving to the left.

b. Motion from Position 2 to Position 3. Call the position where the package returns to its initial position as position 3 (Fig. 4).

Kinetic Energy. Position 2.

$$v_2 = 0 \quad T_2 = 0$$

Position 3.

$$T_3 = \frac{1}{2}mv_3^2 = \frac{1}{2}(60 \text{ kg})v_3^2$$

Work. Since the distances involved are the same, the numerical values of the work of the friction force **F** and of the spring force **P** are the same as before. However, the work of **F** is still negative, whereas the work of **P** is now positive.

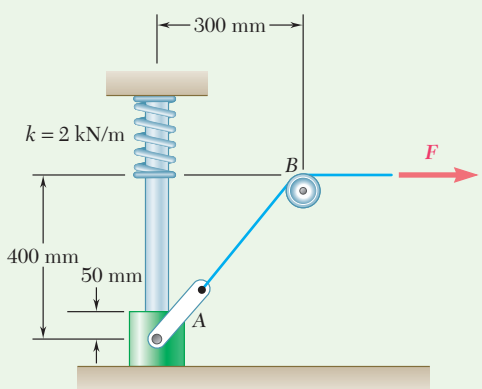
$$U_{2 \rightarrow 3} = -(377 \text{ J})\mu_k + 112.0 \text{ J} = -75.5 \text{ J} + 112.0 \text{ J} = +36.5 \text{ J}$$

Principle of Work and Energy.

$$T_2 + U_{2 \rightarrow 3} = T_3: \quad 0 + 36.5 \text{ J} = \frac{1}{2}(60 \text{ kg})v_3^2$$

$$v_3 = 1.103 \text{ m/s} \quad \mathbf{v_3 = 1.103 \text{ m/s} \leftarrow}$$

REFLECT and THINK: You needed to break this problem into two segments. From the first segment you were able to determine the coefficient of friction. Then you could use the principle of work and energy to determine the velocity of the package at any other location. Note that the system does not lose any energy due to the spring; it returns all of its energy back to the package. You would need to design something that could absorb the kinetic energy of the package in order to bring it to rest.



Sample Problem 13.4

The 2-kg collar **A** starts from rest in the position shown when a constant force $F = 100 \text{ N}$ is applied to the cable, causing the collar **A** to move up the smooth vertical shaft. Neglecting the mass of the frictionless pulley and the spring, determine the speed of **A** when the spring is compressed 50 mm.

STRATEGY: You have information about two positions and are asked to find a speed, so use the principle of work and energy.

MODELING: You have several choices of systems. Two possible systems are shown in Fig. 1.

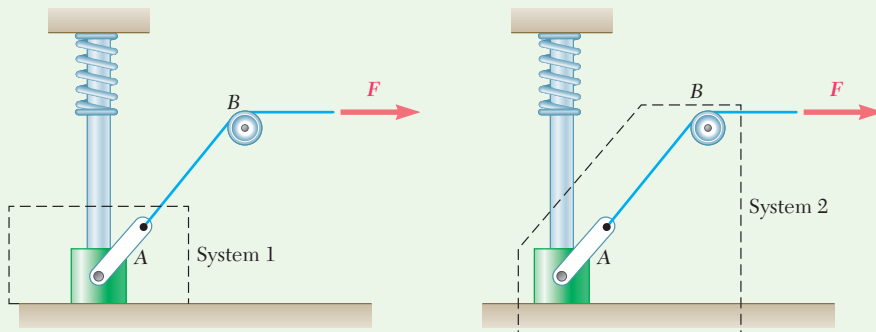


Fig. 1 Possible systems for this problem.

(continued)

Which one should you use? You can solve the problem using either one, but it turns out that some system choices make the problem easier to solve than others. For system 1, the tension in the rope is F , but only the component of F in the direction of the motion does work. This component is continually changing, so calculating the work is difficult. For system 2, the work the force F does is just the magnitude of F (since it is constant) times the distance the force travels horizontally. Therefore, the problem is easiest to solve using system 2.

ANALYSIS: The principle of work and energy is

$$T_1 + U_{1 \rightarrow 2} = T_2 \quad (1)$$

To start, draw the system in the two positions shown in Fig. 2. Since the figure will be very cluttered if you draw the two positions on the same figure, you should draw them side by side.

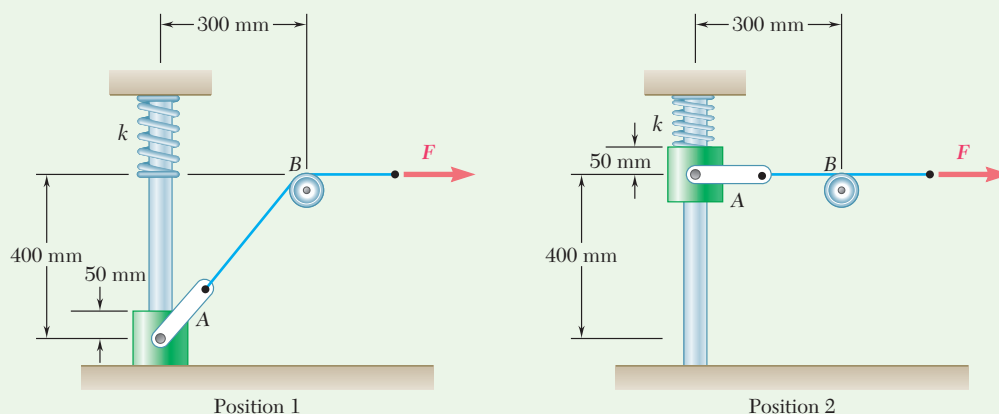


Fig. 2 System in the two positions of interest.

Kinetic Energy. Since the collar is initially at rest, $T_1 = 0$. In position 2, when the upper spring is compressed 50 mm, the kinetic energy is

$$T_2 = \frac{1}{2}mv_2^2 = \frac{1}{2}(2 \text{ kg})v_2^2 = v_2^2$$

Work. As the collar is raised from position 1 to where the spring is compressed 50 mm, the work done by the weight is

$$(U_{1 \rightarrow 2})_g = -mgy_2 = -(2 \text{ kg})(9.81 \text{ m/s}^2)(0.4 \text{ m}) = -7.848 \text{ J}$$

and the work of the spring force is

$$(U_{1 \rightarrow 2})_s = \frac{1}{2}kx_1^2 - \frac{1}{2}kx_2^2 = 0 - \frac{1}{2}(2000 \text{ N/m})(0.05 \text{ m})^2 = -2.50 \text{ J}$$

Finally, you must calculate the work of the 100-N force. In position 1, the length AB is

$$(l_{AB})_1 = \sqrt{(0.4)^2 + (0.3)^2} = 0.5 \text{ m}$$

In position 2, the length AB is $(l_{AB})_2 = 0.3 \text{ m}$. The distance the 100-N force travels through is therefore

$$d = (l_{AB})_1 - (l_{AB})_2 = 0.5 \text{ m} - 0.3 \text{ m} = 0.2 \text{ m}$$

The work done by the 100-N force F is

$$(U_{1 \rightarrow 2})_F = Fd = (100 \text{ N})(0.5 \text{ m} - 0.3 \text{ m}) = 20 \text{ J}$$

Thus, the total work is

$$U_{1 \rightarrow 2} = (U_{1 \rightarrow 2})_g + (U_{1 \rightarrow 2})_s + (U_{1 \rightarrow 2})_F = -7.848 \text{ J} - 2.50 \text{ J} + 20 \text{ J} = 9.652 \text{ J}$$

Substituting these values in the principle of work and energy gives

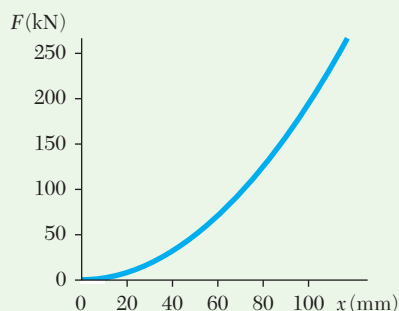
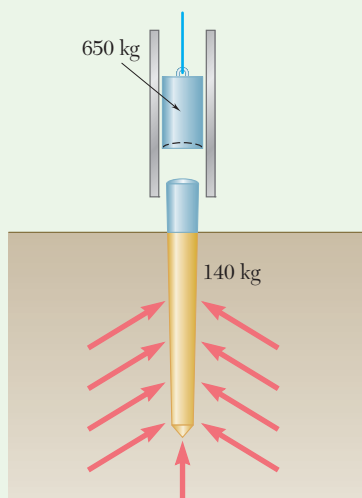
$$T_1 + U_{1 \rightarrow 2} = T_2$$

$$0 + 9.652 = v_2^2$$

$$v_2 = 3.11 \text{ m/s} \quad \blacktriangleleft$$

REFLECT and THINK: What if the force had been only 10 N instead of 100 N? The work would have been a factor of 10 smaller (that is, 2 J), and you would have $v_2^2 = -8.348$, which obviously makes no sense. What does this mean? It means the assumption that the mass will actually reach position 2 is incorrect.

Sample Problem 13.5



The 650-kg hammer of a drop-hammer pile driver falls onto the top of a 140-kg pile. After the impact, the hammer and the pile stick together and have a velocity of 3 m/s. The vertical force exerted on the pile by the ground after the impact is given by $F = 0.02x^2$, where x and F are expressed in mm and kN, respectively. Determine the velocity of the system after it has penetrated 80 mm into the ground.

STRATEGY: You are given a force as a function of displacement and are interested in two positions; therefore, use the principle of work and energy.

MODELING: The system is the hammer and the pile together after the impact. They can be modeled as a single particle. A free-body diagram for this system (Fig. 1) shows that the only two forces that do work are the weight and the force from the ground.

ANALYSIS: The principle of work and energy is

$$T_1 + U_{1 \rightarrow 2} = T_2 \quad (1)$$

Kinetic Energy. The two positions being considered are immediately after the impact and after the system has moved down 50 mm. Since the system is initially traveling at 3 m/s, the initial kinetic energy is

$$T_1 = \frac{1}{2}mv_1^2 = \frac{1}{2}(650 \text{ kg} + 140 \text{ kg})(3 \text{ m/s})^2 = 3555 \text{ J}$$

In position 2, the kinetic energy is

$$T_2 = \frac{1}{2}mv_2^2 = \frac{1}{2}(650 \text{ kg} + 140 \text{ kg})v_2^2 = 395v_2^2$$

Work. As the system moves into the ground, the weight and the resisting force, F , do work. The work the weight does is

$$(U_{1 \rightarrow 2})_g = mgy = (790 \text{ kg})(9.81 \text{ m/s}^2)(0.08 \text{ m}) = 620.0 \text{ J}$$

(continued)



Fig. 1 Free-body diagram after the impact.

The given equation for the force is such that F is in kN when x is expressed in mm. This means that the number in front (that is, the 0.02) has to have the units of kN/mm² for the units to work out. The work of the resisting force is

$$\begin{aligned}(U_{1 \rightarrow 2})_F &= \int_{x_1}^{x_2} F_x dx \\ &= \int_0^{0.05} -(0.02 \text{ kN/mm}^2)x^2 dx = -\left(\frac{0.02}{3} \text{ kN/mm}^2\right)x^3 \bigg|_0^{80} \\ &= -3413 \text{ kN}\cdot\text{mm} = -3413 \text{ J}\end{aligned}$$

Thus, the total work is

$$U_{1 \rightarrow 2} = (U_{1 \rightarrow 2})_g + (U_{1 \rightarrow 2})_F = 620.0 \text{ J} - 3413 \text{ J} = -2793 \text{ J}$$

Substituting the kinetic energies and total work in the principle of work and energy gives

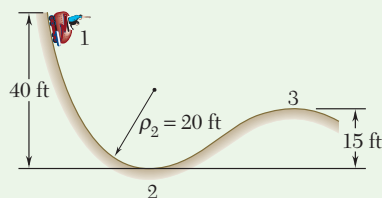
$$\begin{aligned}T_1 + U_{1 \rightarrow 2} &= T_2 \\ 3555 - 2793 &= 395v_2^2\end{aligned}$$

$$v_2 = 1.389 \text{ m/s} \quad \downarrow \quad \blacktriangleleft$$

REFLECT and THINK: To determine how deep the system enters the ground before it stops, you need to set the final kinetic energy equal to zero and make the maximum depth, x_m , unknown. This gives

$$3555 + 790(9.81)x_m - \left(\frac{0.02}{3} \text{ kN/mm}^2\right)x_m^3 = 0$$

Solving this, you find $x_m = 0.0859 \text{ m}$ or 85.9 mm.



Sample Problem 13.6

A 2000-lb roller coaster car starts from rest at point 1 and moves without friction down the track shown. (a) Determine the force exerted by the track on the car at point 2, where the radius of curvature of the track is 20 ft. (b) Determine the minimum safe value of the radius of curvature at point 3.

STRATEGY: Use the principle of work and energy to determine the speed of the car at any location along the track. To determine the force exerted by the track, you need to use Newton's second law. You will need to draw a free-body diagram and kinetic diagram of the car at each position.

MODELING: Choose the car as the system and assume it can be modeled as a particle.

ANALYSIS: Apply the principle of work and energy

$$T_1 + U_{1 \rightarrow 2} = T_2 \quad (1)$$

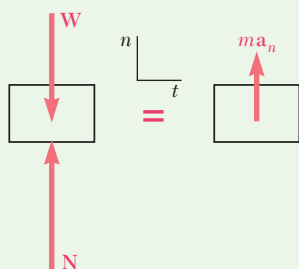


Fig. 1 Free-body diagram and kinetic diagram at point 2.

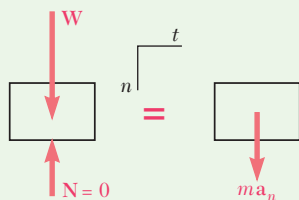


Fig. 2 Free-body diagram and kinetic diagram at point 3.

a. Force Exerted by the Track at Point 2. Use the principle of work and energy to determine the velocity of the car as it passes through point 2.

Kinetic Energy. $T_1 = 0 \quad T_2 = \frac{1}{2}mv_2^2 = \frac{1}{2}\frac{W}{g}v_2^2$

Work. The only force that does work is the weight $\mathbf{W}$. Since the vertical displacement from point 1 to point 2 is 40 ft downward, the work of the weight is

$$U_{1 \rightarrow 2} = +W(40 \text{ ft})$$

Principle of Work and Energy. Substituting these values into Eq. (1) gives

$$\begin{aligned} T_1 + U_{1 \rightarrow 2} &= T_2 & 0 + W(40 \text{ ft}) &= \frac{1}{2}\frac{W}{g}v_2^2 \\ v_2^2 &= 80g = 80(32.2) & v_2 &= 50.8 \text{ ft/s} \end{aligned}$$

Newton's Second Law at Point 2. The acceleration $\mathbf{a}_n$ of the car at point 2 has a magnitude of $a_n = v_2^2/\rho$ and is directed upward. Since the external forces acting on the car are $\mathbf{W}$ and $\mathbf{N}$ (Fig. 1), you have

$$\begin{aligned} +\uparrow \Sigma F_n &= ma_n: & -W + N &= ma_n \\ & & &= \frac{W}{g} \frac{v_2^2}{\rho} \\ & & &= \frac{W}{g} \frac{80g}{20} \end{aligned}$$

$$N = 5W \quad N = 10,000 \text{ lb} \uparrow \quad \blacktriangleleft$$

b. Minimum Value of ρ at Point 3.

Principle of Work and Energy. Applying the principle of work and energy between point 1 and point 3, you obtain

$$\begin{aligned} T_1 + U_{1 \rightarrow 3} &= T_3 & 0 + W(25 \text{ ft}) &= \frac{1}{2}\frac{W}{g}v_3^2 \\ v_3^2 &= 50g = 50(32.2) & v_3 &= 40.1 \text{ ft/s} \end{aligned}$$

Newton's Second Law at Point 3. The minimum safe value of ρ occurs when $\mathbf{N} = 0$. In this case, the acceleration $\mathbf{a}_n$ with a magnitude of $a_n = v_3^2/\rho$, is directed downward (Fig. 2), and you have

$$\begin{aligned} +\downarrow \Sigma F_n &= ma_n: & W &= \frac{W}{g} \frac{v_3^2}{\rho} \\ & & &= \frac{W}{g} \frac{50g}{\rho} \end{aligned}$$

$$\rho = 50 \text{ ft} \quad \blacktriangleleft$$

REFLECT and THINK: This is an example where you need both Newton's second law and the principle of work and energy. Work–energy is used to determine the speed of the car, and Newton's second law is used to determine the normal force. A normal force of $5W$ is equivalent to a fighter pilot pulling $5g$'s and should only be experienced for a very short time. For safety, you would also want to make sure your radius of curvature was quite a bit larger than 50 ft.

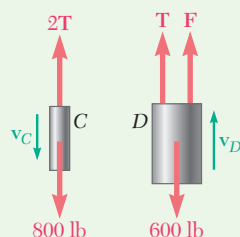
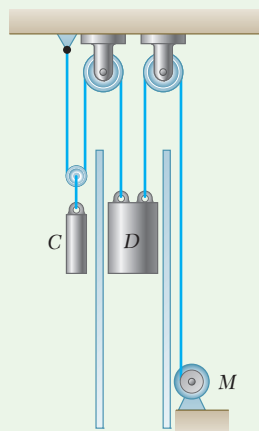


Fig. 1 Free-body diagrams for C and D.

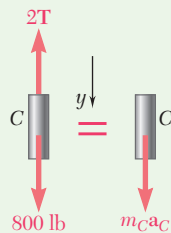


Fig. 2 Free-body diagram and kinetic diagram for C.

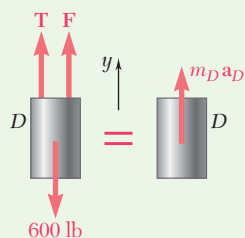


Fig. 3 Free-body diagram and kinetic diagram for D.

Sample Problem 13.7

The dumbwaiter D and its load have a combined weight of 600 lb, whereas the counterweight C weighs 800 lb. Determine the power delivered by the electric motor M when the dumbwaiter (a) is moving up at a constant speed of 8 ft/s, (b) has an instantaneous velocity of 8 ft/s and an acceleration of 2.5 ft/s^2 , where both are directed upward.

STRATEGY: This problem requires you to use the definition of power. You will need to use Newton's second law to determine the tensions in the two cables.

MODELING: Define two separate systems, one for body C and one for body D , and model them as particles. Assume the pulley is weightless and frictionless.

ANALYSIS: The force $\mathbf{F}$ exerted by the motor cable has the same direction as the velocity $\mathbf{v}_D$ of the dumbwaiter, so the power is equal to Fv_D , where $v_D = 8 \text{ ft/s}$. To obtain the power, you must first determine $\mathbf{F}$ in each of the two given situations.

a. Uniform Motion. You have $\mathbf{a}_C = \mathbf{a}_D = 0$; both bodies are in equilibrium (Fig. 1).

$$\text{Body C:} \quad +\uparrow \Sigma F_y = 0: \quad 2T - 800 \text{ lb} = 0 \quad T = 400 \text{ lb}$$

$$\text{Body D:} \quad +\uparrow \Sigma F_y = 0: \quad F + T - 600 \text{ lb} = 0$$

$$F = 600 \text{ lb} - T = 600 \text{ lb} - 400 \text{ lb} = 200 \text{ lb}$$

$$Fv_D = (200 \text{ lb})(8 \text{ ft/s}) = 1600 \text{ ft}\cdot\text{lb/s}$$

$$\text{Power} = (1600 \text{ ft}\cdot\text{lb/s}) \frac{1 \text{ hp}}{550 \text{ ft}\cdot\text{lb/s}} = 2.91 \text{ hp} \quad \blacktriangleleft$$

b. Accelerated Motion. You have

$$\mathbf{a}_D = 2.5 \text{ ft/s}^2 \uparrow \quad \mathbf{a}_C = -\frac{1}{2}\mathbf{a}_D = 1.25 \text{ ft/s}^2 \downarrow$$

The equations of motion are obtained using Figs 2 and 3.

$$\text{Body C:} \quad +\downarrow \Sigma F_y = m_C a_C: \quad 800 - 2T = \frac{800}{32.2}(1.25) \quad T = 384.5 \text{ lb}$$

$$\text{Body D:} \quad +\uparrow \Sigma F_y = m_D a_D: \quad F + T - 600 = \frac{600}{32.2}(2.5)$$

$$F + 384.5 - 600 = 46.6 \quad F = 262.1 \text{ lb}$$

$$Fv_D = (262.1 \text{ lb})(8 \text{ ft/s}) = 2097 \text{ ft}\cdot\text{lb/s}$$

$$\text{Power} = (2097 \text{ ft}\cdot\text{lb/s}) \frac{1 \text{ hp}}{550 \text{ ft}\cdot\text{lb/s}} = 3.81 \text{ hp} \quad \blacktriangleleft$$

REFLECT and THINK: As you might expect, the motor needs to deliver more power to produce accelerated motion than to produce motion at constant velocity.

SOLVING PROBLEMS ON YOUR OWN

In the preceding chapter, you solved problems dealing with the motion of a particle by using the fundamental equation $\mathbf{F} = m\mathbf{a}$ to determine the acceleration $\mathbf{a}$. By applying the principles of kinematics, you could then use $\mathbf{a}$ to determine the velocity and displacement of the particle at any time. In this section, we combined $\mathbf{F} = m\mathbf{a}$ and kinematic relationships to obtain an additional principle called the **principle of work and energy**. This eliminates the need to calculate the acceleration and enables you to relate the velocities of the particle at two points along its path of motion. To solve a problem using work and energy, you need to follow these steps:

1. Compute the work of each of the external forces. The work $U_{1 \rightarrow 2}$ of a given force $\mathbf{F}$ during the finite displacement of a particle from A_1 to A_2 is defined as

$$U_{1 \rightarrow 2} = \int \mathbf{F} \cdot d\mathbf{r} \quad \text{or} \quad U_{1 \rightarrow 2} = \int (F \cos \alpha) ds \quad (13.2, 13.2')$$

where α is the angle between $\mathbf{F}$ and the displacement $d\mathbf{r}$. The work $U_{1 \rightarrow 2}$ is a scalar quantity and is expressed in ft·lb or in·lb in the U.S. customary system of units and in N·m or joules (J) in the SI system of units. Note that the work done is zero for a force perpendicular to the displacement ($\alpha = 90^\circ$). Negative work is done for $90^\circ < \alpha < 180^\circ$ and in particular for a friction force, which is always opposite in direction to the displacement ($\alpha = 180^\circ$).

The work $U_{1 \rightarrow 2}$ can be easily evaluated in the following cases that you will encounter.

a. Work of a constant force in rectilinear motion [Sample Prob. 13.1]

$$U_{1 \rightarrow 2} = (F \cos \alpha) \Delta x \quad (13.3)$$

where α = angle the force forms with the direction of motion

Δx = displacement from A_1 to A_2 (Fig. 13.3)

b. Work of the force of gravity [Sample Probs. 13.2 and 13.6]

$$U_{1 \rightarrow 2} = -W\Delta y \quad (13.4')$$

where Δy is the vertical displacement of the center of gravity of the body of weight W . Note that the work is positive when Δy is negative, that is, when the body moves down (Fig. 13.4).

c. Work of the force exerted by a linear spring [Sample Probs. 13.3 and 13.4]

$$U_{1 \rightarrow 2} = \frac{1}{2}kx_1^2 - \frac{1}{2}kx_2^2 \quad (13.6)$$

where k is the spring constant and x_1 and x_2 are the elongations of the spring corresponding to the positions A_1 and A_2 (Fig. 13.5).

d. Work of a gravitational force

$$U_{1 \rightarrow 2} = \frac{GMm}{r_2} - \frac{GMm}{r_1} \quad (13.7)$$

for a displacement of the body from $A_1(r = r_1)$ to $A_2(r = r_2)$ (Fig. 13.6).

2. Calculate the kinetic energy at A_1 and A_2 . The kinetic energy T is

$$T = \frac{1}{2}mv^2 \quad (13.9)$$

where m is the mass of the particle and v is the magnitude of its velocity. The units of kinetic energy are the same as the units of work, that is, ft·lb or in·lb if you use U.S. customary units and N·m or joules (J) if you use SI units.

3. Substitute the values for the work done $U_{1 \rightarrow 2}$ and the kinetic energies T_1 and T_2 into the equation

$$T_1 + U_{1 \rightarrow 2} = T_2 \quad (13.11)$$

You will now have one scalar equation that you can solve for one unknown. Note that this equation does not yield the time of travel or the acceleration directly. However, if you know the radius of curvature ρ of the path of the particle at a point where you have obtained the velocity v , you can express the normal component of the acceleration as $a_n = v^2/\rho$ and obtain the normal component of the force exerted on the particle by using Newton's second law.

4. We introduced power in this section as the time rate at which work is done as $P = dU/dt$. Power is measured in ft·lb/s or *horsepower* (hp) in U.S. customary units and in J/s or *watts* (W) in the SI system of units. To calculate the power, you can use the equivalent formula

$$P = \mathbf{F} \cdot \mathbf{v} \quad (13.13)$$

where $\mathbf{F}$ and $\mathbf{v}$ denote the force and the velocity, respectively, at a given time [Sample Prob. 13.7]. In some problems [see, e.g., Prob. 13.47], you will be asked for the *average power* that you can obtain by dividing the total work by the time interval during which the work is done.

Problems

CONCEPT QUESTIONS

13.CQ1 Block A is traveling with a speed v_0 on a smooth surface when the surface suddenly becomes rough with a coefficient of friction of μ causing the block to stop after a distance d . If block A were traveling twice as fast, that is, at a speed $2v_0$, how far will it travel on the rough surface before stopping?

- $d/2$
- d
- $\sqrt{2}d$
- $2d$
- $4d$

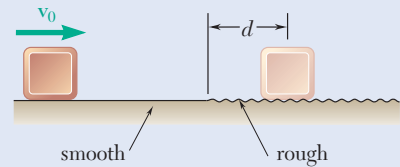


Fig. P13.CQ1

END-OF-SECTION PROBLEMS

- 13.1** A 400-kg satellite is placed in a circular orbit 6394 km above the surface of the earth. At this elevation, the acceleration of gravity is 4.09 m/s^2 . Knowing that its orbital speed is 20 000 km/h, determine the kinetic energy of the satellite.
- 13.2** A 1-lb stone is dropped down the “bottomless pit” at Carlsbad Caverns and strikes the ground with a speed of 95 ft/s. Neglecting air resistance, (a) determine the kinetic energy of the stone as it strikes the ground and the height h from which it was dropped. (b) Solve part a assuming that the same stone is dropped down a hole on the moon. (Acceleration of gravity on the moon = 5.31 ft/s^2 .)



Fig. P13.2

- 13.3** A baseball player hits a 5.1-oz baseball with an initial velocity of 140 ft/s at an angle of 40° with the horizontal as shown. Determine (a) the kinetic energy of the ball immediately after it is hit, (b) the kinetic energy of the ball when it reaches its maximum height, (c) the maximum height above the ground reached by the ball.
- 13.4** A 500-kg communications satellite is in a circular geosynchronous orbit and completes one revolution about the earth in 23 h and 56 min at an altitude of 35 800 km above the surface of the earth. Knowing that the radius of the earth is 6370 km, determine the kinetic energy of the satellite.
- 13.5** In an ore-mixing operation, a bucket full of ore is suspended from a traveling crane which moves along a stationary bridge. The bucket is to swing no more than 10 ft horizontally when the crane is brought to a sudden stop. Determine the maximum allowable speed v of the crane.
- 13.6** In an ore-mixing operation, a bucket full of ore is suspended from a traveling crane which moves along a stationary bridge. The crane is traveling at a speed of 10 ft/s when it is brought to a sudden stop. Determine the maximum horizontal distance through which the bucket will swing.

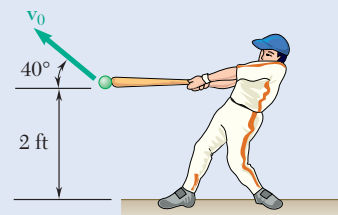


Fig. P13.3

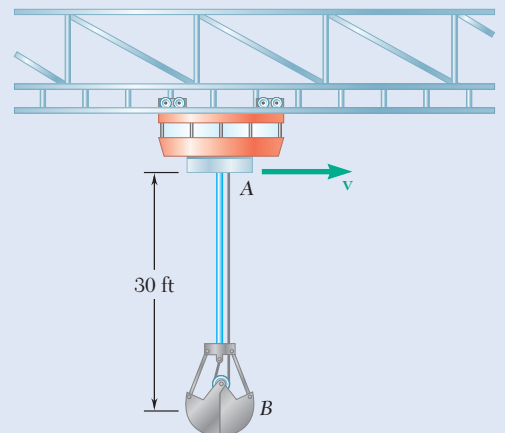


Fig. P13.5 and P13.6

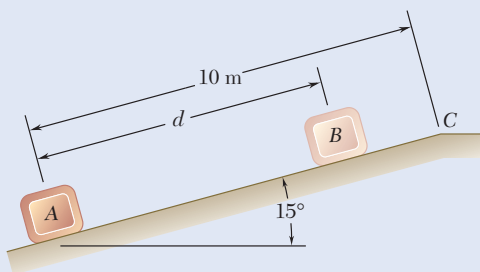


Fig. P13.9

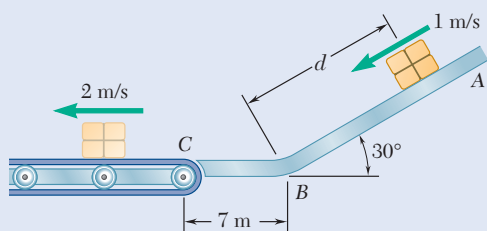


Fig. P13.11 and P13.12

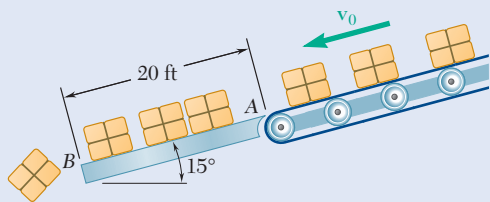


Fig. P13.13 and P13.14

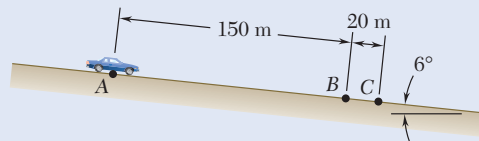


Fig. P13.8

- 13.7** Determine the maximum theoretical speed that may be achieved over a distance of 110 m by a car starting from rest assuming there is no slipping. The coefficient of static friction between the tires and pavement is 0.75, and 60 percent of the weight of the car is distributed over its front wheels and 40 percent over its rear wheels. Assume (a) front-wheel drive, (b) rear-wheel drive.
- 13.8** A 2000-kg automobile starts from rest at point A on a 6° incline and coasts through a distance of 150 m to point B. The brakes are then applied, causing the automobile to come to a stop at point C, which is 20 m from B. Knowing that slipping is impending during the braking period and neglecting air resistance and rolling resistance, determine (a) the speed of the automobile at point B, (b) the coefficient of static friction between the tires and the road.
- 13.9** A package is projected up a 15° incline at A with an initial velocity of 8 m/s. Knowing that the coefficient of kinetic friction between the package and the incline is 0.12, determine (a) the maximum distance d that the package will move up the incline, (b) the velocity of the package as it returns to its original position.
- 13.10** A 1.4-kg model rocket is launched vertically from rest with a constant thrust of 25 N until the rocket reaches an altitude of 15 m and the thrust ends. Neglecting air resistance, determine (a) the speed of the rocket when the thrust ends, (b) the maximum height reached by the rocket, (c) the speed of the rocket when it returns to the ground.
- 13.11** Packages are thrown down an incline at A with a velocity of 1 m/s. The packages slide along the surface ABC to a conveyor belt which moves with a velocity of 2 m/s. Knowing that $\mu_k = 0.25$ between the packages and the surface ABC, determine the distance d if the packages are to arrive at C with a velocity of 2 m/s.
- 13.12** Packages are thrown down an incline at A with a velocity of 1 m/s. The packages slide along the surface ABC to a conveyor belt which moves with a velocity of 2 m/s. Knowing that $d = 7.5$ m and $\mu_k = 0.25$ between the packages and all surfaces, determine (a) the speed of the package at C, (b) the distance a package will slide on the conveyor belt before it comes to rest relative to the belt.
- 13.13** Boxes are transported by a conveyor belt with a velocity v_0 to a fixed incline at A where they slide and eventually fall off at B. Knowing that $\mu_k = 0.40$, determine the velocity of the conveyor belt if the boxes leave the incline at B with a velocity of 8 ft/s.
- 13.14** Boxes are transported by a conveyor belt with a velocity v_0 to a fixed incline at A where they slide and eventually fall off at B. Knowing that $\mu_k = 0.40$, determine the velocity of the conveyor belt if the boxes are to have zero velocity at B.

- 13.15** A 1200-kg trailer is hitched to a 1400-kg car. The car and trailer are traveling at 72 km/h when the driver applies the brakes on both the car and the trailer. Knowing that the braking forces exerted on the car and the trailer are 5000 N and 4000 N, respectively, determine (a) the distance traveled by the car and trailer before they come to a stop, (b) the horizontal component of the force exerted by the trailer hitch on the car.



Fig. P13.15

- 13.16** A trailer truck enters a 2 percent uphill grade traveling at 72 km/h and reaches a speed of 108 km/h in 300 m. The cab has a mass of 1800 kg and the trailer 5400 kg. Determine (a) the average force at the wheels of the cab, (b) the average force in the coupling between the cab and the trailer.

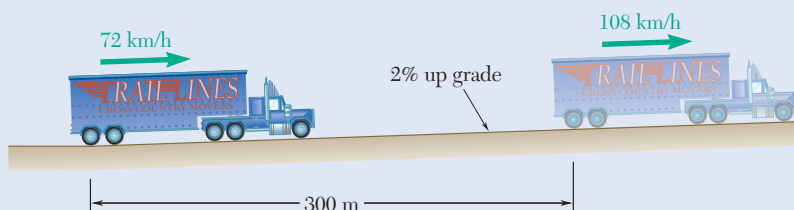


Fig. P13.16

- 13.17** The subway train shown is traveling at a speed of 30 mi/h when the brakes are fully applied on the wheels of cars B and C, causing them to slide on the track, but are not applied on the wheels of car A. Knowing that the coefficient of kinetic friction is 0.35 between the wheels and the track, determine (a) the distance required to bring the train to a stop, (b) the force in each coupling.

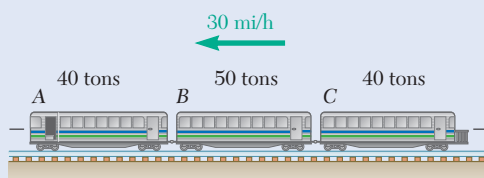


Fig. P13.17 and P13.18

- 13.18** The subway train shown is traveling at a speed of 30 mi/h when the brakes are fully applied on the wheels of car A, causing it to slide on the track, but are not applied on the wheels of cars B or C. Knowing that the coefficient of kinetic friction is 0.35 between the wheels and the track, determine (a) the distance required to bring the train to a stop, (b) the force in each coupling.

- 13.19** Blocks A and B weigh 25 lb and 10 lb, respectively, and they are both at a height 6 ft above the ground when the system is released from rest. Just before hitting the ground, block A is moving at a speed of 9 ft/s. Determine (a) the amount of energy dissipated in friction by the pulley, (b) the tension in each portion of the cord during the motion.

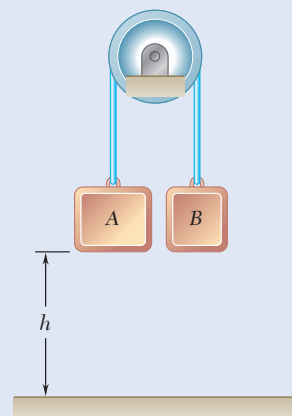


Fig. P13.19

- 13.20** The system shown is at rest when a constant 30-lb force is applied to collar B . (a) If the force acts through the entire motion, determine the speed of collar B as it strikes the support at C . (b) After what distance d should the 30-lb force be removed if the collar is to reach support C with zero velocity?

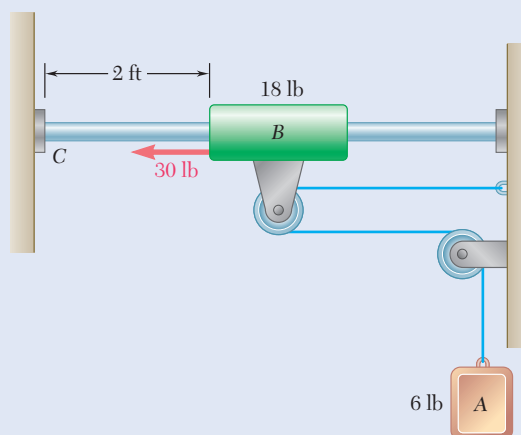


Fig. P13.20

- 13.21** Car B is towing car A at a constant speed of 10 m/s on an uphill grade when the brakes of car A are fully applied causing all four wheels to skid. The driver of car B does not change the throttle setting or change gears. The masses of the cars A and B are 1400 kg and 1200 kg, respectively, and the coefficient of kinetic friction is 0.8. Neglecting air resistance and rolling resistance, determine (a) the distance traveled by the cars before they come to a stop, (b) the tension in the cable.

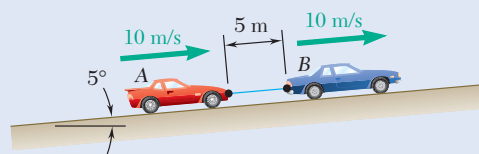


Fig. P13.21

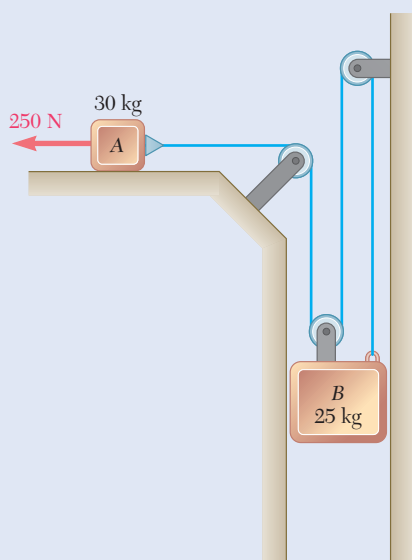


Fig. P13.22 and P13.23

- 13.22** The system shown is at rest when a constant 250-N force is applied to block A . Neglecting the masses of the pulleys and the effect of friction in the pulleys and between block A and the horizontal surface, determine (a) the velocity of block B after block A has moved 2 m, (b) the tension in the cable.
- 13.23** The system shown is at rest when a constant 250-N force is applied to block A . Neglecting the masses of the pulleys and the effect of friction in the pulleys and assuming that the coefficients of friction between block A and the horizontal surface are $\mu_s = 0.25$ and $\mu_k = 0.20$, determine (a) the velocity of block B after block A has moved 2 m, (b) the tension in the cable.

13.24 Two blocks *A* and *B*, of mass 4 kg and 5 kg, respectively, are connected by a cord that passes over pulleys as shown. A 3-kg collar *C* is placed on block *A* and the system is released from rest. After the blocks have moved 0.9 m, collar *C* is removed and blocks *A* and *B* continue to move. Determine the speed of block *A* just before it strikes the ground.

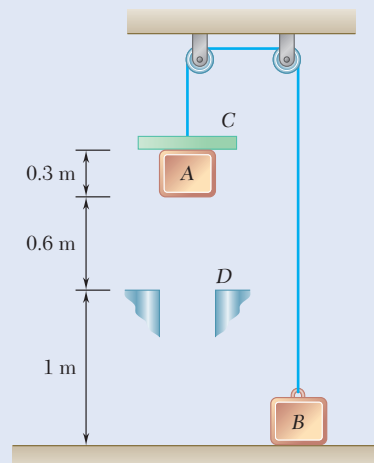


Fig. P13.24

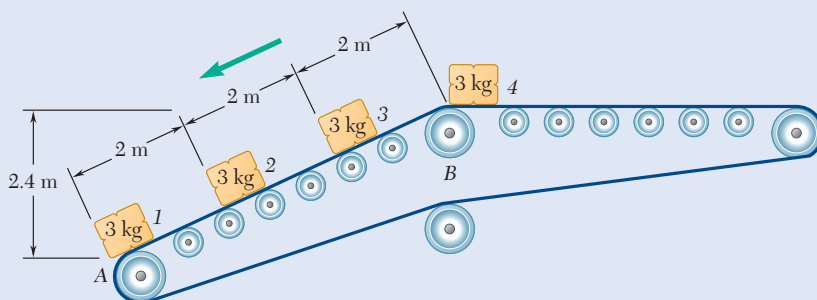


Fig. P13.25

13.26 A 3-kg block rests on top of a 2-kg block supported by, but not attached to, a spring of constant 40 N/m. The upper block is suddenly removed. Determine (a) the maximum speed reached by the 2-kg block, (b) the maximum height reached by the 2-kg block.

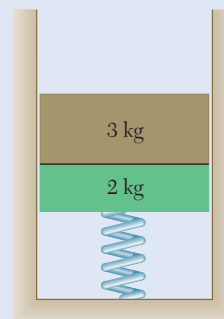


Fig. P13.26

13.27 Solve Prob. 13.26, assuming that the 2-kg block is attached to the spring.

13.28 People with mobility impairments can gain great health and social benefits from participating in different recreational activities. You are tasked with designing an adaptive spring-powered shuffleboard attachment that can be utilized by people who use wheelchairs. Knowing that the coefficient of kinetic friction between the 15 ounce puck *A* and the wooden surface is 0.3, the maximum spring displacement you desire is 6 inches, and that you want the puck to travel at least 30 ft/s, determine (a) the spring constant *k*, (b) how far the athlete should pull back the spring to make the puck come to rest after 34 ft.

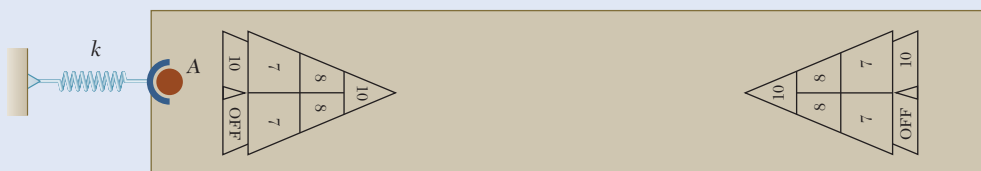


Fig. P13.28

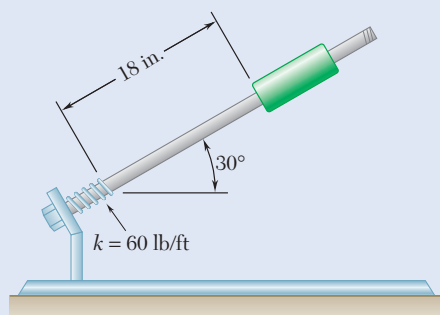


Fig. P13.29

13.29 A 7.5-lb collar is released from rest in the position shown, slides down the inclined rod, and compresses the spring. The direction of motion is reversed and the collar slides up the rod. Knowing that the maximum deflection of the spring is 5 in., determine (a) the coefficient of kinetic friction between the collar and the rod, (b) the maximum speed of the collar.

13.30 A 10-kg block is attached to spring *A* and connected to spring *B* by a cord and pulley. The block is held in the position shown with both springs unstretched when the support is removed and the block is released with no initial velocity. Knowing that the constant of each spring is 2 kN/m, determine (a) the velocity of the block after it has moved down 50 mm, (b) the maximum velocity achieved by the block.

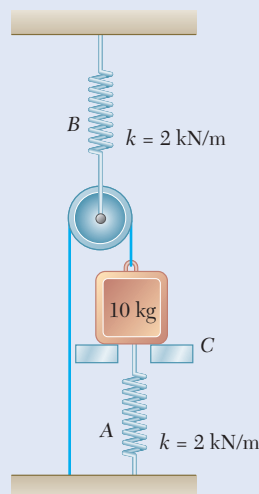


Fig. P13.30

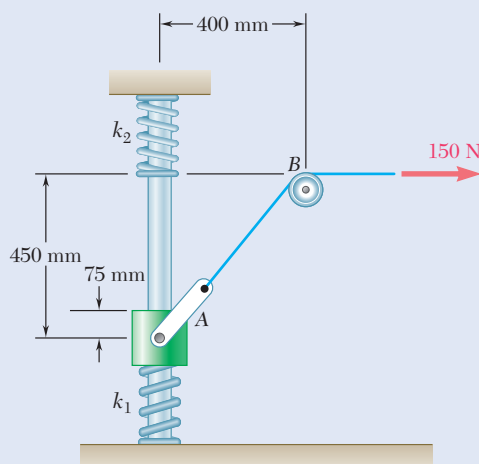


Fig. P13.31

13.31 A 5-kg collar *A* is at rest on top of, but not attached to, a spring with stiffness $k_1 = 400$ N/m when a constant 150-N force is applied to the cable. Knowing *A* has a speed of 1 m/s when the upper spring is compressed 75 mm, determine the spring stiffness k_2 . Ignore friction and the mass of the pulley.

13.32 A piston of mass m and cross-sectional area A is in equilibrium under the pressure p at the center of a cylinder closed at both ends. Assuming that the piston is moved to the left a distance $a/2$ and released, and knowing that the pressure on each side of the piston varies inversely with the volume, determine the velocity of the piston as it again reaches the center of the cylinder. Neglect friction between the piston and the cylinder and express your answer in terms of m , a , p , and A .

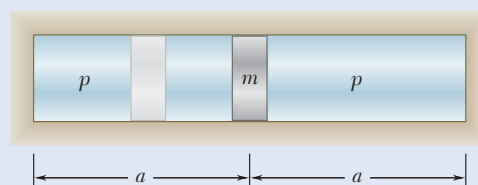


Fig. P13.32

- 13.33** An uncontrolled automobile traveling at 65 mph strikes squarely a highway crash cushion of the type shown in which the automobile is brought to rest by successively crushing steel barrels. The magnitude F of the force required to crush the barrels is shown as a function of the distance x the automobile has moved into the cushion. Knowing that the weight of the automobile is 2250 lb and neglecting the effect of friction, determine (a) the distance the automobile will move into the cushion before it comes to rest, (b) the maximum deceleration of the automobile.

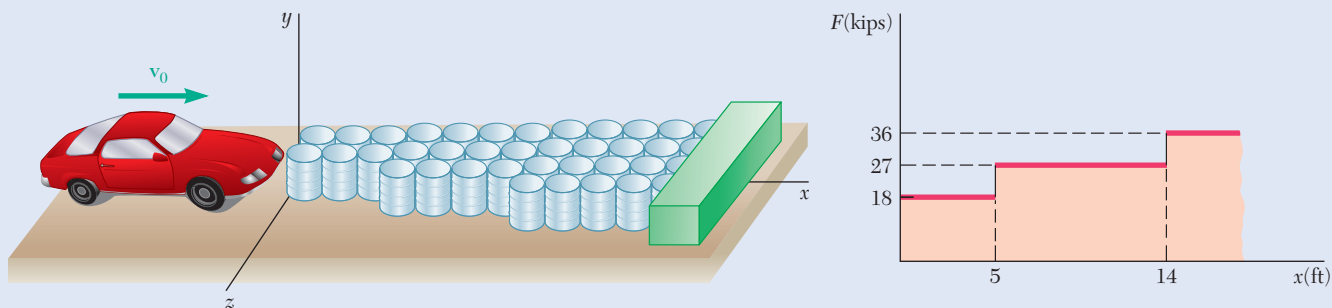


Fig. P13.33

- 13.34** Two types of energy-absorbing fenders designed to be used on a pier are statically loaded. The force-deflection curve for each type of fender is given in the graph. Determine the maximum deflection of each fender when a 90-ton ship moving at 1 mi/h strikes the fender and is brought to rest.



Fig. P13.34

- 13.35** Nonlinear springs are classified as hard or soft, depending upon the curvature of their force-deflection curve (see figure). If a delicate instrument having a mass of 5 kg is placed on a spring of length l so that its base is just touching the undeformed spring and then inadvertently released from that position, determine the maximum deflection x_m of the spring and the maximum force F_m exerted by the spring, assuming (a) a linear spring of constant $k = 3$ kN/m, (b) a hard, nonlinear spring, for which $F = (3 \text{ kN/m})(x + 160x^3)$.

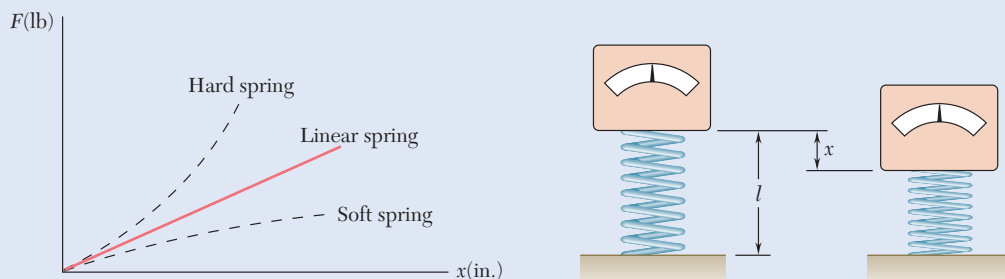


Fig. P13.35

- 13.36** A meteor starts from rest at a very great distance from the earth. Knowing that the radius of the earth is 6370 km and neglecting all forces except the gravitational attraction of the earth, determine the speed of the meteor (a) when it enters the ionosphere at an altitude of 1000 km, (b) when it enters the stratosphere at an altitude of 50 km, (c) when it strikes the earth's surface.

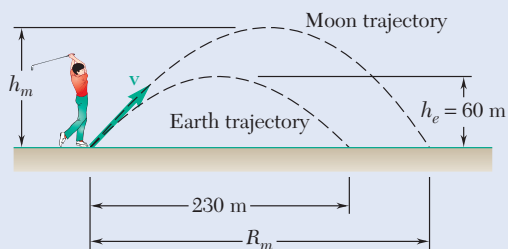


Fig. P13.38

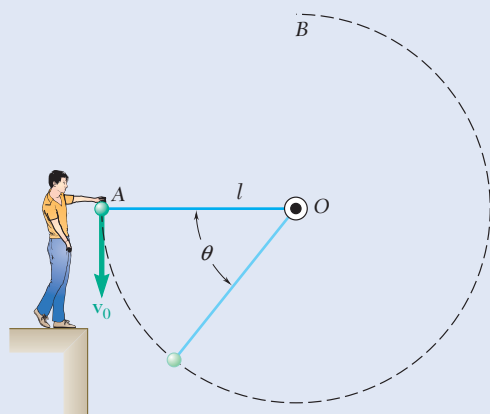


Fig. P13.39 and P13.40

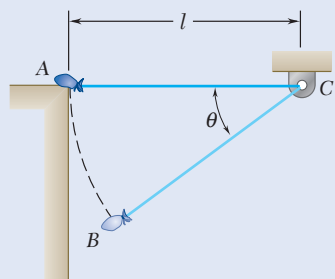


Fig. P13.41

13.37 Express the acceleration of gravity g_h at an altitude h above the surface of the earth in terms of the acceleration of gravity g_0 at the surface of the earth, the altitude h , and the radius R of the earth. Determine the percent error if the weight that an object has on the surface of the earth is used as its weight at an altitude of (a) 0.625 mi, (b) 625 mi.

13.38 A golf ball struck on earth rises to a maximum height of 60 m and hits the ground 230 m away. How high will the same golf ball travel on the moon if the magnitude and direction of its velocity are the same as they were on earth immediately after the ball was hit? Assume that the ball is hit and lands at the same elevation in both cases and that the effect of the atmosphere on the earth is neglected, so that the trajectory in both cases is a parabola. The acceleration of gravity on the moon is 0.165 times that on earth.

13.39 The sphere at A is given a downward velocity v_0 of magnitude 5 m/s and swings in a vertical plane at the end of a rope of length $l = 2$ m attached to a support at O. Determine the angle θ at which the rope will break, knowing that it can withstand a maximum tension equal to twice the weight of the sphere.

13.40 The sphere at A is given a downward velocity v_0 and swings in a vertical circle of radius l and center O. Determine the smallest velocity v_0 for which the sphere will reach point B as it swings about point O (a) if AO is a rope, (b) if AO is a slender rod of negligible mass.

13.41 A bag is gently pushed off the top of a wall at A and swings in a vertical plane at the end of a rope of length l . Determine the angle θ for which the rope will break, knowing that it can withstand a maximum tension equal to twice the weight of the bag.

13.42 A roller coaster starts from rest at A, rolls down the track to B, describes a circular loop of 40-ft diameter, and moves up and down past point E. Knowing that $h = 60$ ft and assuming no energy loss due to friction, determine (a) the force exerted by his seat on a 160-lb rider at B and D, (b) the minimum value of the radius of curvature at E if the roller coaster is not to leave the track at that point.

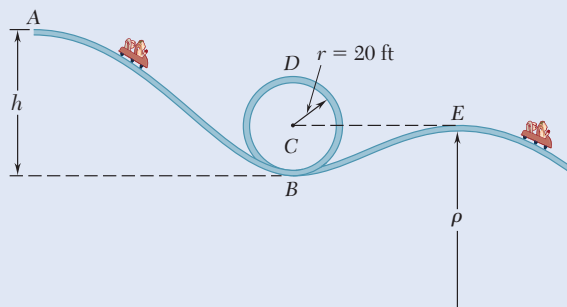


Fig. P13.42

13.43 In Prob. 13.42, determine the range of values of h for which the roller coaster will not leave the track at D or E , knowing that the radius of curvature at E is $\rho = 75$ ft. Assume no energy loss due to friction.

13.44 A small block slides at a speed v on a horizontal surface. Knowing that $h = 0.9$ m, determine the required speed of the block if it is to leave the cylindrical surface BCD when $\theta = 30^\circ$.

13.45 A small block slides at a speed $v = 8$ ft/s on a horizontal surface at a height $h = 3$ ft above the ground. Determine (a) the angle θ at which it will leave the cylindrical surface BCD , (b) the distance x at which it will hit the ground. Neglect friction and air resistance.

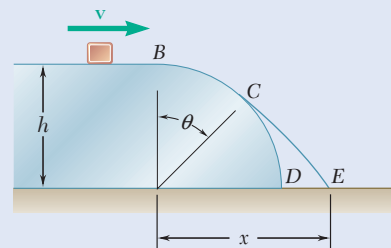


Fig. P13.44 and P13.45

13.46 A chair-lift is designed to transport 1000 skiers per hour from the base A to the summit B . The average mass of a skier is 70 kg and the average speed of the lift is 75 m/min. Determine (a) the average power required, (b) the required capacity of the motor if the mechanical efficiency is 85 percent and if a 300-percent overload is to be allowed.

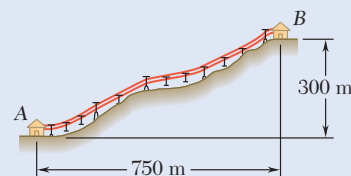


Fig. P13.46

13.47 It takes 15 s to raise a 1200-kg car and the supporting 300-kg hydraulic car-lift platform to a height of 2.8 m. Determine (a) the average output power delivered by the hydraulic pump to lift the system, (b) the average electric power required, knowing that the overall conversion efficiency from electric to mechanical power for the system is 82 percent.

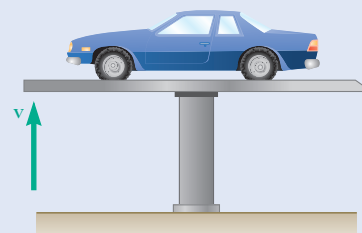


Fig. P13.47

13.48 The velocity of the lift of Prob. 13.47 increases uniformly from zero to its maximum value at mid-height in 7.5 s and then decreases uniformly to zero in 7.5 s. Knowing that the peak power output of the hydraulic pump is 6 kW when the velocity is maximum, determine the maximum lift force provided by the pump.

13.49 (a) A 120-lb woman rides a 15-lb bicycle up a 3-percent slope at a constant speed of 5 ft/s. How much power must be developed by the woman? (b) A 180-lb man on an 18-lb bicycle starts down the same slope and maintains a constant speed of 20 ft/s by braking. How much power is dissipated by the brakes? Ignore air resistance and rolling resistance.

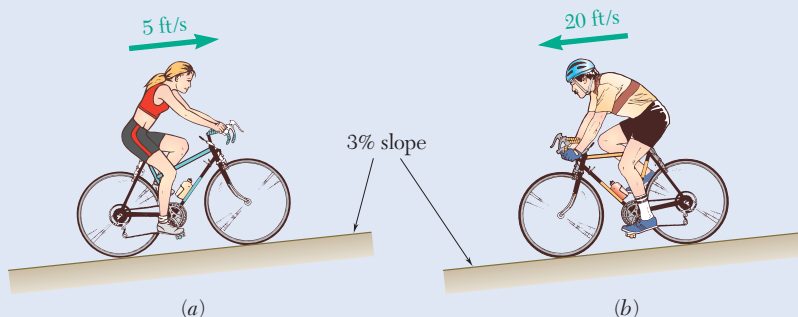


Fig. P13.49

- 13.50** A power specification formula is to be derived for electric motors which drive conveyor belts moving solid material at different rates to different heights and distances. Denoting the efficiency of a motor by η and neglecting the power needed to drive the belt itself, derive a formula (a) in the SI system of units for the power P in kW, in terms of the mass flow rate m in kg/h, the height b and horizontal distance l in meters and (b) in U.S. customary units, for the power in hp, in terms of the material flow rate w in tons/h, and the height b and horizontal distance l in feet.

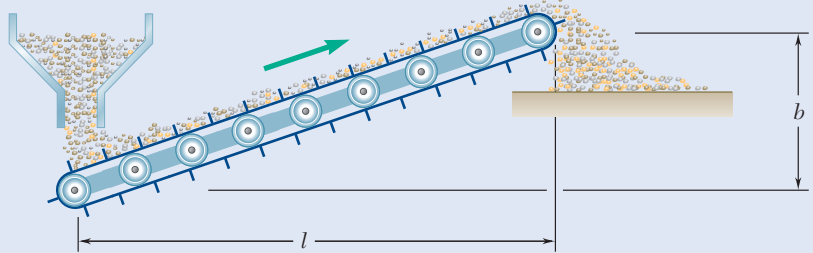


Fig. P13.50

- 13.51** A 1400-kg automobile starts from rest and travels 400 m during a performance test. The motion of the automobile is defined by the relation $x = 4000 \ln(\cosh 0.03t)$, where x and t are expressed in meters and seconds, respectively. The magnitude of the aerodynamic drag is $D = 0.35v^2$, where D and v are expressed in newtons and m/s, respectively. Determine the power dissipated by the aerodynamic drag when (a) $t = 10$ s, (b) $t = 15$ s.

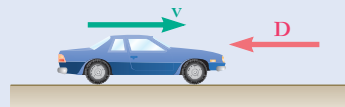


Fig. P13.51 and P13.52

- 13.52** A 1400-kg automobile starts from rest and travels 400 m during a performance test. The motion of the automobile is defined by the relation $a = 3.6e^{-0.0005x}$, where a and x are expressed in m/s^2 and meters, respectively. The magnitude of the aerodynamic drag is $D = 0.35v^2$, where D and v are expressed in newtons and m/s, respectively. Determine the power dissipated by the aerodynamic drag when (a) $x = 200$ m, (b) $x = 400$ m.

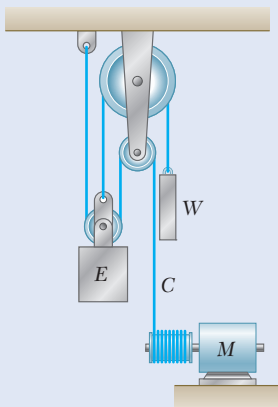


Fig. P13.54

- 13.53** The fluid transmission of a 15-Mg truck allows the engine to deliver an essentially constant power of 50 kW to the driving wheels. Determine the time required and the distance traveled as the speed of the truck is increased (a) from 36 km/h to 54 km/h, (b) from 54 km/h to 72 km/h.
- 13.54** The elevator E has a weight of 6600 lb when fully loaded and is connected as shown to a counterweight W of weight of 2200 lb. Determine the power in hp delivered by the motor (a) when the elevator is moving down at a constant speed of 1 ft/s, (b) when it has an upward velocity of 1 ft/s and a deceleration of 0.18 ft/s^2 .

13.2 CONSERVATION OF ENERGY

The principle of work and energy is useful for solving many different types of engineering problems. However, in many engineering applications, the total mechanical energy remains constant, although it may be transformed from one form into another. This is known as the principle of conservation of energy. To formulate this principle, we must first define a quantity known as potential energy. (Some of the material in this section was considered in Sec. 10.2B.)

13.2A Potential Energy

Let's consider again a body of weight $\mathbf{W}$ that moves along a curved path from a point A_1 of elevation y_1 to a point A_2 of elevation y_2 (Fig. 13.4). Recall from Sec. 13.1A that the work done by the force of gravity $\mathbf{W}$ during this displacement is

$$U_{1 \rightarrow 2} = -(W y_2 - W y_1) = W y_1 - W y_2 \quad (13.4)$$

That is, we obtain the work done by $\mathbf{W}$ by subtracting the value of the function $W y$ corresponding to the second position of the body from its value corresponding to the first position. The work of $\mathbf{W}$ is independent of the actual path followed; it depends only upon the initial and final values of the function $W y$. This function is called the **potential energy** of the body with respect to the **force of gravity $\mathbf{W}$** and is denoted by V_g . We have

Gravitational potential energy on earth

$$U_{1 \rightarrow 2} = (V_g)_1 - (V_g)_2 \quad \text{where } V_g = W y \quad (13.16)$$

where y is measured from an arbitrary horizontal datum where the potential energy is zero by definition. Note that if $(V_g)_2 > (V_g)_1$, that is, **if the potential energy increases** during the displacement (as in the case considered here), **the work $U_{1 \rightarrow 2}$ is negative**. On the other hand, if the work of $\mathbf{W}$ is positive, the potential energy decreases. Therefore, the potential energy V_g of the body provides a measure of the work that can be done by its weight $\mathbf{W}$. Also note that the *change* in potential energy—not the actual value of V_g —is involved in formula (13.16). For this reason, the level, or datum, from which we measure the elevation y can be chosen arbitrarily. Finally, note that potential energy is expressed in the same units as work, i.e., in joules we use if SI units and in ft·lb or in·lb if we use U.S. customary units.

This expression for the potential energy of a body with respect to gravity is valid only as long as we can assume the weight $\mathbf{W}$ of the body remains constant, i.e., as long as the displacements of the body are small compared with the radius of the earth. In the case of a space vehicle, however, we need to take into consideration the variation of the force of gravity with the distance r from the center of the earth. Using the expression obtained in Sec. 13.1A for the work of a gravitational force, we have (Fig. 13.6)

$$U_{1 \rightarrow 2} = \frac{GMm}{r_2} - \frac{GMm}{r_1} \quad (13.7)$$

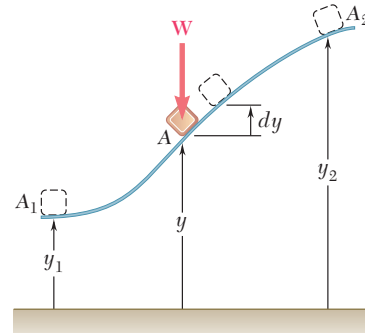


Fig. 13.4 (repeated)

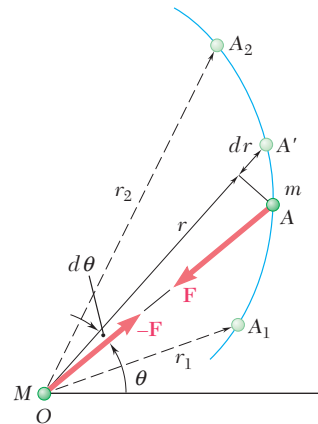


Fig. 13.6 (repeated)

Therefore, we can obtain the work of the force of gravity by subtracting the value of the function $-GMm/r$ corresponding to the second position of the body from its value corresponding to the first position. Thus, the expression that we use for the potential energy V_g when the variation in the force of gravity cannot be neglected is

Gravitational potential energy in space

$$V_g = -\frac{GMm}{r} \quad (13.17)$$

Taking the first of the relations of Eq. (12.27) into account, we can write V_g in the alternative form

$$V_g = -\frac{WR^2}{r} \quad (13.17')$$

where R is the radius of the earth and W is the value of the weight of the body at the surface of the earth. When using either of the relations in Eqs. (13.17) or (13.17') to express V_g , the distance r should, of course, be measured from the center of the earth.[†] Note that V_g is always negative and that it approaches zero for very large values of r .

Consider now a body attached to a spring and moving from a position A_1 , corresponding to a deflection x_1 of the spring, to a position A_2 , corresponding to a deflection x_2 of the spring (Fig. 13.5). Recall from Sec. 13.1A that the work of the force $\mathbf{F}$ exerted by the spring on the body is

$$U_{1 \rightarrow 2} = \frac{1}{2}kx_1^2 - \frac{1}{2}kx_2^2 \quad (13.6)$$

That is, we obtain the work of the elastic force by subtracting the value of the function $\frac{1}{2}kx^2$ corresponding to the second position of the body from its value corresponding to the first position. This function is denoted by V_e and is called the **potential energy** of the body with respect to the **elastic force $\mathbf{F}$** . We have

Elastic potential energy

$$U_{1 \rightarrow 2} = (V_e)_1 - (V_e)_2 \quad \text{with } V_e = \frac{1}{2}kx^2 \quad (13.18)$$

where $x = L_{\text{stretched}} - L_{\text{unstretched}}$, or the deflection of the spring from its undeformed position. Note that, during the displacement from A_1 to A_2 , the work of the force $\mathbf{F}$ exerted by the spring on the body is negative and that the potential energy V_e increases. We can use formula (13.18) even when the spring is rotated about its fixed end (Fig. 13.10a). The work of the elastic force depends only upon the initial and final deflections of the spring (Fig. 13.10b).

We can use the concept of potential energy when forces other than gravity forces and elastic forces are involved. Indeed, it remains valid as long as the work of the force considered is independent of the path followed by its point of application, as this point moves from a given position A_1 to a given position A_2 . Such forces are said to be **conservative forces** or **path-independent forces**. We next consider their general properties.

[†]The expressions for V_g in Eqs. (13.17) and (13.17') are valid only when $r \geq R$; that is, when the body considered is above the surface of the earth.

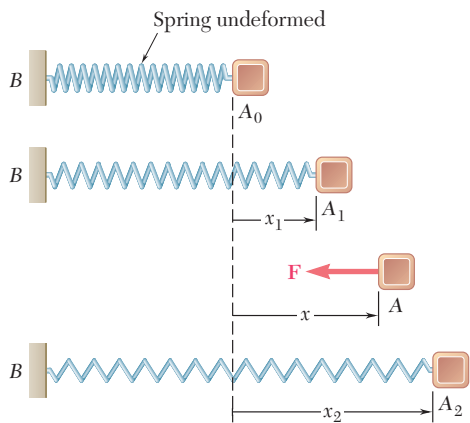


Fig. 13.5 (repeated)

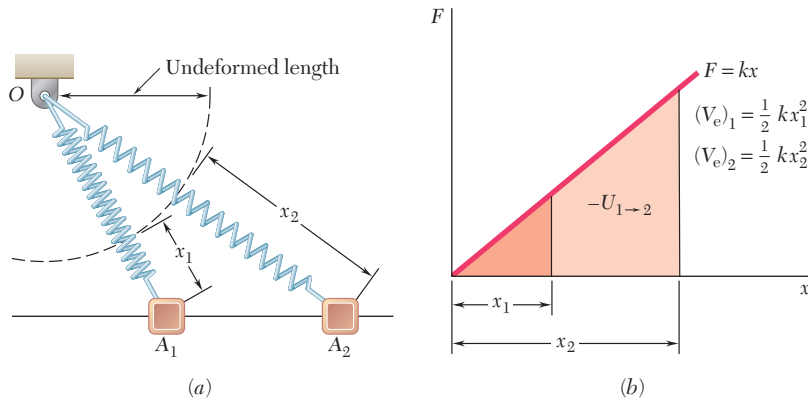


Fig. 13.10 (a) The equation for potential energy of a spring force is valid if the spring stretches when rotated about a fixed end; (b) the work of the elastic force depends only on the initial and final deflections of the spring.

*13.2B Conservative Forces

As indicated in the preceding section, a force $\mathbf{F}$ acting on a particle A is said to be **conservative** if its work $U_{1 \rightarrow 2}$ is **independent of the path followed by the particle A as it moves from A_1 to A_2** (Fig. 13.11a). We then have

$$U_{1 \rightarrow 2} = -(V(x_2, y_2, z_2) - V(x_1, y_1, z_1)) = V(x_1, y_1, z_1) - V(x_2, y_2, z_2) \quad (13.19)$$

or for short,

$$U_{1 \rightarrow 2} = V_1 - V_2 \quad (13.19')$$

The function $V(x, y, z)$ is called the **potential energy**, or **potential function**, of $\mathbf{F}$.

Note that if A_2 is chosen to coincide with A_1 —that is, if the particle describes a closed path (Fig. 13.11b)—we have $V_1 = V_2$ and the work is zero. Thus for any conservative force $\mathbf{F}$, we can write

$$\oint \mathbf{F} \cdot d\mathbf{r} = 0 \quad (13.20)$$

where the circle on the integral sign indicates that the path is closed.

Let us now apply Eq. (13.19) between two neighboring points $A(x, y, z)$ and $A'(x + dx, y + dy, z + dz)$. The elementary work dU corresponding to the displacement $d\mathbf{r}$ from A to A' is

$$dU = V(x, y, z) - V(x + dx, y + dy, z + dz)$$

or

$$dU = -dV(x, y, z) \quad (13.21)$$

Thus, the elementary work of a conservative force is an **exact differential**.

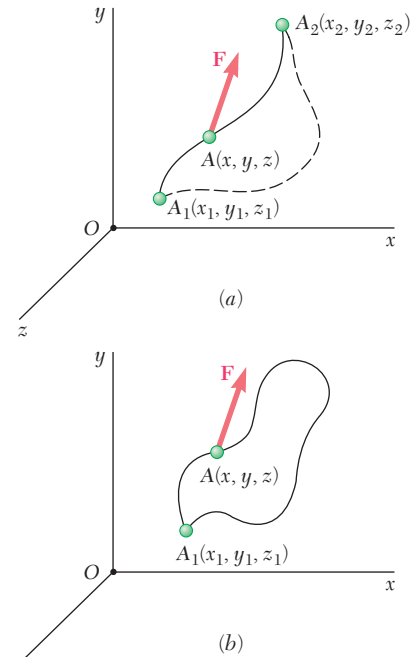


Fig. 13.11 (a) The work of a conservative force acting on a particle is independent of the path of the particle; (b) if the particle travels a closed path, the work of a conservative force is zero.

If we substitute the expression obtained in Eq. 13.1'' for dU in Eq. (13.21) and recall the definition of the differential of a function of several variables, we have

$$F_x dx + F_y dy + F_z dz = -\left(\frac{\partial V}{\partial x}dx + \frac{\partial V}{\partial y}dy + \frac{\partial V}{\partial z}dz\right)$$

from which it follows that

$$F_x = -\frac{\partial V}{\partial x} \quad F_y = -\frac{\partial V}{\partial y} \quad F_z = -\frac{\partial V}{\partial z} \quad (13.22)$$

It is clear that the components of $\mathbf{F}$ must be functions of the coordinates x , y , and z . Thus, a necessary condition for a conservative force is that it depends only upon the position of its point of application. The relations in Eq. (13.22) can be expressed more concisely if we write

$$\mathbf{F} = F_x\mathbf{i} + F_y\mathbf{j} + F_z\mathbf{k} = -\left(\frac{\partial V}{\partial x}\mathbf{i} + \frac{\partial V}{\partial y}\mathbf{j} + \frac{\partial V}{\partial z}\mathbf{k}\right)$$

The vector in parentheses is known as the **gradient of the scalar function** V and is denoted by **grad** V . We thus have for any conservative force

$$\mathbf{F} = -\text{grad } V \quad (13.23)$$

The relations in Eqs. (13.19) to (13.23) are satisfied by any conservative force. It can also be shown that if a force $\mathbf{F}$ satisfies one of these relations, $\mathbf{F}$ must be a conservative force.

13.2C The Principle of Conservation of Energy

We saw in the preceding two sections that we can express the work of a conservative force, such as the weight of a particle or the force exerted by a spring, as a change in potential energy. When a particle moves under the action of conservative forces, the principle of work and energy stated in Sec. 13.B can be expressed in a modified form. Substituting for $U_{1 \rightarrow 2}$ from Eq. (13.19') into Eq. (13.10), we have

$$V_1 - V_2 = T_2 - T_1$$

or

**Conservation
of energy**

$$T_1 + V_1 = T_2 + V_2 \quad (13.24)$$

Formula (13.24) indicates that when a particle moves under the action of conservative forces, **the sum of the kinetic energy and of the potential energy of the particle remains constant**. The sum $T + V$ is called the **total mechanical energy** of the particle and is denoted by E . So far, we have discussed two types of potential energy: gravitational potential energy, V_g , and elastic potential energy, V_e . Therefore, another way to write Eq. (13.24) is

$$T_1 + V_{g1} + V_{e1} = T_2 + V_{g2} + V_{e2} \quad (13.24')$$

Consider, for example, the pendulum analyzed in Sec. 13.1C that is released with no velocity from A_1 and allowed to swing in a vertical plane (Fig. 13.12). Measuring the potential energy from the level of A_2 , that is, placing our datum at A_2 , we have at A_1

$$T_1 = 0 \quad V_1 = Wl \quad T_1 + V_1 = Wl$$

Recalling that at A_2 the speed of the pendulum is $v_2 = \sqrt{2gl}$, we have

$$T_2 = \frac{1}{2}mv_2^2 = \frac{1}{2} \frac{W}{g} (2gl) = Wl \quad V_2 = 0$$

$$T_2 + V_2 = Wl$$

Thus, we can check that the total mechanical energy $E = T + V$ of the pendulum is the same at A_1 and A_2 . Whereas the energy is entirely potential at A_1 , it becomes entirely kinetic at A_2 , and as the pendulum keeps swinging to the right past A_2 , the kinetic energy is transformed back into potential energy. At A_3 , $T_3 = 0$ and $V_3 = Wl$.

Because the total mechanical energy of the pendulum remains constant and its potential energy depends only upon its elevation, the kinetic energy of the pendulum must have the same value at any two points located at the same height. Thus, the speed of the pendulum is the same at A and at A' (Fig. 13.12). We can extend this result to the case of a particle moving along any given path, regardless of the shape of the path, as long as the only forces acting on the particle are its weight and the normal reaction of the path. The particle of Fig. 13.13, for example, which slides in a vertical plane along a frictionless track, has the same speed at A , A' , and A'' .

The weight of a particle and the force exerted by a spring are conservative forces, but **friction forces are nonconservative, or path-dependent, forces**. In other words, *the work of a friction force cannot be expressed as a change in potential energy*. The work of a friction force depends upon the path followed by its point of application; and whereas the work $U_{1 \rightarrow 2}$ defined by Eq. (13.19) is positive or negative according to the sense of motion, the work of a friction force, as we noted in Sec. 13.1C, is always negative. It follows that when a mechanical system involves friction, its total mechanical energy does not remain constant but decreases. The energy of the system, however, is not lost; it is transformed into heat, and the sum of the *mechanical energy* and of the *thermal energy* of the system remains constant.

Other forms of energy also can be involved in a system. For instance, a generator converts mechanical energy into *electrical energy*; a gasoline engine converts *chemical energy* into mechanical energy; a nuclear reactor converts *mass* into thermal energy. If all forms of energy are considered, the energy of any system can be considered as constant, and the principle of conservation of energy remains valid under all conditions.

If we express the work done by non-conservative forces as $U_{1 \rightarrow 2}^{\text{NC}}$, we can express Eq. (13.2) as

$$T_1 + V_{g_1} + V_{e_1} + U_{1 \rightarrow 2}^{\text{NC}} = T_2 + V_{g_2} + V_{e_2} \quad (13.24'')$$

Note that if $U_{1 \rightarrow 2}^{\text{NC}}$ is zero, then the expression reduces to the conservation of energy equation of Eq. (13.24').

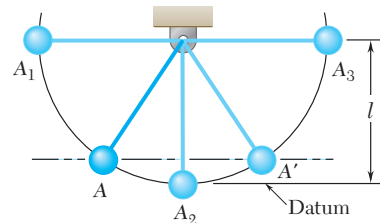


Fig. 13.12 The motion of a pendulum is easily analyzed using conservation of energy.

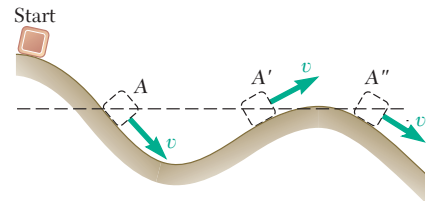


Fig. 13.13 A particle moving along a frictionless track has the same speed every time it passes the same elevation.



Photo 13.2 The potential energy of the roller coaster car is converted into kinetic energy as it descends the track.



Photo 13.3 Once in orbit, Earth satellites move under the action of gravity, which acts as a central force.

13.2D Application to Space Mechanics: Motion Under a Conservative Central Force

We saw in Sec. 12.2B that when a particle P moves under a central force $\mathbf{F}$, the angular momentum $\mathbf{H}_O$ of the particle about the center of force O is constant. If the force $\mathbf{F}$ is also conservative, there exists a potential energy V associated with $\mathbf{F}$, and the total energy $E = T + V$ of the particle is constant. Thus, when a particle moves under a conservative central force, we can use both the principle of conservation of angular momentum and the principle of conservation of energy to study its motion.

Consider, for example, a space vehicle of mass m moving under the earth's gravitational force. Let us assume that it begins its free flight at point P_0 at a distance r_0 from the center of the earth with a velocity $\mathbf{v}_0$ forming an angle ϕ_0 with the radius vector OP_0 (Fig. 13.14). Let P be a point of the trajectory described by the vehicle; we denote by r the distance from O to P , by $\mathbf{v}$ the velocity of the vehicle at P , and by ϕ the angle formed by $\mathbf{v}$ and the radius vector OP . Applying the principle of conservation of angular momentum about O between P_0 and P (Sec. 12.2B), we have

$$r_0 m v_0 \sin \phi_0 = r m v \sin \phi \quad (13.25)$$

Recalling the expression in Eq. (13.17) for the potential energy due to a gravitational force, we apply the principle of conservation of energy between P_0 and P , obtaining

$$T_0 + V_0 = T + V$$

$$\frac{1}{2} m v_0^2 - \frac{GMm}{r_0} = \frac{1}{2} m v^2 - \frac{GMm}{r} \quad (13.26)$$

where M is the mass of the earth.

We can solve Eq. (13.26) for the magnitude v of the velocity of the vehicle at P when we know the distance r from O to P . Then we can use Eq. (13.25) to determine the angle ϕ that the velocity forms with the radius vector OP .

We can also use Eqs. (13.25) and (13.26) to determine the maximum and minimum values of r in the case of a satellite launched from P_0 in a direction forming an angle ϕ_0 with the vertical OP_0 (Fig. 13.15). We obtain the desired values of r by making $\phi = 90^\circ$ in Eq. (13.25) and eliminating v between Eqs. (13.25) and (13.26).

Note that applying the principles of conservation of energy and of conservation of angular momentum leads to a more fundamental formulation of the problems of space mechanics than does the method indicated in Sec. 12.3B. It also results in much simpler computations in all cases involving oblique launchings. Although you must use the method of Sec. 12.3B when the actual trajectory or the periodic time of a space vehicle is to be determined, the calculations will be simplified if you first use the conservation principles to compute the maximum and minimum values of the radius vector r .

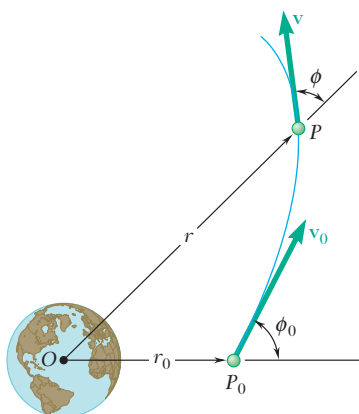


Fig. 13.14 A space vehicle moving from P_0 to P under the earth's gravitational force.

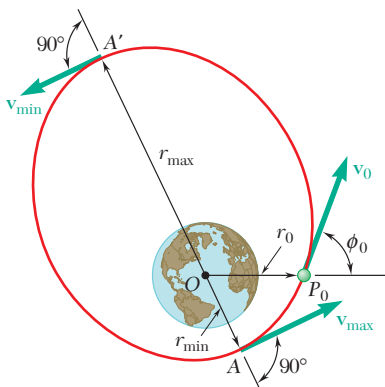


Fig. 13.15 A space vehicle launched from point P_0 into an orbit around the earth.

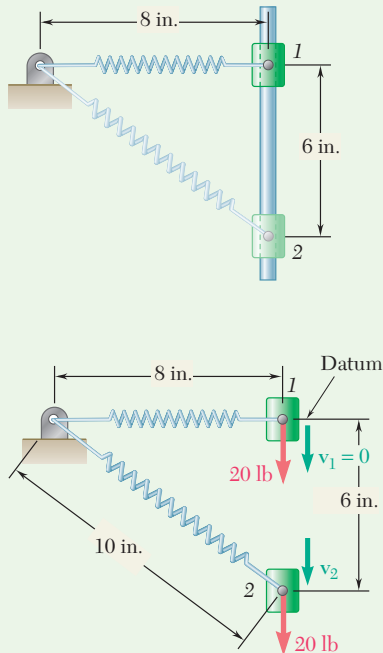


Fig. 1 The system in position 1 and position 2.

Sample Problem 13.8

A 20-lb collar slides without friction along a vertical rod as shown. The spring attached to the collar has an undeformed length of 4 in. and a spring constant of 3 lb/in. If the collar is released from rest in position 1, determine its velocity after it has moved 6 in. to position 2.

STRATEGY: You are given two positions and want to determine the velocity of the collar. No non-conservative forces are involved, so use the conservation of energy.

MODELING: For your system, choose the collar and the spring. You can treat the collar as a particle.

ANALYSIS: Conservation of Energy. Applying the principle of conservation of energy between positions 1 and 2 gives

$$T_1 + V_{g1} + V_{e1} = T_2 + V_{g2} + V_{e2} \quad (1)$$

You need to determine the kinetic and potential energy at these positions.

Position 1. Potential Energy. The elongation of the linear spring (Fig. 1) is

$$x_1 = 8 \text{ in.} - 4 \text{ in.} = 4 \text{ in.}$$

This gives

$$V_{e1} = \frac{1}{2} k x_1^2 = \frac{1}{2} (3 \text{ lb/in.}) (4 \text{ in.})^2 = 24 \text{ in}\cdot\text{lb} = 2 \text{ ft}\cdot\text{lb}$$

Choosing the datum as shown, you have $V_{g1} = 0$.

Kinetic Energy. Since the velocity at position 1 is zero, $T_1 = 0$.

Position 2. Potential Energy. The elongation of the spring is

$$x_2 = 10 \text{ in.} - 4 \text{ in.} = 6 \text{ in.}$$

so you have

$$V_{e2} = \frac{1}{2} k x_2^2 = \frac{1}{2} (3 \text{ lb/in.}) (6 \text{ in.})^2 = 54 \text{ in}\cdot\text{lb} = 4.5 \text{ ft}\cdot\text{lb}$$

$$V_{g2} = W y_2 = (20 \text{ lb}) (-6 \text{ in.}) = -120 \text{ in}\cdot\text{lb} = -10 \text{ ft}\cdot\text{lb}$$

Kinetic Energy.

$$T_2 = \frac{1}{2} m v_2^2 = \frac{1}{2} \frac{20}{32.2} v_2^2 = 0.311 v_2^2$$

Conservation of Energy. Substituting into Eq. (1) gives

$$\begin{aligned} T_1 + V_{g1} + V_{e1} &= T_2 + V_{g2} + V_{e2} \\ 0 + 0 + 2 \text{ ft}\cdot\text{lb} &= 0.311 v_2^2 + (-10 \text{ ft}\cdot\text{lb}) + (4.5 \text{ ft}\cdot\text{lb}) \\ v_2 &= \pm 4.91 \text{ ft/s} \end{aligned}$$

$$v_2 = 4.91 \text{ ft/s} \downarrow \quad \blacktriangleleft$$

REFLECT and THINK: If you had not included the spring in your system, you would have needed to treat it as an external force; therefore, you would have needed to determine the work. Similarly, if there was friction acting on the collar, you would have needed to use the more general work–energy principle to solve this problem. It turns out that the work done by friction is not very easy to calculate because the normal force depends on the spring force.

Sample Problem 13.9

A 2.5-lb collar is attached to a spring and slides along a smooth circular rod in a vertical plane. The spring has an undeformed length of 4 in. and a spring constant k . The collar is at rest at point C and is given a slight push to the right. Knowing that the maximum velocity of the collar is achieved as it passes through point A, determine (a) the spring constant k , (b) the force exerted by the rod on the collar at point A.

STRATEGY: Since you have two positions and are given information about the speed, use the conservation of energy. To find the force, you need to use Newton's second law.

MODELING: For the conservation of energy portion of the problem, model the collar as a particle and use it and the spring as your system. When using Newton's second law, use the collar as your system.

ANALYSIS: Conservation of Energy. Position 1 is when the collar is at point C, and position 2 is when it is at point A (Fig. 1).

Applying conservation of energy between positions 1 and 2 gives

$$T_1 + V_{g1} + V_{e1} = T_2 + V_{g2} + V_{e2} \quad (1)$$

Position 1. Because the system starts from rest, $T_1 = 0$, and since the spring has an unstretched length of 4 in., you know $V_{e1} = 0$. Putting the datum at A gives.

$$V_{g1} = (2.5 \text{ lb})(7/12 \text{ ft}) = 1.4583 \text{ ft}\cdot\text{lb}$$

Position 2. From geometry, the distance from the pin to A is $\sqrt{(3 \text{ in.})^2 + (7 \text{ in.})^2} = 7.616 \text{ in.}$ Therefore, the elongation of the linear spring (Fig. 1) is $x_2 = 7.616 \text{ in.} - 4 \text{ in.} = 3.616 \text{ in.} = 0.3013 \text{ ft.}$ You know $V_{g2} = 0$ because the datum is at position 2. You also know

$$V_{e2} = \frac{1}{2} k x_2^2 = \frac{1}{2} k (0.3013 \text{ ft})^2 = 0.04539k$$

$$T_2 = \frac{1}{2} m v_2^2 = \frac{1}{2} \left(\frac{2.5 \text{ lb}}{32.2 \text{ ft/s}^2} \right) v_2^2 = 0.03882 v_2^2$$

Substituting these expressions into Eq. (1) gives

$$0 + 1.4585 + 0 = 0.03882 v_2^2 + 0 + 0.04539k \quad (2)$$

You have two unknowns in this equation, so you need another equation. In the problem statement, you are also given that the collar has a maximum velocity at point A. Therefore, the tangential acceleration must be zero at A, and you should use Newton's second law to get additional equations. The system now includes only the collar; the spring applies an external force to the system. A free-body diagram and kinetic diagram for the collar at position 2 are shown in Fig. 2. Applying Newton's second law in the t -direction gives

$$+\uparrow \Sigma F_t = 0 = kx_2 \sin \theta - W \quad \text{or} \quad k(0.3013 \text{ ft})(3/7.652) - 2 \text{ lb} = 0$$

Solving for k ,

$$k = 21.06 \text{ lb/ft} \quad \blacktriangleleft$$

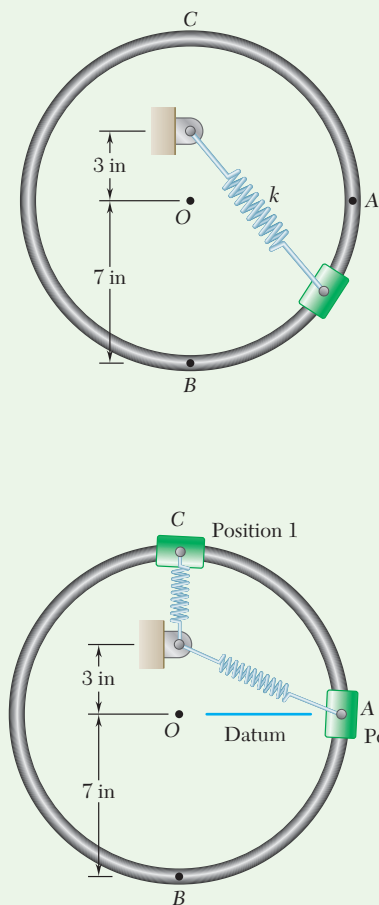


Fig. 1 The system in the two positions of interest.

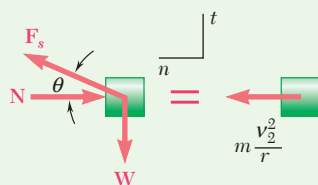


Fig. 2 Free-body diagram and kinetic diagram for the collar at point A.

Force Exerted by the Rod: Substituting this value of k into Eq. (2) gives $v_2 = 3.597$ ft/s. Applying Newton's second law in the n -direction gives

$$\leftarrow \Sigma F_n = m \frac{v_2^2}{r} \quad kx_2 \cos \theta - N = \frac{mv_2^2}{r}$$

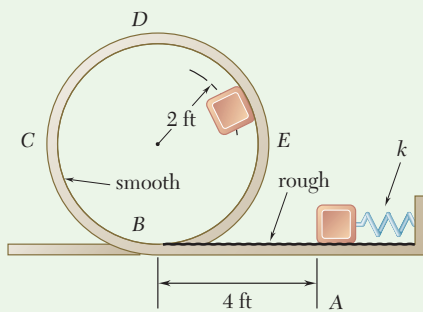
Solving for N and substituting in values provides

$$\begin{aligned} N &= kx_2 \cos \theta - \frac{mv_2^2}{r} \\ &= (21.06 \text{ lb/ft})(0.3013 \text{ ft})(7/7.616) \\ &\quad - \frac{(2.5 \text{ lb}/32.2 \text{ ft/s}^2)(3.597 \text{ ft/s}^2)}{(7/12 \text{ ft})} \end{aligned}$$

$$N = 4.11 \text{ lb} \quad \blacktriangleleft$$

REFLECT and THINK: When the collar is pushed to the right, its speed increases until it reaches point A, and then it begins to decrease. The minimum speed occurs when the collar is at B, since the only forces are in the normal direction; that is, no forces act in the tangential direction. Therefore, the acceleration in the tangential direction is zero, indicating a minimum speed.

Sample Problem 13.10



A 0.5-lb pellet is pushed against the spring at A and released from rest. It moves 4 ft along a rough horizontal surface until it reaches a smooth loop. The coefficient of kinetic friction along the rough horizontal surface is $\mu_k = 0.3$, and the spring is initially compressed 0.25 ft. Determine the minimum spring constant k for which the pellet will travel around BCDE and always remain in contact with the loop.

STRATEGY: You are given two positions and a non-conservative force is present, so use the work-energy principle. Also, for the pellet to remain in contact with the loop, the force N exerted on the pellet by the loop must be equal to or greater than zero. Therefore, you also need to use Newton's second law.

MODELING: Choose the pellet as your system and model it as a particle. A free-body diagram and kinetic diagram for the pellet when it is at point D are shown in Fig. 1.

ANALYSIS:

Newton's Second Law. Applying Newton's second law in the normal direction and setting $N = 0$ gives you

$$\begin{aligned} +\downarrow \Sigma F_n &= ma_n: & W &= ma_n & mg &= ma_n & a_n &= g \\ a_n &= \frac{v_D^2}{r}: & v_D^2 &= ra_n = rg & & & & = (2 \text{ ft})(32.2 \text{ ft/s}^2) = 64.4 \text{ ft}^2/\text{s}^2 \end{aligned}$$

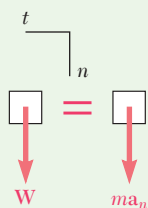


Fig. 1 Free-body diagram and kinetic diagram for the pellet at point D.

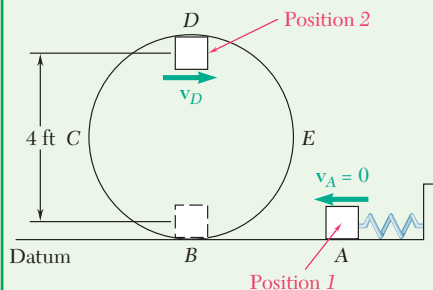


Fig. 2 The system at the positions of interest.

This is the minimum speed of the pellet at D in order for it to remain in contact with the path.

Work–Energy. Choose the system to be the pellet and the spring. Apply the principle of work–energy between positions 1 and 2 (Fig. 2)

$$T_1 + V_{g_1} + V_{e_1} + U_{1 \rightarrow 2}^{\text{NC}} = T_2 + V_{g_2} + V_{e_2} \quad (1)$$

You need to determine the kinetic and potential energy at positions 1 and 2 and the work done by friction.

Position 1. Potential Energy. The elastic potential energy is

$$V_{e_1} = \frac{1}{2}kx^2 = \frac{1}{2}(k)(0.25 \text{ ft})^2 = 0.03125k$$

Choosing the datum at A , you have $V_{g_1} = 0$.

Kinetic Energy. Since the pellet is released from rest, $v_A = 0$ and $T_1 = 0$.

Position 2. Potential Energy. The spring is now undeformed; thus $V_{e_2} = 0$. Since the pellet is 4 ft above the datum, you have

$$V_{g_2} = Wy_2 = (0.5 \text{ lb})(4 \text{ ft}) = 2 \text{ ft} \cdot \text{lb}$$

Kinetic Energy. Using the value of v_D^2 obtained above, you have

$$T_2 = \frac{1}{2}mv_D^2 = \frac{1}{2} \frac{0.5 \text{ lb}}{32.2 \text{ ft/s}^2} (64.4 \text{ ft}^2/\text{s}^2) = 0.5 \text{ ft} \cdot \text{lb}$$

Work. Since the normal force is equal to the weight on a horizontal surface, you find the work that friction does to be

$$U_{1 \rightarrow 2}^{\text{NC}} = -\mu_k Nd = -0.3(0.5 \text{ lb})(4 \text{ ft}) = -0.6 \text{ ft} \cdot \text{lb}$$

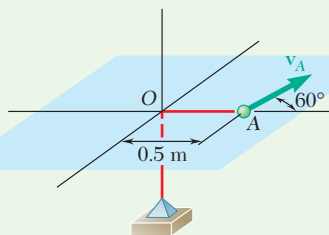
Work–Energy. Substituting these values into Eq. (1) gives you

$$\begin{aligned} T_1 + V_{g_1} + V_{e_1} + U_{1 \rightarrow 2}^{\text{NC}} &= T_2 + V_{g_2} + V_{e_2} \\ 0 + 0 + 0.3125k - 0.6 \text{ ft} \cdot \text{lb} &= 0.5 \text{ ft} \cdot \text{lb} + 2 \text{ ft} \cdot \text{lb} + 0 \end{aligned}$$

You can solve this for k .

$$k = 99.2 \text{ lb/ft} \quad \blacktriangleleft$$

REFLECT and THINK: A common misconception in problems like this is assuming that the speed of the particle is zero at the top of the loop, rather than that the normal force is equal to or greater than zero. If the pellet had a speed of zero at the top, it would clearly fall straight down, which is impossible.



Sample Problem 13.11

A sphere of mass $m = 0.6 \text{ kg}$ is attached to an elastic cord of constant $k = 100 \text{ N/m}$, which is undeformed when the sphere is located at the origin O . The sphere may slide without friction on the horizontal surface and in the position shown its velocity $\mathbf{v}_A$ has a magnitude of 20 m/s . Determine (a) the maximum and minimum distances from the sphere to the origin O , (b) the corresponding values of its speed.

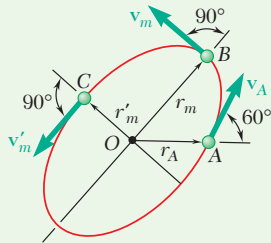


Fig. 1 The particle at locations A, B, and C.

STRATEGY: The force exerted by the cord on the sphere passes through the fixed point O , so use conservation of angular momentum. Also, you are interested in the speed at two locations, and no non-conservative forces act on the sphere. You can therefore use conservation of energy.

MODELING: Choose the sphere, which can be modeled as a particle, as your system.

ANALYSIS:

Conservation of Angular Momentum About O . At point B , where the distance from O is maximum (Fig. 1), the velocity of the sphere is perpendicular to OB and the angular momentum is $r_m m v_m$. A similar property holds at point C , where the distance from O is minimum. Expressing conservation of angular momentum between A and B , you have

$$\begin{aligned} r_A m v_A \sin 60^\circ &= r_m m v_m \\ (0.5 \text{ m})(0.6 \text{ kg})(20 \text{ m/s}) \sin 60^\circ &= r_m (0.6 \text{ kg}) v_m \\ v_m &= \frac{8.66}{r_m} \end{aligned} \quad (1)$$

You have one equation and two unknowns, v_m and r_m . Therefore, you need to use conservation of energy to get a second equation.

Conservation of Energy.

At Point A.

$$\begin{aligned} T_A &= \frac{1}{2} m v_A^2 = \frac{1}{2} (0.6 \text{ kg}) (20 \text{ m/s})^2 = 120 \text{ J} \\ V_A &= \frac{1}{2} k r_A^2 = \frac{1}{2} (100 \text{ N/m}) (0.5 \text{ m})^2 = 12.5 \text{ J} \end{aligned}$$

At Point B.

$$\begin{aligned} T_B &= \frac{1}{2} m v_m^2 = \frac{1}{2} (0.6 \text{ kg}) v_m^2 = 0.3 v_m^2 \\ V_B &= \frac{1}{2} k r_m^2 = \frac{1}{2} (100 \text{ N/m}) r_m^2 = 50 r_m^2 \end{aligned}$$

Apply the principle of conservation of energy between points A and B :

$$\begin{aligned} T_A + V_A &= T_B + V_B \\ 120 + 12.5 &= 0.3 v_m^2 + 50 r_m^2 \end{aligned} \quad (2)$$

a. Maximum and Minimum Values of Distance. Substituting for v_m from Eq. (1) into Eq. (2) and solving for r_m^2 , you obtain

$$r_m^2 = 2.468 \text{ or } 0.1824 \quad r_m = 1.571 \text{ m}, r'_m = 0.427 \text{ m} \quad \blacktriangleleft$$

b. Corresponding Values of Speed. Substituting the values obtained for r_m and r'_m into Eq. (1), you have

$$\begin{aligned} v_m &= \frac{8.66}{1.571} & v_m &= 5.51 \text{ m/s} \quad \blacktriangleleft \\ v'_m &= \frac{8.66}{0.427} & v'_m &= 20.3 \text{ m/s} \quad \blacktriangleleft \end{aligned}$$

REFLECT and THINK: This problem is similar to problems dealing with space mechanics; instead of the gravitational central force acting on an orbiting body, you have the spring force acting on the sphere. It can be shown that the path of the sphere is an ellipse with center O .